Life Science Research Handbook

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About this handbook

This handbook gives advice on scientific reasoning and professional practice in life science research. For humans it might still be unbearable to read, due to slop prose! But maybe it can inspire a more complete handbook in the future.

Chapter 1

Research strategy and project design

Choosing a scientific problem

Choose a question you want to spend time answering. Consider what you could learn and whether you can do the work with the time and support available. Comparing a few possible projects can help you decide [1].

What would you learn?

Describe what you want to find out. “Single-cell analysis of regeneration” names a method and a topic, but leaves the question open. A more specific question would be: “Why does injured tissue recover more slowly in older animals? Are fewer repair cells present, or do those cells divide less?” You can then decide which measurements would help answer it.

Read the literature. What is already established, in which systems and under what conditions? What is being assumed? Which observation remains unexplained?

A gap in the literature can suggest a project, but explain what answering the question would add to our understanding [2]. For example, a treatment that works in cultured cells may fail in intact tissue. Testing it in tissue would establish whether the effect occurs there too. A project can also be valuable without testing an explanation: describing which cells appear during recovery may be a useful first step when little is known about the process.

Your own interest matters too. Notice the questions you keep returning to and the work you enjoy doing. You will spend time on routine tasks as well as discoveries, so consider whether the day-to-day work appeals to you [1].

Compare realistic alternatives

Compare projects you could undertake within the available time. For each one, describe the question and how you would begin answering it. Identify anything you would need to obtain or establish first, such as access to samples or a reliable assay. This makes it easier to compare the work involved as well as the scientific interest.

Alon proposes comparing interest with feasibility. A question that interests you more and is easier to investigate is usually the better choice. The harder decision is between an ambitious question that may take years to answer and a smaller project you could complete sooner. Consider how much time and support you have. If you need to develop a new method first, will enough time remain to investigate the question? [1].

Check the assumptions that could change your choice. If a project depends on a collaborator’s samples, ask whether they can provide them, when they would arrive and whether you can use them for the proposed study. Until you have an agreement, sample access remains uncertain.

Consider the work needed to establish the method. If it is new to you or your laboratory, find out who can help and allow time to learn it. You may need a pilot experiment to check that the assay measures what you need before you can use it to answer the biological question.

In this fictional example, a researcher trained in live-cell imaging has eighteen months of funded work and a persistent interest in how living systems recover after stress. Two projects appeal to them.

Comparison: two recovery problems

A: Why do cultured cells resume growth at different times after nutrient restoration?

The first task is to distinguish cells that resume division from those that remain arrested or die. Explaining these different responses would require follow-up experiments. The laboratory already has the culture and imaging methods. The researcher still needs to check whether they can change the medium repeatedly, follow cell divisions and assess viability in the same experiment.

B: What explains differences in recovery after injury across species?

The researcher would need access to new cultures and a way to compare injury and recovery across species. They do not yet know how long this would take to set up.

Before choosing, the researcher needs to check whether A's imaging assay can distinguish division, arrest and cell loss. For B, they need to establish which cultures are available and how recovery could be compared across species. These checks could show that one project is much harder to carry out than it first appeared.

Time, experience and support

For a first project, it can help to choose a question you can begin investigating with established methods and regular guidance. That gives you time to learn how to do research without also having to develop every method. A less predictable project may be reasonable when you have more experience, longer funding or colleagues who can share the work [1].

When you feel out of your depth, ask what you need to learn. You may be missing background knowledge or a technique that someone can teach you. But you may also have reached a question that nobody knows how to answer yet. Schwartz describes this difference between learning established knowledge and doing research [3]. Reading and discussion can help you distinguish the two; an unknown answer alone is no reason to abandon a question.

Set aside time to read, discuss possible questions and check feasibility before committing. Alon's laboratory gives new researchers three months for this, though he acknowledges that funding and deadlines can limit the time available [1]. Agree with your supervisor how much time you can spend exploring and when you need to choose.

If your broad topic is assigned, ask your supervisor which decisions are still open. You may be able to choose the specific question or the system in which to study it, even when the overall aim is fixed.

Making the choice

Write down why you chose the project and what you learned about its feasibility. Note the main risk and what would make you reconsider. As Alon describes, research may lead to a more interesting question than the one you started with [1].

The fictional researcher contacts the imaging facility and the potential collaborator. The facility can support repeated imaging and medium changes for A. For B, the collaborator can provide two species, but an additional species is not expected for at least six months. The teams have not yet agreed how to measure recovery across species.

Decision record: choose a defined recovery problem

Chosen question: Why do cultured cells resume growth at different times after nutrient restoration?

Reason to pursue it: I want to understand why some cells recover while others remain arrested. I have the imaging experience and local support to begin studying this.

Information checked: The facility can support repeated imaging and medium changes. I still need to test whether the assay can distinguish resumed division, continued arrest and cell loss.

Alternative set aside: The cross-species comparison interests me, but I cannot yet obtain all the cultures or compare their recovery reliably. I judge that setting this up would leave too little of my eighteen months to investigate the question. I will discuss a longer-term project with the collaborator.

Risk accepted: Developing the assay may take longer than expected. I need a way to assess viability, because a cell that does not divide may be arrested or dead.

Condition for reconsideration: If the assay cannot distinguish division, arrest and cell loss, I will need to change the measurement or narrow the question. If cultures and a suitable comparative assay become available for B, I will reconsider that project.

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Developing ideas

An observation you cannot explain can give you the beginning of a research idea. Explore possible explanations and look for a way to test them. The question may change as you learn more.

Yanai and Lercher call exploratory thinking *night science* and systematic testing *day science*, following François Jacob's distinction. You can move between the two as an experiment suggests a new idea or an idea becomes ready to test [3].

Start with an observation

Write down what drew your attention and why it is surprising. Keep the observation separate from the explanation. "The processed images look brighter near the colony edge" reports something you can inspect. "Cells at the edge activate the pathway" adds claims about cells, measurement and mechanism. Those claims may become ideas to investigate, but they should not replace the starting observation.

You can explore data before choosing a hypothesis to test. You still need to understand what the assay measures and how the data were produced. Familiarity with the biology helps you recognize an unexpected observation; familiarity with the method helps you judge whether the procedure could have produced it [7].

Try different views of the data. A distribution can reveal variation hidden by an average; individual trajectories can show whether samples change in the same way over time. Check whether a pattern coincides with how or when samples were prepared and measured. If the pattern appears in processed data, compare them with the raw observations to see whether processing created it. Shoresh and Wong distinguish plots used to explore data from plots used to explain a finding already identified [2]. Choosing a figure or table explains what different displays can reveal.

A specific hypothesis can focus attention so narrowly that other features go unnoticed. But searching many views of a dataset can also turn up patterns by chance. Keep a record of the plots and analysis choices that led to an idea, including any observations you excluded. Identify it as an idea developed from those data: the pattern that suggested it does not also provide an independent test of it [4].

Connections with other fields

An idea from another field may help explain an unfamiliar pattern. If a signal changes with distance from a colony's edge, for example, you might ask whether a substance reaches the center less readily. Transport theory could help you develop that possibility [5].

Check the assumptions before applying a model from another system. In the colony example, ask which substance the model could describe and whether differences in its availability could change the reporter signal. A model might assume that concentrations have reached equilibrium, whereas your experiment observes a changing supply. Read how the model was derived, and ask someone familiar with it to help assess whether its assumptions fit your system.

An informal phrase such as "the colony's center is being starved" can suggest an idea before you know which substance might be scarce. Before designing a test, spell out the proposed explanation: "Perhaps a substance supplied from outside the colony reaches the center at a lower concentration, causing the reporter to be dimmer there." You would then need to find out which substance could do this and how it would affect the reporter. Yanai and Lercher describe how metaphors can help generate ideas that later need more precise wording [6].

Several other explanations remain possible. Lower fluorescence could mean less reporter per cell, different reporter chemistry or fewer cells in the imaged region. It could also arise from how the microscope records light. The nutrient explanation needs to be considered alongside these alternatives.

Discussing unfinished ideas

Discuss the unfinished idea with someone who can help you develop it. A colleague familiar with the assay may notice a measurement problem; someone from another field may suggest an explanation you had not considered. Choose a person or group with whom you can think aloud before you have a defensible answer [9]; [11].

Give a tentative idea time to develop before rejecting it. Ask what would have to be true for it to work, or whether a narrower version could explain the observation. Then examine it critically: does it conflict with known evidence, and what alternatives have you overlooked? Yanai and Lercher recommend this constructive exchange without requiring colleagues to agree that an idea is true [9].

Developing a spatial question

The following fictional conversation concerns processed fluorescence images of bacterial colonies that appear brighter near their edges. No mechanism or quantitative difference has been established.

> **Conversation: from a plausible story to alternatives**
>
> **Researcher:** Perhaps cells near the edge get more nutrients, so the reporter is brighter there.
>
> **Colleague:** Then we should look at how brightness changes with distance from the edge. Do we see the same pattern in different colonies?
>
> **Researcher:** We could look for that. But the reporter might also behave differently where oxygen is more available. We have not checked its chemistry under these conditions.
>
> **Colleague:** And are we sure the pattern follows the colony? If illumination is uneven, its position in the image could matter. Colony thickness might change the recorded signal too.
>
> **Researcher:** Let's first check the raw images. Are the bright regions near colony edges, or do they occur in the same part of the microscope's field of view? If the pattern follows the colonies and remains after checking acquisition and processing, we can investigate nutrient access and reporter chemistry.

After a discussion, note how the question changed, what you still disagree about and who contributed each suggestion. Include alternatives you do not favor, so that your notes do not preserve only your preferred explanation.

Questioning assumptions

When an observation does not fit, examine the observation and the expectation together. An apparent contradiction may reflect a technical problem, a difference in conditions or a hidden assumption about the biology. Apply the same scrutiny to an attractive finding. Gosling and Noordam warn both against dismissing data that disagree with a hypothesis and against mistaking systematic error or ordinary variation for a discovery [1].

State the assumption you need to check. In the colony case, you might have assumed that a brighter region contains more reporter per cell. But a thicker region may contain more cells along the imaging path. Before interpreting brightness as a difference between cells, you would need to account for how many cells contribute to the signal.

When you are stuck, ask whether the facts you already have can explain the observation or whether you need to look elsewhere. You may need to learn about a process outside your field, as in the transport example. Or you may have imposed a restriction without noticing it, such as assuming that brightness must reflect a change inside individual cells. Yanai and Lercher's puzzle framework offers these as different ways to reconsider a problem; you may not know in advance which one will help [10].

Try developing the strongest contrary account of your preferred idea. What would a careful critic say you had overlooked? Which observation would be difficult to explain under your account? Yanai and Lercher suggest imagining a future paper that challenges the project as a way to expose assumptions. Use that exercise to generate alternatives, then assess whether they actually fit the evidence [8].

What to check next

Choose the next step by asking what you need to learn before designing an experiment. If you do not yet understand the reporter, read about its chemistry and check how it behaves under your conditions. If the observation is reliable but several explanations fit, work out what each would predict. You do not need a complete explanation before beginning, but you should be able to say what the next check would tell you.

For the colony example, the discussion produces the following note.

> **Idea note: does the spatial pattern belong to the sample?**
>
> **Starting observation:** Processed fluorescence images appear brighter near colony edges. This is an impression from images, not a validated estimate of reporter abundance per cell.
>
> **Developed idea:** A substance supplied from outside the colony might reach the center at a lower concentration and affect the reporter signal. We could examine how brightness changes with distance from the edge.
>
> **Alternatives to retain:** Reporter chemistry may vary with local conditions; colony thickness may alter the recorded signal; illumination or background subtraction may create a pattern. We have not established which explanation applies.
>
> **Next check:** Inspect raw images, acquisition metadata and processing settings. Establish whether the pattern exists before processing and follows colony position rather than position in the field of view. Check what the reporter can indicate under these conditions.
>
> **Decision after that check:** If acquisition or processing creates the pattern, investigate that cause before interpreting it biologically. If those checks leave a pattern that follows the colony, compare the predictions of the remaining explanations. We still need to determine what the reporter measures before choosing a biological test.

For the next experiment, compare the predictions of the remaining explanations.

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Connecting questions, models and experiments

You may need to describe a pattern before trying to explain it: when does the response appear, how much does it vary, and does it recur? Once you have possible explanations, choose a comparison for which they predict different results. Then check whether your measurement can detect that difference.

Defining the question

The question states what you want to learn. A hypothesis offers a possible answer; a model describes how the proposed explanation works and what it assumes. A prediction states what you would expect to observe if the explanation and its assumptions hold. A statistical test can assess a difference between conditions, but that difference may have several biological explanations. [5]; [4].

Describing a response over time can be worthwhile before you have a mechanism to test. Decide which population you want to describe, what you will measure and how you will sample it. To investigate an unexpected pattern, you also need to understand how the method could produce it. Yanai and Lercher argue for moving between exploration and hypothesis testing as the work develops. Neither must always come first, and both depend on assumptions about the system and its measurements. [7].

Separate the observation from the explanation. Higher reporter fluorescence in dense cultures is an observation; communication between cells is one possible explanation. A question such as “Do these cells communicate?” leaves many choices hidden. Which cells, which response, under which conditions, and what would count as communication? A narrower question might ask whether something in medium conditioned by dense cultures increases a recipient population’s reporter response after a depleted nutrient has been restored.

“The cells warn their neighbors” may help you imagine a mechanism. To test it, ask whether a factor released by one population changes the response of another. You then need conditions that distinguish the factor’s action from other changes in the medium [6].

Keep the biological model and the measurement assumptions distinct. The biological model describes what produces the response. The measurement assumptions explain why your readout represents that response: whether fluorescence reflects promoter activity, whether a tag changes protein behavior, or whether a sample represents the population you intend to study. A precise measurement of the wrong proxy can produce a reproducible result without answering the biological question [2].

Comparing predictions

Write a small set of plausible models, including an explanation that challenges your preferred one. For each, ask what should happen under the proposed intervention and why. A statement such as “expression will change” is rarely enough. Direction, timing, dependence on another factor, or the shape of a response may distinguish models that agree about an endpoint. If their predictions overlap, change the conditions or measure a different feature before treating the experiment as a test between them [8].

A model can start as a verbal mechanism or a sketch. Mathematics becomes useful when it exposes an assumption or yields a prediction that the verbal account leaves vague. For example, two mechanisms may both predict an increase but differ in how the response depends on concentration. Check whether those predicted differences exceed the variation and resolution of the proposed assay. Adding equations does not help if the competing models remain indistinguishable at the conditions you can measure [5].

Explaining existing data is useful work. It can expose missing mechanisms and suggest the next comparison. However, a model adjusted to reproduce a dataset has not thereby passed an independent predictive test. Mark which observations shaped it and identify a prediction to examine with new data or data kept separate from model development. Theory can guide an experiment, and an unexpected observation can prompt a better theory; neither must come first in every project [5]; [7].

Example: a proposed test of conditioned medium

This fictional bacterial strain carries a promoter reporter. At a fixed sampling time, background-corrected fluorescence per viable cell is higher after exposure to cell-free medium conditioned by dense cultures than after fresh medium. A nutrient, N, is measured to be lower in the conditioned medium. These observations establish neither a signal nor its mechanism.

Consider two simple models. In A, N depletion alone causes the increase. In B, an extracellular factor released by the donor cells causes it, N depletion has no effect on its own, and the factor acts independently of N over the tested range. Compare the following conditions in recipient cultures started at the same density. “Baseline” means no increase relative to fresh medium; these are predictions, not results.

Proposed condition	Model A: N depletion	Model B: N-independent factor
Fresh medium	Baseline	Baseline
Untreated conditioned medium	Increased response	Increased response
Fresh medium with N lowered to the measured conditioned-medium level	Increased response	Baseline
Conditioned medium with N restored to the fresh-medium level	Baseline	Increased response

Both models predict an increase in untreated conditioned medium, so that comparison cannot distinguish them. They disagree about the two conditions in which N is changed. These test only the two explanations specified here. If the response increases in both, neither simple model accounts for the full pattern. Multiple causes or an interaction could be involved, but the result alone would not establish that two mechanisms coexist.

Assumptions and controls

Ask what else your intervention changes. Deleting a gene may induce compensation; adding a compound may affect growth, uptake or the reporter itself. A loss-of-function phenotype can support a role for the gene in the response, without showing how it acts. Likewise, reproducing a behavior with selected components shows what they can do together under those conditions. It does not establish that the intact organism uses the same mechanism [2].

In the medium example, restoring N must leave the proposed extracellular factor available and active for the table’s B prediction to hold. If N changes the factor’s stability or uptake, loss of the response after restoration could fit either explanation. Conversely, nutrient depletion can alter recipient physiology and reporter production. Dividing fluorescence by cell count addresses cell number, but does not automatically correct those effects. Keep the fluorescence, viable-cell counts and growth measurements available alongside the ratio.

Choose controls that check a specific possible problem. A negative control can reveal background or an effect of handling; a positive control can show that the assay responds in the medium being tested. Neither shows whether an unidentified factor survived a manipulation. If you cannot check that the factor remains active after restoring N, a loss of response will leave its possible role unresolved.

Designing the experiment

Specify what is independently sampled or assigned to treatment. Many cells from one culture describe that culture more precisely, but do not replace independent cultures when the question concerns variation between cultures. Split samples can provide useful paired comparisons; preserve that pairing in the analysis. State the direct contrast that answers the question. A significant response in one condition and a non-significant response in another do not establish that the responses differ [4].

Keep treatment groups comparable in handling, timing and measurement. Calibrate the assay and consider how a systematic error could survive repeated runs. A second method is useful when it challenges a different possible error; repeating the same proxy with a new label may preserve the same weakness [1]; [4].

Suppose all treated cultures are measured on Monday and all controls on Tuesday. A difference then combines treatment with whatever changed between days. Measuring more cells on each day can make that difference look precise without separating its causes. If both conditions can be measured on each day, the within-day comparison is more informative. Random allocation reduces systematic assignment differences; blinding addresses a different problem, such as scoring borderline cells more generously when you expect a treatment to work. Neither repairs an assay that measures the wrong quantity. [4]; [1].

Experimental plan for the fictional medium comparison

Question: Does N depletion account for the reporter increase, or does activity remain after N restoration?

Manipulations: Prepare the four medium conditions in the prediction table. Verify N concentration after preparation. Match pH, added volume and vehicle, exposure time and recipient starting density.

Measurement: At the specified sampling time, measure background-corrected reporter fluorescence and viable-cell count separately, then calculate fluorescence per viable cell. Track growth to identify physiological differences that complicate the ratio.

Sampling and allocation: Grow independent donor cultures and split each conditioned-medium preparation between untreated and N-restored conditions. Use independent recipient cultures and distribute all four conditions across experimental batches. Preserve donor pairing and recipient identity; extra aliquots are not extra donor cultures.

Assay checks: Include a reporter-free background control and a validated reporter-inducing positive control in each medium matrix. Verify that the recipient population remains viable. These checks address detection and recipient competence; they leave the preservation of an unknown factor unresolved.

Analysis and decision: Estimate N-lowered fresh medium versus fresh medium, N-restored conditioned medium versus untreated conditioned medium, and N-restored conditioned medium versus fresh medium. Retain the paired structure for the conditioned-medium comparison. Choose replication after assessing variation and the smallest response difference that would separate the predictions. Fix the primary comparisons, sampling time and exclusion criteria before the discriminating experiment; record additional analyses as exploratory.

Assumption that could defeat the test: N restoration may interfere with the factor. If that cannot be excluded, an A-like pattern will remain compatible with an N-dependent extracellular mechanism. A B-like pattern would motivate identifying the active component and challenging its action with a different approach; it would not establish direct promoter regulation.

When you cannot check the key assumption. Suppose the factor is still unidentified, so you cannot measure its activity independently after restoring N. You could develop that measurement before running the comparison. Alternatively, you could run the comparison now to find out whether restoring N changes the reporter response. A loss of response would then leave two explanations open: N depletion caused the increase, or restoring N interfered with a factor that caused it. Decide whether learning about the reporter response alone is worth the experiment, or whether you need a different way to distinguish the explanations. Repeating the same ambiguous comparison more precisely will not settle which explanation applies.

Interpreting negative and inconclusive results

A result can challenge your preferred model only if the manipulation worked and the assay could detect the predicted response. Check these points whether the result agrees or disagrees with your expectation. A failed control, a failed manipulation and an imprecise estimate each require a different next step [3].

Possible outcome	What it establishes	Next action
The manipulation and assay checks pass; the estimated response is precise enough to exclude the increase predicted by the model	Evidence against that prediction under the tested conditions	Revise the model or its stated domain; develop a new discriminating prediction
The positive control fails in the experimental matrix	The absent response cannot distinguish biological absence from detection failure	Diagnose the assay or matrix before interpreting the biological comparison
The manipulation did not achieve its intended change	The proposed intervention was not tested	Repair or replace the manipulation
The checks pass, but uncertainty allows both little response and the predicted increase	The data do not resolve the relevant difference	Improve precision or choose a contrast with better separated predictions

A non-significant test alone cannot tell these situations apart. Define the response magnitude that matters for the prediction, then assess the estimate and uncertainty against it. A sufficiently precise result may constrain an effect of that size; it does not prove an exact zero or absence in every condition. Select any equivalence or other absence-testing procedure with its assumptions and decision threshold before using it to interpret the outcome. See showing observations and uncertainty for estimates and intervals [4].

Revising the explanation

If the medium comparison follows B's predictions, something other than simple N depletion is needed to explain the response. The experiment would not identify the factor or show how it affects the reporter. Those questions would still need to be investigated.

An unexpected pattern may suggest a model you had not considered. Record which explanation arose after seeing the results, then test a prediction with observations you did not use to develop it. Check whether a technical problem could explain a contradictory result. If the measurements hold up, reconsider the explanation rather than changing analyses or rejecting observations until they fit it [7]; [8].

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Planning and reviewing a research project

Plan the work in stages. Decide what you can do now and which later steps depend on the results. Estimate the time and resources each stage needs, then revise the plan as you learn.

Scope and milestones

Define what the next stage should establish. “Understand heat stress” is too broad to plan. “Check whether this imaging method can measure division timing after a heat pulse” gives the first stage a specific purpose. The method may work even if the heat pulse does not change division timing.

Work out what you can do with the time, equipment and help available. Check sample access and any commitments already made to collaborators. If a stage depends on an instrument booking, confirm the booking before scheduling work that requires it. Do the same for other resources you are counting on [1].

Agree on scope with the people who will do and support the work. Include what this stage will leave unanswered. A first experiment may establish a response without identifying its mechanism; a technical pilot may establish whether a response can be measured at all. These are different milestones. For the scientific design itself, use connecting questions, models and experiments.

Feasibility checks

Check early for problems that would prevent the later work. Can you obtain the essential material? Can the assay distinguish the outcomes you want to compare? If these checks take little time or expense, do them before committing to the larger experiment [2].

Define what the pilot needs to establish. For an imaging experiment, you might check whether cells remain visible long enough to measure division and whether the control cells behave as expected. Decide what would justify proceeding and which failures would require changes to the method. A small pilot may establish that the assay works while providing too few independent observations to estimate the biological effect precisely.

Not every important uncertainty has a cheap test. A larger commitment may still be worthwhile, but consider what it would teach and which other work it would displace.

An unexpected observation may suggest a better question. Yanai and Lercher distinguish this evolution of a research programme from the design of each study within it: the programme can change direction while its experiments retain clear comparisons. Before following a new direction, decide which existing work to stop or postpone and whether the change affects commitments to collaborators. Keep exploratory work distinguishable from tests of predictions specified beforehand. [6].

Scheduling the work

Break the near-term work into tasks, with a responsible person and a clear endpoint for each. Tasks taking one or two weeks can be a useful starting scale; unfamiliar procedures may need smaller steps. Plan distant stages more broadly and add detail as you learn what they will require [1].

Identify which tasks must wait for others. You can sometimes check an analysis script on existing data while preparing new samples. But two experiments that could run independently may still need the same microscope or the same person’s time. Check these conflicts before scheduling them together.

Estimate elapsed time as well as hands-on work. Culture growth, deliveries, training, access queues and feedback can determine the completion date even when they require little active work. Use records of comparable tasks to improve estimates, and name uncertainty where a method is new.

Include data inspection, documentation and interpretation; a schedule that ends at acquisition has left part of the research unplanned [1].

A short table or shared document may be enough. For a larger project, a timeline can help you see which tasks overlap and where a delay would hold up other work.

Example: plan a feasibility stage before expanding

In this fictional project, Mira asks whether a short heat pulse changes the time to first division in yeast after return to standard temperature, compared with mock-handled cultures. Six weeks are available for the initial project stage. The imaging method has not yet been shown to retain identifiable cells through the observation period. These are local planning constraints, not recommended durations.

First two weeks — establish feasibility. Before imaging, Mira and her supervisor will agree on the observation window, the evidence needed to trust track identity, and the amount and pattern of cell loss that would prevent a timing comparison. Mira will use two confirmed pilot sessions to check handling, retention and event detection in both heat-pulsed and mock cultures, including whether mock cells can divide. She will inspect the raw movies alongside the tracks and record cells lost from view. The milestone is a documented judgment about whether the assay can answer the timing question, including unresolved limitations.

Support and preparation. Leo has agreed to one training session before imaging and a technical review afterward. Mira can prepare the tracking workflow on existing practice movies while waiting for the first session. The practice movies can reveal software problems; they cannot establish that the new experimental setup works.

Decision at the end of week two. Mira and her supervisor will review feasibility before committing the remaining four weeks. If tracking and handling are adequate, they will settle the biological sampling and analysis plan, confirm further instrument access and begin the comparison. If a specific technical problem remains, they will assess a bounded repair. If the method cannot answer the question within the available resources, they will reconsider the approach rather than collect a larger uninterpretable dataset.

Work held back. No additional conditions or mechanistic assays will be added during the pilot. Later imaging access is provisional, so the six-week allocation is not a promise of a completed biological answer.

Reviewing the plan

For an extended project, start with a monthly review, as Gosling and Noordam suggest. Compare progress with the previous plan, discuss why work took a different course and agree on the next goals [3]. Use a shorter interval when a decision cannot wait until the next month.

Review sooner if an essential control fails, access is lost or new evidence changes the question. Routine troubleshooting can usually stay within the current plan.

Revise the plan with the people affected. Record what will change and what will be postponed. If the remaining work no longer fits the time available, agree on a smaller scope, a later deadline or additional help. When a deadline slips repeatedly, address the cause of the delay before setting another date [1].

What have you learned?

Begin with the current question and what you agreed to do at the previous review. Explain any change of direction. Show what you learned, including control results and technical problems that limit the conclusions. Mention changes in access or support that affect what you can do next.

Explain what the work establishes, with a link to the experiment or analysis record. “We completed four imaging runs” says how much was done. “We can follow cells through the required period in one condition but not the other” explains why the planned comparison is not yet possible.

Compare this account with the previous plan. Was an estimate unrealistic? Did access disappear? Did the experiment answer a different question than expected? Or did useful new evidence change the aim? A regular comparison of planned and actual work can reveal recurring obstacles that a list of achievements conceals [3].

Understanding setbacks

A result that disagrees with the preferred explanation is not necessarily a failed experiment. If the relevant controls, measurement and comparison are sound, that disagreement can constrain the explanation. If the control fails or the measurement cannot distinguish the possibilities, the experiment may leave the question unresolved. KamounLab makes this distinction central to judging progress [4].

Even a technically successful experiment can be inconclusive. An uncertain estimate may allow both a meaningful effect and little or no effect. A negative finding may apply only to the tested dose, time, system or detection limit. Before treating it as evidence against a model, ask what that model actually predicted under those conditions. The fuller reasoning belongs in connecting questions, models and experiments and showing observations and uncertainty.

If work has stopped because samples, equipment or help are unavailable, discuss whether and when access can be restored. The scientific question may still be worth answering, but you may need another way to study it or a different project while you wait [1].

When discussing a setback, ask what prevented progress and what help would make a difference. If a control is missing, agree on how to establish it. If equipment is unavailable, resolve access or change the plan. Assess the work without treating the setback as a judgment of the researcher [3].

Deciding what to do next

Decide whether the next period of work is worth its time and cost. Having already spent months on a project is not sufficient reason to continue. Raj recommends setting dates to reconsider the project and seeking outside advice when there always seems to be “one more” experiment to do [5].

Before adding an experiment, ask what it could resolve and how the answer would affect the plan. If every possible outcome leads to the same next step, the experiment may not be needed for that decision. It could still be necessary to document a method or complete an agreed obligation; be clear about that purpose.

Compare credible alternatives. Continuing may be justified when the method is sound and the remaining question is valuable. A bounded repair may be preferable when one identifiable technical problem blocks an otherwise useful comparison. Changing direction can make sense when the evidence points to a more informative question. Pausing may be appropriate when a necessary resource cannot be obtained soon enough. Closure can be reasonable when the remaining contribution no longer warrants its cost [1].

If another group publishes on the same question, read what their study establishes before deciding to stop. It may answer your question, leave a limitation you could investigate or make independent replication more valuable. Assess a new project just as carefully before replacing the current one.

When seeking advice, explain the current plan and the strongest alternative. Say what each could establish and what work it would take. Choose someone who knows enough of the science to challenge your assumptions, and explain the constraints they may not know about [5].

A constrained choice

The next step can remain debatable even after everyone agrees on the evidence. In the example below, repairing the assay preserves the original question but consumes time that could support a different project. The decision depends on how much that question matters and whether the proposed repair is credible.

Example: repair the measurement before expanding the study

This fictional review follows a two-week microscopy pilot within a six-week project allocation. Mira asks whether a short heat pulse changes time to first division in yeast after return to standard temperature, compared with mock-handled cultures. The pilot was intended to establish feasibility. The following outcomes are introduced for this review exercise; they are not real experimental findings.

Evidence reviewed. Temperature handling met the pilot's checks. Mock-handled cells divide while visible, but cells frequently leave the field in both conditions, preventing reliable assignment of first-division times. The raw movies show the losses. The current records do not establish whether loss is unrelated to division timing or treatment.

Scientific conclusion. The heat-pulse question remains unanswered. A cell lost from view cannot be counted as a cell that did not divide. Comparing only the cells that stayed would require assumptions the pilot has not justified.

Options. Expanding the current assay would repeat the tracking problem. A specialist has identified a possible mounting change that can be tested in one further week; its effects on growth and observation still need checking. Measuring bulk growth is another possible project, but it would answer a different question from the timing of individual first divisions.

Decision. Suspend the biological expansion and spend one week on the mounting test. Mira will run it with the imaging specialist in one confirmed core session. Before the test, they will agree on the observation window and the criteria for trustworthy tracks and acceptable cell loss. They will inspect retention and event detection in both conditions, check whether mock-handled cells can divide under the revised mounting conditions, and assess whether losses could select cells by treatment or division timing. Retaining cells is not sufficient if the mounting prevents normal division or remaining losses still make the timing comparison unjustifiable.

Next review. Mira and her supervisor will review the records at the end of that week. If the revised assay supports an interpretable comparison, they will replan the remaining three weeks and confirm access before expanding. If tracking remains inadequate, they will pause this imaging route and assess a different approach. The review does not assume the repair will work or promise a completed biological result within the original allocation.

The one-week repair is a defensible choice because it targets an identified problem with confirmed help. It is not the only defensible choice. Mira could instead decide that learning about bulk growth is worth more than preserving the single-cell question within this short allocation. That would be a change of question, not an alternative measurement of the same outcome. If the specialist or core session becomes unavailable, the case for the repair weakens. A failed repair would not by itself show that choosing it was unreasonable with the information available.

For a pause or closure, record what the work established, where the data and materials can be found, and what would make reopening worthwhile. Agree on who will complete outstanding commitments. If the project continues, update its plan to reflect the decision. When the same obstacles affect several projects, consider reviewing how the group works.

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Chapter 2

Reading the literature

Keeping up with the literature

Choose a few ways to find papers on your current questions and wider interests. If screening new papers takes all your reading time, reduce the number of alerts or narrow their searches.

Choosing what to follow

Write down the question, method or research area you currently need to follow. “Single-cell biology” is a broad interest. A more focused question is how to distinguish a change in the abundance of a cell type from a change in gene expression within that type. You can search for methods that make this distinction while still reading more widely about single-cell biology.

How closely you read depends on what you need from the paper. In a new field, an article may help you learn the vocabulary and find earlier work. If you plan to adopt its method, examine the procedure, the assumptions it relies on and the evidence that it works under conditions like yours [1].

Alerts help you notice new work. Before designing an experiment or making a claim in a manuscript, also search for earlier studies that address the question. Reviews can help you find them; read the original studies when you need to assess their methods or rely on their findings [2].

Finding relevant papers

A topic search can find papers across journals, but it depends on the terms you know to search for. Following a few journals or research groups may reveal work you would otherwise miss. Suggestions from colleagues and seminars can also take you beyond your usual searches [3].

Start with a query that describes the biological question or method, then inspect what it returns. Try alternative terms used by authors, including names for the system, intervention or measurement. Check whether it finds a few papers you already know are relevant. If an important paper is absent, ask why: missing terminology, a narrow condition in the query or a database that does not cover the material. An experienced colleague or research librarian can help with terminology and search strategy [2].

Use alerts or feeds for searches and publications you want to follow, or revisit their pages regularly. Check what each subscription adds: perhaps one finds new applications of your assay while another follows a neighboring field. When several alerts return the same paper, keep one record.

When you find an important paper elsewhere, check why your searches missed it. A different term or an additional source may help. Even several well-chosen searches can miss relevant work; adding subscriptions is useful only if they help fill a gap you have identified.

Choosing what to read

Use a paper’s title and abstract to decide whether it addresses a question that interests you. Look at a relevant figure or method passage if the connection is unclear. Assessing its conclusions will take closer reading.

Decide what to do with the paper:

- **Read for a current question.** Name what you need to learn. A result that could change your experimental plan, a method you may use or a paper you must explain to others warrants a focused reading slot.
- **Keep for a named later use.** Save the reason to return: background for an introduction, a method for a possible extension or a study to revisit when a particular result appears. Keep these records separate from the active reading queue.
- **Leave it.** If you cannot name a reason to return to it, you need not save it. This is a decision about your reading priorities, not a judgment of the paper’s quality.

Reading a selected paper may reveal a method or concept you need to learn first. Allow time for that background reading, and keep your note labeled as a screening note until you have examined the evidence [1]. Understanding a research paper explains how to follow the connection between a paper's experiments and its conclusions.

Saving papers and notes

Keep papers and their notes together in a searchable reference manager or document collection. Include the authors, title, year and a stable link or identifier so you can reopen the source [2].

For a screened paper, note why you saved it and what you have read. After closer reading, add the finding you expect to use and where its supporting evidence appears. Keep your interpretation and unanswered questions distinct from what the authors report. If you highlight a passage, add a note when you might otherwise forget why it mattered.

Use labels you are likely to remember when the paper becomes relevant again: a project question, assay, organism or unresolved problem. Before relying on the record in a new plan or manuscript, reopen the source and check the relevant passage. Check which version you retained and whether a later version or correction changes the point. When several records bear on the same question, move from individual notes to comparing and synthesizing papers.

Making time to read

Reserve recurring time for discovery and selected reading. Weekly screening is a useful starting point, with frequency adjusted to the pace of the field and the decision ahead. Ulitsky emphasizes regular screening and warns that too many subscriptions can make researchers abandon the routine altogether [3].

If your reading queue keeps growing, check what is filling it. Revise searches that mostly return irrelevant papers and remove overlapping subscriptions. If the papers are relevant but you cannot read them all, choose which questions need attention now or set aside more reading time. During a busy experimental period, you may need to pause broad alerts while following work needed for an imminent decision. Return to wider interests when time allows.

Example: a routine for a method decision

In this fictional example, a researcher is deciding whether to pilot a new cell-segmentation method on crowded fluorescence images. During assay setup, they can give literature work 90 minutes each week. They initially reserve 20 minutes for screening and 70 for selected reading and notes; these times describe their available week, not a recommended reading rate.

Discovery: a focused method-topic search and a broader imaging-journal stream, with occasional suggestions from colleagues.

Current reading: paper A describes a method intended for crowded fluorescence images. Its abstract suggests a possible fit. The researcher assigns it the reading question "Does the evaluation include image conditions similar to ours, and how were errors assessed?"

Orientation: review B explains segmentation-evaluation metrics. The researcher keeps it for the pilot plan, beginning with the section needed to understand paper A rather than treating the whole review as an immediate assignment.

Later interest: paper C concerns a different tissue. It goes into a background collection to revisit if the project expands to that tissue, outside the active method-reading queue.

No further reading: paper D addresses microscope calibration. It is related to imaging, but the researcher has no current calibration question and the paper does not address segmentation. They leave it outside the reading collection.

> The initial note attached to A reads:
>
> **Purpose:** decide whether this method merits a pilot on our images.
>
> **Reading status:** abstract and one figure inspected; methods and evaluation not yet assessed.
>
> **Next check:** read the evaluation Methods and the crowded-image example in Figure 3. Identify the reference annotation, image conditions and types of error counted. The abstract's claimed relevance is still unverified.
>
> The incoming alerts double during a busy assay week. The researcher removes duplicate notices and pauses the broad journal stream. They keep the method-topic search and use the reading slot to continue A, postponing optional background reading. Before committing to the pilot, they also search for earlier evaluations of the method. If understanding A requires more background reading than fits this week, they postpone the decision.

Review the routine after using it for a few weeks. Can you find a saved paper when you need it? Which alerts led to useful reading, and which mostly added notices? Adjust the searches and reading time accordingly.

References

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Understanding a research paper

You may understand every word in a paper yet be unsure why an experiment supports its conclusion. Work through that connection using the figures, captions and relevant methods. Aim to explain what the study found and which questions remain open.

What do you need from the paper?

Decide what you need from this paper: orientation in an unfamiliar field, evidence for a particular claim, a method you might adopt or preparation for a discussion. That purpose changes where to spend time. For orientation, the Introduction can identify the question and useful earlier work. To adopt a method, concentrate on the procedure, its validation and conditions of use. To rely on a biological conclusion, follow the experiments supporting it, including controls and supplementary material [1].

Identify the article type as well. A review can orient you and lead to original studies; its account of an experiment is not a substitute for reading that experiment when your conclusion depends on it. A methods paper needs evidence about performance and limits. An original research article presents observations and an argument about what they mean. The workflow below concentrates on that argument.

Begin with enough of the abstract and Introduction to state the question and proposed answer. Then move between Results, figures and methods. Reading the figures first can work when you know the system; reading each Results passage alongside its figure can work when you need more context. Neither order is compulsory. Look up an unfamiliar term or follow a cited method when it blocks your understanding of a consequential step, rather than allowing every unfamiliar word to start a separate literature search [1].

For an overview, it may be enough to understand the question, approach and main finding. Before adopting a method, check how its performance was assessed and whether that assessment covers the samples you plan to use. Before citing a proposed mechanism as established, read the experiments that distinguish it from other explanations. Note anything you still need to check.

Following the argument

State the research question in your own words, then identify the comparison intended to answer it. What changed between the groups or conditions? What was measured? Why would that measurement help answer the question?

Follow the main claim through its supporting experiments. One may establish that the measurement works, another show a response and another test an alternative explanation. The order of Results usually expresses this logic rather than the order in which the work happened. Ask what each experiment adds and which later conclusion depends on it [3].

Write an observation before accepting the interpretation: *reporter fluorescence was lower* describes a measurement; *the treatment suppresses a specific pathway* proposes why it changed. Results headings and figure titles can contain interpretations too. Check whether the panels support them [1].

Example: what a reporter measurement establishes

In this fictional paper, cultured cells receive inhibitor Q or vehicle. At six hours, Figure 1A shows lower background-corrected whole-well reporter fluorescence with Q. The authors propose that Q suppresses the reporter through a specific molecular target. The material considered so far does not identify viable-cell number or fluorescence per cell.

Part of the argument	What the reader can establish
Question	Does Q suppress reporter activity through the proposed target?
Observation	Whole-well fluorescence is lower with Q than with vehicle at six hours.
Unresolved inference	Total signal depends on contributing cells and their signal. This measurement alone does not distinguish fewer cells from lower signal per cell, or establish target specificity.

To assess the proposed mechanism, look for measurements of cell number and signal per cell, then for experiments testing whether Q acts through the proposed target. The whole-well measurement alone cannot answer those questions.

Figures and methods

For each panel supporting the conclusion, establish what its points or other marks represent. A point may be a cell, a culture, an animal or an average across several measurements. Read the axes, units, group labels and legend. Check whether values are totals, proportions or normalized measurements. For a fold change, identify the baseline used to calculate it [1].

Use the caption to find the relevant method. Follow how samples were selected and measured, then how those measurements became the plotted values. For a normalized result, find the denominator; for an analysis that excludes observations, find the exclusion criteria. These details may be in supplementary methods.

In the reporter example, background correction does not account for how many viable cells contribute to the signal. The measurement also tells you only what happened at six hours, not when the response began or how long it lasts.

Ask whether the display supplies the evidence claimed. A representative image can show a phenotype, but to assess how common it is, you need to know how samples were selected and how the phenotype was counted. A smooth curve may be a fitted model rather than additional observations. For an unfamiliar figure type, consult specialized displays and explanatory diagrams, then return to the paper's caption and methods to check how it was constructed.

If a necessary detail remains missing, name it specifically. "The caption does not identify whether points are cultures or repeated measurements from one culture" is a question you can resolve. "The experiment is unreliable" goes beyond what that absence establishes.

Assessing the evidence

Examine controls by the alternative explanation they address. In a before-and-after experiment, a change may reflect elapsed time or repeated measurement as well as the intervention. A suitable concurrent control helps separate these effects. In the reporter example, vehicle supplies a treatment comparison, but it does not by itself settle cell number, assay interference or target specificity. Read what the controls test and what they leave open [2].

Also identify the independent units behind the comparison. Repeated images from one culture can describe that culture more thoroughly; they do not create additional independently treated cultures. Check how the analysis handles this dependence and whether the replication supports the scope of the claim. If the paper says the treatment works differently in two conditions, look for a direct comparison of those effects. A significant result in one condition and a non-significant result in the other does not establish that they differ [2].

Read the estimated effect together with its uncertainty. Identify what error bars represent: standard deviation describes spread among observations; standard error or a confidence interval concerns precision of an estimate. To compare groups, look for an analysis of their difference rather than judging whether their separate error bars overlap [4]. See showing observations and uncertainty for an explanation of these intervals.

A non-significant result alone does not establish that an effect is absent. Check the range of effects compatible with the estimate and its uncertainty interval. If that range includes both no change and the change predicted by a proposed mechanism, the measurement does not resolve the prediction. A narrower interval may rule out an effect as large as predicted without establishing an exact zero. Base that judgment on the analysis the paper reports [2].

Distinguish an association, the effect of an intervention and evidence for its mechanism. Observational covariation can have several explanations. Manipulating a treatment strengthens some inferences, but the treatment may act through routes other than the proposed one. Ask which experiment distinguishes those routes before describing the mechanism as established [2].

Judge a descriptive study by the description it supports. An atlas need not identify a mechanism to be useful, but its coverage depends on which specimens, conditions and measurements it includes. Ask whether the sampling supports the claimed population and whether the method could miss the feature being described. A missing mechanism and an inadequate measurement are different limitations. [1].

Apply the same scrutiny to findings you agree with and those you doubt. Would you ask for the same control if the result supported your expectations? Distinguish a missing explanation of a method from evidence that the method was used incorrectly. Likewise, a study confined to one cell model may answer its stated question while leaving open whether the finding applies elsewhere [1].

Taking useful notes

Close the paper and explain the question, main evidence and conclusion in a few sentences, including any important limits. Then reopen it to check your account and add the relevant figure, table or method locations. If you cannot explain a step, reread it or discuss it with someone who understands the method [1].

Keep the paper's citation or stable link with the note. Include the experimental system, treatment and measurement time when these define the finding you plan to use. In the reporter example, the note needs to say that the measurement is whole-well fluorescence in cultured cells at six hours. Separate what the authors conclude from what you have checked yourself.

Example: a short reading note

> The paper asks whether inhibitor Q suppresses reporter activity through its proposed target. In cultured cells, Q produces lower background-corrected whole-well fluorescence than vehicle at six hours (Fig. 1A). The authors interpret it as target-specific suppression, but I have not yet found viable-cell counts or per-cell signal measurements. Replication and estimate precision also need checking in the caption and Methods. Before using the paper as evidence for the mechanism, I need to look for cell-normalized measurements and experiments testing target specificity.

Suppose you then find an additional panel reporting fewer viable cells with Q. Update the note:

> The additional viable-cell-count panel shows fewer viable cells with Q. Cell number may therefore contribute to the lower whole-well fluorescence. The findings still do not establish whether signal per viable cell changes or whether the proposed target mediates the response. I will look for a per-cell measurement, check how it was obtained and examine whether specificity experiments distinguish the proposed mechanism from other effects of Q.

Fewer viable cells alone do not establish how much of the fluorescence difference is explained by cell number.

If you are preparing a journal review, see writing the report for organizing your comments.

References

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Comparing and synthesizing papers

Before deciding whether papers agree, check what each study measured and under which conditions. An increase in RNA and no detectable increase in protein may both be accurate observations. To explain what the studies show together, you need to keep that difference in view.

Defining the review question

State the question you want the papers to answer. “Does this pathway matter?” could mean a role in development, a molecular response or survival after treatment. “Does inhibiting the pathway reduce cell survival in this model?” is specific enough to guide paper selection. A developmental study may suggest a mechanism to investigate, but it does not directly answer the survival question.

Decide which systems, interventions and outcomes belong in the comparison before selecting papers by their conclusions. Record how you searched, when you searched and why you excluded papers; note papers you could not obtain. A reader should be able to tell whether your account covers a few selected studies or a search intended to find all the evidence on the question [3].

A review can orient you to the concepts and point toward original work. For a conclusion that depends on a particular experiment, inspect that experiment’s paper and relevant supplementary material. Keep a review’s account separate from your assessment of the primary evidence. Before describing an unresolved question, check both recent work and older studies that the current discussion may have overlooked [3].

Comparing methods and evidence

Compare the studies side by side, using the same fields for each. Record where you found each detail so you can return to the figure, table or Methods passage when a difference needs checking.

Record	What to establish
Question and system	Which biological question, organism, cell type, population or state does the study address?
Intervention and comparator	What changes, at what dose or intensity, and relative to what control? Is the study observational or experimental?
Timing	When does the intervention occur, and when is the outcome measured?
Measurement	What quantity is measured, with what assay, normalization and units?
Replication	What are the independent biological units? Which measurements repeat within them?
Finding and uncertainty	What is the effect estimate or observed pattern? How precise is it, and what analysis supports it?
Inference and limits	What conclusion follows, and what alternative explanation or missing control remains?

Do not fill an absent detail with an assumption. Mark it as unreported or not yet checked and return to the source if it affects comparability. These fields extend the distinction between question, approach, observation and interpretation used by Carey and colleagues [1].

Check what each study means by "pathway inhibition." A drug may only partly inhibit its target and may affect other targets too. Deleting a gene produces a different perturbation. Check how long each treatment acts before treating the experiments as equivalent.

Also check how the measurements were normalized. Total fluorescence per well depends on how many cells are present as well as their fluorescence. A measurement divided by the number of viable cells accounts for cell number differently; the two values cannot be compared directly. For fold changes, identify the baseline in each study before comparing the reported effects.

Evaluate the evidence for the particular claim you need. A strong observational association may establish a relationship while leaving its cause unresolved. An intervention can narrow causal explanations, but its controls and possible additional effects still matter. Likewise, many measurements from a few cultures do not provide the same biological replication as independently initiated cultures [2].

Check whether papers reuse a cohort, dataset or experimental series. Analyses of the same observations may answer different questions, but they do not show that a finding recurs in a new sample. Record which papers share data. When studies address different questions or measurements, compare them in separate groups and explain how the groups relate.

Repeating an experiment in independent samples tests whether the finding recurs. Using a different method may help determine whether it depends on a weakness of the first assay. Check what changed in each additional study: repeating the same assay can repeat its bias, and a second method needs its own validation.

Explaining differences between studies

First ask whether the studies disagree about the same quantity under comparable conditions. An increase in mRNA and an unchanged protein estimate are distinct observations. They can both be accurate; neither alone establishes how RNA changes lead to protein changes. Similarly, a short perturbation and a long depletion can reveal different parts of a response.

Compare the estimated effects and their uncertainty, not just their significance labels. A statistically significant result in one study and a non-significant result in another do not establish that their effects differ. The second estimate may be too imprecise to distinguish no effect from an effect as large as the first study reports. To claim that the effects differ, you need an analysis of that difference; judging overlap or nonoverlap of their separate confidence intervals is not a general substitute [2]. See showing observations and uncertainty for the distinction between separate estimates and an estimate of their difference.

When studies of the same outcome still give different findings, examine how their methods differ. Start with differences that could affect the measurement or the biological response. For example, a longer exposure might allow a response to develop. That explanation predicts a change over time that could be tested. Finding a difference in exposure time does not by itself establish why the results differ.

If several explanations remain possible, say which ones and what evidence would distinguish them. Apply the same scrutiny to findings you favor and those you doubt. An unexplained disagreement belongs in the synthesis even when you cannot yet resolve it [4].

Example: compare the evidence behind different conclusions

The following three study summaries are fictional. They ask whether ligand L changes marker M in cultured epithelial cells. All use the same stated cell model, ligand dose, exposure conditions, vehicle comparator and two-hour endpoint. Independently initiated cultures are the biological replication units. The values are authored study reports, not results calculated from a dataset. Fold changes compare L with vehicle; CI denotes the reported confidence interval.

Study	Outcome	Fold change	95% CI
A	M mRNA	1.4	1.2–1.6
B	M protein abundance	1.0	0.9–1.1
C	M mRNA	1.3	0.9–1.7

A fold change of 1 means equal abundance. Study A supports increased mRNA. Study C's estimate is in the same direction but is less precise, allowing no change as well as an increase. Its interval crossing 1 does not show that it contradicts A. No direct comparison of the study effects is supplied.

B measures a different outcome. Its protein estimate is near 1 with the stated interval; this does not refute A's mRNA finding or prove that protein abundance cannot respond. A delay between the RNA and protein responses is one possible explanation, but these reports contain no later measurement that tests it.

Suppose you need to choose a follow-up now. If you need to know whether the RNA increase recurs, repeat the two-hour RNA measurement. If you need to know whether protein abundance rises later, measure it at later times. The reports alone do not tell you which question to prioritize, and they leave the delay explanation untested.

Writing the synthesis

Organize each paragraph around a question that the studies help answer. You might first explain whether the outcome changes, then discuss the evidence for a mechanism or for a response in other systems. Within each paragraph, show how the findings relate: one study may establish an observation while another tests whether it occurs under different conditions [3].

State what the studies establish together and explain which findings support that answer. Keep the relevant conditions and uncertainty beside the conclusion. "These results are consistent with a delayed protein response" allows other explanations; "the protein response is delayed" asserts that this explanation has been established. If the studies still conflict, identify what remains unresolved.

For the fictional studies, the completed synthesis could read:

Synthesis of studies A–C

Ligand L increased marker M mRNA after two hours in the epithelial-cell model examined by study A. Study C estimated an increase of similar magnitude, although its uncertainty interval also allowed no change. The mRNA findings therefore do not establish a disagreement between these studies. Study B measured protein abundance and reported an estimate near no change at the same endpoint. Together, the studies support an mRNA response at two hours more clearly than a protein response at that endpoint. Whether the mRNA change produces a later protein increase remains unresolved; measurements of both outcomes over time would help distinguish that possibility from a response confined to mRNA abundance.

Check each conclusion against the original studies. In the example, the measurements concern a particular marker in one cell model at two hours; they do not establish a general effect on gene expression or a response at later times. Ask a colleague to check your interpretation and identify missing details that could change it [3].

A synthesis for a whole field needs a search and appraisal design suited to that purpose. Quantitative pooling also requires comparable quantities and an analysis that accounts for their uncertainty and dependence.

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Chapter 3

Data visualization

Choosing a figure or table

Choose a display for the comparison you want to make. Try alternatives while exploring the data: a pattern hidden by a group average may become visible when you plot the individual observations. [4].

Decide what readers should notice first, then choose the layers that make it visible. A group comparison may need both a readily visible median and the observations around it; a time course may need only a few clearly labeled solid lines. More colors, symbols and layers are useful only when they make a needed distinction easier to read. The glossary below names the available forms; it is not a list of mutually exclusive choices.

Design the comparison

Start with a question a reader should be able to answer from the figure: which treatment has the larger response, how consistently a response occurs across experiments, or where a protein lies within a cell. These questions need different evidence and different arrangements. A visually attractive figure can still make the intended comparison difficult. Wong describes design as solving a communication problem under known objectives and constraints, rather than adding decoration at the end. [14].

Identify what each observation represents and which variables must remain visible. For a treatment applied to cultures, cells within a culture and independent cultures are different levels of observation. For a time course, the elapsed time and the order of measurements both matter. Put the main comparison on a common positional scale where possible; use labels, grouping and color to carry the remaining distinctions. The chapter on observations and uncertainty develops the choice of observational unit. [2]; [3].

Compare a few plausible arrangements before refining one. An overlaid time course makes treatment differences at each time easy to compare; aligned small multiples make each trajectory easier to follow when many curves cross. Neither is universally better. Sketches or quick plots can expose this tradeoff before you spend time adjusting fonts and colors. Keep the data and the intended reading size the same while comparing designs. [14]; [1].

Then establish a visual hierarchy. Give the main comparison enough space, make its labels easy to find, and keep guides and secondary annotations quieter than the data. A summary can deserve more emphasis than the surrounding observations when readers need to compare centers; the observations still need to remain available for assessing variation. Inspect the result at its intended size and ask someone to make the comparison without your explanation. Their answer tests whether the arrangement works. [6].

Duarte offers a useful way to separate the visual roles: background, data and emphasis. Background guides orient the reader; data remain readable; emphasis directs attention to the comparison. A small contrast change can be enough, and changing the emphasis need not remove the surrounding evidence. The book's presentation examples are practitioner guidance, with an explicit warning against oversimplifying substantive data for scientific audiences. A standalone scientific figure also needs context that a narrated slide might supply in speech. [15].

Visual glossary

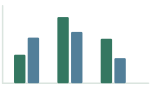
Amounts



Bars



Dot plot

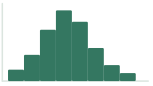


Grouped bars

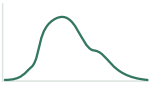
Distributions



Individual points



Histogram



Density

Distribution summaries



Boxplot



Violin



Cumulative distribution (ECDF)

Matching



Connected points

Relationships



Scatterplot



Hexbin



Bubble plot

Time or dose

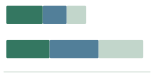


Time-course lines

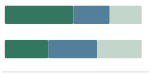


Small multiples

Composition



Stacked bars



Proportional stacked bars

Matrices, networks and sets



Heatmap



Network

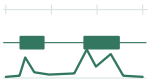


Venn diagram

Sets and genome positions



UpSet



Genomic tracks

Exact values



Table

Choose and compare displays

Reader's task	Useful starting choice and qualification
Compare amounts	**Dots or bars** make positions or lengths easy to compare. Bars need a zero baseline. If differences among observations matter to the question, show them alongside the group means.
Inspect distributions	Use **individual points** to retain observations, a **histogram** for counts in intervals, or an **ECDF** for the fraction at or below each value. Boxplots summarize the median and middle half; density curves and violins smooth the shape, so check these views against the observations.
Compare matched observations	**Connected points** show changes within matched samples that separate group summaries lose. [3].
Examine relationships	Use a **scatterplot** for two numerical variables. When overlapping points obscure how many observations occupy a region, try transparency or a **hexbin** display. A hexbin plot shows counts in bins, so individual observations can no longer be identified.
Follow time or dose	**Lines with observations** emphasize progression. Position values according to their actual spacing; use aligned small multiples when several curves become difficult to follow.

Reader's task	Useful starting choice and qualification
Show composition	**Stacked bars** show totals and their components; **proportional stacked bars** show each component's fraction of the total. Middle segments begin at different positions, so separate dots or bars are often easier for comparing one component across samples.
Inspect matrices or intersections	A **heatmap** reveals patterns across a matrix, with ordering and scaling affecting what stands out. For numerous set intersections, use **UpSet**; simple Venn or Euler diagrams can suffice for two or three sets.
Retrieve exact values	Use a **table** with units and enough digits for the intended comparison, without implying more precision than the measurements support. A plot is usually quicker for seeing a trend or ranking; a small plot with value labels can also support exact lookup.

For fuller comparisons and examples, see Wong [2], Wilke [1], Lex and Gehlenborg [13], and Jambor [10]. The catalogs by Holtz and Healy [11] and Ribecca [12] show further display options.

Explore alternatives

Inspect the observations before settling on a summary. Datasets with similar means and spreads can have quite different patterns; a boxplot, for example, can hide two clusters. If the data include several treatments or sample types, examine them separately as well as together. [4]; [5].

Change histogram bin widths and density smoothing to see whether an apparent feature persists. Coarse bins merge nearby features, while fine bins can make small fluctuations prominent. Check a violin's shape against the observations. [1].

Change the display when the question changes. A density helps locate modes; an ECDF makes the fraction below a threshold readable without choosing bins. A Q–Q plot instead compares quantiles with a specified distribution or another sample. Its reference and line convention need explanation, and apparent agreement is a diagnostic rather than proof of the generating model. These views complement one another because they expose different features. [1].

Categories or a time course?

The same fifteen readings appear in both displays below: one reporter culture per strain measured at 0, 1, 2, 4 and 8 hours. The grouped bars place the strains side by side at each time. In the aligned line panels, it is easier to follow Cedar's early rise and decline and Hazel's later rise. The horizontal spacing preserves elapsed time, including the longer interval between the final two measurements. Lines connect the recorded observations; they do not supply measurements between them. [7]; [1].

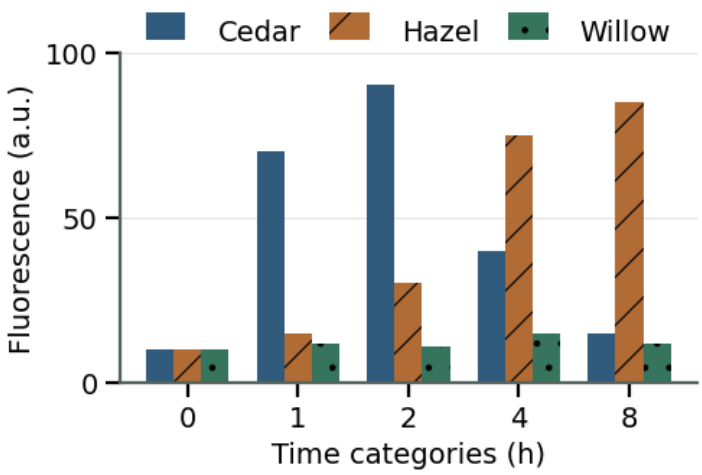


Figure 1: Grouped bars treat the five observation times as categories.

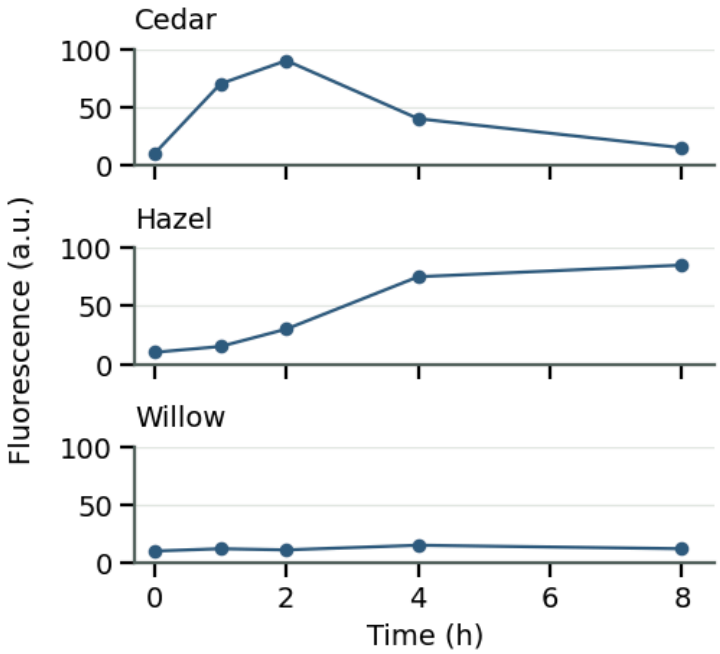


Figure 2: Aligned panels show each strain on the same time and fluorescence scales.

All plotted values are illustrative; the glossary icons are schematic.

Ordering and scales

Sort categories by value when readers need to identify the largest and smallest amounts. Keep an established sequence, such as developmental stages, when that order helps explain the result. Put the quantities readers need to compare next to one another, using a shared axis or aligned panels. [1].

Use the same axis scales across panels when comparing response sizes. If a smaller response becomes difficult to see, add a clearly labeled expanded view beside the full-range comparison. Giving each panel its own scale can make small and large fluctuations look alike. A linear axis uses equal distances for equal differences; a logarithmic axis uses equal distances for equal ratios, which helps compare positive values spanning orders of magnitude. [1]; [8].

Choose the baseline according to the encoding. A bar's length represents its value, so truncating its baseline changes the apparent ratio of lengths. Points and lines are read by their positions; a narrower, clearly labeled range can reveal small changes without making a false length comparison. Keep the full range available when it changes the biological interpretation. A logarithmic axis cannot contain zero: retain a zero-dose control in a clearly separated control panel or use another justified scale. Moving zero to an arbitrary small positive dose would misrepresent the experiment. [1].

If color identifies treatment and shape identifies genotype, try separate panels for each genotype when the symbols become difficult to distinguish. Keep treatment colors and axes consistent across panels. For spatial structures, a three-dimensional view may be needed; compare it with projections when overlap obscures the relationship. [1]; [9].

Ask a colleague to answer the intended question from the display and caption; a different answer may expose an ambiguous grouping or scale. [6].

Before settling on a display

- Can readers find the groups being compared and see which observations are matched?
- Does the caption or legend identify what each mark and summary represents, including units and any denominator?
- Do the order and axis scales make the intended differences visible without exaggerating them?
- Does a view of the individual observations reveal anything hidden by aggregation, smoothing or overlap?

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Showing observations and uncertainty

A figure of repeated measurements can show both how much the observations vary and how precisely you have estimated their mean or a treatment effect. These are different quantities, so label the points, summaries and intervals that represent them. [1].

Identify the observations

Choose what the points show

Plot individual cells when the question concerns cell-to-cell variation. To compare responses across cultures or experimental repeats, show a summary for each culture or repeat. Both can appear in one figure, using smaller marks for cells and larger, distinct marks for sample summaries. [2].

Keep the two levels distinct: the spread of cells within one treated culture shows variation within that culture, not how the treatment response varies across independent cultures. Likewise, repeated instrument readings show technical variation. [2].

Name the quantity you want to estimate before choosing a summary. An average across all measured cells gives cultures with more measured cells more influence; an average of culture means gives each culture equal influence. Those quantities need not agree when cell counts differ. Neither weighting should be selected merely because it produces a cleaner plot. Use the summary that answers the scientific question and agrees with the analysis, and show the culture structure so readers can see what supports it. For complex unbalanced designs, a display of culture means is useful context but does not by itself specify the appropriate analysis. [2].

Keep samples distinguishable

When several measurements come from each sample, show which belong together. Use separate panels or distinguish the samples within a plot. Add a summary for each sample if the comparison concerns differences between samples. Pooling all cells into one cloud hides whether a treatment response occurred in every experiment or was driven by one. [2].

Cells within cultures and culture summaries

Each condition has three independently prepared cultures, with six cell measurements per culture. Small circles show cells; the larger diamond shows that culture's mean. All plotted values are invented.

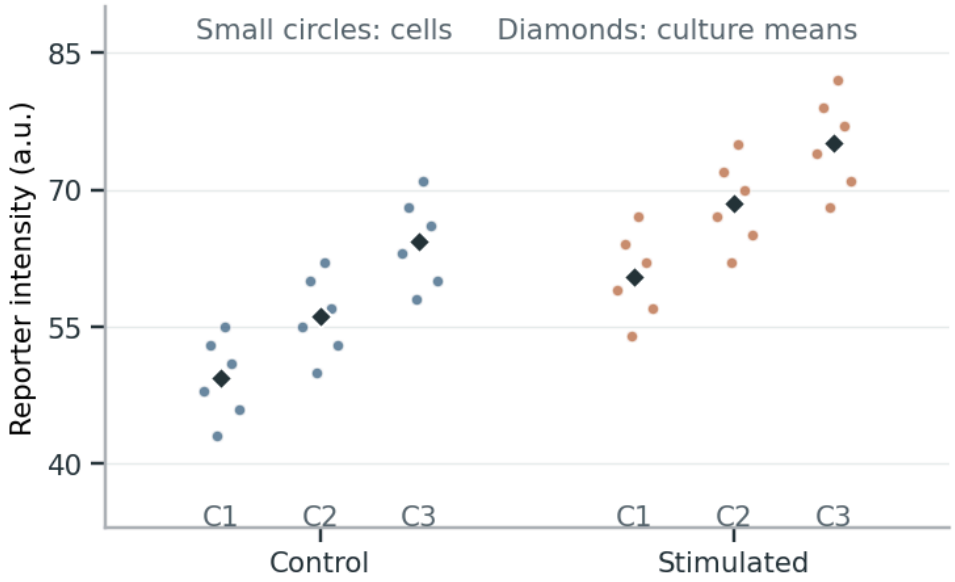


Figure 3: Cells form six labeled culture groups within two conditions. Small circles show cell-level reporter intensity and diamonds show the mean of each culture.

The horizontal displacement is **jitter**: it separates overlapping marks without changing their reporter-intensity values. When treatment is applied to cultures, measuring more cells describes each culture more thoroughly but does not add independently treated cultures. Cells can instead be independent units in a design that assigns treatment to cells individually and supports that independence. [2].

Show variation and matching

Show observations and summaries together

For repeated continuous measurements, a mean-only bar conceals the distribution, unusual values and overlap between groups. Points reveal observations, but a cloud of points can make the center difficult to compare. Show a clearly identifiable mean or median alongside them when that comparison matters. A bar remains useful when its height is the quantity being reported, such as a count or total amount. [3]; [4].

Choose complementary layers: a boxplot makes medians and quartiles easy to compare, a violin shows estimated distribution shape, and points show the observations supporting those summaries. A light violin with a narrow box and small circles is often useful when all three matter. Keep the summary visible; use a quieter fill and place points beside or over it without covering the median. A simpler box-and-points plot is enough when density shape adds little. [1]; [3].

When points overlap, use horizontal jitter without changing measured values, smaller marks or partial transparency. For many observations, a histogram, density or cumulative distribution may communicate more than a dense cloud. If only a sample of points is drawn, disclose the sampling and calculate summaries from the intended full dataset. [3].

A boxplot can hide separate clusters. A histogram or violin can reveal them, but bin width or smoothing can also create or erase apparent peaks. Assess support separately for each group. For a very small sample, keep all points and a named descriptive center, and omit the smooth violin; a box can also be unhelpful when only a few values determine its quartiles. Keep useful boxes and violins for the better-supported groups. One sparse group does not require a dots-only figure for everyone. [3]; [1].

Label each group's sample size and the unit counted. State the violin's width convention: equal maximum widths help compare shapes but do not show how many observations support each one. Count-scaled widths make sample-size differences visible but can make a small group difficult to inspect. Check density smoothing and boundaries, especially with tied or bounded values. No universal sample-size cutoff guarantees a reliable shape. [1]; [3].

A median describes the middle of the ordered values and is less affected by extremes; a mean includes their contribution to the average. Choose according to which quantity you want to report, then keep that choice consistent between the analysis and the figure. Showing the distribution helps readers see why the mean and median differ. [1].

A visible center does not require an inferential interval. With four observations, points and a labeled median can honestly describe the observed sample even when there is too little information to support a smooth density or a useful population estimate. If you add an interval, its calculation needs a defined estimand, sampling unit and method; the plotting layer cannot supply those missing assumptions. [1]; [2].

Different sample sizes, readable centers

All observations are invented sample-level measurements. The same median stroke identifies the center in every group. The sparse group retains its four observations; the other groups add boxes and light violins. Every observation remains visible beside its summary.

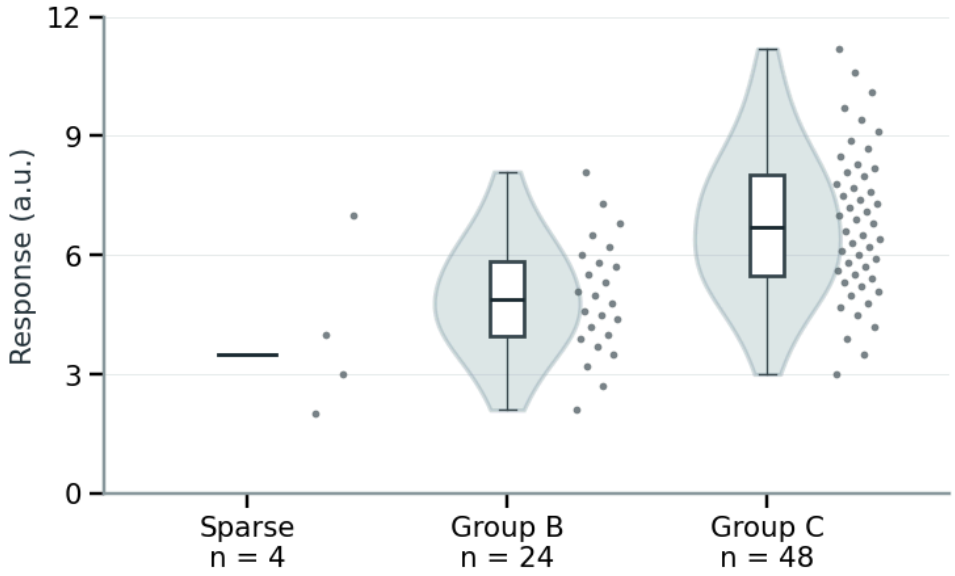


Figure 4: Points show individual samples; dark strokes show medians. Boxes span the middle 50%, with whiskers reaching observations within 1.5 interquartile ranges. Violins use Scott smoothing, end at the observed extrema and have equal maximum widths, so their widths do not encode sample size. No confidence intervals are shown.

Identical boxplots, different distributions

The two invented datasets below each contain 41 protein-abundance measurements. Their boxplots are identical, although one dataset is evenly spread and the other contains two clusters.

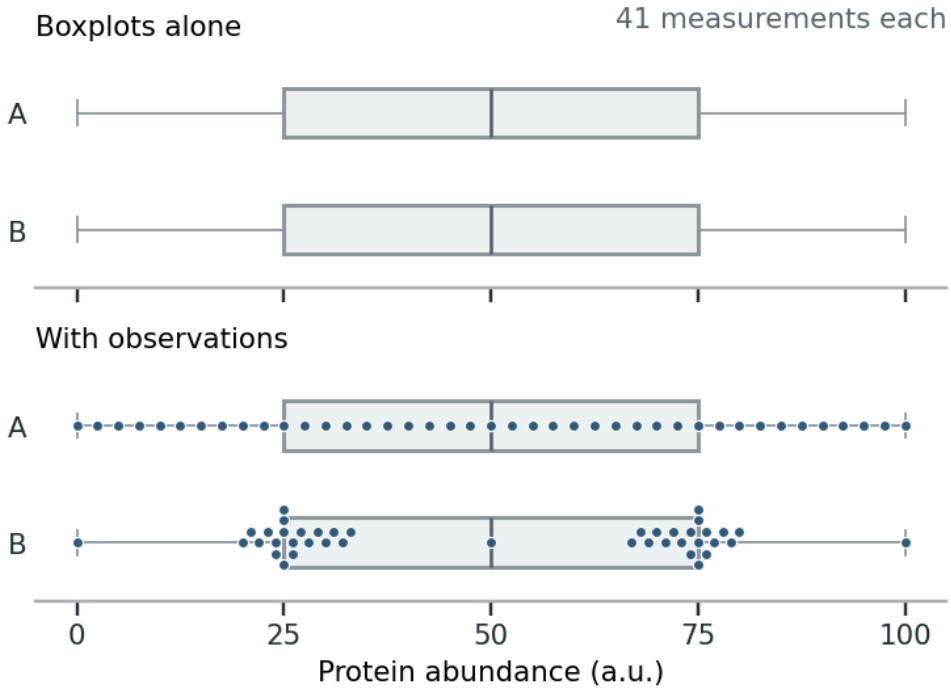


Figure 5: Adding observations reveals the different distributions. Both boxes span 25–75, with a median of 50 and whiskers at 0 and 100. Boxes show the middle 50%; whiskers extend to the most extreme observations within 1.5 interquartile ranges.

Use a boxplot to compare medians and quartiles; add observations or a distribution display when clusters and gaps matter. Retain points beyond the whiskers: the whisker rule does not establish that they are errors. [4]; [6].

Connect paired measurements

Connect measurements from the same sample before and after treatment, or control and treated measurements from the same experimental repeat. The lines show the direction and size of each change. Keep the original pair identities; connecting separately sorted values would create false pairs. [2]; [3].

Example: showing a response across experiments

In eight independent cell-culture experiments, each culture was split into control and stimulated conditions. The percentage of reporter-positive cells was measured in both. The two plots show the same illustrative measurements.

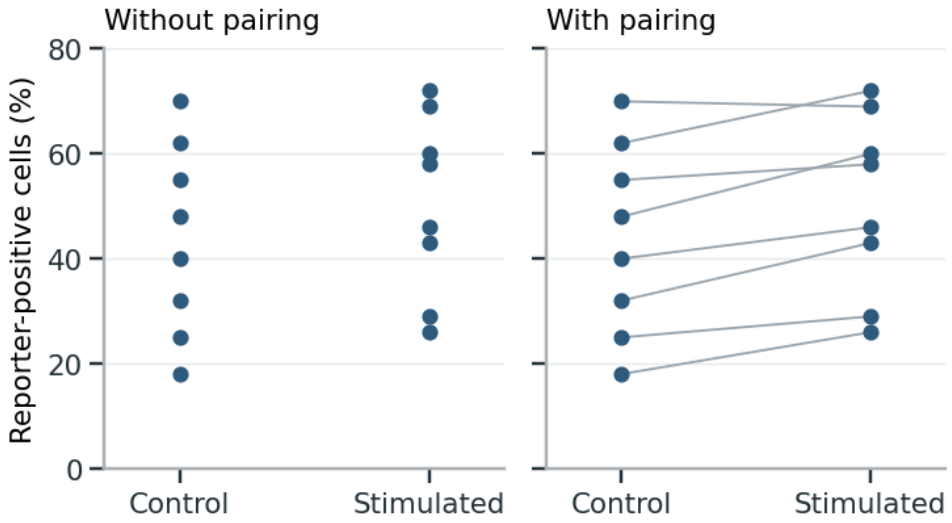


Figure 6: Connecting the pairs reveals an increase in seven experiments and a slight decrease in one.

The two groups overlap substantially. Without the lines, it is difficult to see that stimulation usually increased the response within an experiment.

To compare the changes themselves, plot the treated-minus-control difference for each pair, as in the next figure. If many lines overlap, separate subsets into small panels or show the distribution of changes. [3].

Show estimates and uncertainty

Choose the interval for the question

An **error bar** can show standard deviation, standard error or a confidence interval. Use standard deviation to summarize variation among observations; use a confidence interval to show the precision of an estimated mean or effect. Specify which quantity the bars show and identify the independent units in the caption. [1]; [2].

Quantity	What it describes	What to specify
Standard deviation (SD)	Spread of observations around their mean.	Which observations: cells, sample means or paired differences.
Standard error (SE)	How much an estimate would vary across repeated samples.	The estimate and the independent units used in the calculation.
Confidence interval (CI)	Precision of an estimate, using a method with a stated confidence level.	The estimate, confidence level and calculation method.
Credible interval	A range containing a stated probability for a parameter under a Bayesian model.	The probability level, model and prior.

[1].

For independent observations, the standard error of the mean is estimated as SD divided by the square root of n. More independent samples usually give a more precise mean, even when the observations remain just as variable. Counting dependent measurements as independent samples makes the SE misleadingly small. [1]; [2].

In the culture example, cells from the same culture share its treatment and conditions. To estimate a treatment effect, you can analyze culture summaries or use an analysis that accounts for this dependence among cells. Treating all the cells as independently treated samples would exaggerate the replication and make the treatment estimate appear too precise. [2]; [6].

Example: variation and precision

Subtract the control percentage from the stimulated percentage in each of the eight experiments above. The resulting changes range from −1 to 12 percentage points, with a mean of 6.6.

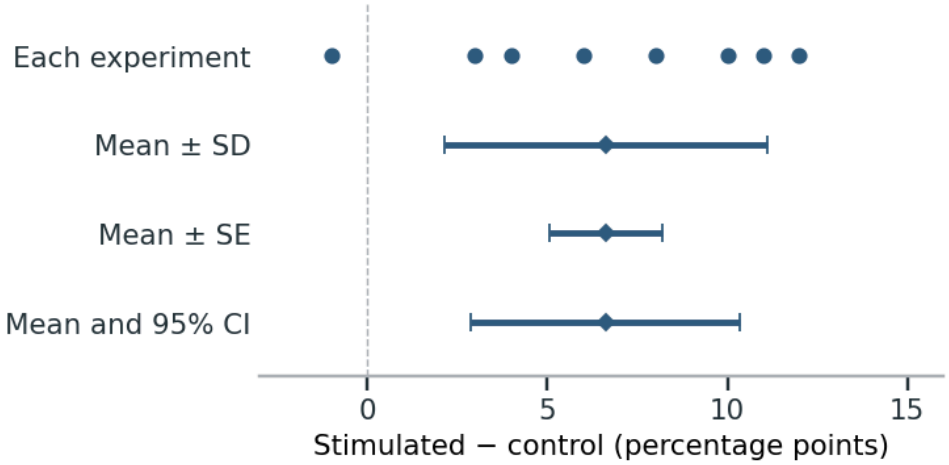


Figure 7: The same eight changes give different intervals: SD describes their spread; SE and the confidence interval describe precision of the mean.

The SD is 4.5 percentage points; the SE is 1.6. The 95% CI for the mean change is 2.9–10.4 percentage points. It was calculated as the mean ± 2.365 SE, using a Student-t interval with seven degrees of freedom. This calculation assumes independent experiments and approximately normally distributed experimental differences. Use the SD to describe how much the changes vary, and the CI to report how precisely their mean is estimated.

The values in the paired and interval figures are invented teaching observations.

The "95%" describes the confidence-interval method: across repeated samples, 95% of the intervals would contain the population mean if the assumptions hold. It does not mean that 95% of individual measurements lie inside the interval. A Bayesian credible interval instead gives a probability for the parameter under its model and prior. [1].

Choose how much of the uncertainty to show. A labeled interval is compact; a density with an interval can reveal asymmetry or multiple plausible regions. Frequency or quantile-dot displays can make probability more concrete for some audiences, provided the denominator and meaning of each dot are clear. Extra marks help only when readers can interpret the added information. For fitted curves, distinguish uncertainty in the mean response from prediction uncertainty, and pointwise intervals from a band with simultaneous coverage. [1].

Plot the effect being compared

Show the estimated treatment effect and its interval alongside the observations. For a difference, mark zero; for a ratio, mark one. Label the direction and units, such as "stimulated – control, percentage points." [5]; [1].

Calculate the interval for the comparison itself. For paired data, use an analysis that preserves pairing. For an adjusted estimate, use the model's effect estimate and interval, labeled separately from the raw measurements. Overlap between two groups' error bars is not a reliable test of their difference. [1]; [2].

Compare the interval with the size of effect your question concerns. If it includes both no change and an increase of the size predicted by the proposed biological explanation, the estimate does not distinguish those possibilities. Reporting only “not significant” would hide that uncertainty. A small P value, in turn, does not show that an effect is large enough to matter biologically. [5]; [6].

An interval describes uncertainty under the analysis assumptions. If every treated sample was measured in one batch and every control in another, even a narrow interval cannot tell you whether treatment or batch caused the difference. Check the design and controls before interpreting the effect as causal. [6].

Align the display with the analysis

Keep transformations visible

Place observations, summaries and intervals on the same labeled scale. A logarithmic axis changes their positions, not how the mean was calculated. If you display a log-scale difference as a ratio, back-transform both the estimate and its interval. [1].

Normalizing each treated value to its paired control makes every control equal to one and hides the original control variation. Show the unnormalized measurements when that variation matters, and explain how it was included in the analysis. Connecting paired measurements may make normalization unnecessary, as in the paired-measurement example. [2].

Smoothing and detrending also change what readers see. A trailing moving average can lag a response; a centered window loses information at the ends, and a flexible smoother can erase or introduce apparent features. Compare plausible settings and retain meaningful departures. Additive residuals show absolute departures, whereas ratios can show departures relative to an appropriate trend. Fitting on a log scale changes error weighting, so choose it for the analytical question rather than a smoother-looking line. A decomposition remainder is not automatically random noise. [1].

Show which observations enter the comparison

Use a distinct symbol for displayed observations excluded from the analysis, and explain it in the caption. If an extreme value compresses the rest of the plot, add a zoomed view while retaining the full-range view. [3]; [6].

Arrange groups and panels around the comparisons being analyzed. If the question is whether a treatment works differently in two groups, show the difference between their effects and its uncertainty. Separate significance labels for the two groups do not show whether their effects differ. [3]; [6].

Check the figure and caption

Use the caption to explain how to read the figure. Define the points, lines, summaries and intervals. For a boxplot, specify the median, quartiles and whisker rule; whiskers may show the range or use a rule based on the interquartile range. [4].

Give exact sample sizes for each group and level—for example, cultures and cells per culture—and the number of complete pairs for a paired comparison. “Biological replicates” or “n = 3–8” alone is insufficient. [2]; [3].

State whether summaries use raw, transformed or normalized values. Account for missing and excluded observations, with reasons. Identify any adjustment for multiple comparisons and which comparisons it covers. Name the analysis and refer to Methods for fuller details. See Methods, results and figure legends for legend structure. [3].

Caption for the interval comparison

Changes in reporter-positive cells after stimulation in eight independent experiments. Circles show stimulated-minus-control differences, one per experiment. Diamonds show the mean; bars indicate ±1 SD, ±1 SE or the two-sided 95% Student-t confidence interval (7 degrees of freedom), as labeled. Each experiment included both conditions; no observations were excluded.

Before finalizing the figure

- Define each mark and give sample sizes at the relevant levels.
- Show individual measurements, grouping and pairing where needed.
- Name the summary and interval, including its level and calculation method.
- Check units, comparison direction, transformations and exclusions against the analysis.
- Show the effect's magnitude and uncertainty, not only its significance.

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Color, symbols and accessibility

Color can identify groups, show how a quantity varies or draw attention to a selected observation. First decide whether color helps: labels, position and separate panels may already make the distinction clear. When color adds information, choose its role and visual strength deliberately. [1]; [3].

Decide what color should show

For a plot of one series, a single dark color is usually enough. Several groups can also share a neutral color when labeled positions separate them. Use distinct treatment colors when they help readers follow treatments within or across panels, and keep those assignments consistent. Avoid giving every bar or point a different color when its label or position already identifies it. [9]; [8].

To highlight one group, give it an accent color and label it. Keep the other observations visible in a subdued color. If readers should compare all groups equally, avoid making one much brighter or more saturated than the others: it can attract attention for reasons unrelated to the data. [2]; [1].

Color is useful for finding patterns in a large matrix, but less reliable for judging small numerical differences. The same colored square can look different against light and dark surroundings. If readers need to compare a few values precisely, put them on a common axis or print the values rather than asking readers to estimate them from a color key. [3]; [5].

Match the scale to the data

Groups, magnitudes and deviations

Use a **categorical palette** when color identifies groups without a numerical order, such as cell types or treatments. Its colors should be distinguishable without implying a ranking. A pale-to-dark progression instead suggests order and is better suited to quantities such as dose. Established palettes, including Okabe–Ito, are useful candidates, not a required visual style. A more subdued palette may suit large areas or a quiet figure, provided essential distinctions and contrast survive in the actual plot. Accessibility does not require bright, highly saturated colors. [4]; [2]; [1].

Use a **sequential scale** for values that run from low to high. Lightness should progress in one direction so that the order remains visible. A perceptually balanced scale also aims to give equal numerical steps comparable visible changes; monotonic lightness alone does not ensure that balance. This can be a single-hue gradient or a multihue palette such as cividis or viridis. Consider which values should stand out against the background: dark marks are prominent on white, while light marks stand out on a dark background. [5]; [7].

Use a **diverging scale** to distinguish values above and below a reference, such as no change or a specified target. Give the two sides different hues and make the reference neutral. No change is zero for a log fold change and one for a ratio. Set that reference explicitly; the middle of the observed range may represent a different value. [5]; [7].

The signs of the stored numbers do not determine the palette. Ratios of 0.5 and 2 are both positive, but lie on opposite sides of no change. The same numbers as absolute concentrations may simply run from low to high. For ratios, decide whether equal additive departures or equal fold departures should have comparable emphasis. On a $\log_2$ ratio scale, 0.5 and 2 become −1 and +1; label that transformation so readers can recover the meaning. [5]; [1].

The quantity determines the reference

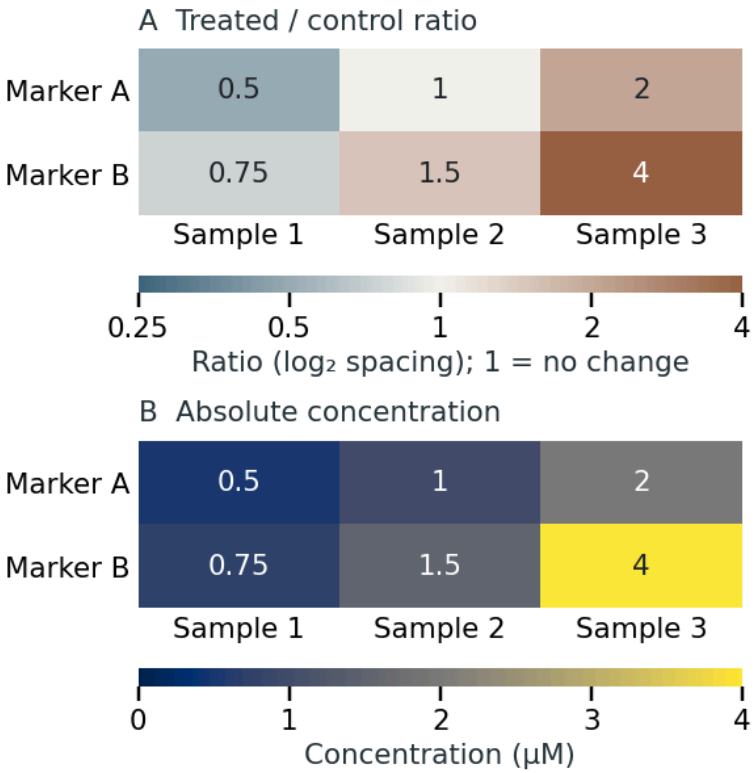


Figure 8: Identical invented numbers represent ratios in A and concentrations in B. The ratio scale centers on one and gives twofold increases and decreases equal color strength. The concentration scale runs from low to high without implying a target concentration. Cell labels retain the original values.

Three common color encodings

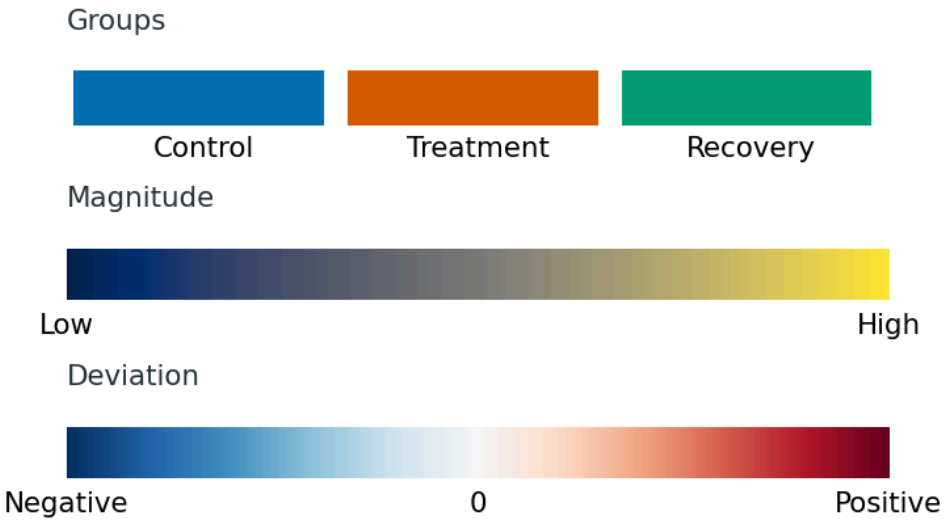


Figure 9: Categories use separate colors; magnitudes follow an ordered gradient; deviations use two directions from a reference. Shown here: three Okabe–Ito colors, cividis and RdBu.

Avoid rainbow scales for ordinary quantitative data. Their uneven changes in hue and lightness can create apparent boundaries where the values change smoothly, while making other differences difficult to see. Selecting a well-designed scale is easier than constructing one by eye. [2]; [1].

Two less common choices are useful when the data require them. A **cyclic scale** has matching ends, so angles near 0° and 360°, or phases immediately before and after a cycle boundary, look similar. A

discrete ordered scale divides values into labeled ranges. Use it when those ranges matter; use a continuous gradient when variation within them matters. Show the boundaries of discrete ranges in the key. [7].

Set the reference and limits

Use the same color limits for panels whose values you want readers to compare. Otherwise, a color indicating a small response in one panel may indicate a much larger response in another. Within a single panel, you can show the full possible range or a narrower range that makes small differences easier to see. Show the chosen limits in the colorbar. [5].

Full possible range or local range?

The same invented percentages appear twice. On the full 0–100% scale, the 42–58% values have little color contrast. On the stated 40–60% local scale, their differences are easier to see. The narrow scale uses its strongest colors for values still near 50%, so readers need the colorbar to interpret their size. Use shared limits if these values must be compared with other panels.

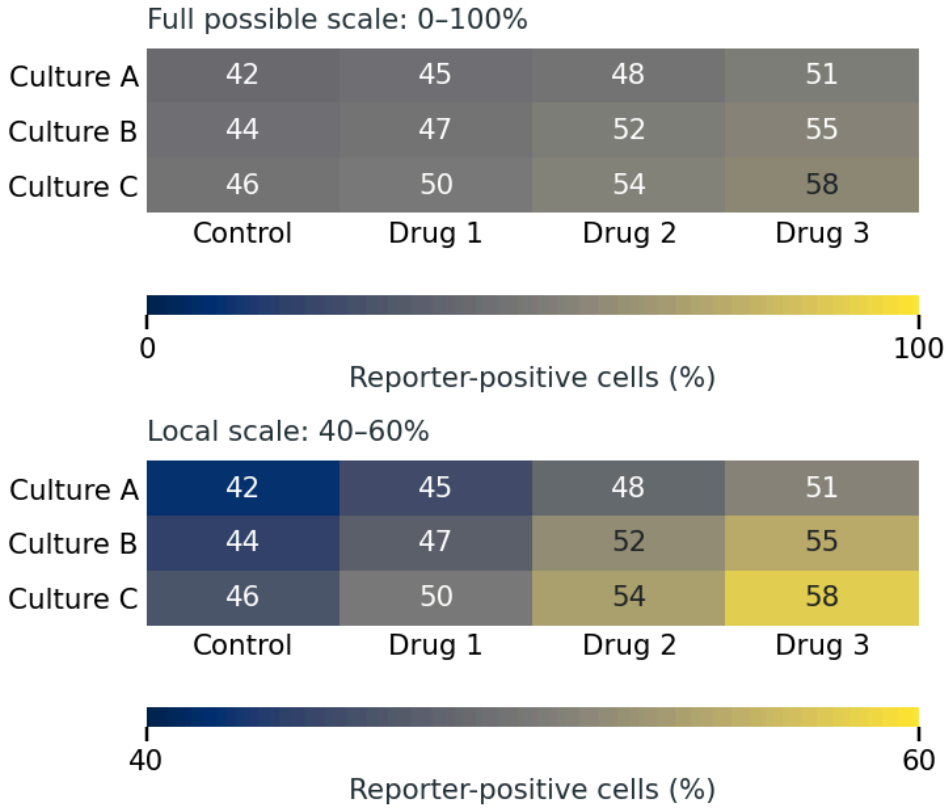


Figure 10: Two identical heatmaps of reporter-positive percentages. The upper colorbar spans zero to one hundred percent and the lower spans forty to sixty percent; all cells retain their printed values.

For a diverging scale, equal departures from the reference should receive comparable color strength. With values from −1 to +3, stretching both ends to equally strong colors would give −1 as much emphasis as +3. A balanced −3 to +3 scale avoids that distortion, even though part of its negative range is unused. [7].

Center the scale on the biological reference

These invented $\log_2$ fold changes range from −1 to +3. Zero means no change. Both panels show the same values.

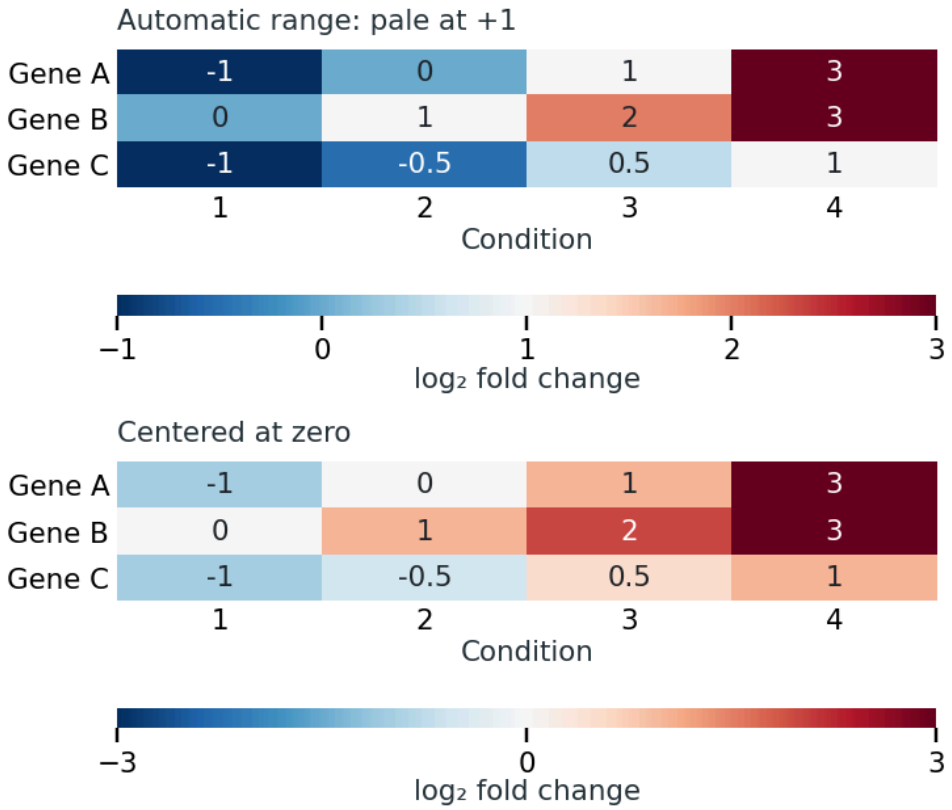


Figure 11: Automatic minimum-to-maximum scaling puts the pale color at +1. Centering the scale on zero correctly distinguishes decreases from increases and gives equal-sized changes comparable emphasis.

Label what the colorbar measures, including units and transformations: for example, "$\log_2$ fold change" or "row-standardized expression." If values beyond a limit all receive the same end color, indicate that saturation in the key. Give missing values a separate labeled appearance so they cannot be mistaken for zero or the lowest measurement. [5]; [1].

Keep series identifiable without unnecessary variation

A useful starting style for a few ordinary trajectories is solid lines, with circles where markers help show measured positions. Label lines near their ends when there is room. If the labels and paths make every series traceable, there is no need to give each one a different dash pattern and symbol as well. For dense continuous curves, markers may add clutter without information. These are starting choices, to be judged in the actual figure. [6]; [1].

Use another encoding when it does a job: dashes might distinguish a forecast from observations, and shapes might distinguish a second factor or help identify overlapping groups in monochrome. Explain that meaning in the labels or key. When several curves cross, check that a reader can follow each path; an endpoint label alone may be insufficient. Try separate panels before adding increasingly complex dash and marker combinations. Refer to groups by name in the caption, rather than only as "the red line." [6]; [7]; [1].

Choose shapes that remain distinct at the final size. Circles and squares are usually easier to distinguish than several similar polygons. Adjust their sizes so they appear equally prominent; equal numerical marker sizes do not necessarily give equal visual weight. Open circles can reveal overlapping boundaries, whereas large filled symbols can merge into a solid patch. If overlap is

severe, changing symbols alone will not solve it: separate groups into panels or use a distribution display. [6].

Color and shape can identify the same treatment when that redundancy improves reading. They can instead encode two variables, such as treatment and genotype. Each combination then represents a different group. Separate panels for genotype can make those combinations easier to follow while retaining consistent treatment colors. In the facets below, panel titles replace the need for genotype-specific markers and dashes. [6]; [1].

Separate panels for each genotype

Separate WT and KO panels make the treatment response easier to follow within each genotype. The combined view is useful for comparing genotypes directly, but readers must distinguish both colors and symbols. Both arrangements show the same invented values.

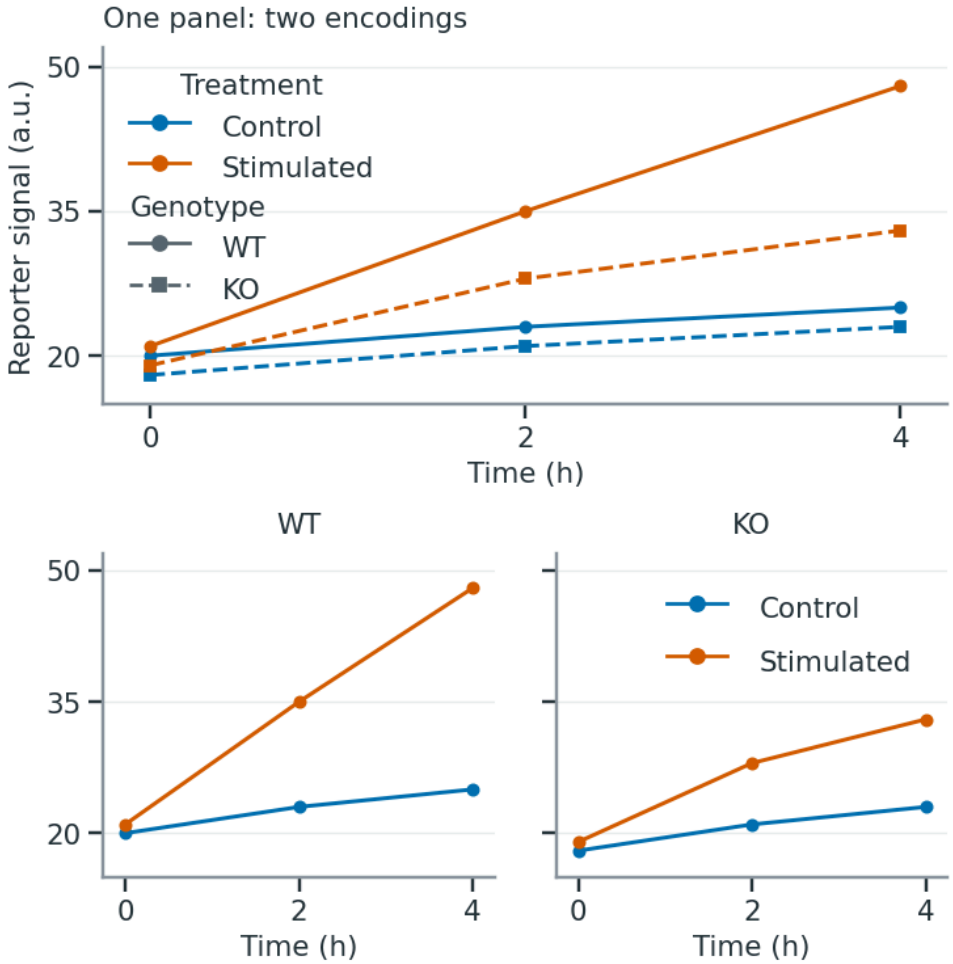


Figure 12: A line plot with four series uses color for treatment and marker and line style for genotype. Below it, two panels labelled WT and KO show the same treatment trajectories with consistent blue and vermilion colors.

Keep series identifiable without hue

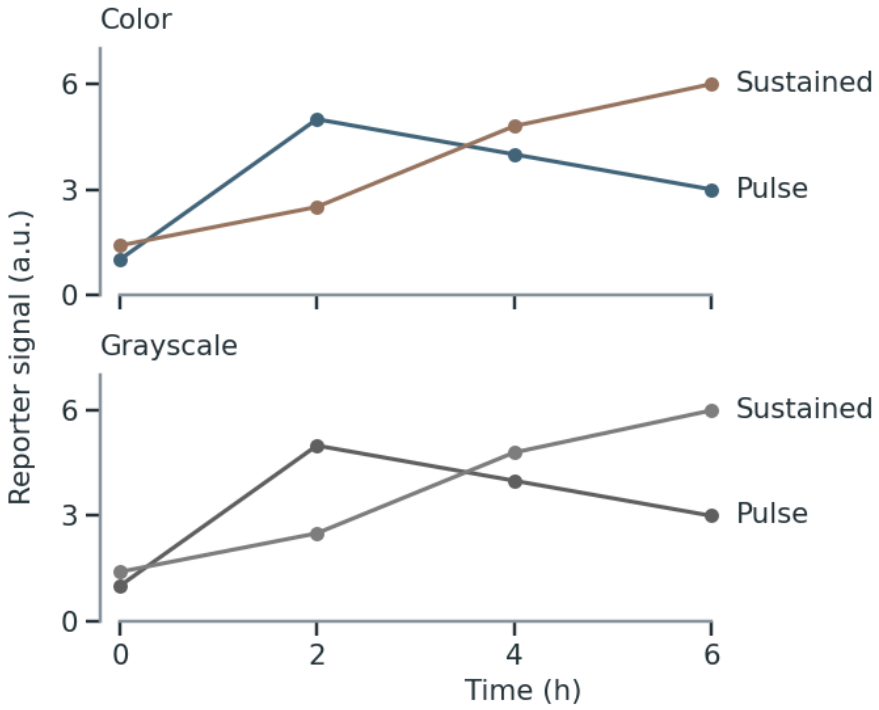


Figure 13: The same invented reporter trajectories in subdued color and grayscale. Both use solid lines and circles; direct labels and distinct gray levels identify the two simple paths.

Make marks readable on their background

A palette that works in large swatches may fail on thin lines or small points. Inspect the plot at its intended reading size. Darken pale marks, increase their size or weight, or use a more distinct shape if they disappear. Large filled areas often need less saturation than small marks; intense fills can overwhelm nearby labels and grid lines. [2]; [1].

Check colors against the background and against neighboring marks. Yellow may be clear on a dark background but nearly invisible on white. Transparency also changes the displayed color through blending, so inspect overlapping regions as well as isolated points. Keep essential labels dark enough to read; a label need not inherit the color of the series it names. [8]; [2]; [1].

For choosing colors for fluorescence channels and showing merged images, see Biological images. [4].

Check accessibility and consistency

Inspect the finished figure under simulations of common red–green and blue–yellow color-vision deficiencies. Test whether the groups remain distinguishable, whether a quantitative gradient still has a clear order, and whether labels and small marks remain visible. If a distinction disappears, change the palette or supply it through labels, shapes or separate panels. Check the exported figure, not just the palette preview. [4]; [1].

Grayscale reveals weak contrast and whether identification depends entirely on hue. A symmetric diverging scale will often give negative and positive values similar gray levels: it preserves distance from the center but loses direction. If that direction must survive monochrome reproduction, add signed values, explicit labels or separate views. [7]; [1].

Treat these checks separately. A figure can preserve contrast in grayscale yet lose a distinction under a particular color-vision deficiency; a palette can separate hues yet leave thin marks too faint against the page. Simulations help identify failures but do not establish that every reader can use the figure. Judge the needed comparisons in the finished display, including its labels and arrangement. [4]; [8].

Check the figure alongside the rest of the paper. Do recurring groups keep their colors and symbols? Do comparable panels use the same scale and reference? Check that keys, direct labels and captions agree, including the labels for missing values and highlighted observations. [8].

Before finalizing the colors and symbols

- Does each encoding identify a group, show a quantity or mark a deliberate highlight?
- Would fewer hues, shapes or dash patterns make the figure easier to follow without losing a needed distinction?
- Do the palette type, reference and limits match the quantity being shown?
- Can readers distinguish the important groups without relying on hue alone?
- Are marks and labels readable at final size, including in grayscale and color-vision simulations?
- Do recurring colors and symbols mean the same thing across panels?

The teaching graphics use invented values or schematic encodings.

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Biological images

Show images in which readers can inspect the feature you discuss, alongside its measurements when available. Prepare the figure from copies and preserve the images used for analysis.

First decide what readers should compare: shape and location, signal intensity, or the output of an image-analysis procedure. A display adjustment that reveals a faint structure can remove an intensity difference. A clean segmentation overlay can make boundaries visible without establishing that they are correct. Choose the presentation for the claim, then retain the original pixels, acquisition information and quantitative analysis needed to assess it. [2].

Select and frame images

Choose fields from the images used in the analysis that reflect the reported result. When a phenotype varies substantially, show several examples spanning that variation rather than one especially striking cell. Place them beside the corresponding quantitative results. [2].

A tissue overview shows how structures are arranged; a closer view can reveal cell shape or where a marker lies within a cell. If readers need both, pair the overview with an enlarged region. Check that the feature remains visible at the figure's final size before deciding how many images fit in a panel. [1].

Crop without losing context

Crop empty margins or irrelevant regions, but retain neighboring structures that affect the interpretation. Mark the exact enlarged region on the overview and place the detail beside it. An inset within the overview saves space only when it does not cover relevant data. Use a visible gap between separate images so their edges cannot be mistaken for a continuous field. [1]; [2].

Keep the image's proportions when resizing. If rotating images makes anatomical comparisons easier, use a consistent orientation and label it. Rotation by an arbitrary angle resamples pixels in many image editors; make measurements on the analysis images before preparing rotated copies for the figure. [2].

Calibrate each view

Add a scale bar with its length and unit. Use the image's spatial calibration; an objective label such as "40×" does not specify the size of a feature in a resized figure. Put the bar where it contrasts with the background without covering the feature of interest. When no suitable area exists, place it just outside the image. [1]; [2].

An enlarged crop needs its own scale indication. Keep a scale bar attached to its image when resizing, or regenerate it from the final calibration. [1]; [2].

Show channels

For a single fluorescence channel, start with grayscale. Dark blue or red against black can hide fine structures that are readily visible in grayscale. An inverted grayscale display can help with thin structures on a light page; keep the choice consistent across comparable panels. [3]; [1].

The same structures under different color mappings

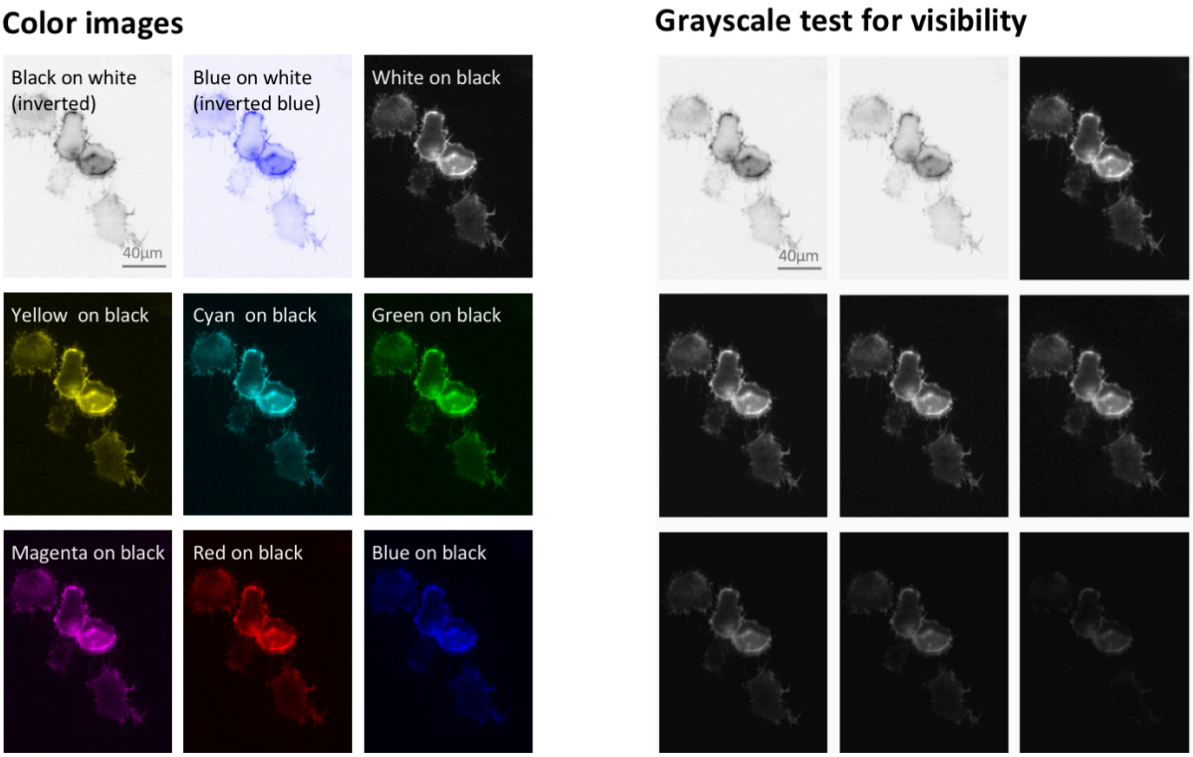


Figure 14: Actin in Dictyostelium discoideum expressing LifeAct-GFP, shown with several color mappings and corresponding grayscale previews. All views have the same spatial scale; the source's scale bar is 40 µm.

Figure 6 reproduced unchanged from Jambor and colleagues (2021), © the authors, CC BY 4.0. Compare the fine extensions around the cells, rather than only their bright centers.

Combine an overlay with separate channels

Use a merged image to show where labeled structures lie relative to one another. Include separate grayscale channels when overlap or unequal brightness makes either component difficult to inspect. Keep the field, crop and orientation identical across the component images and merge. Label channels with their markers or targets, and use those labels consistently throughout the figure. [4]; [1].

Choose display colors for visibility rather than to match fluorophore names. Cyan and magenta are useful starting choices for two channels; assess the actual merge under color-vision simulations. A palette that separates bright objects may still hide a weak channel. Separate grayscale panels let readers inspect that channel without relying on the composite color. [4]; [1].

Keep colors assigned to the same targets across conditions. Do not compare the abundance of two targets from their apparent brightness in different colors: color mapping, staining and acquisition affect that appearance. If you make a claim about how much two markers overlap, show how overlap was measured. The color of a merge alone does not establish its extent. [4]; [2].

Intensity ranges and processing

Choose the displayed minimum and maximum while inspecting both the image and its intensity histogram. A range that is too wide can make real structures nearly invisible. A range that is too narrow clips dim pixels to black or bright pixels to white, concealing differences within those regions. Start with a linear mapping and retain the features needed for the comparison. [2].

For images intended to compare signal intensity, use the same display range and processing for the corresponding channel across conditions. Automatic contrast applied separately to each image can make a weak signal look as bright as a strong one. [2].

Shared and independent intensity ranges

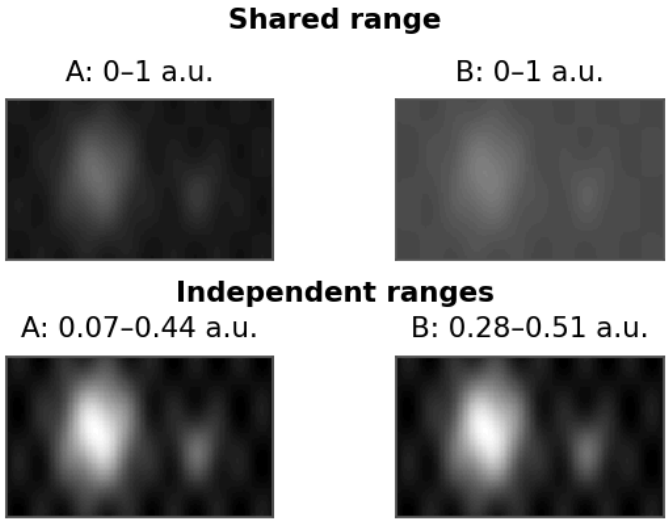


Figure 15: Synthetic calibration pattern shown with a shared 0–1 a.u. display range (top) and independently stretched ranges (bottom). The pattern is not microscopy data.

Independent stretching makes both patterns look alike despite their different intensity ranges. It helps inspect their structure but removes the brightness comparison.

When the purpose is to compare structure and the images differ greatly in brightness, separately adjusted views may reveal details that a common range hides. Identify those adjustments in the caption, and use the quantitative measurements to compare intensity. Keep each view's range visible when the numerical intensity scale is important. [2].

If acquisition settings differ or are unknown, do not infer an abundance difference from displayed brightness. A morphology comparison can still be useful if the structures are adequately resolved and the display adjustments are disclosed. An intensity comparison needs compatible acquisition and a justified quantitative analysis; forcing both images onto the same color scale cannot create that comparability.

Make transformations visible

Gamma adjustment and other nonlinear mappings change the relative display of weak and strong intensities. If one is needed to reveal structure, identify it and provide a linearly mapped comparison. Include a labeled intensity scale for pseudocolor or nonlinear displays. [2].

Filtering, deconvolution and learned image restoration can also change the appearance of structures. Identify these operations in the caption and describe their settings in Methods. Preserve the input images so readers can inspect them before restoration. State which processing was used for measurements and which adjustments were made only for display. [2].

Connect images to measurements

When a result comes from image analysis, show an example input image. If segmentation determines which objects or regions were measured, add outlines or masks showing those boundaries. Keep an unobstructed image alongside a dense overlay. Include difficult cases when errors, such as merging neighboring objects, could change the counts or measurements. [2]; [1].

Example: inspect the segmentation before counting

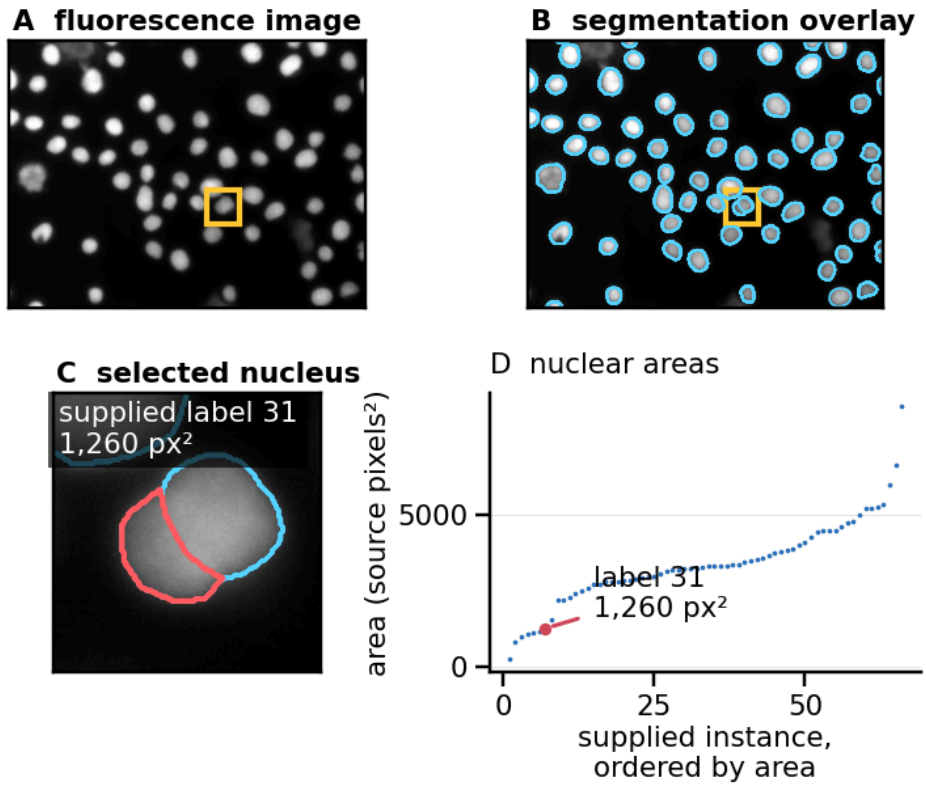


Figure 16: A real fluorescence image, its supplied expert instance boundaries, a labeled boundary crop, and the source-pixel areas of the supplied instances.

The crop shows two touching nuclei assigned separate labels in the supplied expert annotation. The red outline identifies the nucleus represented by the red point in the area plot (1,260 source pixels²). The 66 areas are calculated from the original mask; this example uses the supplied annotation without an independent accuracy assessment.

Adapted from the CC0 S-BIAD634 BioImage Archive dataset, An annotated fluorescence image dataset for training nuclear segmentation methods, by Taschner-Mandl and colleagues (2023); raw image and supplied expert annotation normal_39. S-BIAD634 is released under CC0 1.0.

In the accompanying quantitative plot, preserve the grouping of cells or fields within samples. Many cells from one culture do not show how the finding varies between cultures. The chapter on observations and uncertainty explains how to display this grouping and choose summaries. [5].

Labels and annotations

Use a few arrows, outlines or direct labels to identify the structures relevant to the comparison. Place labels beside small features rather than on top of them. Distinguish annotation types by shape or line style when they have different meanings; color alone can be difficult to distinguish against a multicolored image. If numerous labels obscure the image, provide a separate annotated copy. [1].

For a time series, put elapsed time on each frame and keep the framing and scale consistent. Label the anatomical orientation of a section and identify any projection, so readers know which view of the specimen they are seeing. [2].

In the caption, identify the specimen and tissue, markers or stains, channel colors, scale and annotations. Explain any inset or projection. State whether images have separate intensity adjustments and identify processing that changes their appearance, as described above. Put acquisition and analysis details in Methods, including the software, version, parameters and manual selections needed to reproduce the workflow. [1]; [2].

Verify the image figure

Keep the original images and metadata, the analysis inputs and outputs, and the assembled figure as separate files. Record which source image produced each panel and which crop, channel mapping and adjustments were applied. Share the images used in figures and quantification through an appropriate repository where possible, together with the workflow and example inputs and outputs. [2].

Inspect the exported figure at publication size. Check that the crop markers still point to the correct regions, scale bars remain attached to their images, weak structures are visible and annotations do not cover the evidence. [1].

Final image check

- The selected fields show the reported phenotype and its variation.
- Crops retain necessary context; enlarged regions and physical scales are clear.
- Each channel is identifiable, and the merge does not hide essential structures.
- Images used for intensity comparisons share display ranges; separate or nonlinear adjustments are disclosed.
- Readers can see the input image and the objects or regions used for image-derived measurements.
- The exported panels can be traced to preserved images and processing records.

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Specialized displays and explanatory diagrams

Explain how to read the display. In a heatmap, label the quantity shown by color and explain how the rows and columns are ordered. In a pathway diagram, identify the molecules or processes and explain what the arrows mean.

Data displays

Heatmaps

Use a heatmap to compare patterns across many genes, samples or conditions. Decide whether color should show abundance, a ratio to a reference or a standardized value. With row standardization, each gene is centered and scaled separately. The colors then show which samples have high or low values for that gene; they cannot tell readers which gene is more abundant. Name the quantity on the color key. [1]; [2].

Arrange rows and columns to expose the comparison. Keep time points in chronological order and doses in concentration order. Group samples by known conditions when those conditions are the focus; cluster them when the question is which samples have similar profiles. Apply the same order to accompanying annotations and related panels. A sample-group strip beside a heatmap can show whether an apparent cluster follows treatment, batch or another recorded characteristic. [1]; [2].

In hierarchical clustering, branches can rotate without changing their relationships, and neighboring rows on the page need not be close relatives in the tree. Retain the dendrogram when those relationships matter. State the transformation, distance measure and clustering method in the caption or Methods. [1].

Ordering rows while preserving time

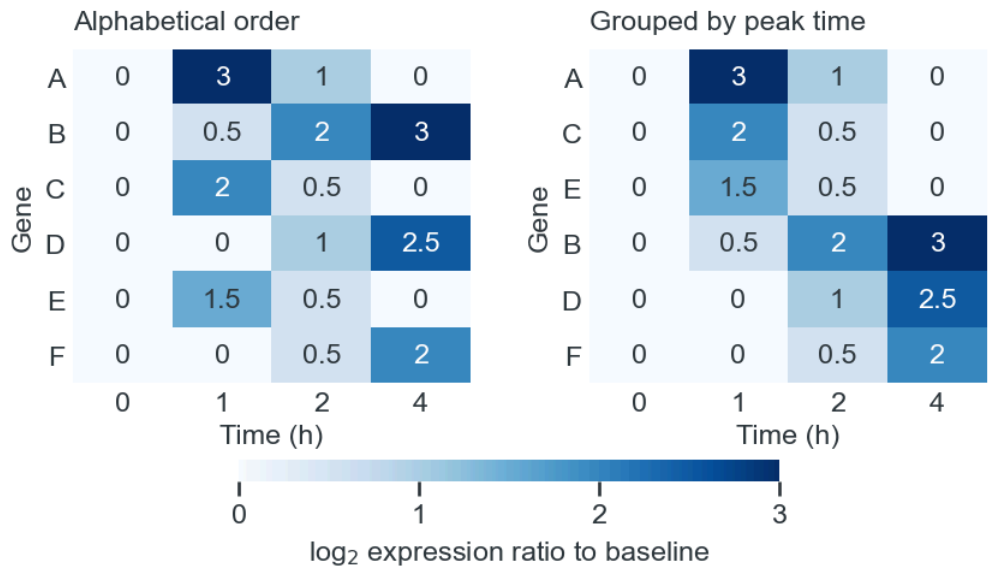


Figure 17: The same six invented gene profiles in alphabetical order and grouped by the time of their maximum. Both heatmaps retain columns at 0, 1, 2 and 4 hours and an identical zero-to-three color scale.

Equal column widths do not represent equal elapsed time here. Use a line plot with proportional time spacing when readers need to compare rates.

Inspect the value distribution before fixing the color limits. A few extreme values can leave most cells nearly the same shade. A transformation may make differences among the less extreme values easier to see. Another option is to assign values above a chosen limit the same endpoint color; label the limit and make the extreme values available so readers can identify them. Use the color-scale guidance to choose a sequential or diverging palette. For close comparisons between a few profiles, aligned line plots are easier to read than small color differences. [2]; [1].

Genomic tracks and rearrangements

Use aligned tracks when readers need to compare signals at the same locus. Show the genomic interval and coordinate scale, identify the reference assembly, and align gene models and measurements to the same positions. Tracks being compared for signal magnitude need compatible units and the same vertical scale. A track rescaled to its own maximum can show its shape, but its height cannot then be compared directly with a neighboring track. [3]; [4].

Shared and separate track ranges

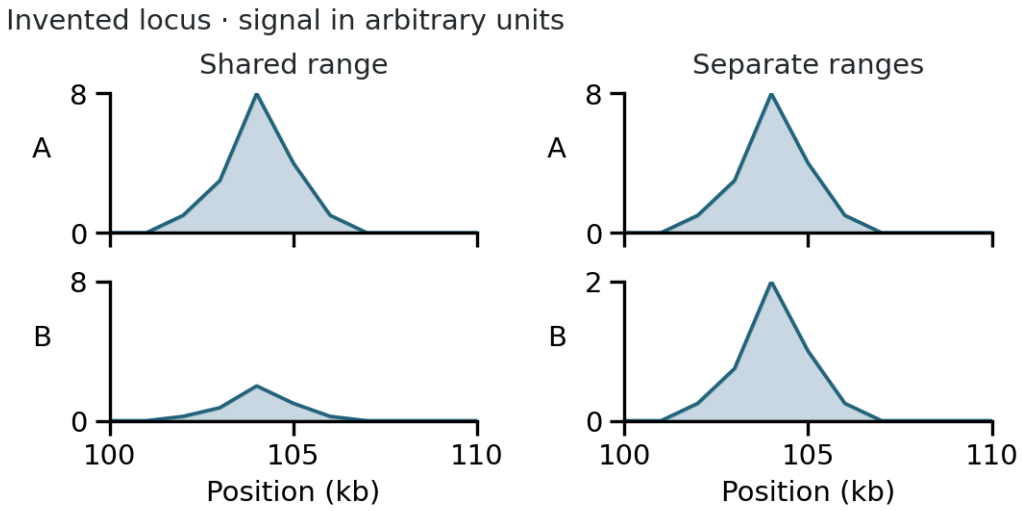


Figure 18: Track B has one quarter of A's signal at every position. Shared ranges show that difference; separate ranges give the peaks the same apparent height. Both columns contain the same invented values and genomic positions.

A chromosome-wide view can hide a narrow peak because many bases contribute to each displayed bin. Use a locus view to show local detail, adding an overview if readers also need to locate it on the chromosome. Mark any breaks where intervening sequence has been removed. To compare regions around a feature such as a transcription start site, align them at that feature and orient them consistently. The horizontal coordinate then describes distance from the feature rather than absolute position in the genome. [3].

When there are too many tracks to read, a heatmap can fit them into less space while retaining a row for each sample. Exact signal magnitudes will be harder to judge from color. Averaging tracks reduces clutter too, but hides differences between samples; an inferred state also replaces measured values with a summary. Keep the individual tracks available if those differences affect the interpretation. For a few profiles, an overlay may work, provided filled areas and blended colors do not obscure one another. [4].

For structural variation, decide whether readers need to locate breakpoints or understand the rearranged sequence. Arcs connect the joined positions; arranging the genome in a circle brings distant loci closer, although the arcs can become crowded. To show an inversion or a new sequence order, use sequence blocks with their orientation marked, or a dot plot comparing reference and variant coordinates. In the dot plot, corresponding segments form lines whose direction shows their orientation. [5].

Networks

Use a node-link diagram when readers need to follow a path or identify a node's neighbors. Define what a connection represents: physical interaction, coexpression, regulation and sequence adjacency are different relationships. Add arrowheads only when direction is part of the data. Try alternative layouts, then retain one that makes the relevant paths and groups easy to follow. Unless positions have a defined biological meaning, proximity on the page is a layout choice. [6]; [2].

For dense networks, an adjacency matrix avoids overlapping edges. The same nodes appear along both axes, and a marked cell records a connection between its row and column nodes. Placing related nodes together can reveal groups with many connections within them and fewer between them. To follow a chain through several nodes, use a node-link diagram instead. If you show only part of a network, explain how you selected it. [6].

For a directed network, label which axis contains the source and which contains the target; a row-to-column connection is different from its reverse. Explain whether an empty cell means an observed absence, an untested pair or simply no recorded connection. Keep missing or untested entries distinct when the dataset provides that information. A compact matrix improves lookup only if the reader can recover the relationship represented by each cell.

Two views of the same connections

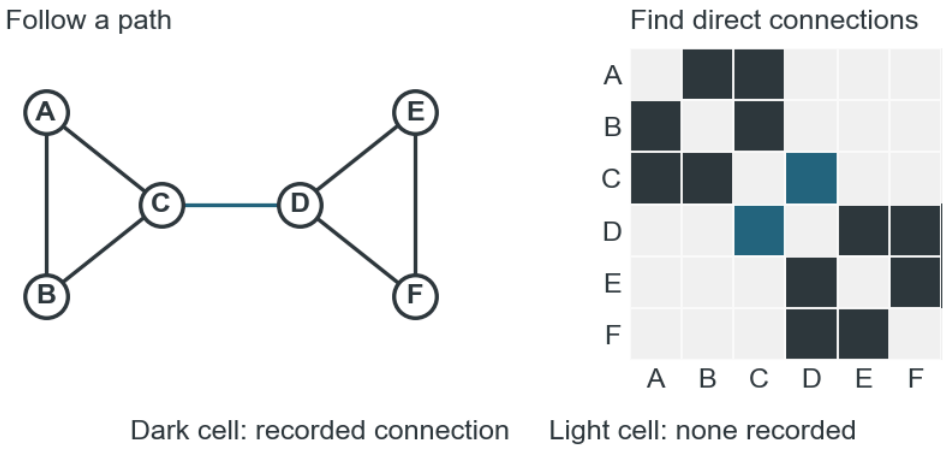


Figure 19: An undirected six-node network beside its adjacency matrix. The C–D bridge is blue in both views; other recorded edges are dark gray.

Follow the path from A through C and D to F in the node-link view; use row C of the matrix to find its direct neighbors. A light cell means no connection is recorded in this invented dataset.

When adding expression or activity to a network, keep node positions fixed across conditions and use color to show the measured quantity. Put each condition in a separate panel so readers can compare states side by side. With animation, they must remember earlier frames. [7].

Set intersections

For a few simple overlaps, a labeled Euler or Venn diagram can be sufficient. Use UpSet when there are many sets or when readers need to compare intersection sizes. Its bars provide a common baseline for counts, and the membership matrix below identifies the set combination for each bar. Include set totals when the size of an intersection needs that context. Order intersections by size or by the combinations relevant to the question. [8].

Distinguish exclusive combinations from inclusive overlap. "A and B only" excludes members of C; "A and B" may include them. In an exclusive UpSet display, active dots identify the included sets and inactive dots exclude the others. A heatmap of pairwise overlaps can show overlap counts, but it does not identify which elements make up a three-way intersection. Define the counting rule, and identify omitted or filtered intersections. [8].

Reading an exclusive intersection

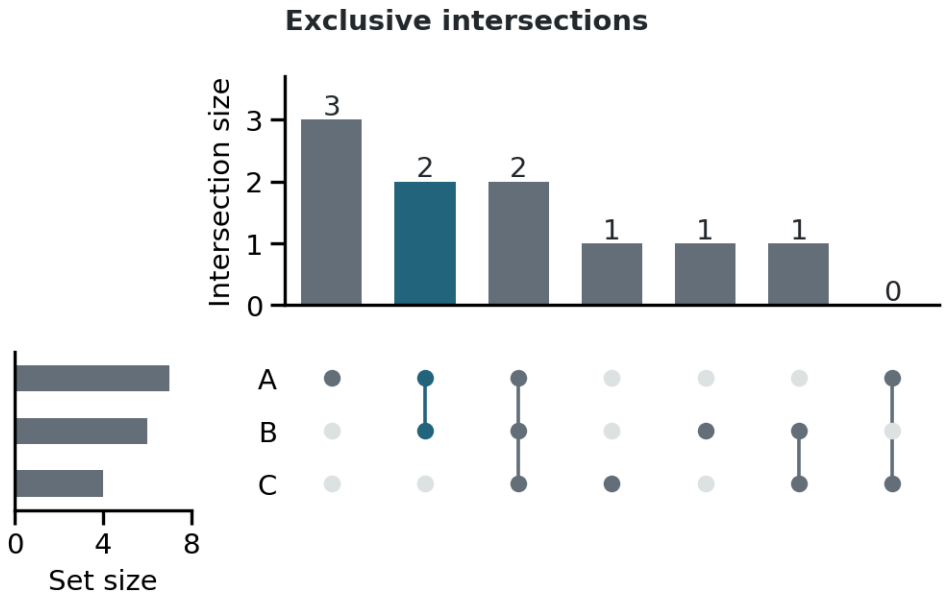


Figure 20: Each upper bar counts the combination marked directly below it. The blue bar counts two members in A and B but not C; the next bar counts two members in all three sets. Thus the inclusive overlap of A and B is four. Side bars show each set's total membership.

Multivariable displays

For a multifactor experiment, assign the main comparisons to rows and columns before adding color, size or shape. For example, a grid can put cell types in rows and treatments in columns, with the same signaling diagram in each cell. Measured responses then appear at their corresponding pathway nodes. This is useful when the pathway explains the results; a conventional plot is simpler when readers only need to compare magnitudes. [9].

Use a scatterplot matrix to inspect pairwise relationships among variables. Use parallel coordinates when the profile of an individual sample across several variables matters: each sample becomes a line joining its values on parallel axes. Axis order and scaling affect the patterns, so keep comparable measurements on comparable scales and label any separate scaling. Too many overlapping profiles or repeated categorical values can make individual lines impossible to follow. [10].

Cycles and trajectories

To compare repeated cycles, align corresponding time intervals, then examine when the response begins, when it peaks or how long it lasts. Keep elapsed-time units when duration matters. Rescaling cycles to a common phase can clarify their shape, but hides differences in cycle length. Aligned linear plots make precise comparisons easier than radial plots. When a plot's axes already show other quantities, connect successive observations and indicate their order to show a trajectory through time. Use separate panels for selected times when readers need to compare positions at those times. [11].

When responses span very different ranges, small multiples can show each pattern more clearly. Use shared scales for magnitude comparisons. If individual scales are necessary, make the ranges conspicuous and consider retaining a common-scale overview beside the enlarged detail. [12].

Explanatory diagrams

Show the experimental design

Sketch the starting material, the main operations and the resulting measurements. Keep the same objects recognizable from one step to the next, and label the operation that changes them. If a sample splits into control and treatment branches, align corresponding steps so the difference between

branches is easy to locate. Show collection points on a proportional timeline when their spacing matters; otherwise a sequence of labeled steps is sufficient. [13].

Include biological structures when they help explain the procedure, such as a compartment boundary that locates a reaction. Use a text label if readers might not recognize a pictogram. A schematic heatmap can stand for an analysis step, but label it as a schematic so it cannot be mistaken for a result. [14]; [13].

Experimental sequence and biological explanation

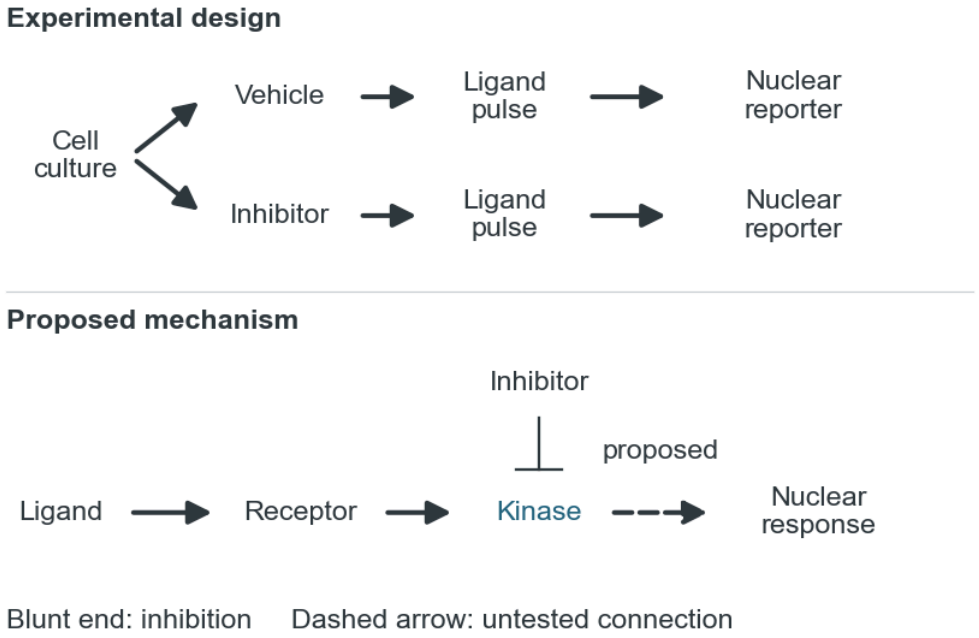


Figure 21: A hypothetical split-culture experiment and a separate signaling diagram. Experimental arrows connect culture, intervention, ligand pulse and reporter measurement. Mechanism arrows connect ligand, receptor, kinase and a proposed nuclear response.

The experimental arrows mark a sequence of operations; the mechanism arrows represent biological effects. The dashed edge identifies the untested connection.

Mechanisms and pathways

Define each type of connection in the figure or legend. A plain line can show an association, an arrow a directed process and a blunt end inhibition. To connect a label to an object, use a line without an arrowhead and keep it visually distinct from the biological connections. Otherwise, readers may mistake it for another reaction. [15].

Arrange the main pathway in a consistent reading direction, with branches and feedback loops routed around it. Place nodes on a simple grid and use consistent connection points to reduce crossings. Group components by compartment or function where that grouping helps explain the process. Keep node shapes consistent rather than stretching each one to fit its label. [16].

Preserve spatial relationships when they are part of the explanation: a membrane crossing, a cortical layer or the location of a synaptic contact may be essential. When location is irrelevant, move nodes to make the connections clearer. Avoid drawing precise anatomical routes when only connectivity is known. Mark proposed or indirect steps explicitly, using a defined dashed line or a short label. [17].

Check what the drawing implies beyond its labels: a location chosen for convenience can assert a specificity that the evidence does not establish. Rough sketches help expose such assumptions before polishing. [19]. For example, placing a protein inside a nucleus asserts more than saying that it is intracellular. Making one object much larger can suggest abundance or importance even when size is only a drawing convenience. Use a less specific arrangement or explain schematic size and placement

when those properties are unknown. Likewise, an arrow from an observed binding event to gene expression needs its own evidence; the binding result alone does not establish that causal step.

Overviews and graphical abstracts

Sketch a sequence for a process, parallel views for a comparison, or a focused scene for a spatial mechanism. Reserve circular layouts for cycles. Compare rough alternatives before polishing the artwork. [18].

A graphical abstract about a finding should show the finding, not just the workflow used to obtain it. A methods contribution may instead need the distinctive operation and the capability it enables. Include a result plot only when readers can understand it at the displayed size. A procedural overview can use a simplified symbol for a measurement step; a quantitative result still needs enough labeling to identify its quantities and groups. [13]; [18].

Emphasize the main action and simplify the surroundings. Keep illustrations consistent in detail, stroke and color meaning. Label unfamiliar objects directly, replace decorative icons with text, and remove repeated steps that add no information. If using an icon library, check the selected asset's attribution and publication conditions. [14]; [18].

Show a rough version to a colleague before refining it. Ask what they notice first, where they start reading and what each arrow means. Then inspect the final version at its actual display size: a diagram that works across a full page may need fewer elements when used as a thumbnail. [18].

Final display check

- Can readers make the intended comparison or follow the main path?
- Are values, ordering, scale changes and omitted data identifiable?
- Do positions, connections and symbols mean what the figure suggests?
- Are proposed relationships distinguishable from established ones?
- Are the essential labels readable at the intended size?

Original illustrations with invented data.

References

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Preparing figures for publication

A figure can look clear on a large editing canvas and become difficult to read when it is reduced to a journal column. Assemble it at its intended publication width. Arrange panels around the comparisons readers need to make, then adjust labels, spacing and graphic weights at that size. [1].

Set the publication dimensions

Check the journal's figure widths before arranging panels. Set the canvas to its intended single-column or full-width size, and allow enough height for the content. At that size, you can judge whether labels remain legible and panels have enough space. [10]; [1].

The complete figure canvas includes panels, labels, keys and margins. Within each plot, the data rectangle bounded by the axes can be substantially smaller. A sparse comparison with short labels may therefore need much less space than a plotting program's default canvas. Final-size balance matters: readable text should accompany readable marks, rather than overwhelm a tiny plotting area. [1].

As a practical starting approach, try a compact data rectangle while keeping the intended final-size typography and line weights stable. Replot at the candidate dimensions rather than shrinking a finished graphic, which also shrinks its text. Shorter physical axes do not require narrower numerical limits or removal of observations. Check whether the changed aspect ratio alters the apparent slopes or shapes; retain matched geometry where panels depend on it for comparison. This approach combines the final-size and layout principles here; it does not prescribe a universal minimum panel size.

For example, three groups with short names and sparse observations may fit comfortably in a small panel with 7 pt labels. Long condition names or dense paired trajectories may require more width, height or another arrangement even at the same text size. Stop compressing before points, intervals or curves become hard to distinguish. Labels that merely avoid touching are not sufficient evidence that a plot is readable.

Arrange panels

Sketch the panel arrangement before polishing individual plots. Include space for axis labels, panel letters, keys and annotations. A few rough alternatives make it easier to compare a row, a grid or a larger central panel with smaller supporting panels. [2]; [3].

Compare alternatives using the same data and intended reading size. Look first at what attracts attention: the relevant comparison should be easier to find than the grid, key or decoration. Remove unnecessary formatting variation, but keep useful summaries and explanatory labels. A restrained figure can still show rich evidence; restraint concerns how that evidence is organized and emphasized. [9]; [1].

Build that hierarchy from the scientific question. Choose a common-axis comparison when readers need to judge small numerical differences; separate views when several simultaneous encodings make them search for a combination of color, size and shape. In a process figure, keep an entity's appearance consistent from one step to the next. Repeated labels, boxes and leaders can often be replaced by a shared heading, alignment or whitespace. Check after each simplification that the necessary identities, comparisons and sequence remain clear. [9]; [4].

Group comparisons

Place panels that readers should compare beside one another. For a treatment comparison across several assays, give each assay a row and each condition a column. If the question instead concerns several measurements within each condition, group those measurements together. Use row and column headings to make the grouping clear. [4]; [1].

A figure can also explain a sequence: an experimental design, its results and a summary. Arrange these in a consistent reading order, usually left to right and then down. The layout need not reproduce the order in which experiments were performed. Give a complex plot or image enough room to be read; panels need not all have the same dimensions. [3]; [9].

Group the same panels in two ways

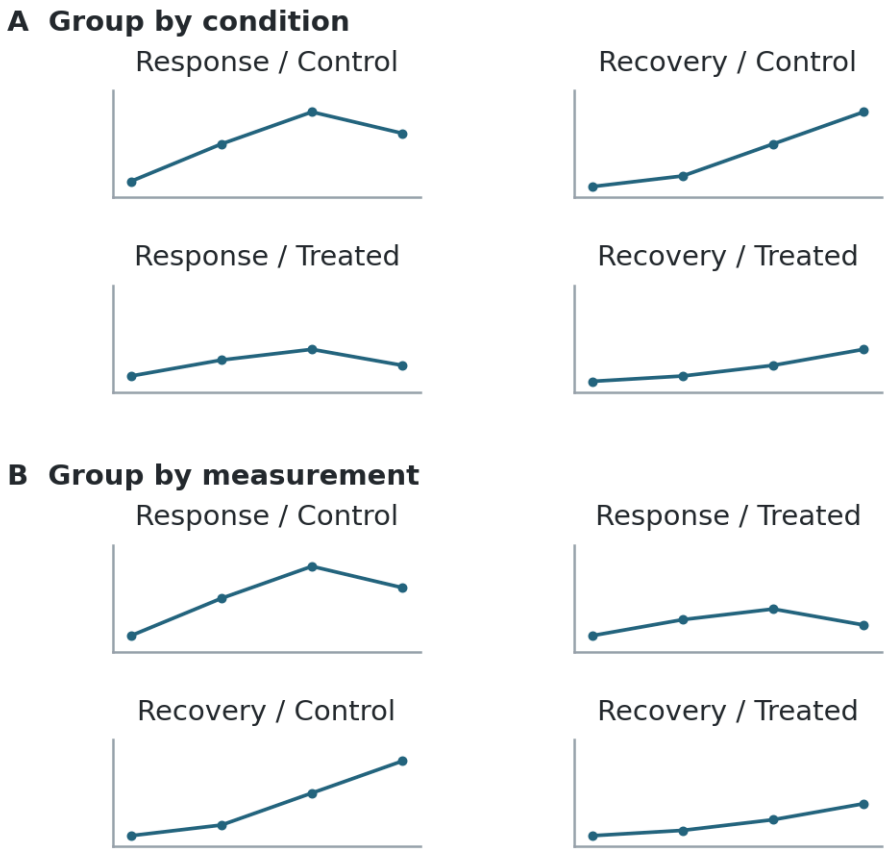


Figure 22: The same four schematic plots in two arrangements: A groups measurements within each condition; B places control and treated side by side for each measurement.

Use B to compare treatment effects on each measurement; use A to examine response and recovery together within each condition.

Allocate space

Use alignment guides to line up panel edges and keep gaps consistent. Leave more space between distinct groups of panels than between panels belonging together. The spacing can show which panels belong together without enclosing every group in a box. [5].

Allow for long labels and keys when deciding panel widths. If they become cramped, change the arrangement or give the figure more space before reducing the text. Preserve the proportions of images and exported plots when resizing them. Stretching one dimension changes the apparent shapes in images and slopes in plots. [1]; [5].

Align axes and annotations

Align plotting areas

Align the plotting areas of comparable panels: their baselines, axis endpoints and tick positions. Aligning the outer edges of imported graphics may leave the actual axes offset because their label

margins differ. Use the same axis lengths, ranges and ticks when readers need to compare magnitudes across panels. [1]; [6].

Different ranges can help when the aim is to examine the shape of each response and their magnitudes differ greatly. In that case, show the scale on every panel and avoid a shared axis label that suggests a common range. Do not remove observations merely to make panels fit the same scale. [1]; [9].

Keep axes and grid lines lighter than the data, but visible. A few horizontal guides help readers compare heights across a wide plot; a reference line at zero or at no change may be more useful than a full grid. [6]; [1].

Use concise tick labels and put their common unit or multiplier in the axis title. Avoid repeating a long number prefix at every tick. For aligned panels with an identical scale, one clearly shared axis can replace repeated labels. Keep separate labels where there is any doubt about which scale applies. [6].

Place labels and keys

Give panel letters the same size, weight and position throughout the figure. Keep them distinct from the panel content without making them the most prominent marks. Short panel titles can identify an assay or condition; a row or column heading can replace repeated wording. [1].

In a densely labeled diagram, arrange labels around the graphic and use short leader lines where a label's target would otherwise be ambiguous. Keep the leaders from crossing or covering important features. [7].

Keep group names and ordering consistent across panels. A color or symbol should identify the same group wherever it appears. Use one shared key when its entries mean the same thing everywhere; keep it near the panels it explains. Put the information needed to identify a curve or image on the figure, and the longer methodological explanation in the figure legend. [1]; [9].

For a small table within a figure, align text to the left and comparable numbers to the right or by their decimal points. Use consistent precision and put units in the headings. Space between columns and a rule beneath the header are often sufficient; subtle row shading can help readers follow a wide table. [1].

Set text and graphic weights

Use one typeface and a small set of sizes: one for ordinary labels and a distinct treatment for panel letters or headings. Reserve bold and italic for a clear purpose, including the scientific conventions for gene names and mathematical notation. When combining plots from different programs, standardize their typography after placing them at the intended size. [8]; [1].

Judge text, symbols and line weights together. Large axis labels can dominate a plot whose data points are small and hard to see. Thick lines can obscure nearby observations; very thin ones can disappear on reduction. Check subscripts, superscripts, error-bar caps and dashed lines as well as the main labels. [1]; [9].

Practical starting settings

These are practical manuscript starting preferences, not universal legibility thresholds or journal requirements. Set them at the final publication size, then adjust for the typeface, mark density, reading medium and current destination specifications. A figure that remains difficult to read needs more space or less simultaneous detail even when it meets these numbers. [10]; [1].

Element	Starting setting
Ordinary labels, ticks and keys	Arial or Helvetica, 7 pt; avoid text below 6 pt.
Panel letters	8 pt, bold and upright.
Plot lines and axes	At least 0.5 pt; avoid hairlines below 0.25 pt. Increase weight where the line must stand out.
Sequence labels	A monospaced font, such as Courier, when character alignment matters.
Figure dimensions	Set the canvas to the journal's single-column or full-width dimensions before arranging panels. Let the content determine height.

Resizing a plot also resizes its text and marks

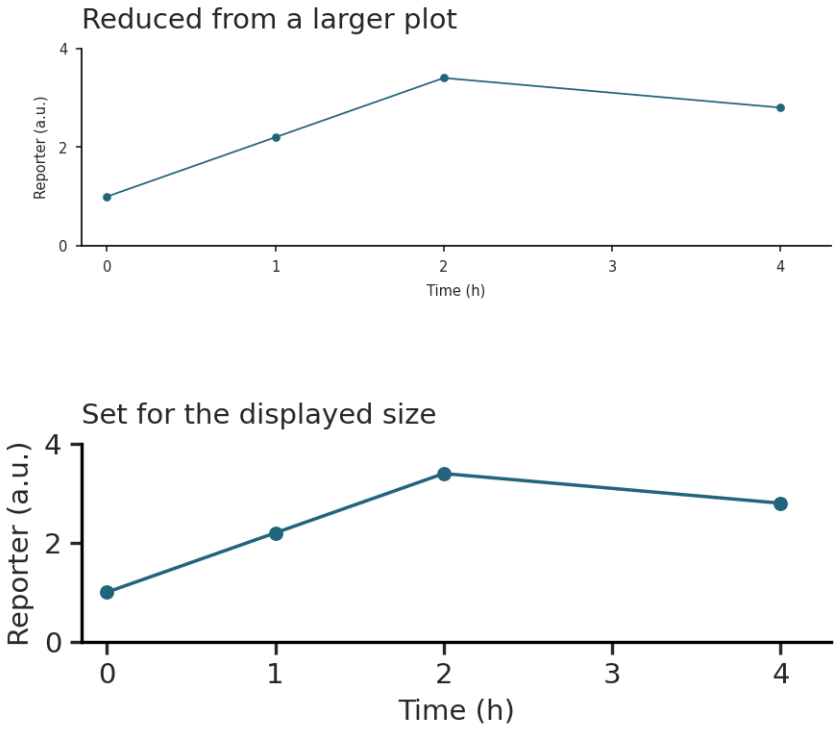


Figure 23: The same invented reporter trace at the same panel size. The upper plot retains text and marks from a larger graphic reduced by half; the lower plot is set for the displayed size.

Set the plotting canvas to the required width before export, or account for the reduction when choosing sizes. If readable labels leave too little room for the data, enlarge the panel, rearrange the figure or split it. Judge this at reading size rather than zoomed in. [1].

Check the intended journal's current figure instructions for permitted dimensions, type sizes and line weights. A slide usually needs larger text, stronger lines and fewer simultaneous details than a manuscript figure, because the audience sees it briefly and from a distance. Prepare a separate slide version where necessary. [9].

Export and inspect

Use a font with the required Greek and mathematical characters, and check for substitutions in the exported file. Keep RGB while designing unless the destination specifies otherwise; inspect any required color conversion before submission. [10].

Choose formats and resolution

Keep plots, text and diagrams as vector elements when the destination accepts them. They can be resized without pixelating and remain easier to edit. Photographs and microscopy images remain raster images even when placed inside a PDF or SVG. A dense scatterplot may be easier to display as a raster layer with vector axes and labels. [1].

For raster export, use a lossless format such as PNG or TIFF for figures containing text, lines or sharp boundaries. JPEG compression can produce visible artifacts around these features. Saving a low-resolution screenshot as a PDF does not restore vector lines or recover image detail. [1].

Keep an editable master and export submission copies from it. Where plots are produced by code, correct labels or data in that code and regenerate them. Use the assembly file for arrangement and annotation, rather than manually moving data points or redrawing results. Embed fonts when the format permits, and reopen the export to check for substituted characters, missing symbols or changed spacing. [1]; [9].

Raster resolution depends on the number of pixels and the size at which the image is placed. Calculate it from:

Pixels needed = final width in inches × required pixels per inch.

For example, a 4-inch-wide raster figure requires 1,200 pixels across at 300 pixels per inch. Use the destination's required resolution in this calculation. [1].

Inspect transparency, gradients and any raster layers in the exported file. Retain the original images and editable figure when a journal requires a flattened submission copy. [1]; [9].

Check the exported file

Open the exported figure at its intended physical size, ideally also in the assembled manuscript. Inspect it closely for clipping, missing marks and raster artifacts, then return to normal reading size. Ask a colleague to identify the main comparison and explain the panels without your spoken guidance; revise any grouping or labels they misread. [9]; [1].

Before submission

- Related panels sit together, with a clear reading order and enough room for labels.
- Comparable plots have aligned axes and consistent scales; any differences are explicit.
- Panel letters, group names, keys and the caption agree.
- Text, symbols and lines remain readable at the final size; color and accessibility have been checked.
- The exported file has no clipped labels, substituted fonts or degraded images and meets the destination's specifications.

References

Original teaching graphics; the reporter trace is invented.

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Chapter 4

Writing a paper

Planning the paper

Which finding should the paper lead with, and which experiments support it? A working title and a short statement of the contribution can help you decide. Check them against the results before developing the figure plan and manuscript outline.

Define the contribution

Identify the reader and question

The paper needs a question to answer or a problem for the new method to solve. Write that question with the intended readers in mind: what do they already know, and what background will they need to understand it? [1]; [3]; [2]. The relevant literature should establish what is known, what remains unresolved and why resolving it matters. [12]; [5]; [6].

State what the study establishes

Write a working title and a one- or two-sentence contribution statement that names the main finding and what it adds to existing knowledge. Specify the system or conditions when they limit the conclusion. [1]; [4]. Distinguish the observation from its interpretation: a measured response may support a functional role without identifying the mechanism. Choose the wording accordingly. [11].

A computational method and the biological insight it enables may belong in one paper. If two groups of results answer independent questions, consider separate papers. [1]; [2].

Build the argument from the evidence

Connect claims to results

List the proposed claims in a table, with the supporting result and planned figure or panel beside each one. Add the assumptions on which the claim depends and any limits to its interpretation. Include results that challenge it and questions that remain unresolved. [3]; [7]; [8]. For a quantitative claim, include the effect estimate and its uncertainty. Note whether the analysis tested a planned prediction or suggested an explanation during exploration. [11].

Example: defining the contribution

Brake is a fictional bacterial defence system inspired by the antiviral gene-synthesis discovery of [15]. Genome annotation reveals a reverse transcriptase and an RNA template, but no conventional defence-effector gene. The researchers ask what this system produces and how it restricts phage infection.

Working title: An RNA-templated gene arrests bacterial growth and restricts phage propagation

Working contribution: Brake assembles a protein-coding gene from an RNA template. Its protein product arrests bacterial growth and restricts phage propagation, revealing a route to defence in which the functional gene must first be assembled.

Claim inventory

Proposed claim	Supporting analysis/result	Planned figure/ panel	Assumptions	Interpretation limits
Brake restricts phage propagation.	Lower infectious phage yield with intact Brake than with empty vector or inactive reverse transcriptase.	Figure 1B–C	Matched infections and comparable titre measurements.	Establishes restriction in the tested host.
The RNA template supplies a protein-coding DNA product.	Repeat-junction sequences, DNA measurements and protein detection; template-deletion control.	Figure 2A–D	Product identification distinguishes newly assembled DNA from the construct.	The infection signal that activates the pathway is unknown.
The protein mediates growth arrest and defence.	Blocking translation preserves DNA production but removes growth arrest and phage restriction; separate protein expression restores both.	Figure 3A–C	The translation change leaves upstream RNA and DNA production intact.	The protein's molecular target is unknown.
Activity differs among tested phages.	Restricted output of phages A and B; phage C remains productive.	Figure 4A	Infection conditions permit each phage to replicate in the control.	Three phages do not establish the full host–phage range.
Recovery after infection remains unresolved.	Infected cells stop growing during observation; subsequent recovery was not measured.	Figure 4B	Tracking captures the infected cells of interest.	Growth arrest does not establish survival or irreversible death.

The third row identifies the decisive control. If blocking translation also stopped DNA production, loss of defence would not distinguish a protein effect from a DNA effect.

Resolve gaps and conflicting findings

An incomplete argument may need a clearer explanation, a different analysis or additional evidence. Use the claim inventory to locate the missing step before planning more experiments. In Brake, checking that DNA production survives the translation-disrupting change is necessary to attribute defence to the protein. Identifying the protein's molecular target would address a further question. [1]; [4].

When findings disagree, examine the conditions, measurements and analytical choices. If the disagreement persists, revise the interpretation and discuss it in the main text, even when the detailed displays are in supporting material. [2]; [3].

How to resolve incomplete arguments

[1]; [4]; [2]; [11].

Problem	How to recognize it	Next action
Missing explanation	The evidence is present, but a link in the reasoning is unstated.	Explain why the comparison answers the question. Note the missing explanation beside the relevant claim, then include it in the manuscript's paragraph outline.
Missing essential evidence	The claim depends on an untested assumption or missing control.	Obtain the necessary evidence or narrow the claim.
Contradictory findings	Results support different answers to the same question.	Check conditions, measurements and analyses. If disagreement remains, include it and revise the interpretation.
Inconclusive findings	The estimate or assay is too uncertain to answer the question.	State what remains unresolved. If the central claim needs that answer, improve the evidence or narrow the claim.

Arrange the figures

Give each figure a purpose

Group results from the claim inventory around the question each figure will answer. Write its title early: a finding-based title helps you judge whether the panels support one conclusion, while a descriptive title suits a design or unresolved comparison. [7]; [8]; [3].

For an unfamiliar procedure or a comparison across several stages, include a schematic showing the starting material, important steps and measurement. Align corresponding stages when comparing conditions. Use recognizable schematic symbols and keep them distinct from the images or plots reporting observations. [13].

Plan the comparisons within each figure. Put the treatment and reference condition together; use a separate control panel when you need to establish that the intervention worked. Pair an illustrative observation with a population summary when readers need to judge how typical it is. Keep group names consistent across figures. Rough plots and panel notes are sufficient for this stage; refine the layout once the scientific content is settled. [5]; [7].

Choose the figure sequence

Order figures by the questions they answer. First establish the problem or observation; then present the tests needed to explain or resolve it. Arrange panels within each figure so the relevant comparisons can be read together. [1]; [6].

Example: planning the figures

The experiment log starts with a growth defect during protein expression, followed by phage-yield assays, sequence analysis, DNA measurements and tests of the translation-disrupted construct. The figure plan changes that order: first establish defence, then explain how it works.

Figure 1 — Brake restricts phage propagation

Does Brake reduce phage output?

- **A:** Introduce the defence region and the control constructs.
- **B:** Compare infectious phage A yield with empty vector, intact Brake and inactive reverse transcriptase.
- **C:** Show host growth during infection alongside uninfected controls.

Figure 2 — Brake produces a protein-coding DNA intermediate

What does the reverse transcriptase make?

- **A:** Map the RNA template and the coding sequence created across DNA repeat junctions.
- **B:** Compare DNA products before and during infection.
- **C:** Show expression of the encoded protein during infection.
- **D:** Test the RNA-template deletion.

Figure 3 — The Brake protein mediates growth arrest and defence

Which product carries out the defence?

- **A:** Introduce the translation-disrupting change and verify that DNA production remains intact.
- **B:** Show loss of growth arrest and phage restriction.
- **C:** Test restoration by expressing the protein separately.

Figure 4 — Brake responses across phage challenges

How do the tested phages differ in restriction and infected-cell growth?

- **A:** Compare phage output for A, B and C.
- **B:** Track growth of infected cells, retaining the observation window in the display.

Starting with protein expression would leave readers wondering why that protein was tested. Placing it after product identification makes it a test of the proposed effector.

Decide what belongs in supporting material

Keep the evidence essential to the main contribution in the paper, including controls on which its interpretation depends. Supporting material can hold extended analyses and additional checks. A result from another system belongs in the main paper when the claim depends on its generality. [4]; [7].

Read the figure titles in sequence and ask a colleague to explain the argument from the figures and provisional legends. If a figure needs several unrelated explanations, split it; if two displays establish the same point, consider combining them. [7]; [5].

Develop the outline

Section purposes

Build the manuscript outline around the figure plan, using the section purposes below. [6]; [1].

Section overview

Component	Purpose
Title and abstract	Identify the contribution and summarize the question, approach, main findings and meaning.
Introduction	Establish the relevant knowledge, explain the unresolved question and introduce the study.
Methods	Describe the design, materials, procedures and analyses needed to assess and reproduce the work.
Results	Present a logical sequence of findings, supported by figures, tables and the controls needed to interpret them.
Discussion	Answer the research question, relate the findings to prior work, assess alternatives and limitations, and explain the contribution.
Supporting material	Supply additional methods, data and analyses that support the paper while keeping its main argument easy to follow.

Adjust the outline to the journal's format: Methods may appear later, Results and Discussion may be combined, or a separate conclusion may be required. [6]; [9].

Start the outline with the working title and contribution statement, then place the figure findings under Results. Work outward from those findings: list the background needed to understand them, the procedures to describe in Methods and the interpretations to examine in Discussion. [1]; [7].

Check the journal's length and display limits before developing the manuscript outline. Use a rough word and figure budget to allocate space. [14].

Plan paragraph purposes

Outline what each paragraph will establish and which results or sources it will draw on. A figure may require several paragraphs if readers need to follow distinct steps in the reasoning. [1]; [8]. For Results, specify the finding and the comparison that supports it; leave sentence-level development to drafting. See building a Results paragraph. [5]; [6]; [12].

Read the paragraph statements in order. Does the reader have enough information to understand each new comparison? Combine repeated points and divide paragraphs that try to explain several things at once. Introduce each experiment through the question it addresses; you need not repeat the motivation for the whole study. [12].

Example: developing the outline

Manuscript outline for the Brake study

Introduction

1. Explain why understanding bacterial defence requires identifying the product that inhibits infection.
2. Introduce the puzzle: Brake contains a reverse transcriptase and an RNA template but no recognizable effector gene. Explain why identifying its products could resolve this gap.
3. State the question: what does Brake produce, and how does that product restrict phage infection?

Results

1. Intact Brake reduces infectious phage A output relative to the controls; host growth measurements distinguish infection-associated effects from growth without infection (Figure 1).
2. RNA-derived DNA repeat junctions create a coding sequence. Use the sequence analysis and RNA-template deletion control to explain the product's origin (Figure 2A, D).
3. The DNA product is present before infection; double-stranded DNA and the encoded protein are detected during infection (Figure 2B–C).
4. Preventing translation removes defence despite continued DNA production; separate protein expression restores growth arrest and restriction (Figure 3).
5. Activity varies among the three tested phages, and infected-cell tracking shows growth arrest without establishing recovery (Figure 4).

Methods

1. Identify the host, defence constructs, phages and infection conditions.
2. Describe DNA-product identification, sequence analysis and protein measurements.
3. Explain growth measurements and infectious-titre assays, including independent cultures, technical measurements and matching by experimental day.
4. Describe cell tracking and the analyses behind each comparison.

Discussion

1. Explain how RNA-templated gene assembly supplies a defence protein, connecting product identification with the genetic tests.
2. Contrast this organization with a defence protein encoded by a conventional genomic gene. Identify the literature needed to support that comparison in an actual manuscript.
3. Explain why growth arrest leaves cell recovery unresolved and why the protein's molecular target requires further work.
4. Close with the contribution: a functional defence gene can emerge from RNA-templated DNA assembly rather than exist as a conventional coding sequence in the genome.

Figure 2 supports two paragraphs: one identifies the DNA product; the next connects protein production to infection. The following paragraph tests the protein's function. Product detection alone would not establish that role.

You can plan Methods alongside Results, noting software versions, analysis choices and experimental details while they are available. Check that every reported analysis has a corresponding method. [7]; [2].

Choose where to start drafting

If the comparisons are settled, starting with Results and Methods lets you describe work you already know well. If you are unsure what the paper contributes, try writing the title and Introduction first. If you have many results but no clear order, return to the figure plan. These are different ways into the

draft; the finished paper's reading order need not determine where you start. See Drafting and revising for starting a passage and organizing writing sessions. [9]; [5]; [3].

Plan the writing with coauthors

Agree the plan and responsibilities

Share the contribution statement, rough figures and manuscript outline before dividing the writing. Ask coauthors to challenge the claims and identify missing evidence or alternative interpretations. Agree on the main argument before authors develop separate sections. [4]; [3].

Assign a writer and a scientific checker for each part, set dates and choose who will integrate the manuscript. That person should reconcile terminology, transitions and repetition across contributions. Keep one current version with recoverable history, and record decisions that affect other sections. [10]; [3].

Assign data and metadata preparation early, allowing time for repository queries and corrections alongside the writing. [12].

Agree authorship and credit

Discuss who should be an author and the proposed order before drafting, then revisit the agreement when contributions change. Use the target journal's criteria and relevant disciplinary or institutional guidance. Record the reasons for the order, including any shared-author designation, so that later discussion can refer to the work each person actually did. [10]; [16].

For a journal following ICMJE, a substantial contribution to the research alone does not complete the authorship requirements. Contributors must also help draft or critically review the paper, approve the version for publication and accept responsibility for addressing concerns about the work. Give eligible contributors the manuscript and enough time to participate; do not exclude them by withholding that opportunity. [16].

Prepare the contribution statement with the authors. If the journal uses CRediT, its roles describe the work performed; they do not decide who qualifies as an author or the order of names. Check the statement with each contributor, agree who will handle correspondence with the journal, and identify other contributions to acknowledge. [17]; [16]. For a worked credit decision and guidance when contributions are disputed, see Contributions, authorship and order.

Organize feedback

Plan an early review of the contribution and figures, then a later review of the written explanation. Ask a domain expert to challenge the inference and a colleague less familiar with the project to identify missing context. Agree these roles and review dates with the coauthors. The chapter Drafting and revising explains how to request and resolve their feedback. [1]; [2]; [8].

Before drafting

- Can a reader state the question, contribution and conditions without an oral explanation?
- Can each retained claim be traced to supporting evidence, with important assumptions and limits identified?
- Can readers find the controls needed to interpret each finding, including comparisons that remain inconclusive?
- Do paragraph purposes connect the figures to the section plan, with each reported analysis assigned a method?
- Have coauthors agreed who will write and check each section and how contributions will be credited?

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Titles, abstracts and introductions

A reader scanning search results needs to recognize what the paper is about and what it adds. The title, abstract and Introduction explain this at increasing depth. Draft the title and abstract from the main contribution identified in Planning the paper, then revise them as the manuscript develops. [5]; [6].

Three different jobs

Component	What the reader needs	What to include
Title	Is this study relevant to me?	The subject and a precise finding or scope.
Abstract	What question was answered, and what was learned?	Enough context, approach, results and interpretation to understand the contribution independently.
Introduction	Why ask this question, and how will the study address it?	Selected background, the unresolved problem and a clear account of the study's purpose.

The examples continue the fictional Brake study, inspired by Tang and colleagues (2024) [15]. Brake assembles an RNA-derived gene whose protein product arrests bacterial growth and restricts phage propagation.

Choose an informative title

Identify the subject and contribution

Build the title around the biological subject and the main finding. For a methods paper, name the capability the method provides. Include the study system or condition when leaving it out would make the finding appear more general than it is. [1]; [2].

Use recognizable search terms and spell out unfamiliar abbreviations; secondary technical details can wait for the abstract. Replace phrases such as "new insights" with the finding itself. Choose a verb that matches the evidence. "Is associated with" reports a relationship; "activates" claims an effect, and "directly activates" makes a further claim about how it occurs. [3]; [4]; [2].

Choose a descriptive or finding-based title

A finding-based title works when one conclusion captures the paper's main contribution; a descriptive title may better suit a paper that characterizes several related aspects of a system or resource. Draft both, then ask whether the promised finding or scope is what the paper actually delivers. [3]; [1].

Example: choosing a title

Brake produces an RNA-derived DNA intermediate whose protein product arrests bacterial growth and restricts phage output. The experiments identify the protein's role but leave its molecular target and the recovery of infected cells unresolved.

Too vague: Characterization of the Brake defence system

The system's name tells an unfamiliar reader little about the biological finding.

Overclaiming: An RNA-templated gene saves bacteria from viral infection

The study measures phage restriction and growth arrest, not recovery of individual infected cells.

Finding-based: An RNA-templated gene arrests bacterial growth and restricts phage propagation

Use this title when the protein's origin and function are the paper's central findings.

Descriptive: RNA-templated gene assembly in the Brake antiviral defence system

This suits a paper organized around several properties of the system. It conveys less of the functional result, so the abstract needs to supply it promptly.

Draft several titles and ask a colleague what each leads them to expect. Revise any wording that suggests a different contribution from the paper's. [1]; [2].

Write an abstract that stands alone

Connect the question, findings and implication

An abstract should explain the question, what the study found and why that finding matters. Include enough background to understand the question and enough of the approach to understand the evidence. The Nature summary-paragraph guide illustrates one way to arrange these elements. [1].

The Nature summary-paragraph structure

Stage	Purpose	Sentences in Nature's guide
1. Broad context	Orient scientists outside the immediate field.	1–2
2. Specific background	Supply the knowledge needed to understand this study.	2–3
3. Problem	Identify the unresolved question.	1
4. Main result	State the central answer clearly.	1
5. What the result adds	Explain the finding through evidence and comparison with prior knowledge.	2–3
6. Wider meaning	Connect the answer back to the larger scientific problem.	1–2

Nature (2019) [9] also allows an optional 2–3 sentences of broader perspective when the editor judges that they improve accessibility. The allocations describe that guide; use the target journal's current word limit and format.

Draft the question and answer first, then add the background an unfamiliar reader needs. [6]. Test whether the opening sentence helps explain the question: if deleting it leaves the abstract just as clear, it may be too general. [7]; [11].

Give the main finding enough space for its supporting comparison. Briefly introduce an unfamiliar experiment before stating its result. For a familiar comparison, begin with the finding and include the method in the same sentence. [8]; [1]. Nature's annotated example introduces its assay with the supporting results, without a separate Methods block. [9].

End by explaining what the finding adds. A biological study might identify how a process works; a methods study might make a previously difficult measurement possible. Keep this closing statement within what the study demonstrated. [9]; [4].

Example: giving an abstract a clear argument

Both drafts describe the same fictional Brake study. The numerical comparison is the illustrative dataset developed in the reporting chapter.

Draft organized around procedures

> We tested the bacterial defence system Brake using phage infections, DNA sequencing and protein-expression experiments. Intact Brake reduced infectious phage A yield 79-fold relative to the empty-vector control. Repeated RNA-derived DNA was detected, and infection was accompanied by double-stranded DNA and protein production. A translation-disrupting change preserved DNA production but removed growth arrest and defence. Separate protein expression restored both. Phages A and B were restricted, whereas phage C remained productive. These experiments addressed how a defence region encoding a reverse transcriptase and an RNA template, without an annotated effector gene, inhibits infection.

Revised abstract

Some bacterial defence regions encode a reverse transcriptase and an RNA template but lack an identifiable effector gene. How these regions inhibit infection is unclear. Here we show that the Brake defence system assembles an RNA-derived gene whose protein product arrests bacterial growth and restricts phage propagation. Intact Brake reduced infectious phage A yield 79-fold relative to the empty-vector control, with activity differing among three tested phages. Sequencing identified a coding sequence spanning junctions in the repeated DNA product; double-stranded DNA and the encoded protein were detected during infection. Blocking translation preserved DNA production but removed growth arrest and defence, both of which were restored by expressing the protein separately. These findings reveal a defence pathway whose effector gene is assembled from an RNA template.

The revised opening explains why the gene's origin is a puzzle before presenting the answer. The effect's magnitude remains; the full interval and replication details appear in Results.

Select details that make the result understandable

Select details that establish what was compared and what the comparison showed. Name the system and any method central to the contribution; summarize routine procedures with the finding. [3]; [11]. Give the principal effect's magnitude and reference group, adding uncertainty or sample information when interpretation or the journal's format requires it. A percentage without its reference group, or an improvement without its comparator, leaves the result ambiguous. [4]; [9].

Include a limitation if leaving it out would misrepresent the answer. For Brake, stating that activity differed among the three tested phages prevents the abstract from implying that Brake restricts every phage. Secondary controls and exploratory findings can remain in Results. [4]; [1]. Read the abstract without the paper beside it: can readers identify the methods and groups, and understand its terms and abbreviations? Use the same names throughout. [12]; [4].

Adapt to the format and word limit

Check the article's word limit, headings and required reporting elements before polishing. For a structured abstract, distribute the argument under the journal's headings, commonly Background or Objectives, Methods, Results and Conclusions. Make sure the Results answer the stated objective. [5].

If the draft is too long, first cut background that does not lead to the question, repeated claims and secondary procedural detail. Then combine related statements, checking that the main comparison remains clear. Avoid saving words through unfamiliar abbreviations or overloaded sentences. [6]; [4]. Use the structural outline to check whether background has crowded out the findings. Studies linking writing features with citation counts do not establish that changing abstract length makes a paper clearer or more influential. [14].

Some journals also request a separate significance statement for a broader audience. Explain the problem and the value of the advance with less technical detail, following the journal's instructions. Keep the approach and principal findings in the research abstract.

Give a methods paper its own emphasis

A methods abstract needs to explain what researchers can do with the method and how well it performs. Identify the limitation it addresses, such as an unreliable measurement or a time-consuming procedure, then explain what it changes. Give the benchmark or application that demonstrates its performance, including the conditions under which it was tested. Keep the method's name and recognizable technical terms prominent. If the study also makes a biological discovery, explain how the method enabled it. [10].

Build the introduction around the question

Select the background the reader needs

Use the sequence below to select the background and plan the paragraphs. [1]; [12].

A common introduction structure

Stage	Purpose
1. Scientific context	Establish the process or problem and why understanding it matters.
2. Relevant knowledge	Explain the findings, ideas or methods on which the study builds. Select what the reader needs for the next step.
3. Unresolved question	Show what remains uncertain or inadequate, and why resolving it would matter. State the question or hypothesis the study addresses.
4. Present study	Introduce the approach and, where appropriate, the principal answer and contribution. Prepare readers for the Results.

Combine these functions when they form one point: a paragraph describing a method can also explain the limitation your study addresses. Start a new paragraph when the point changes. [1]; [12].

Choose background in relation to the question, rather than trying to summarize the whole field. Explain what remains unresolved and what an answer would change. For a methods paper, this means identifying the measurement that is difficult or unreliable and why researchers need it. [13]; [4].

Use the literature fairly and precisely

Organize previous work by the points it establishes, using reviews for orientation and original studies for specific findings. Where disagreement motivates the study, check whether the earlier experiments asked comparable questions under comparable conditions. Place citations beside the claims they support, splitting a sentence if different references support different parts. [5]; [4].

Before claiming priority, search alternative terminology and examine the closest studies. If priority remains uncertain, describe the specific advance. [4]; [13].

Connect paragraphs and introduce the study

Connect paragraphs through the scientific reasoning. After describing a method's limitation, explain how the proposed method addresses it. After identifying an unexplained observation, introduce the question it raises. Reuse the same scientific terms so readers can follow the connection. In the Brake example, the missing effector leads to a question about the DNA product, followed by tests of the protein's function. [1]; [4]. A closing interpretation can explain why the next step is needed. Omit a closing summary that merely repeats the paragraph's opening. [12].

End with the study's question and approach, usually followed by its main finding. Follow the article's conventions if it calls for an aims-focused ending. Include a section-by-section roadmap only if the paper's organization needs explanation. [13]; [3]; [12].

Example: introducing a biological question

This short introduction develops the question raised by Brake's apparently missing effector gene. A real manuscript would need background citations and more context about the system.

> Bacterial defence relies on molecules that interrupt phage replication. A protein-coding gene provides a starting point for identifying and testing a defence effector. When a defence region contains no recognizable effector gene, the origin of the protective molecule remains unresolved.
>
> Brake contains a reverse transcriptase and a conserved RNA template, but no annotated effector gene. Its reverse transcriptase suggests that an RNA-derived DNA product may be involved. What does the reverse transcriptase produce, and how could that product contribute to defence?
>
> Here we combine sequence analysis with genetic perturbations to determine how Brake restricts phage infection. We find that repeated RNA-derived DNA forms a continuous coding sequence whose protein product arrests bacterial growth and restricts phage propagation. Preventing translation abolishes defence while preserving DNA production; expressing the protein separately restores it. These findings reveal a defence pathway in which the functional gene is assembled from an RNA template.

The final paragraph includes the genetic tests because detecting the product alone would not establish its role in defence. Phage range and the numerical effect can remain in the abstract and Results. The absence of an annotated effector gene is part of the fictional study's starting analysis; a real manuscript would cite the source of that annotation.

Check the opening against the paper

Read the title, abstract and final Introduction paragraph together, then compare them with Results. Check that they describe the same contribution and that the evidence answers the question posed. [4]; [1]. Trace numerical statements back to their comparisons, including the population, condition and reference group. [5].

Some repetition between the abstract and Introduction is necessary because each must make sense independently. Within either section, remove explanations that do the same job twice and check that each paragraph advances the argument. [12]; [4].

Ask a colleague from the broader field to explain the question and answer after reading the opening. Revise where their understanding differs from the intended meaning. [7]; [1].

Final opening checklist

- The title identifies the subject and gives a supported finding or clear scope.
- The abstract explains the question, approach, main result and implication independently.
- Quantitative statements name the comparison, and shortened passages retain the conditions needed to interpret the findings.
- The introduction supplies the background needed for a specific, worthwhile question.
- Citations support the precise statements made, and prior work is distinguished from this study.
- Paragraphs lead into one another, and the ending prepares readers for the Results.
- Terminology and claim strength agree across the opening, figures and main text.
- The chosen article format is satisfied, and an unfamiliar reader can explain what was learned.

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Methods, results and figure legends

Draft Methods from the experimental records and analysis code, Results from the figures, and legends alongside each display. As you write, check that all three describe the same samples, measurements and comparisons. A change to the analysis may require a correction in each place. [1]; [2].

The examples continue the fictional Brake study, inspired by Tang and colleagues (2024) [12].

Plan what readers need

Where details belong

Place details according to the question they answer. [3]; [4].

Where the information belongs

Component	Main question	Information to provide
Methods	How was the evidence obtained?	Design, materials, procedures, measurements, processing and analysis choices needed to evaluate and reproduce the work.
Results	What did the study find?	The question or comparison, principal finding, quantitative support and qualifications that affect its interpretation.
Figure legend	What am I looking at?	Figure topic, panel descriptions, experimental context, visual encodings and statistical details needed to read the display.

The same treatment may need three different descriptions: how it was applied in Methods, what it changed in Results, and which marks represent it in the figure legend. Give readers the context they need at each point, while describing the full protocol once. [2]; [5].

Choose how much detail to include

State which population was sampled and what was compared, even for a specialist audience. Define scores developed for the study. Before describing an unfamiliar analysis, explain the question it answers so readers know why you used it. [3]; [6].

Use prose for procedures and reasons, tables for repeated fields, and a schematic when several inputs follow different paths into an analysis. Keep exact settings in Methods. [4].

Supporting material can hold detailed procedures and extended analyses. Keep the evidence essential to the main conclusion accessible in the paper, with precise references to supporting displays and controls. [3].

Write reproducible Methods

Organizing Methods

Begin with the design and materials, then describe measurement, processing and analysis. Within a procedure, preserve the order of operations; across procedures, group shared methods so readers encounter each protocol once. [7]; [8]. Open each subsection with its purpose or material, explaining unusual choices that affect interpretation before describing the steps. [4].

Write completed work in the past tense, choosing the subject that makes the procedure easiest to follow. Active sentences often clarify responsibility; passive wording works when the sample or procedure is the natural subject. [6]; [4]. Have contributing authors check the account against their records and final code, including the steps where data passed between collaborators. [1]; [2].

Include the details that determine the outcome

Identify the materials or datasets and their sources, inclusion criteria, conditions and controls. Give laboratory settings needed to repeat the procedure, and identify datasets by accession or durable reference and relevant release. A table can collect reagent or dataset identities; the prose should explain how they were used. [2]; [7].

Consult study-specific reporting requirements for additional details. Obtain details of the required ethics approvals and compliance information from the responsible investigators, identifying the samples or procedures they cover. [2]; [7].

Describe how samples were allocated, when measurements were taken and how randomization or blinding operated when used. If a safeguard was absent, report that accurately and assess its effect on the conclusion. [9].

For an established protocol, cite the version used and describe what you changed. The phrase “with minor modifications” does not tell someone repeating the work which steps to follow. [7]; [2]. For computational procedures, report filtering, reference annotations, normalization and thresholds because they can affect the result. Routine file-handling operations usually need no description. [2]; [4].

Example: describing a matched comparison

The Brake study compares infectious phage output across three bacterial genotypes. The values and design details below are invented for teaching. This excerpt covers titre calculation and analysis; the host, constructs, infection conditions and collection protocol would be specified elsewhere in Methods.

Infectious-titre analysis

We compared empty-vector cultures, cultures carrying intact Brake and cultures carrying Brake with a catalytically inactive reverse transcriptase. On each of six experimental days, one independently started culture of each genotype was infected with phage A and sampled at the same endpoint.

For each culture, we estimated infectious titre in plaque-forming units per millilitre (PFU/mL) from technical duplicate plaque counts, accounting for the plated volumes and dilution factors. We combined the duplicate measurements into one culture-level estimate.

The primary comparison was intact Brake versus empty vector, matched by experimental day. We divided the intact-Brake titre by the empty-vector titre within each day and took the log10 of that ratio. We estimated the mean log10 ratio and its two-sided 95% confidence interval using a Student t distribution with five degrees of freedom. We exponentiated the estimate and interval limits to obtain a geometric mean titre ratio. Fold reduction was its reciprocal, with the interval limits inverted accordingly. The inactive reverse-transcriptase measurements were displayed as a mechanistic control.

What to notice. Each day contributes one matched contrast. Duplicate plates improve the measurement of a culture; they do not double the sample size. The analysis is specified for the between-genotype effect, rather than inferred from separate error bars on each group.

Describe the analysis from input to inference

Describe how the input becomes the reported result. Identify the input data and version, then explain processing in order, naming the software and settings that affect the output. For genomic analyses, give the reference assembly. State what the final comparison or model estimates. Readers should be able to follow that account in Methods without having to reconstruct it from the code. [2]; [6].

Explain selection and exclusion before presenting findings from the selected data. State how samples, genes, cells or regions entered the analysis and how many remained where counts affect interpretation. Distinguish prespecified decisions from choices made after inspecting the results. A subset selected because it shows the desired effect cannot also supply an independent test of that effect. Report exploratory analyses as exploratory and describe any separate data used for confirmation. [9].

Name the statistical comparison, not merely the software command. Describe the outcome, explanatory variables and relevant pairing or grouping; state how uncertainty was estimated and how multiple testing was handled when many comparisons were made. If different panels use different analyses, make those assignments clear. [7]; [2]; [9].

Identify the data and code in durable repository records, subject to applicable access restrictions. Prepare those records early and specify the version or retrieval date for changing datasets. [2]; [6].

Make the sampling and replication clear

Explain which samples were independently collected or treated and which measurements came from each. Hundreds of cells from one culture describe that culture more thoroughly, but do not establish consistency across independent cultures. Individual cells can be independent treatment units when the design assigns them that way and supports their independence. [10].

Give sample sizes with their meaning: independent cultures, repeated readings or analysis runs. Where the study has more than one level, report the counts and their relationship, including exclusions. In the Brake example, there are six cultures per genotype, each measured on duplicate plates, and the primary effect is estimated from six contrasts matched by day. [10]; [9].

Preserve that structure in both the analysis and its description. State whether the comparison uses sample summaries or a model that accounts for measurements nested within samples. Identify pairing when corresponding control and treatment observations belong to the same experimental run. A display can show individual observations and sample summaries together, provided the legend distinguishes them and identifies the unit used for statistical inference. [10].

Present results as an argument

Order findings by what they establish

Introduce each result before relying on it. In the Brake study, identifying the protein-coding DNA product prepares the experiment that tests the protein's role. Presenting that test first would leave readers unsure why the protein was chosen. [3]; [6].

Use finding-based Results headings when one statement covers the evidence. "The Brake protein mediates phage restriction" is more informative than "Protein experiments." A descriptive heading suits a resource, procedure or unresolved comparison. Read the heading with the figure title to check that the written claim and display answer the same question. [2].

One figure may support several paragraphs, and a claim may draw on several figures. Refer to the panels that support each finding and give the values needed to explain it. [2]; [3].

Build a paragraph around a finding

Build the paragraph around a finding, bringing together observations that support it. Briefly introduce the question and approach for an unfamiliar experiment; begin with the finding when the comparison is already clear. Name the treatment, comparison and outcome, then give the evidence, panel reference and any necessary qualification. Keep the full procedure in Methods. [8]; [2].

When introducing the next experiment, explain what you still needed to find out after the previous result. You do not need to repeat the paper's broad motivation. [4]; [6]. Explain what each result supports where readers need that connection to follow the next experiment. Reserve extended literature comparisons and broader implications for Discussion. If Results and Discussion are combined, still distinguish the observations from your proposed explanations. [3]; [4].

A closing interpretation is useful when it connects several observations or motivates the next experiment. If the paragraph has already made and supported its point, move on without restating it. [6].

Report magnitude, comparison and uncertainty

State what changed, relative to what and by how much, with units and the scale of comparison. Give the important difference or proportion explicitly. The display or underlying data can carry the remaining values. [6]; [3].

Report the estimate together with appropriate uncertainty and, where used, the statistical test result. A small p-value cannot convey the magnitude of an effect, and "significant" alone does not tell readers its direction or biological importance. Distinguish variability among observations from uncertainty in an estimated effect. Make clear whether an interval describes a group mean, a between-group contrast or another quantity. [7]; [9]; [10].

A claim that two effects differ needs a direct comparison between them. A significant response in one condition and a non-significant response in another can occur even when their estimated effects are similar, because the estimates have different precision. Separate significance tests do not answer the comparative question. [9].

Name what was measured before giving its biological interpretation. In the Brake example below, the measurement is endpoint infectious titre; the interpretation concerns phage restriction. [6].

Example: reporting a quantitative finding

Endpoint infectious titres were compared within each experimental day, as described in the Methods above.

> Brake reduced infectious phage A output. Across six experimental days, the geometric mean of the day-matched titre ratios indicated a 79-fold reduction with intact Brake relative to the empty-vector control (95% confidence interval, 55–116-fold; Figure 1B). Cultures carrying the inactive reverse transcriptase retained high phage titres. These results support a requirement for reverse-transcriptase activity in phage restriction.

What to notice. The interval belongs to the fold reduction. The final sentence gives a local interpretation of the control; it does not identify which downstream product causes defence. That question motivates Figures 2 and 3.

A lower endpoint titre is not automatically a smaller burst size per infected cell. Keep the result named after the quantity measured.

Negative and inconclusive findings

Report results that disagree with the prediction. State whether they provide evidence against it or are too imprecise to assess it. Also report controls that leave an alternative explanation unresolved. [9]; [3].

Give the estimated effect and uncertainty even when a test is non-significant. If you want to claim that any difference is too small to matter scientifically, specify how small it would need to be and whether the design and analysis can rule out larger differences. An inconclusive control leaves open the explanation it was intended to test. [9].

When an unexpected result prompts exploration, explain how that analysis arose and label it accordingly. Report what it establishes and which questions remain unresolved, preserving the distinction from the original test. [9].

Figure captions and legends

The figure caption, often called a figure legend, is the text beneath or alongside the display. Here, "legend" refers to that text rather than the key inside a plot. Include the conditions, label definitions and statistical details needed to understand the figure without searching the main text. [5]; [2].

Identify what each panel shows

Write the legend while looking at the figure. Begin with a concise title identifying its subject or principal finding, then describe the panels in their labeled order. Explain what readers are looking at before interpreting it: a schematic needs its relationships explained; an image needs the specimen, displayed signal and relevant scale; a quantitative plot needs the measured quantity and comparison. Add a brief finding when the journal permits and the title has not already conveyed it. [5]; [11].

Label figures directly where practical, using the legend to define remaining symbols, abbreviations and encodings. Describe shared conditions once and identify panel-specific exceptions. Use the same names for conditions in the legend and display. [5]; [2].

Make the statistical display interpretable

Identify sample sizes, what each plotted point represents, the summary statistic and the meaning of error bars or intervals. Distinguish individual observations from independent-sample summaries. [2]; [10].

State which comparison each test or annotation refers to. Include the test and relevant sidedness, pairing and multiple-comparison information where applicable, following the journal's reporting format. For box plots, define the boxes, central line, whiskers and any separately plotted points. For normalized data, explain the reference used for normalization if it is needed to interpret the values. [2].

Keep the full analysis in Methods, but define the displayed samples and error bars in the legend. Shorten a long legend by describing shared procedures once. If the panels require unrelated explanations, consider whether they belong in separate figures. [5]; [2].

Example: a panel and its legend

This is the quantitative panel from Figure 1 in the Brake figure plan. It uses the same illustrative data as the Methods and Results passages above.

Original teaching figure; all plotted values are illustrative.

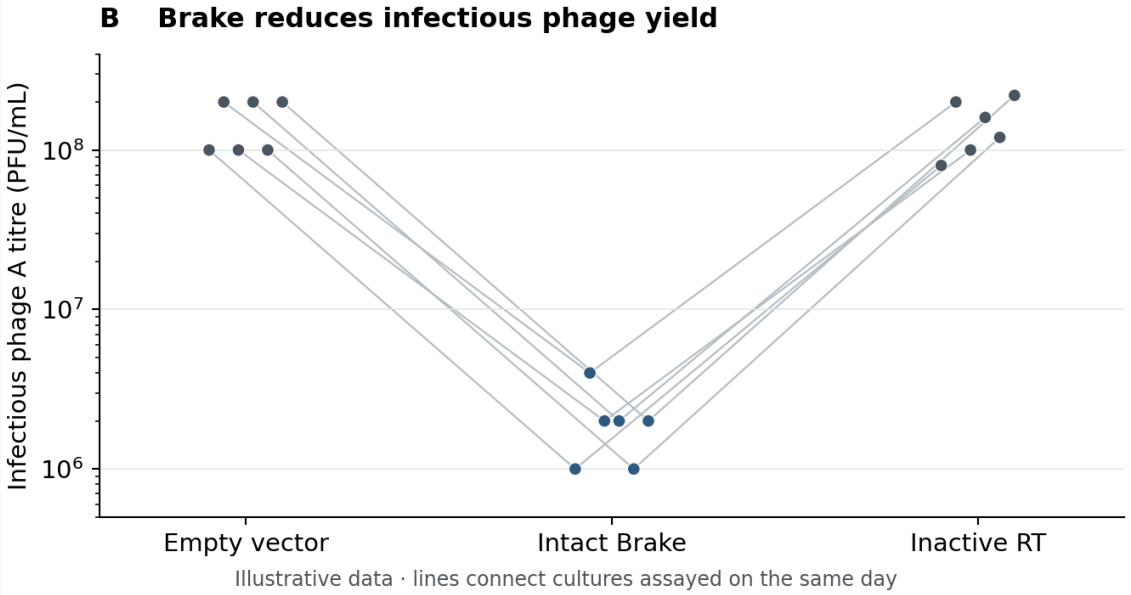


Figure 1B. Brake reduces infectious phage A yield. Endpoint titres from cultures carrying empty vector, intact Brake or Brake with a catalytically inactive reverse transcriptase (inactive RT). Each point represents one independently started culture; lines connect the three cultures assayed on the same day. Six days were measured, with technical duplicate plaque counts combined for each culture. The vertical axis is logarithmic. The geometric mean titre was 79-fold lower for intact Brake than for empty vector (95% confidence interval, 55–116-fold), calculated from the six day-matched log10 titre ratios.

A legend for the complete Figure 1 would also describe the construct schematic and growth curves in panels A and C.

Adapt the explanation to the publication stage

An initial legend can explain the figure's takeaway to reviewers; some journals retain that style, while others require shorter descriptions for publication. [11]. Use a finding-based title when it accurately summarizes the figure and the journal permits it, or a descriptive title otherwise. When shortening the legend, move necessary procedural or interpretive detail to the appropriate Methods or Results passage while retaining what readers need to decode the display. [5]; [11].

Keep figures and legends together in a review copy when the submission format allows it. To test a legend, first give it to a colleague without the figure and ask them to sketch the panels they expect. Compare that sketch with the figure: a missing comparison, unexplained image or ambiguous panel identity tells you what to clarify. Then inspect the figure and legend together to check the labels and statistical details. [11].

Check reporting across text and figures

Check each result against its figure, legend, Methods and underlying analysis, following the same samples and comparison through to the reported numbers and interpretation. [2]; [3]. Contributing

authors should verify technical details; a second reader can identify missing explanations or unclear figure labels that the authors overlook. [1]; [2].

Make the final pass on the assembled manuscript and supporting files, checking figure letters, units, terminology and statistical labels. Test data and code links, and carry each correction through all affected files. [1]; [6].

Reporting checklist

- Each result states a clear comparison and points to its supporting evidence.
- Methods identify the materials, design, procedures and analysis choices needed to follow the work.
- Selection, exclusions and exploratory decisions are reported where they affect inference.
- Sample counts distinguish independent units from measurements within them.
- Quantitative claims include the relevant magnitude and uncertainty, with suitable statistical support.
- Findings that limit the conclusion remain visible; inconclusive evidence retains its uncertainty.
- Legends identify the panels, conditions, encodings, summaries and statistical comparisons.
- Text, figures, legends, Methods and supporting files agree, and data/code access has been checked.

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Discussion and conclusions

The Results tell readers what happened; the Discussion asks what those findings establish. Begin with the study's answer, then explain how it relates to previous work, where it remains uncertain and what follows from it.

The examples continue the fictional Brake study: an RNA-derived gene produces a protein that arrests bacterial growth and restricts phage propagation.

Opening and structure

Start with an answer the evidence supports

Answer the Introduction's question in the opening paragraph, using the few findings needed to support that answer. State the conditions in which it holds. Save detailed comparisons and controls for the passages that discuss them. [1]; [2]; [3].

If the study answers only part of the question, say what was resolved and what evidence is missing for the rest. A study may show that a protein mediates an effect without identifying its molecular target; the unknown target limits the mechanistic explanation, while the functional conclusion can stand. [5]; [4].

Give the argument a clear order

The following sequence is a starting point; limitations may belong within an interpretation or comparison rather than in a separate block. [1]; [6].

A working structure for the Discussion

1. **Answer:** State the central finding and the essential evidence behind it.
2. **Interpretation and comparison:** Explain the major findings alongside the most relevant previous work. Show what agrees, differs or becomes clearer.
3. **Limits and alternatives:** Evaluate the explanations the evidence can distinguish and the conditions under which the answer remains uncertain.
4. **Implications:** Explain what the findings change and which unresolved question is worth pursuing next.
5. **Closing contribution:** Leave the reader with a concise statement of what the study adds.

Follow the Results order when it supports the same argument. Group experiments that jointly answer one question, and give an unexpected finding early attention when it changes the interpretation. Use subheadings for a long Discussion; a short one may read better without them. [9]; [10].

A combined Results and Discussion can interpret each finding as it appears, then draw the findings together at the end. With separate sections, report new measurements in Results and their procedures in Methods; the Discussion can then concentrate on what they mean. [7]. Some formats permit a preliminary observation within Discussion, but it still needs enough methodological and reporting detail to be evaluated. Place any result essential to the central conclusion in Results. [6]; [10].

Explain findings alongside previous work

Comparing findings

Start with the studies that motivated the question, adding work needed to interpret the findings and relevant papers published during the project. The Introduction may only have needed their central conclusions; here, a particular comparison may require closer attention to the evidence. [10]; [5]. Read the relevant Results and Methods, noting the system, intervention, measured outcome and conditions before deciding whether the findings agree or conflict. [4].

Explain how your finding confirms, extends or conflicts with earlier work. Distinguish confirmation under similar conditions from extension to a new setting. [5]; [3].

Compare studies through their effect estimates and uncertainty, alongside experimental context and analysis choices. Different significance labels do not establish different effects; see comparing effects. Likewise, similar directions do not establish close agreement when estimates are too imprecise to distinguish substantially different effects. [8].

Differences in conditions may suggest why studies disagree, but they do not by themselves establish the explanation. Identify the evidence that would distinguish it from alternatives, and leave the conflict unresolved if that evidence is unavailable. Apply the same care to contrary and supportive findings: cite the specific result, and discuss methodological limitations where they affect the comparison. [5]; [4].

Build paragraphs around an interpretation

Build each paragraph around one point. A paragraph explaining a mechanism needs the evidence for it; a paragraph about disagreement needs the findings being compared. Give readers any unfamiliar background before asking them to follow the interpretation. Add a closing consequence or next question when there is something further to explain. [7]; [4].

As you move between your results and published work, make clear which experiment established each finding. Name the biological subject and use consistent terminology so that a phrase such as "this finding" does not leave readers choosing between several studies. [3]; [4]. The paragraph should explain what the evidence establishes together; a sequence of paper summaries that does not affect the interpretation is usually unnecessary. [5]; [6].

Evaluate limitations and alternatives

Explain why one interpretation is better supported

Consider plausible alternatives that would change the conclusion, and compare their predictions with the observations. Explain which control distinguishes the alternatives and where more than one explanation remains possible. An intervention may establish that a protein contributes to a response while leaving open how it acts. [5]; [4]; [8].

Unexpected findings require the same scrutiny. Examine the conditions and possible measurement problems before developing a biological explanation, and identify explanations developed afterward as hypotheses generated by the result. [8].

Explaining limitations

Explain what each limitation prevents you from concluding. Testing one cell model leaves generality uncertain; an assay that cannot distinguish two mechanisms leaves the mechanism unresolved. Qualify the affected claim and state what the checks did resolve. [1]; [9]; [2]; [5].

Instead of simply noting a small sample, explain whether the estimate is too imprecise to distinguish effects that matter. For a method tested under limited conditions, state which uses remain untested. [6]; [4].

Inconclusive results

Use the estimates and uncertainty described in the reporting chapter to explain what an inconclusive result leaves open. If the uncertainty includes effects large enough to change the interpretation, that uncertainty belongs in the conclusion. Establishing a practically negligible effect requires a suitable design and analysis, such as a justified equivalence analysis; a non-significant test alone cannot establish it. [8].

Example: connecting mechanism, limits and implications

Here RNA-templated defence-gene assembly is a new discovery, as in the main Brake case.

> Brake assembles a functional defence gene from an RNA template. Sequence analysis identified a coding sequence spanning DNA repeat junctions. Genetic tests established the encoded protein's role: blocking translation preserved DNA production but removed defence, and separate protein expression restored it. The effector therefore becomes available through gene assembly rather than expression of a conventional genomic coding sequence.
>
> These experiments establish the protein's role without identifying its molecular target. Growth arrest could result from interference with a host process needed for both cell growth and phage replication, but the current measurements do not identify that process. Determining the target would help explain how arrest is initiated and why activity differs among the tested phages.
>
> The fate of infected cells is a separate question. They stopped growing during observation, but we did not measure subsequent recovery. Reduced infectious output therefore establishes phage restriction without showing that individual cells survive the infection. Recovery measurements would distinguish reversible arrest from a lasting loss of growth capacity.
>
> These findings show that a functional defence gene can emerge in an RNA-derived DNA product. Searching those products for coding sequences could reveal effectors missed by conventional genome annotation.

The molecular target and cell recovery need different follow-up experiments, so they receive separate paragraphs.

Example: comparing a follow-up with its precedent

In a follow-up version of the Brake paper, RNA-templated antiviral gene synthesis is already known, and the study tests the effector's function in another system. Tang and colleagues (2024) [11] serves as real prior work. This is also the premise of the second cover letter.

Question	Published DRT2 study	Fictional Brake study
What connects gene assembly to defence?	RNA-derived DNA encodes Neo, whose expression causes growth arrest.	A translation-disrupting change preserves DNA production but removes defence; separate protein expression restores it.
What happens to arrested cells?	The study includes viability and recovery measurements.	Recovery after infection was not measured.
What remains unresolved?	The protein's molecular function requires further investigation.	The molecular target of the Brake protein is unknown.

RNA-templated gene synthesis was previously established as a route to bacterial defence in DRT2 systems. [11]. Our Brake experiments identify the protein encoded by the assembled gene as an effector of defence in this system. Blocking translation preserves DNA production but removes defence, and separate protein expression restores it. These results distinguish the protein's contribution from a direct effect of the DNA product alone. Whether Brake-arrested cells recover remains unresolved: recovery was measured in the DRT2 study but was not tested here.

Develop implications and conclusions

Explain what follows from the finding

Explain how the finding changes an explanation, prediction or measurement capability. The implication should follow from the result: identifying a previously unrecognized source of protein-coding genes, for example, gives researchers a reason to examine material that conventional annotation overlooks. [3]; [7]. For a method, distinguish a newly available capability from better performance on an existing task. A performance claim needs a specified comparator and benchmark, together with the conditions tested. [4]; [6].

Use a model or schematic to connect findings when it clarifies the explanation. Distinguish established relationships from proposed links in the labels and legend. Explain the evidence for each important connection and a testable prediction for a hypothetical one. [6]; [4].

Identify the next question to resolve

Identify the most important remaining question and explain how its possible answers would change the interpretation. [1]; [9]. If that answer is essential to the present conclusion, obtain the evidence now or narrow the claim. [2]; [4].

If you propose an explanation that the study has not tested, show how it follows from the results or previous work and what evidence could challenge it. For a proposed practical application, identify the test still needed before recommending that use. [5]; [4].

Finish with what the study changes

Close with the contribution and its most important implication, retaining any qualification essential to its meaning. The ending should draw out what the findings change, without recounting the experiments again. [7]; [3]. Use a separate Conclusion when the format requires it or a substantial Discussion benefits from a final synthesis. A short Discussion may already have the ending it needs. [2].

Check the interpretation

Read the Introduction's question and the Discussion's opening and ending together. If the answer changed during writing, revise the framing while preserving the distinction between the original prediction and interpretations developed afterward. [3]; [8]. Then follow the argument paragraph by paragraph: check claims against results or publications, retain important alternatives, and use the paragraph openings to locate repetition or abrupt changes of subject. [4]; [5].

Before asking for sentence edits, ask a colleague to identify the main answer, unresolved issue and contribution. Where their account differs from yours, find the missing or misleading step in the reasoning. [1].

Before finishing the Discussion

- Does the opening give a supported answer to the question in the Introduction?
- Do the main comparisons explain what the study adds, including relevant disagreement?
- Can readers follow the evidence favoring the interpretation over plausible alternatives?
- Does each important limitation explain what cannot yet be concluded or recommended?
- Can readers distinguish an established mechanism from a proposed one, and see what inconclusive findings leave open?
- Would the proposed future work extend the findings, or is it needed to support a claim the paper already makes?
- Does the ending leave a clear contribution without repeating the experiment summary?

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Drafting, reviewing and revising a paper

Writing can reveal that you have not explained why an experiment supports a conclusion. Resolve such problems before polishing sentences. As the draft develops, check the evidence, arrange the results so readers can follow the reasoning, and then edit the paragraphs and sentences. New findings or feedback may require you to revisit earlier decisions.

The examples use the fictional Brake study, inspired by Tang and colleagues (2024) [12].

Starting a draft

Choose a passage you can develop

If the question and figures are settled, start with a Results passage whose comparison you understand well. Explain the finding and draft the corresponding Methods from the records. [1]; [2].

If you have results but are unsure how they fit together, draft a working title and a short statement of what the study adds. Sketch the question and background for the Introduction. Then check the figures: do they support the answer you have proposed? [3].

State the paragraph's purpose, then develop it with the figures, analysis outputs and sources beside you. Mark missing information precisely—for example, "confirm which samples enter this comparison"—and continue where you can. [4]; [2].

Make writing sessions manageable

Choose a task you can tackle in the available writing time: draft one result, revise a transition or check a Discussion claim. If you stall, make the task smaller. Allow a longer session when you need to reread the whole argument. [5]; [2].

If sentence polishing prevents progress, explain the finding aloud or write an informal version before refining the terminology. Correct errors that block your thinking, but leave equivalent wording choices for later. [5]; [6].

When stuck, ask why. If you do not understand a result, inspect it or discuss it with a collaborator. If only the opening sentence is troublesome, draft the evidence and return to it. A break may help when concentration has faded. A shared writing session or an agreed date for sending a draft can help you return to the work. [5]; [2].

Read for the question and answer

Read the draft through before making extensive local edits. If you have just finished writing it, a break can help you notice what the page actually says. Record places where the explanation stops making sense, but continue far enough to see whether the missing information appears later. [8]; [14].

Write a brief account of the paper's question, answer and decisive evidence. Include the condition or unresolved issue that most limits the answer. Then compare that account with the title, abstract, final Introduction paragraph and Discussion. These passages can differ in detail, but they should identify the same contribution. An abstract that promises a mechanism while the Results establish an association creates a substantive discrepancy, even if each section reads smoothly on its own. [4]; [9].

Ask whether the answer addresses the question the Introduction led readers to expect. Sometimes the experiments have resolved a narrower question, or an unexpected finding has become the main contribution. In that case, revise the framing without presenting an explanation developed afterward as the original prediction. If the answer is supported but difficult to recognize, identify the missing connection before deciding to collect more data. [14]; [15].

Check what the paper adds to existing knowledge. If the contribution depends on differing from an earlier study, reread that study's relevant result and method. Did it use the same system,

measurement and intervention? A difference may explain why the findings disagree, or show that your work extends a finding to another setting. See comparing with previous work. [9].

Ask a colleague who is less immersed in the project to explain the paper's question and answer. If their account differs from yours, reread the passage that led to it and check whether the problem lies in the explanation or the evidence. [4]; [2].

Checking claims against the evidence

Start with the claims needed for the paper's main conclusion. Follow each claim through the cited panels and legend to the Methods, supporting material and analysis. What was measured, what was compared, and why does that comparison support the claim? Pay particular attention when the conclusion combines several experiments or proposes a biological explanation for a measurement. [14]; [9].

A legend may omit the meaning of *n* even though the analysis uses the correct units. Check the analysis before deciding whether the problem is an unclear description or repeated measurements treated as independent samples. [15]; [16].

Reconstruct the comparison

For a quantitative claim, identify the measured quantity, reference group, sampled or treated units and any pairing or nesting. Then check which observations entered the analysis and how the reported effect and uncertainty were calculated. A label such as "six replicates" does not explain whether the six are independently treated cultures, measurements within a culture or matched experimental runs. The appropriate unit follows from the design and the intended inference. [16].

Follow the same samples through exclusions, transformations and summaries. If the reported counts differ between a figure and Methods, establish whether different subsets were intended and whether the distinction is explained. Reconcile the reported number with the final analysis output rather than choosing whichever version appears more plausible. For a ratio or fold change, check its direction and denominator as well as its magnitude. See reporting quantitative findings. [9]; [17].

Examine whether the analysis answers the comparison stated in the prose. A significant change in one condition and a non-significant change in another do not establish that the changes differ. Likewise, an imprecise estimate may remain compatible with a biologically substantial effect. If a claim depends on the absence of that effect, a non-significant test alone cannot supply the required evidence. [15].

Test the interpretation against an alternative

Choose a plausible alternative that would change an important conclusion and identify what distinguishes it from the preferred explanation. For an intervention experiment, ask what else the intervention changes and which control separates those effects. If the two explanations predict the same observed result, that experiment cannot choose between them. [15]; [9].

Include findings that challenge the preferred explanation. Check whether the difference reflects conditions, samples or analytical choices; if it persists, decide how it limits the answer. Preserve the distinction between a prediction tested as planned and an explanation suggested by inspecting the results. In particular, selecting observations because they show an effect and then using them to confirm that effect can make apparently strong support circular. [15].

A study can establish that a protein is required for a phenotype without identifying its molecular target. But if the perturbation also removes an upstream product, the loss of the phenotype could result from either change. Further evidence would be needed to distinguish them. Decide which unresolved question affects the conclusion you intend to retain, and update the claim inventory accordingly. [9]; [4].

Figures and legends

Read the figures and legends independently of the main text, then compare the inference they invite with the analysis actually performed. Visual grouping can suggest a comparison the authors never tested; connecting points can imply pairing that did not occur. Check whether the display reveals the variation needed to assess the summary and whether exclusions or overlapping points hide relevant observations. [17].

For a schematic, follow the material or process from beginning to end. Check that symbols, colors and names remain consistent and that the sequence agrees with Methods. [18].

For light-microscopy results, compare the selected image with the range of observations it is meant to illustrate. Check that readers can identify the specimen, channels and scale, and trace image processing into the reported measurements. Example input and output data can expose a segmentation or processing problem that a polished final image conceals. Use the applicable Schmied and colleagues (2024) [19] checklist to investigate missing information; it does not establish that the experiment itself was well designed.

Inspect the assembled figures at their intended reading size for unreadable labels and inconsistent colors. [9].

Deciding what to revise

For each concern, name the passage, explain the problem and state what needs checking. Replace "Strengthen the Discussion" with a task such as "The abstract infers survival from growth arrest; check whether recovery was measured and revise the claim accordingly." [14].

If the evidence supports the claim but the explanation is incomplete, clarify the comparison or explain what a control rules out. Put any needed background before the result that depends on it. If the analysis is wrong, correct it and reassess the claim before rewriting the conclusion. If essential evidence is absent, obtain it or narrow the claim. Another interesting question can remain future work when its answer is not needed for the conclusion. [4]; [15]; [9].

Example: assessing the complete Brake draft

An early draft of the fictional Brake study, inspired by Tang and colleagues (2024) [12], describes RNA-templated gene assembly, growth arrest and phage restriction. Its abstract additionally claims survival, although recovery was not measured. The numerical result uses the handbook's illustrative dataset.

Draft packet

The title is *An RNA-templated gene arrests bacterial growth and restricts phage propagation.* The Introduction asks what the reverse transcriptase produces and how that product contributes to defence. Figures 1–4 respectively establish phage restriction, identify the protein-coding DNA product, test the protein's function and examine activity across the tested phages.

Abstract, final sentence

> By arresting growth during infection, Brake enables infected bacteria to survive phage attack.

Figure 1B, display and legend excerpt

The plot shows endpoint infectious phage A titres for empty vector, intact Brake and inactive reverse transcriptase. Lines join observations across the three genotypes. The Results report a

79-fold reduction for intact Brake relative to empty vector, with a 95% confidence interval of 55–116-fold.

Endpoint infectious titres for the three genotypes. Six replicates were measured in duplicate; lines connect matched measurements.

Figure 3, Results excerpt

We next disrupted translation of the Brake protein. DNA production was retained (Figure 3A), but growth arrest and phage restriction were lost (Figure 3B). Expressing the protein separately restored both phenotypes (Figure 3C).

Only in the Discussion does the draft explain that retaining DNA production is necessary to distinguish the protein's contribution from an effect of losing the upstream DNA product.

Checks against the full manuscript and records

The author checks the cell-tracking records and Methods behind Figure 4B: growth arrest was observed, but subsequent recovery was not measured. The phage assays establish restriction of A and B; C remains productive. None of these findings measures whether an individual infected cell survives.

For Figure 1B, the culture log records one independently started culture per genotype on each of six experimental days. Technical duplicate plaque counts were combined into one titre per culture. The analysis already compares intact Brake with empty vector within each day, using six log10 titre ratios to estimate the mean and a two-sided Student *t* confidence interval with five degrees of freedom. Converting back to a ratio and taking the reciprocal reproduces the reported fold reduction and interval. The Methods give this analysis; the legend fails to identify its units clearly.

For Figure 3, the author checks the construct description and product measurements against the panels. The translation-disrupting change preserves the RNA and DNA products, and separate protein expression restores arrest and restriction. The control needed for the protein inference is present. Its purpose becomes apparent too late in the written explanation.

Author's assessment

The checked evidence supports an RNA-templated route to a protein effector of growth arrest and phage restriction. The repairs are specific: remove the unsupported survival claim, clarify the six day-matched cultures and comparison in Figure 1B, and move the protein-versus-DNA rationale before the Figure 3 results. These repairs require no repeat experiment on the records checked here. The protein's molecular target and recovery of arrested cells remain unresolved.

Prioritized revision plan

Work and responsibility	Completion criterion
1. Settle the claim. The lead author and coauthors retain gene assembly, growth arrest and phage restriction as the contribution. Remove the survival inference throughout the manuscript.	Title, abstract and Discussion agree; none infers survival or irreversible death from growth arrest. Recovery remains an explicit unresolved question.

Work and responsibility	Completion criterion
2. Clarify Figure 1B. The analyst verifies the culture log, plotted values and analysis; the figure author clarifies cultures, technical duplicates and day matching. Proceed alongside task 1.	Figure, legend, Methods and Results agree: six cultures per genotype; six day-matched contrasts; 79-fold reduction (95% confidence interval, 55–116-fold).
3. Explain the Figure 3 test. The Results author introduces the protein-versus-DNA question before the perturbation results. The contributing experimentalist checks the explanation against panels 3A–C.	Readers can explain why retained DNA production matters and how complementation supports the protein's role. The wording does not identify an unmeasured molecular target.
4. Review the assembled revision. After the changes are integrated, the lead author circulates one version with the revised figures and supporting material. Technical contributors check their results; a colleague less familiar with the project reconstructs the argument.	The scientific concerns are resolved in the manuscript itself, and no stale claim, inconsistent legend or broken transition remains elsewhere. Any new substantive concern returns to the plan.

Two targeted repairs

Replace the abstract's survival sentence with a conclusion consistent with the paper's discovery:

> These findings reveal a defence pathway whose effector gene is assembled from an RNA template.

Replace the Figure 3 paragraph's first sentence with the question its controls answer:

> We next asked whether the protein encoded by the RNA-derived DNA mediates defence. We disrupted translation while preserving the RNA and DNA products, allowing us to test the protein's contribution without removing the upstream products.

The existing Figure 3 results then provide the answer. Had the translation-disrupting change also prevented DNA production, moving this explanation would not resolve the concern: the evidence would no longer distinguish the protein's contribution from loss of the upstream product.

Revise the argument and organization

From the argument to the paragraphs

Once you know which claims the evidence supports, decide how to explain them. Check the paragraph order before polishing passages that may move or disappear. If rearranging the draft reveals a claim you cannot explain from the evidence, investigate it before continuing. [7]; [4].

Reviewing the paragraph order

Summarize each paragraph's point in one sentence. Read these sentences in sequence to find repeated points, paragraphs trying to do too much and missing steps in the reasoning. Revise the paragraph order before editing individual sentences. [3]; [4].

Example: reorganizing a Results draft

The Brake draft follows the order in which experiments were completed. Protein expression appears before readers know why that protein was tested. The paragraph summaries below expose where the explanation arrives too late.

Paragraphs in the draft

1. Expressing the Brake protein arrests growth.
2. Details of expression-vector construction.
3. Intact Brake restricts phage output.
4. DNA sequencing identifies a coding sequence across repeat junctions.
5. Phage activity differs among the three tested phages.
6. Infection is accompanied by double-stranded DNA and protein production.
7. Blocking translation preserves DNA production but removes defence; separate protein expression restores it.

Revised paragraph plan

1. Establish the defence phenotype and the reverse-transcriptase control (Figure 1).
2. Identify the RNA-derived DNA product and the coding sequence across its repeat junctions (Figure 2A, D).
3. Describe the DNA products and protein expression before and during infection (Figure 2B–C).
4. Test the protein's role using the translation-disrupted construct and separate expression (Figure 3).
5. Define the tested phage range and describe infected-cell growth (Figure 4).

Protein expression now answers a question raised by product identification. Vector-construction details move to Methods. Rewrite the transition into Figure 3 so it asks whether the protein is responsible for the defence already observed.

Once the order is settled, check what each paragraph contains. Keep an explanation or limitation beside the finding it concerns; move detailed procedures to Methods and remove background already established. If successive additions have obscured the paragraph's point, list the facts and qualifications to retain and write it again. Compare the versions for changed meaning. [9]; [4].

To meet a word limit, first remove duplicated explanations, tangential background and details already clear in figures or Methods. Then shorten sentences. Read across each cut to check for lost definitions or premises. [9]; [4].

Rough word and figure budget

For a 3,000-word body, start with about 500 words for the Introduction, 1,700 for Results and 800 for Discussion. For 4,000 words, the corresponding starting allocation is 600, 2,400 and 1,000 words. Shift space between sections as the argument requires. These totals exclude Methods; check the journal's counting rules. Plan around 4–6 main figures and tables combined. [13].

Make each paragraph advance the explanation

Open with the finding when the comparison is familiar; otherwise introduce the question and essential design first. Develop that point with the observations and reasoning it needs. A closing interpretation helps when it connects several observations or leads to the next experiment. If it merely repeats the opening, remove it. [3]; [4]; [9]; [7].

Example: giving a Results paragraph a clear point

Both passages use the illustrative titres from the reporting chapter.

Draft

> We measured phage A titres in empty-vector cultures, intact-Brake cultures and cultures carrying inactive reverse transcriptase on six experimental days. The inactive reverse-transcriptase cultures retained high titres. The matched comparison of intact Brake with empty vector gave a 79-fold reduction in geometric mean titre (95% confidence interval, 55–116-fold; Figure 1B). Brake therefore restricted phage output.

Revision

> Brake reduced infectious phage A output. The geometric mean of the six day-matched titre ratios indicated a 79-fold reduction with intact Brake relative to empty vector (95% confidence interval, 55–116-fold; Figure 1B). Cultures carrying the inactive reverse transcriptase retained high titres, supporting a requirement for reverse-transcriptase activity.

The control follows the main comparison because it addresses whether restriction requires reverse-transcriptase activity. The revision retains the matching by experimental day.

Edit sentences for clarity

Subjects and verbs

Start with the main clause: who or what did what? Direct verbs often make it easier to find—"we measured induction" instead of "we performed an assessment of induction." Keep the subject and verb close, and place modifiers beside what they describe. Long qualifications may need a separate sentence. [9]; [3]; [7].

Check that introductory phrases attach to the right subject. In "Using regulatory analysis, the candidate was identified," the candidate appears to perform the analysis. "Using regulatory analysis, we identified the candidate" makes the actor clear. [9]; [7].

Use a short phrase to identify an unfamiliar subject: "Brake, the candidate defence system, was tested in phage infections." Once it is familiar, omit that reminder. Give longer explanations their own sentence and check that moving a qualifier preserves its meaning. [10].

Choose the subject that continues the paragraph's topic. "We measured induction" attributes an action; a passive construction can keep attention on the sample or process. Read both in context. [3].

Connect sentences and comparisons

Connect each sentence to what the reader has just learned: after introducing the RNA-derived DNA, explain what that DNA encodes. In comparisons, parallel constructions put interventions and effects in corresponding positions, making differences easier to follow. Repeat the relevant terms rather than introducing synonyms that might suggest a different measurement or condition. [3]; [4]; [10].

Example: choosing the focus of a comparison

In the Brake study, preventing protein production removes defence; expressing the protein separately restores it. Choose the subject that connects to the preceding sentence.

Continuing a discussion of defence

> Intact Brake restricted phage propagation. Phage restriction was lost when translation of the Brake protein was blocked and restored when the protein was expressed separately.

Continuing a discussion of the genetic tests

> We used two genetic tests to examine the Brake protein's role. Blocking its translation removed phage restriction; expressing the protein separately restored restriction.

Words such as "however" and "therefore" signal a relationship that the sentences must also explain. Check what is being contrasted or what evidence supports the stated consequence; add the missing connection if necessary. [10].

Keep related findings together when one sentence makes their relationship clear. Split a sentence when competing points hide the subject or comparison. A short sentence can emphasize a conclusion; a longer one can keep a result beside its necessary qualification. [10]; [6].

Read aloud to find awkward phrases and ambiguous pronouns. Then read silently, checking the text against figures and equations. [6]; [9].

Keep terminology and notation stable

Use one term for each scientific concept. Distinguish measurements from inferred quantities, such as RNA abundance and regulatory activity. Agree on terminology with coauthors. [9]; [3].

Use past tense for completed procedures and observations, and present tense for interpretations or figure descriptions where appropriate. Check that tense changes reflect the intended meaning and the journal's conventions. [9].

Define useful abbreviations on first use, including in the abstract and legends where readers may encounter them independently. Spell out terms used only once or twice unless the abbreviation is familiar. Likewise, name quantities before using symbols and define terms beside equations. For ratios, specify the numerator and denominator, including whether they are measurements or means. Check units, normalization and notation against the figures. [9].

Delete phrases such as "It is important to note that." Replace vague results with the direction, comparison and relevant magnitude, including uncertainty where needed. [7]; [9]; [3].

Cut words without cutting meaning

Remove repeated caution but retain qualifications that affect the conclusion. Check that edits preserve the subject, comparison, scope and strength of the original statement. [9].

Example: replacing vague caution with a specific limit

The genetic tests establish a role for the Brake protein, but the study has not identified its molecular target.

Draft

> The results suggest that the Brake protein may play a role in phage restriction, although further work is needed to understand the mechanism.

Revision

> The translation and complementation experiments identify the Brake protein as the defence effector. Its molecular target remains unknown.

"May play a role" understates the genetic evidence. The uncertainty concerns the molecular target, so the revision places it there.

Word searches and readability scores can flag passages to inspect; read those passages before changing them. Studies relating writing features to citation counts do not establish which edits improve understanding. After an edit, check that readers can still identify the comparison and its limitations. [11].

Use coauthor feedback

Ask for the review the draft needs

Match the reader and request to the draft's main problem. A knowledgeable collaborator can check the science; someone less familiar with the project can identify missing context. When the contribution is disputed, ask about the argument and missing evidence before requesting language edits. Once it is stable, ask readers to mark passages they reread and to summarize the contribution. [2]; [8]; [4].

Send one current version with those questions, a response date and a note identifying unfinished material. [4]; [2].

Working through comments

Read the comments together before accepting edits. Group comments about the same passage or claim, then identify what confused the reader. Their proposed wording may not solve the problem, but the confusion still needs attention. [4].

Discuss edits that change scientific meaning with the coauthors responsible for that work, using the sentence and its supporting figure. Ask why the change was proposed before choosing wording. For stylistic alternatives that mean the same thing, choose the version that fits the paragraph and agreed terminology. [2]; [9].

Maintain one working version with recoverable history. Assign an editor to integrate separate files, or agree ownership during simultaneous editing. Check surrounding definitions and transitions after substantial changes, and return unresolved scientific questions to the responsible coauthors. [9]; [2].

Resolving substantive disagreements

If coauthors disagree about a central conclusion, settle what the evidence supports before polishing that conclusion's wording. Investigate possible analysis errors early and update any changed estimate in the figure, Results and abstract. Coauthors can continue work on unaffected sections. [14]; [4].

For a disputed control, state which explanations it would distinguish. An author may reasonably want to finish a paper with a narrower conclusion while another wants to pursue the larger mechanism. Agree first on what the existing evidence supports, then discuss whether the additional question belongs in this paper. Combining their preferred wording will not settle a disagreement about its scope. [15]; [4].

Check whether the authors are answering different questions. An experiment can be technically successful yet leave the proposed mechanism unresolved. If disagreement remains, record the assumption or missing evidence that would settle it. [2]; [4].

Check the revised paper against the original concern

Return to each concern with the revised passage and its evidence beside you. Check whether the change resolves the problem. If a claim has narrowed, read its other appearances; if an analysis has changed, verify every reported value derived from it. [9]; [2].

Local repairs can leave an Introduction promising an abandoned conclusion, a Discussion relying on an old figure or a legend describing an earlier analysis. Read the complete revision for those consequences. [14]; [4]; [2].

Finish and verify the manuscript

Check the final version as a connected document

Check the manuscript against figures, tables and analysis outputs. Verify numbers, units, group names, citations and cross-references. Inspect the rendered output even when citation and cross-reference tools maintain the links. [7]; [3]; [2].

Allow time for repository records, metadata corrections and coauthor approval. Check the journal's submission requirements, then inspect the PDF and figures at their intended size for conversion errors. [9]; [8].

Decide when to stop revising

Continue revising while the contribution is unclear, support is missing or coauthors disagree about the interpretation. Once those issues are resolved, circulate the final version for approval. Use this reading to catch errors; return to structural editing if it reveals a problem with the argument, and leave equivalent wording choices settled. [2]; [8].

Before sending the final version

- Can an unfamiliar reader summarize the contribution and follow the evidence supporting it?
- Do paragraph purposes, transitions and figure references form a coherent sequence?
- Have sentence edits preserved comparisons, terminology, scope and uncertainty?
- Are substantive coauthor comments resolved in one agreed version?
- Have numbers, units, citations, cross-references and supporting files been checked together?
- Does the rendered manuscript satisfy the journal's requirements and have final coauthor approval?

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Chapter 5

Writing a thesis

From thesis plan to finished text

A thesis explains the contribution of your research. Published papers may supply much of the material; the thesis also needs to explain how the projects relate and what their findings establish together.

Choose the thesis scope and structure

Agree the local expectations.

Thesis format and length vary between countries, universities and programmes, and with supervisors' expectations. Check the current requirements and look at recent accepted theses. Before substantial drafting, agree a provisional contents page and length with supervisors, including how much background and general discussion they expect. If you plan to include papers, check the conditions for doing so. [8].

Plan the length as well as the contents.

Check which components count toward the word limit. [7]. Agree provisional lengths for the chapters, allowing more space for those that need fuller explanation. Page counts are harder to compare because figures, spacing and appended papers affect them. Revisit the estimates after drafting a representative chapter.

State the thesis question and contribution.

Write a short statement of what the thesis investigates and what the completed work adds. If the projects do not answer one question, explain the topic or approach that connects them.

Identify how the projects relate. For example, one may test an explanation suggested by another, or provide a method used in a later study. Projects can also examine different aspects of the same problem. An unfinished project can belong if its available findings contribute to the thesis; explain what those findings establish and what remains unresolved.

Develop the agreed contents into headings, figure and table titles, with notes about what each chapter will explain. [1].

Select a format that the programme permits.

Three useful starting patterns are:

A continuous monograph. An introduction establishes the problem, a literature review develops the background, and methods, results and discussion build one sustained account. The review may sit within the introduction, and results and discussion may be combined. This arrangement works when the studies are best understood as parts of one argument. [5].

A monograph organized around studies or themes. A general introduction and, where useful, shared methods precede several research chapters. Each chapter introduces its own question, explains its methods, presents findings and discusses them. A general discussion then brings the studies together. This is a useful option for several related projects, including work that has not become a paper. Whether that work is sufficient for the degree remains a local academic judgment.

An article-based, compilation or cumulative thesis. Papers form part of a larger account that explains their shared background, aims and combined contribution. Connecting chapters may surround the articles, or an integrative account may precede the collected papers. The arrangement and terminology vary, but the intellectual task is similar: explain how the studies fit together and what they establish beyond their individual conclusions. The integrating text needs its own critical argument, even when the papers are already published. [9].

Front matter usually includes the title, declarations, abstract and contents; references and appendices complete the document. Their required order, additional summaries and any separate impact

statement are local choices or requirements. A hybrid structure can be sensible where permitted, but agree its logic before drafting.

Check which parts of published papers you may edit for the thesis. Where changes are permitted, adapt the opening to explain the paper's place in the thesis, update references and remove repeated overview figures that add no new information. Shared methods can reduce repetition, but each chapter still needs enough context to understand its experiments and assess its findings. [1].

Map the completed projects

Map the chapters.

Before drafting the general introduction, list each research chapter's question, main finding and contribution to the thesis. Add the shared background and final synthesis. Check whether readers will understand why one project follows another and which findings the synthesis needs to compare.

Example: connecting three completed projects

In this fictional thesis, three projects examine whether brief heat pulses affect lateral-root emergence in young seedlings.

Chapter	Question or main finding	Role in the thesis
General introduction	Can brief heat exposure alter root branching during seedling development?	Introduce the biological question, relevant literature and the three approaches.
Project 1: greenhouse observations	Seedlings under an afternoon heat regime had fewer emerged lateral roots; primary-root length was similar between groups.	Establish the pattern. Because other greenhouse conditions also differed, Project 2 isolates heat exposure.
Project 2: controlled heat pulses	Heat pulses in sterile culture reduced lateral-root emergence without changing primary-root length.	Establish a heat-pulse effect under controlled conditions. Project 3 tests whether it depends on developmental timing.
Project 3: timing test	Early exposure reduced emergence; later exposure did not produce a comparable reduction. Each exposed group was compared with age-matched controls at the same interval after exposure.	Locate the stronger response in early development under the tested pulse regime. The molecular cause remains unresolved.
General synthesis	How do the controlled effect and timing result help interpret the greenhouse observations?	State the combined answer and distinguish the experimentally established effect from the explanation proposed for the greenhouse pattern.

Give chapters enough local context.

Put broad background and shared methods in a common chapter where the format permits. Give each research chapter a short introduction to its question and enough methodological context to interpret its results. Cross-reference the common account for detail. [1].

A schematic may need to appear again if readers use it to understand a new comparison. A general overview figure usually belongs in the introduction. Revisit the literature for paper-derived chapters: work published since submission may change how you interpret their findings.

Develop the introduction and background

Write an introduction for the whole thesis.

Introduce the research area, explain the unresolved problem and state the thesis contribution early. Use the chapter map to select the background readers need before the first research chapter. Explain the scope of the work and finish with a brief account of how the chapters develop the answer. [2]; [3].

Draft a short introduction early to test the proposed structure. Keep collecting relevant literature and notes, then develop the full account once the research chapters are stable. [1]. For project-level openings and literature transitions, see Titles, abstracts and introductions.

Introduction, background, motivation and significance

These are functions, not mandatory headings. Their order and emphasis depend on the work and the agreed format. [6].

Introduction: orient the reader. Establish the topic, question and route through the thesis. The opening may lead to a separate background chapter or incorporate the fuller literature review.

Background or literature review: establish the basis for the question. Explain the necessary concepts and critically compare what is known, disputed or missing. Include the breadth needed to understand the studies, while connecting each topic to a later question, design choice or interpretation.

Motivation or rationale: explain why the investigation matters. Explain why the question is worth answering and how your approach could help. Saying that little is known leaves the reader to work out why knowing more would be useful.

Aims, questions or hypotheses: specify the task. State what the thesis investigates and how the studies contribute. If a study tests a hypothesis, explain the prediction used to test it. Do not invent prior hypotheses for exploratory or resource-building work.

Contribution and significance: preview what the thesis adds. Name the new evidence, method, resource or understanding and why it matters. Keep this preview proportionate to the completed work; develop the interpretation in the discussion.

Impact: explain possible or demonstrated use. Identify who could use the work and how. Benefits may lie within research itself. Distinguish prospective uses from demonstrated effects, and place this account where it fits the thesis rather than assuming a separate statement is needed. [10].

Build the background around the thesis questions.

Work backward from the chapter map to select the background and arrange it so that the aims follow naturally. Comparing and synthesizing papers explains how to compare evidence and evaluate conflicting accounts.

For the fictional seedling thesis, introduce heat exposure and root development, then explain why lateral-root emergence and primary-root length are different outcomes. Review what is known about their dependence on developmental timing. Because conditions other than heat differed in the greenhouse, the second project tests heat exposure under controlled conditions. The third asks

whether the response differs with the seedlings' age. Together, the studies help identify which effect needs a mechanistic explanation. They may guide later experiments, but do not demonstrate improved crop performance.

Write the methods and results chapters

The principles in Methods, results and figure legends apply to theses too: describe what was done reproducibly, organize findings around questions and comparisons, report uncertainty, and keep prose, figures and legends consistent. A thesis gives these principles more room and may distribute them across several chapters.

Methods explain both how and why. Describe the procedures and analysis, including why you chose them and which assumptions or limitations affect the conclusions. Include validation or method development needed to show that a measurement can answer the research question. Common procedures can appear in a shared account; explain each study's design and modifications in its own chapter. [11].

Results establish what the evidence shows. Introduce each question or comparison, present the finding and direct readers to its support. Include negative or inconclusive findings when they affect the answer. A thesis can accommodate fuller controls, validation and exploratory work than a short article, but each addition should serve the argument. Keep essential evidence in the main account and use appendices for supporting detail. Whether interpretation accompanies each result or follows in a separate discussion, make the distinction between observation and explanation clear.

Bring the findings together

Begin the final synthesis with what the thesis establishes. Use the findings from different chapters to explain that answer: one study may test an explanation suggested by another, or show that a result depends on particular conditions. Discuss disagreements and explain which conclusions remain uncertain. A reader should finish with an understanding of the combined contribution, rather than another summary of each chapter.

Discussion, synthesis and implications

A research chapter's discussion interprets that study; the general discussion asks what the whole thesis establishes. The functions below may form sections or recur within a discussion organized by theme. Use the detailed advice in Discussion and conclusions at both scales.

Overall answer: return to the thesis question. State what can now be concluded, including a partial answer where necessary. Use a brief summary of the decisive findings to support it.

Interpretation and synthesis: explain the relationships. Show whether studies reinforce, qualify or challenge one another. Differences between systems, measurements or conditions may explain an apparent conflict. A methodological chapter may enable a later result rather than provide another test of the same claim.

Comparison with the literature: locate the contribution. Explain how the combined findings change, extend or support existing understanding. Discuss relevant evidence that disagrees, and distinguish what your work establishes from explanations proposed by others. [12].

Limitations and alternatives: explain what remains uncertain. Identify conclusions that depend on an untested assumption or a measurement limitation, and discuss other explanations that still fit the findings. Consider weaknesses shared by the studies: repeating a result does not remove a bias present in every experiment.

Significance and implications: explain what follows. Develop the contribution previewed in the introduction. Separate a change in scientific understanding from a potential application, and state what further evidence the application would require.

Future work: choose the questions that follow. Select a few unresolved questions raised by the findings. For each, explain what the proposed experiment would help you learn.

For the fictional thesis, the central synthesis paragraph could read:

Example: a general-synthesis paragraph

> Brief heat pulses reduced lateral-root emergence in sterile culture, with a stronger response to exposure early in seedling development. Primary-root length remained unchanged, suggesting that the treatment affected root branching rather than uniformly slowing root growth. These controlled experiments provide a possible explanation for the reduced emergence observed under the greenhouse heat regime, where other environmental conditions also varied. The timing result now directs attention to the developmental processes that make early seedlings more sensitive to heat exposure.

The synthesis starts with the controlled result, then uses it to interpret the greenhouse pattern. Its order differs from the chapter map because the paragraph is organized around the answer.

For a complex synthesis, use a table to compare what different projects establish about the same question, including conditions under which their findings differ.

The conclusion

A separate conclusion is useful after a substantial general discussion. State the overall contribution, its scope and the next question that follows from it. Develop detailed interpretations and present essential evidence in the preceding chapters. [4].

Plan the route to submission

Work backward from the submission date.

List what already exists and what each unfinished chapter still needs. Working backward from submission, allow time for final checks and approvals, review of the complete draft, chapter revisions and any unfinished analysis or figures. Include permissions and declarations. Assign each task to a person and agree when it will be ready, leaving time for anyone whose work depends on it.

For work that may not finish, agree when to decide whether to continue and what you need to know to make that decision. Also agree what the thesis will contain if the work stops. You can then complete the document without leaving its scope dependent on an open-ended experiment.

Work item	Finished item	Dependency or decision
Revise the three research chapters	Complete chapter drafts with final figures and local methods	Supervisor comments returned before the general synthesis is revised
Integrate the heat-exposure findings	General-synthesis draft comparing the greenhouse, controlled-pulse and timing results	Research-chapter findings and interpretations settled sufficiently to compare them
Attempt the optional mechanistic follow-up	A reviewed result and analysis, if the measurement and controls support interpretation	At the agreed decision date, assess what has been learned, including negative or inconclusive findings, and decide whether to include the completed work or proceed without this follow-up
Prepare the submission copy	Rendered thesis, declarations, permissions and deposit files	Programme checks and committee review started early enough to finish before submission

Discuss what you would do if a critical step is delayed. You may need to narrow a claim, change the chapter order or revise the submission date. Agree the response with supervisors and update the affected chapters and schedule. [1].

Decide what to do with the last project.

When the last project has not yet become a paper, decide which deadline matters more. Write the paper first if competition, priority or a coauthor commitment requires rapid submission. Write the thesis chapter first if the thesis needs the fuller account and publication timing allows it. Reconsider the choice when the data, competition or graduation deadline changes. [1].

If the selected thesis format requires accepted publications, include that dependency in the submission plan. Finishing the prose cannot substitute for a publication requirement; discuss permitted alternatives early if acceptance is uncertain.

Check the complete document

Read the general introduction, research-chapter openings and final synthesis together. They should name the same question or framework, assign each chapter a distinct role and state compatible conclusions. Then inspect the rendered thesis against the current programme requirements: chapter order, cross-references, references, figure and table numbering, declarations, permissions, acknowledgements and deposit files. Return to the map if this check exposes a chapter that promises or concludes something different.

Before submitting the thesis

- Can a reader state the thesis question or framework, its contributions and their scope from the general introduction and final synthesis?
- Have supervisors agreed the structure, background depth and provisional length within the current programme rules?
- Does the background prepare the reader for the research questions, and are motivation, significance and potential impact explained without inflated promises?
- Does every research chapter contribute a distinct finding, test or boundary to that answer?
- Do article-derived chapters give readers the local context and method detail needed to assess them?
- Does the final synthesis compare findings across projects instead of repeating chapter summaries?
- Does the completion plan include analysis, review, administrative dependencies and a decision for each critical uncertainty?
- Has the rendered document been checked against the current local submission requirements?

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Chapter 6

Communicating research

Explaining research to an audience

The same study can call for very different explanations. A colleague may want to know whether a control distinguishes two mechanisms; someone outside the field may first need to understand why the result is surprising. Before preparing slides or writing a summary, decide who needs the explanation and what they should understand when it ends.

Start with audience and purpose

Find out who will attend and why. For an invited talk, ask the host which departments were invited and what the audience knows about your topic. A room of biologists is not necessarily one audience: some may know the biological problem but not your methods, while others know the methods but not the system. [1]; [3].

Purpose changes the explanation, too. A departmental seminar might introduce an unfamiliar discovery; a meeting with potential collaborators might need to show where their contribution could help solve an open problem. In the second setting, spend more time on the unresolved work and the help you are seeking. Write the intended outcome in one sentence before selecting material. [1].

With a mixed audience, everyone needs access to the question and main finding, even if only some listeners need the technical detail that follows. Introduce an unfamiliar concept when it becomes necessary to understand the next step. A long preliminary glossary asks listeners to remember terms before they know what the terms are for. [2]; [3].

Choose the message and the evidence

Before choosing results, try stating the main finding in a sentence. “Bacterial antiviral defence” names a subject; “Brake uses an RNA template to produce a gene encoding an antiviral protein” makes a claim the explanation can develop. A longer talk may need supporting messages for different parts of that finding, but they should still lead toward the central claim. [3].

That sentence gives you a test for each result: does it support the claim, or rule out a serious alternative? A colleague may need the control that separates two mechanisms, though not every preliminary experiment that led to it. Explain the comparison, measurement and distinguishing test whenever they determine what can be inferred. Procedural detail can wait for discussion or the accompanying paper. [1].

The harder decision may be what to leave out. An additional result that needs several new concepts yet adds little to the conclusion can wait for another occasion. A result that changes the conclusion belongs in the explanation, even if something else must be shortened. For an unfinished study, explain what the current result shows and what question it leaves open. [3]; [5].

Organize the explanation

Begin with a question or observation listeners can understand, then give enough context to show what needs explaining. For unfamiliar work, that may mean introducing the biological process or experimental comparison before stating the finding. [2].

When the finding emerges through several tests, let each result answer a question and raise the next one. In Brake, identifying a coding sequence in the DNA does not by itself show that the encoded protein causes defence. The next question is whether defence survives when translation is blocked but DNA production continues. Framing that comparison before its result lets listeners see why the experiment matters. [4].

For a short update or an audience already familiar with the question, give the conclusion first and work through its support. Successive questions are more helpful when listeners need to see how a

mechanism was resolved. In either order, the reasoning should determine the sequence; the dates on which experiments were performed need not. [1]; [4].

Some relationships are easier to follow when they can be seen. If listeners must keep track of several intermediates or transformations, a simple schematic can show how one leads to another. Introduce its objects and arrows while explaining them, using the same names in speech and in the diagram. Specialized displays and explanatory diagrams covers its design. [6]; [7].

Return to the opening question at the end. Give the answer and explain what remains unresolved. If the audience is interested in an application, say what would still need to be tested before the result could be used that way. [1].

Adapt detail and language

Explain an unfamiliar technical term when you introduce it: “the enzyme that copies RNA into DNA, a reverse transcriptase.” Specialists can usually start with “reverse transcriptase.” Then use the same name throughout, so listeners do not have to work out whether different labels refer to the same thing. [1]; [2].

Vocabulary is only one part of the adjustment. Researchers studying bacterial defence may need the construct comparison and the timing of DNA production. Biologists outside that field may first need to know why producing a gene from RNA is the finding. Both still need the experiment that supports the conclusion. The two explanations below use the fictional Brake study, inspired by Tang and colleagues (2024) [8].

Example: explaining the finding to two audiences

For researchers studying bacterial defence

> Brake posed a puzzle. Its locus has a reverse transcriptase and a conserved RNA template, but no conventional effector gene. The DNA product contains repeated units, with a coding sequence spanning their junctions. We detect single-stranded DNA before infection; during infection, we also detect the double-stranded product and the Brake protein.
>
> But is the protein responsible for defence? In a translation-disrupted construct, DNA production continues, yet growth arrest and restriction of phage A disappear. Expressing the protein separately restores both. That supports an RNA-templated route to a protein effector; it does not identify the protein’s molecular target. Brake also restricts phage B, while phage C remains productive.

For biologists outside the field

> Some bacteria use proteins to restrict viruses called phages. Brake was puzzling because we could not find a conventional gene for its defence protein. Instead, it copies an RNA template into DNA that carries the instructions for that protein.
>
> Was making the DNA enough for defence, or did the system also need the protein? We blocked production of the protein, but the DNA was still made. The bacteria no longer stopped growing, and phage A multiplied again. Supplying the protein separately restored growth arrest and restricted phage multiplication. So Brake builds a functional defence gene from RNA. It restricted two of the three phages we tested; the third still multiplied. We do not yet know what the protein acts on.

> The specialist version can name the DNA intermediates, coding junctions and construct comparison because its listeners can use those details. The broader version asks whether DNA production alone is enough, then gives the same decisive comparison in ordinary language.

An analogy can help listeners grasp an unfamiliar relationship, provided the correspondence is clear: which part of the familiar example stands for which object or process? Keep it brief, then return to the actual system for the evidence. If the analogy implies intention or a mechanism the study has not established, use another explanation. [2]; [5].

Test and refine the explanation

Try the explanation with someone whose background resembles the intended audience's. Ask them to summarize the finding and explain how the main experiment supports it. Where did they need more background, or lose the connection between results? [1]; [2].

Their response tells you what to revise. Someone who knows the terms but cannot explain why the result supports the conclusion needs a clearer comparison or transition. Someone who cannot follow the comparison may need a concept introduced earlier. If the main point disappears among details, remove a secondary result and try again.

Use a short outline to plan the explanation or review its order. The outline below connects the broader Brake version's question, evidence and answer before any slides are designed.

> **Example: an explanation outline for a biology seminar**
>
> **Purpose:** explain how an RNA template can supply an antiviral gene to biologists unfamiliar with this defence pathway.
>
> 1. **The puzzle:** Brake restricts phage propagation, yet its annotated region has no conventional effector gene. Introduce the reverse transcriptase and adjacent RNA template as the clues to where a gene might come from.
> 2. **Where the gene comes from:** show how repeated DNA contains a coding sequence. Connect the template and junction analysis to the observations of DNA and protein production.
> 3. **What causes defence:** compare intact Brake with the translation-disrupted construct. The continued production of DNA makes the loss of defence informative; separate protein expression then restores growth arrest and restriction.
> 4. **What this establishes:** the RNA template supplies a gene encoding the defence protein. Close with the tested phage range and the protein's unresolved molecular target.

Keep more detailed results available for questions about the experiments in the outline.

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Designing a presentation

Before polishing individual slides, arrange the questions and results into a sequence your audience can follow. Work out what listeners need to understand before each experiment appears.

Plan the sequence

Write a short sentence for each main point and place the supporting results beneath it. Arrange these groups so that each result answers a question the audience already understands. Add the background and experimental design needed to interpret it. A research talk will often move from the biological problem through a few connected questions to the answer and its implications. A lab meeting about a proposed experiment may instead compare designs and end with a decision for discussion. [1]; [2].

Sketch each slide with a proposed title, its figure or diagram, and a note about what you will say. Paper, a whiteboard or an unformatted slide list all work. Then read the titles in order. If one finding does not lead clearly to the next question, add the missing explanation or rearrange the slides before working on their layout. [4].

The following four-slide sequence uses the fictional Brake study. It introduces the unfamiliar route before asking listeners to interpret the decisive experiments.

Example: a connected four-slide storyboard

Slide 1: Brake restricts phage propagation although no effector gene is annotated

On screen: A compact locus diagram labels the reverse transcriptase and RNA template; the missing conventional effector is marked as the puzzle.

Spoken transition: “If there is no ordinary effector gene, how does this system restrict phage propagation?”

Slide 2: The RNA template is copied into DNA that encodes a protein

On screen: Introduce the RNA template, then add the single-stranded DNA detected before infection, followed by the double-stranded DNA and protein detected during infection. Keep earlier elements visible and label when each product was observed.

Spoken transition: “This gives us a possible route to defence. Does defence depend on the enzyme that makes this DNA?”

Slide 3: Brake reduces infectious phage yield, and reverse transcription is required

On screen: Show the three-condition culture plot at readable size: empty vector, intact Brake and inactive reverse transcriptase, with matched days retained.

Spoken transition: “Reverse transcription is necessary. Next, we need to test the protein’s role while leaving DNA production intact.”

Slide 4: The encoded protein is required for defence

On screen: Show that disrupting translation leaves DNA production intact but loses defence, and that expressing the protein separately restores defence. Conclude that the protein is required for defence; its molecular target remains unknown.

Spoken transition: “So Brake uses an RNA template to make the gene for a defence protein. The genetic tests establish the protein’s role, but we still do not know its molecular target.”

> The third slide can use the quantitative plot below. The fourth needs the translation-disruption and restoration results: the three-condition culture plot alone cannot establish the protein's role.

Decide which details belong in the main talk and which can be reserved for questions. A control that distinguishes two explanations belongs with the result it supports; an extended protocol can usually wait unless the method itself is under discussion. End with the answer to the opening question and its significance. Keep this conclusion visible during questions, with acknowledgments on the preceding slide or alongside it. [1]; [3].

Allow more time for an unfamiliar experiment than for a short transition. A fixed slides-per-minute target is unreliable when some slides build a diagram and others contain several comparisons. Determine the length by rehearsing the explanation aloud. [5]; [1].

Give each slide a clear message

Use the title to state the finding or question. "Brake reduces the yield of infectious phage A" tells listeners what to look for; "Phage restriction assay" names only the experiment. A question works well when the slide introduces a test or asks colleagues to consider an unresolved choice. Keep the title short enough to read at a glance and specific enough to describe the evidence below it. [3]; [2].

Choose one main message for the slide, then include the comparisons needed to establish it. This may require two related plots, an image and its quantification, or a schematic beside the result. If the slide contains several independent conclusions, split it. Keep plots together when listeners need to compare them directly. [3].

Write the spoken explanation in your notes. On the slide, retain the title, labels and brief annotations that help people interpret the visual. Replace a paragraph describing an experimental sequence with a diagram when the sequence is easier to see than to read. Short text can still be the best choice for a definition, a question or a small set of alternatives; a decorative image adds little. [3]; [4].

Adapt figures for live viewing

Prepare a presentation version of a manuscript figure. Select the relevant panels, enlarge the labels and place brief explanations beside the features you will discuss. Keep the units, controls, observations and uncertainty needed to assess the claim. If these cannot fit legibly, spread the explanation across slides rather than shrinking the complete publication figure. [2]; [4].

For an unfamiliar assay or display, first explain the experimental comparison and how to read the result. A small schematic or an illustration of the expected patterns can help. Distinguish predictions from measurements. For a familiar plot, a brief spoken orientation to the conditions and axes may be enough. [2].

The following slides use the fictional Brake study, inspired by Tang and colleagues (2024) [6]. Brake is a bacterial defence system containing a reverse transcriptase. The experiment compares the infectious yield of phage A from cultures carrying an empty vector, intact Brake or Brake with an inactive reverse transcriptase.

Example: turning a manuscript result into a slide

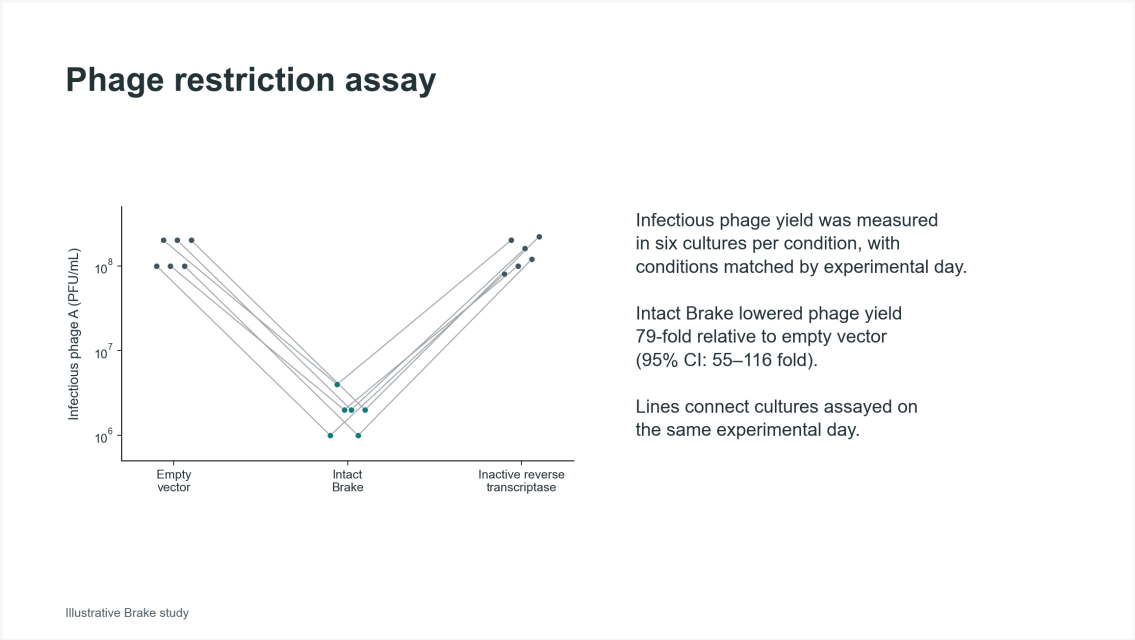


Figure 1: A draft slide with a method-based title, a small three-condition graph and a manuscript-style explanation beside it.

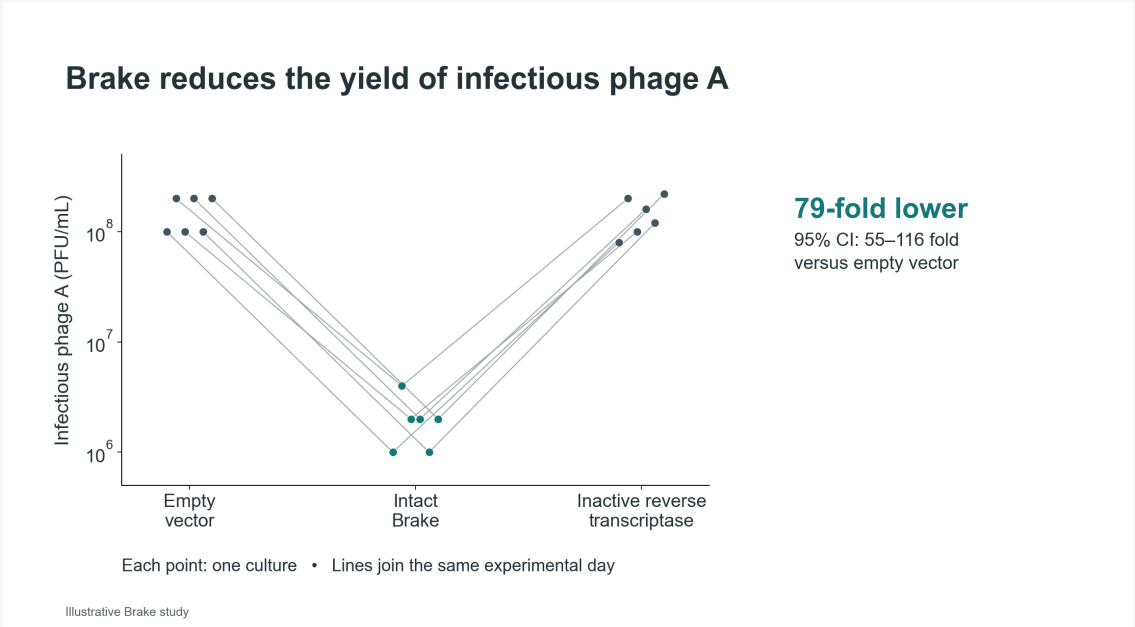


Figure 2: The revised slide enlarges the same observations and states the finding in the title. The annotation reports a 79-fold reduction in infectious phage A yield versus empty vector (95% CI: 55–116 fold).

Both slides retain all observations, day matching and the inactive-enzyme control. The revised slide gives more space to the comparison and replaces the paragraph with labels. The presenter can explain how the cultures were assayed while listeners inspect the result. The control remains visible, although the numerical estimate refers specifically to Brake versus empty vector.

Original teaching figure; all plotted values are illustrative.

Highlight the condition or region you are discussing while keeping the other evidence visible. Across slides, preserve condition colors, axis limits and plot positions when viewers need to compare results. If a scale changes, make the change explicit. [4].

Make the slide readable

Build a simple layout with a consistent title position, generous margins and aligned visual elements. Put labels next to what they describe and separate unrelated groups with space. Give the central figure most of the slide rather than squeezing it between a large institutional banner and a long caption. [4].

Check the smallest essential label at the size and distance the audience will see it. A graph that is readable on your laptop may fail on a projector or in a shared screen. Increase the figure and its labels together; enlarging the title alone does not help. If the slide is still crowded, remove a secondary point or split the explanation. [3].

Use strong text–background contrast and consistent sizes for titles, labels and annotations. Pair color with labels, symbols or line styles so that viewers can distinguish conditions with poor projection or differences in color vision. Avoid placing text directly over a photograph; put it on a solid background if the image interferes with reading. For ways to distinguish groups without relying on color alone, see Color, symbols and accessibility. [3].

Use builds and prepare the shared version

Reveal a complex diagram in stages when you need to explain its parts in sequence. Keep the existing elements in place as new ones appear. A few duplicate slides with additional labels or arrows are often easier to edit and export than a complicated animation. Check every copy after a revision so that an old label or value does not reappear. For a comparison that requires simultaneous inspection, show both alternatives together. [3]; [4].

Use video or animation when motion, timing or a change in state is part of the explanation. Plan where to pause and what to point out. Keep a still image or short sequence that supports the explanation if playback fails, and test the exported deck as well as the editable file. [3].

Credit published figures and ideas on the slide where they appear, and identify collaborators' contributions. A compact author–year reference can point to full details in the notes or reference slides; retain any required image credit. For unpublished work, agree with collaborators what can be shown and shared. [3].

If people will read the deck afterward, add the explanations they would otherwise hear from you in notes or a companion document. Check that the shared file includes those notes or explanations, and that diagrams revealed in stages can still be followed in the exported format. [5]; [3].

Before rehearsal, review the sequence in slide-sorter view. Then display the slides at presentation size to check small labels and crowded figures.

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Delivering a talk and answering questions

Rehearse and revise

Rehearse the complete talk aloud while it is still easy to change. Listen for awkward wording and results whose explanation takes longer than expected. Where the move from one result to the next feels abrupt, check whether you have explained the question the next experiment answers. Revise the transition or reorder the material before trying another run. [3].

Time the complete talk, including the opening, transitions and close. If the run reaches the end of the allotted slot, remove a secondary result, shorten a detour or move a detail to material for discussion. Keep the comparison that makes the central claim credible, and leave the agreed time for questions. [2].

Mark a few timing checkpoints in your notes, and decide which slides you can skip if questions or interruptions use the spare time. Use a visible timer during the talk. If you fall behind, move to the prepared shorter version and preserve the conclusion. [1].

Rehearse the shorter version, including what you will say between the slides that remain. Skipping a result may leave a later reference or explanation without its context.

Practice the beginning and end until you can deliver them without searching for the next sentence. They need not be memorized verbatim. The opening can introduce the question and help listeners follow the first result; the close can state the answer and what remains unresolved. If possible, rehearse with someone outside the immediate project, who is less likely to fill in missing explanations from memory. Ask what they took to be the main claim, which comparison supported it and where they lost the thread. [4].

Rehearsal tool: revise after each complete run

Run	Observe	Revise before the next run
First complete aloud run	Where did a sentence become hard to say or a transition sound like a jump?	Rewrite the spoken link; state the question that the next result answers.
Timed run	Which material puts the close or discussion period at risk?	Cut a supporting detour or reserve it for a question; do not compress the decisive comparison.
Listener run	Can a colleague outside the immediate project identify the claim and explain the key comparison?	Add the missing premise or replace a term that obscures the comparison.
Recovery run	After a planned interruption, can the shorter route still reach the conclusion without dropping its decisive control?	Skip a named secondary result and rehearse the bridging sentence into the next essential comparison.
Delivery check	Can you speak to the displayed evidence without relying on a moving pointer or an unreadable note?	Add a label or annotation, and simplify the spoken cue.

Record one run if you can. Check whether you rush through transitions or spend long stretches speaking to the screen. Notice whether a repeated movement distracts from what you are explaining. Choose a specific change to try in the next run. [3].

Explaining figures aloud

Before explaining a dense figure, tell the audience what to look at. Describe the observation and what you conclude from it, then introduce the next question: “This shows that the DNA is made. The remaining question is whether its encoded protein causes defence.” [3].

Give listeners time to read unfamiliar axes and locate the groups being compared before explaining the result. A pause after the display appears can help. Speak clearly enough for the room or call, at a volume you can sustain. If nerves make you rush, pause at the next transition and take a breath before continuing.

Use a pointer briefly to identify a feature, then hold it still or switch it off. Put essential arrows, labels or staged annotations in the displayed material so the explanation still works without it. Explain the visual material in the order in which you want it inspected, then turn back to the listeners. [3]; [2].

Room, equipment and remote setup

Before an in-person talk, check the room’s display and audio, or ask the host what is available. Confirm how your computer connects and whether you need an adapter. Open the deck you will use and check that the figures are legible from the audience’s seats. Test the slide controls, bring a backup copy and decide how you will continue if the pointer or clicker fails. Confirm whether questions will come during or after the talk and how much time is reserved for them. [3].

For a remote talk, test screen sharing, microphone and camera in the meeting software. Check what a participant can see and hear, and how you would reconnect if the connection drops. Ask the host about captions and question handling. Tell listeners whether to use chat or speak when both are available.

Keep additional slides ready for likely questions, such as a control or methods detail omitted from the main talk. Practice finding them and returning to the slide where you left off. [3].

Answering questions

Let the question finish. If its target is unclear, ask a short clarification or repeat the part you will answer: “Are you asking whether the protein, rather than the DNA intermediate, accounts for phage restriction?” Repeating the question lets the room hear it and gives you a moment to formulate the response. Then answer that part first, using the relevant comparison. [3].

If you do not know the answer, say so. Explain what the study can establish about the question and, when useful, suggest how the unresolved point could be tested. If a challenge reveals a limitation, acknowledge it and explain how it affects the conclusion. If it rests on a false premise, correct that premise before answering. When repeated interruptions prevent you from continuing, ask the chair to help or suggest returning to the discussion in the question period. [1].

When a question requires a separate conversation, say so and offer to continue after the session. If the chair closes the discussion, stop. [2].

The fictional Brake study below is inspired by Tang and colleagues (2024) [5].

Example: a focused answer to a mechanistic challenge

Question: “If the translation-disrupted Brake construct still makes the DNA intermediate, how do you know the DNA does not itself cause phage restriction?”

Answer: “The DNA could still contribute, but this experiment shows that it is not sufficient. When we block translation, the DNA is still produced, yet growth arrest and phage restriction disappear. Supplying the protein separately restores both. That is why we interpret the defence as protein-mediated.”

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Posters

A poster is read in fragments. Someone may notice the title while walking past, approach the central result, ask a question, then leave with a contact or a link.

Choose the content

Decide what you want to discuss: a finding, an unresolved problem or a possible collaboration. Choose content for the people you expect to meet. A poster that will also circulate afterward needs enough explanation to be read without you. [1].

Write the main finding as a sentence, then choose the evidence readers need to assess it. Give the decisive comparison the largest area. Include enough background to explain the question and enough methods to interpret the result. Put secondary analyses and procedural detail in a handout or linked material unless they are needed to evaluate the central finding. [2]; [3].

Identify the presenter, affiliation and contact details. Place them where they are easy to find but do not displace the claim or evidence. [2].

Adapt manuscript figures for the poster: select the relevant panels, enlarge labels and annotate the key comparison. Retain the controls and uncertainty needed to interpret the result. [3].

Arrange the poster

Sketch the reading order before refining fonts or colors. Make the main finding prominent, place related text and figures together, and separate sections with whitespace. Align blocks along a few consistent edges. Use headings or numbers where the sequence would otherwise be unclear. [2]; [3].

The fictional Brake study below is inspired by Tang and colleagues (2024) [5].

Example: a focused poster portion

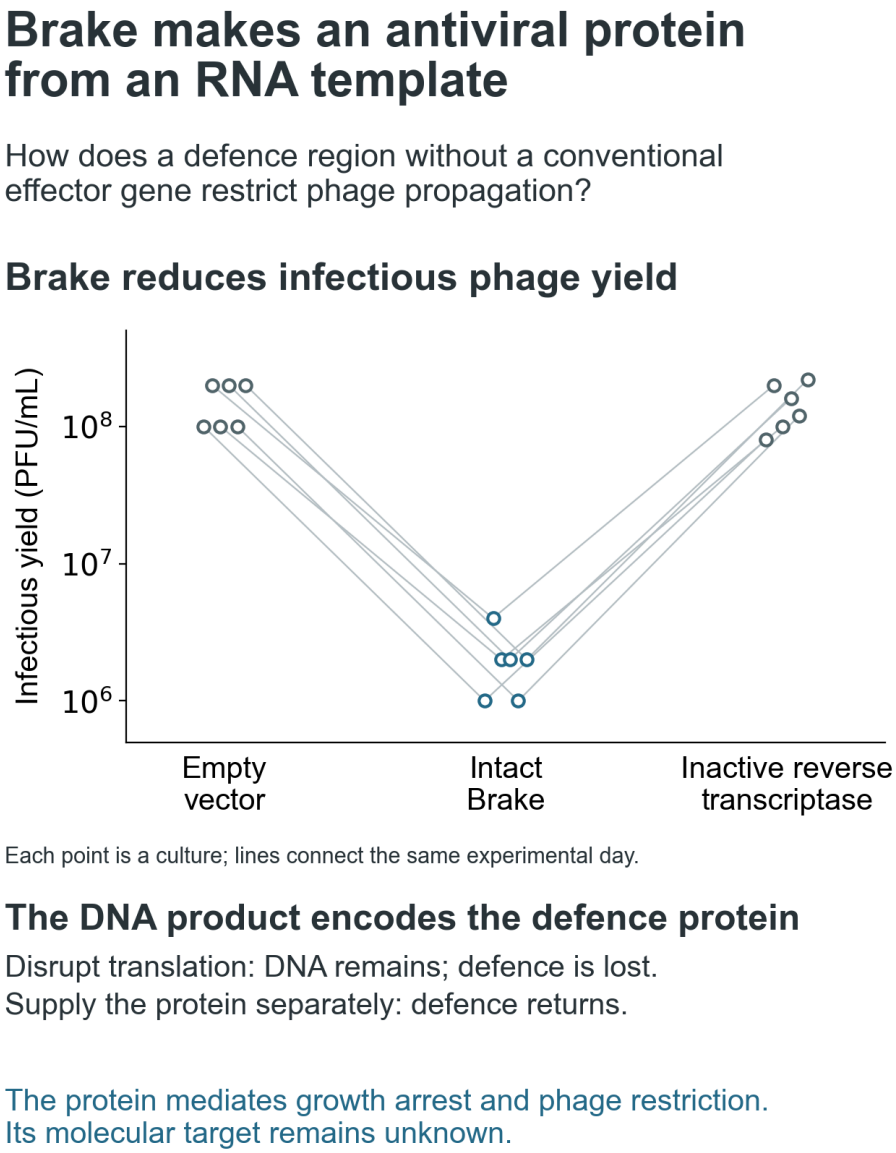


Figure 3: This focused poster portion gives most of its space to the quantitative restriction result, then uses a compact test strip to separate the DNA intermediate from the encoded protein. Original teaching figure.

The quantitative result occupies the largest region. The smaller translation test answers the next mechanistic question without competing with the main plot.

When several results need to be compared, place their evidence blocks beside one another and give each a heading that states its local finding.

Use headings that state the finding where possible: “Translation is required for phage restriction” tells a visitor more than “Results.” Use consistent type sizes for headings and body text, and align related blocks. Color can reinforce a grouping or emphasis, but readers should also be able to identify conditions from labels or symbols. [3]; [2].

Print and digital readability

Check the conference specification before committing to orientation, dimensions, logos, file format or display method. For a conventional A0 print poster, make an early A4 reduction and read it at normal page distance; then inspect the full-size poster in the expected light from roughly 0.5 to 1.5 metres. In both checks, the title, headings, axes, annotations and contact details must be legible. Use the production and export checks in Preparing figures for publication. [2]; [3].

For a digital poster, keep an overview visible so visitors can identify the topic and main finding even if they arrive halfway through an animation or explanation. Check which interactive features the conference platform supports and whether visitors can operate them. An animation can explain a process; a zoomable image can reveal detail. Neither helps if visitors cannot find the comparison or return to the overview. [4].

Use the poster in conversation

Prepare a short orientation: the question, the finding and the decisive evidence. Give it when someone asks what the poster is about, then pause. For the model above: “Brake restricts phage propagation, but its locus has no conventional effector gene. The large plot compares intact Brake with an empty vector and an inactive reverse transcriptase. Infectious phage yield is much lower with intact Brake than with either control. Is the protein encoded by Brake’s DNA needed for defence? We blocked protein production but still detected the DNA. Growth arrest and phage restriction disappeared, and both returned when we supplied the protein separately.” Their next question tells you whether to clarify the method, discuss a limitation or offer deeper material. [1]; [2].

Rehearse with the poster in front of you. For a visitor unfamiliar with the study, start at the question; for a specialist, start at the finding. Point to the relevant evidence as you discuss it. If the conversation becomes lengthy, invite others to join or arrange to continue later. [2]. See Delivering a talk and answering questions for rehearsal and difficult questions.

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Elevator pitches and chalk talks

A short pitch introduces your research and invites a conversation. In a chalk talk, you explain a proposed research program while the audience questions its plans and experiments.

Elevator pitches

Think about what you would like to discuss with this person. If you want advice on a difficult experiment, introduce the problem and explain where you are stuck. If you hope to collaborate, explain how your work connects with theirs. [2].

Start with the scientific question or finding. Introduce your role first if it helps the listener understand the connection. Give enough evidence to make the point credible, then leave room for a reply. You can explain the methods or discuss further results if they ask.

Follow the listener's questions: they may want more technical detail or need some background first. When rehearsing, ask your partner to explain what they understood and what they thought you were asking for. Revise where their account differs from yours. [2].

The fictional Brake study is inspired by Tang and colleagues (2024) [5].

> **Example: a pitch that opens a specialist conversation**
>
> "You work on bacterial defence, so can I run a strange result past you? Brake has no conventional effector gene. Instead, an RNA template supplies DNA that encodes an antiviral protein. When we block translation, the DNA remains but phage restriction disappears; supplying the protein separately restores it. Have you seen infection-linked DNA intermediates play a similar role in other systems?"

Rehearse the pitch with these questions:

Check	Revision when the answer is no
Could this listener repeat the question or contribution without my field's shorthand?	Replace one term with its role in the problem, or supply the missing premise.
Is the evidence enough to make the stated point credible?	Add the decisive comparison; remove a less informative detail.
Does the final sentence give the listener a natural way to respond?	Ask a focused question or name the conversation you would like to have.

Chalk talks

A chalk talk usually presents a proposed research program through an interactive discussion of its questions and experiments. Ask the host about the audience, duration, use of boards or slides, and expected level of detail. Prepare to explain both the first experiments and what you would do with different outcomes. [1]; [3].

Begin with the overall question in terms a scientist outside your field can understand. Sketch the projects before questions take the discussion into detail, then start with the most important one. Explain who would begin the work and when, including which projects can run in parallel. [4]; [3].

For each project, explain the question and the experiment that could answer it. Discuss what different results would lead you to do next and what resources you need to begin. Show how the program develops your own questions, identifying which work you will lead and which depends on continuing collaborations. [1].

Keep an outline of the program in view as questions arrive. Write the project questions and their connections where everyone can refer to them; explain the details aloud. If a question exposes a flaw, address it and revise the plan when warranted. If discussion moves away from the program, offer to return to that question later and continue with the next project. [3]; [4].

Example: a spatial program map for the board

This hypothetical program asks whether a previous inflammatory exposure changes epithelial responses to a later injury. Keep the program question across the top of the board, then place the first two projects side by side because they can begin independently.

How does resolved inflammation change the next epithelial injury response?

Left board — Project 1, starts in year 1	Right board — Project 2, months 9–24
Retained state or changed cell mixture?	**Does exposure history change repair?**
After the cue is removed, compare exposed and control airway organoids. Measure cell composition, expression and accessibility.	Re-injure matched organoids. Measure barrier repair and inflammatory output.
Branch: if no retained difference appears, narrow the proposed scope and decide whether another exposure condition tests a distinct, justified question.	**Branch:** if no functional difference appears, do not call the earlier molecular pattern a memory response; ask what narrower biological conclusion the result supports.

↓ Continue only if the first projects establish a reproducible functional difference

Project 3 — What sustains that difference? Months 18–36, if warranted. Nominate candidates from Projects 1 and 2, perturb a candidate and repeat the injury-response test.

Board margin: residual cue? transferable factor? changed cell composition? Write these confounds where the room can return to them. Keep access to primary cultures, the cue-clearance assay and single-cell measurements beside the project that needs each resource.

Keep the top question and the two parallel projects visible while sketching detailed comparisons below them. Discuss staffing and access alongside the experiment that depends on each resource. The conditional third project should be physically downstream, so the board does not imply that a mechanism will be pursued regardless of the first results. A clear overview matters because questions may interrupt before every experiment has been discussed. [4]; [3].

Example: respond by sharpening the program

Question: "How will you tell a cell-state change from residual cue or a different mixture of epithelial cells?"

Response: "I would first measure the residual stimulus with an assay sensitive below the concentration that affects these cells. Transferring conditioned medium to untreated cultures would test for a soluble factor, whether residual stimulus or a new secretion. We would also compare the same cell types using lineage and single-cell measurements. A difference that remains would justify testing for a persistent cell state; the re-injury experiments would establish whether the exposure history changes function."

Rehearse with interruptions

Run a mock session with close scientific colleagues and people less familiar with the topic. Ask them to interrupt and challenge your reasoning, including whether the work is feasible and distinct from your former supervisor's program. Check that they can read the board from their seats. Afterward,

identify where they lost track of the program. Revise experiments that could not distinguish the proposed explanations, and be ready to explain how you would obtain the equipment or expertise they need. [1]; [4].

Prepare more experimental detail than you expect to present. In a quiet session, use it to develop a comparison or discuss alternatives. In a lively session, return to the overview between exchanges so that questions about one project do not consume the whole talk.

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Chapter 7

Writing grants and fellowships

Choosing an opportunity and planning the application

Before drafting an application, check whether you can apply and whether the funder supports the work you want to do. Prepare an initial budget to see whether the available funding would cover it [2].

Eligibility and fit

Start with a short list of plausible opportunities. Your institution's research office, colleagues and funders' previous awards can help you find schemes that support similar work. Read the current call before using a colleague's successful application as a model: the scheme's scope or requirements may have changed. Previous awards cannot tell you how likely your application is to succeed [2].

Check eligibility. Read the current call's requirements for applicants, host institutions and partners. If a condition is unclear, ask your research office or the funder's contact and keep their answer with the application instructions.

Read what the funder wants to support. A fellowship may place substantial weight on your training, while a pilot grant may support testing whether a method works. Explain how your proposed work meets that purpose. Where the funder welcomes enquiries, send a brief project description and ask about the point you are unsure of [2]; [1].

Application requirements

Read the application instructions and assessment criteria together. Check where the application asks you to address each criterion and how it will be weighted [2].

List the required documents and portal fields, who will prepare them and when they are needed. Record the applicable length limits and use the supplied templates from the start. Allow time for another person to approve a letter or support statement before its due date. Check whether portal registration or institutional authorization must be arranged before submission; finishing the text may not be the last step.

Clarify questions that could change the proposed work before drafting in detail. For example, if institutional costs count towards the award limit, less money remains for the experiments. If a partner has not agreed to provide samples, you cannot yet rely on that access. Ask the funder or research office about an unclear rule, and the partner about their commitment. Check for official amendments to the instructions during preparation [2].

Preparing a budget

Start with the work you propose to do. Estimate the staff time and materials needed, and ask facilities or service providers for quotes. If you plan to use existing equipment, check that time on it is available and include any running charges. Note what the institution or partners have agreed to provide.

Work with your finance or grants office to check which costs the funder allows. They can also establish whether institutional overheads count towards the funding limit. Direct costs pay for the project itself; indirect costs cover shared support, such as administration and building services. Include the applicable charges before deciding whether the project is affordable [2]; [1].

Check whether the requested funding and confirmed contributions cover the work. If essential costs exceed that support, reduce the scope, secure an agreed contribution or choose another opportunity. Removing a necessary cost from the budget leaves the work unfunded. Request what the project needs, even if it is below the maximum award. Eisenstadter advises examining a foundation's usual

award scale and explaining each expense; her observations concern smaller foundations and do not establish that cheaper proposals generally win [3].

The following fictional case concerns a researcher who wants to use spatial RNA profiling in plant tissue. A usable preservation and processing workflow has not yet been established; no pilot measurements or biological findings are available. Both opportunities and all limits below are invented for the example.

Opportunity decision: establish the workflow first

Chosen opportunity: A nine-month methods pilot with a maximum total request of €35,000.

Scientific fit: The pilot supports establishing a workflow and deciding whether it can support the planned biological study. It does not require us to promise the larger seasonal comparison within nine months.

Information checked: The research office confirmed applicant and host eligibility, the institutional submission route and that service charges are permitted. The facility supplied a quote and indicated capacity conditional on an award. Finance estimates €29,000 for staff time, services, consumables and the permitted institutional charge. That total includes €3,000 of institutional costs within the €35,000 cap. Existing space and equipment are institutionally supported; no equipment purchase is requested.

Alternative: A two-year biological project could fund a seasonal comparison, with a maximum total request of €180,000. It is also eligible and relevant. However, access to external seasonal material remains unresolved, and the workflow is untested. We cannot yet show how we would obtain and process the samples needed for that comparison.

Decision: Prepare the pilot application. Reconsider the larger study once sample access is agreed and we have evidence that the workflow can support the seasonal comparison.

Condition for reconsideration: If the facility cannot provide the required service within the proposed period, or revised costs leave essential work unfunded, revisit the pilot's scope or choose another opportunity before completing the application.

The pilot proposal still needs to specify how the workflow will be assessed and what different results would mean for its use in the biological study.

Contributors and reviewers

Identify who will lead the proposal, who will supply scientific or technical contributions, who will cost it and who is authorized to submit it. Agree what each person will provide and when. Confirm what help the research office offers; support may include interpretation of the scheme, budget checks, editing or submission, and it differs between institutions. If no such office is available, identify experienced colleagues and the person who can authorize the application rather than assume those tasks will happen [2].

Ask each partner what they will contribute and whether they can provide it during the proposed project. If a collaborator will supply samples or run an assay, agree the work and timing before describing that support in the application. Use collaboration, authorship and credit to agree scientific responsibilities and credit.

Invite readers before you need the review. A reader familiar with the method can examine feasibility and missing controls; someone outside the immediate field can test whether the question, rationale and contribution are understandable. Give them the call's criteria and ask for specific criticism, with an agreed return date. Send a complete draft early enough to discuss a major objection and revise the science or scope [3].

Scheduling and submission

Begin with the institution's internal deadline and the funder's cutoff, including the specified time zone and who must complete each submission step. Work backwards: when must the package be ready for authorization, the costs agreed and reviewers' comments returned? Set the drafting schedule around those dates. Request quotes and support statements early enough for the people supplying them to respond. Scientific writing and costing can proceed in parallel; send changes in the proposed work to the person preparing the budget.

Choose lead times from the work and people involved. Eisenstadter suggests that two months before a deadline may not be too early to finish a draft for review, and allowing a week or two away from it can expose buried information. The institutional committee example in *Making the Right Moves* also begins at least two months before its NIH deadline. These illustrate room for thinking and revision; they are not a universal timetable [3]; [1].

If time is short, check whether the necessary access, review and authorization can still be arranged. A smaller proposal may be possible if it answers a worthwhile question with resources already available. But if cutting the work removes the experiment needed to answer that question, consider a later round. Waiting delays the research and may mean missing this opportunity; weigh that cost against submitting a project that no longer answers the question or still lacks essential support.

The fictional pilot applicant has seven weeks until the funder cutoff. The research office requires the final package five working days earlier and has agreed to begin checks before then. The applicant and contributors agree the following plan; its future tasks have not yet been completed.

Application plan: review before authorization

Seven weeks before cutoff — choose and assign. The applicant records the eligibility and fit checks above, confirms the current instructions with the research office and invites two readers. The facility and finance contact agree dates for support text and final costing.

Five weeks before — make the proposal assessable. The applicant completes a scientific outline and first full account of the workflow work, including how its performance will be assessed. Finance prepares the initial budget; the applicant sends draft access and support statements to their owners.

Four weeks before — obtain scientific review. A methods reader examines measurement, feasibility and risks. A colleague outside the immediate field reads for purpose and rationale against the pilot's criteria. Both return comments within the agreed week; the applicant discusses substantial objections with them.

Three weeks before — reconcile revisions and support. The applicant resolves comments, revises scope where needed and sends changes to finance and the facility. Final costing and support text are agreed against the revised work.

Two weeks before — assemble for institutional checks. The applicant provides the narrative, budget, applicant information and required support documents to the research office. The office checks the assembled package and requests corrections before its internal deadline.

At least five working days before — complete authorization. The applicant delivers the corrected final package. The research office completes the required authorization and confirms who will submit.

Target three working days before — submit and confirm. The authorized submitter sends the application through the specified system. The applicant and submitter check validation messages and receipt, resolve correctable problems before cutoff, and retain the submitted version and confirmation.

Before submission, reconcile the assembled application with its commitments. The narrative, budget and support documents should describe the same scope, people and access. Inspect the generated application or entered fields where the system makes them available: an uploaded file does not establish that every required component has been accepted. Confirm receipt through the applicable system rather than treating a saved draft as a submission.

Retain the submitted version, its costing and the instructions used. Consider how you would proceed if funding were declined or delayed.

References

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Writing the proposal

A proposal explains why a question is worth pursuing and how you plan to answer it. Give reviewers enough detail to assess the experiments and judge whether your team can carry them out. Explain what you could learn from different outcomes.

Use the current call and application instructions to determine the required sections and supporting material.

Research question and significance

Put the objective near the beginning, before a detailed history of the field or description of techniques. Introduce the unresolved problem, say which part you will address, and explain why the answer matters. If a hypothesis organizes the proposal, describe how you will test it and what would count against it [5]; [6].

Make clear what is already known and what you propose to find out. Support claims about the field with evidence, identify hypotheses, and use future tense for planned experiments. Explain exactly what preliminary results establish. Images might demonstrate that you can measure a response without yet showing which mechanism causes it.

Significance explains why the answer matters. Innovation explains how the question or approach differs from existing work. If you propose a new instrument, explain which comparison it makes possible and why that comparison matters. A familiar method may be sufficient to answer an important unresolved question. Compare the proposal with specific prior work before calling it innovative. If you suggest a practical application, explain what further work would be needed: an experiment in a controlled system rarely establishes a therapy or field application by itself [6]; [3].

Aims and experiments

State what each aim will establish. “Determine whether a response depends on treatment order” gives the experiment a purpose; “perform imaging” leaves the question unanswered. Then describe the comparison and explain how the measurements and analysis will answer it. Include the controls needed to distinguish explanations and identify which samples or experimental runs provide independent replication [2].

Explain how the aims depend on one another. If a later experiment tests the cause of a response you have not yet established, say what must be learned before it can begin. Describe what useful work remains if that response does not appear. Aims can depend on earlier results, provided reviewers can judge the risk and the alternatives [6].

For each aim, explain what different results would mean. A precise result inconsistent with the hypothesis may answer the question. Failure to deliver the treatment or measure the response leaves the proposed test incomplete. Plan for both possibilities [5].

Feasibility and alternatives

Separate scientific uncertainty from technical uncertainty. Whether a mechanism operates is the question; whether you can impose the intended condition, obtain the samples or measure the endpoint is a feasibility issue. Show what you have already checked. Preliminary work may establish that a method works; an access agreement documents who has agreed to provide the samples and on what terms. Where a procedure remains untested, describe how you will assess it, how much time this will take and what happens if it fails. Burchardi’s account of an economics ERC panel emphasizes access to data and a design capable of answering the question. His examples illustrate feasibility concerns, rather than requirements for every funding scheme [1].

Explain what an alternative method would let you learn. If automated scoring fails, manual scoring may still allow a comparison at the final observation time. But observations made by hand may be too infrequent to test a hypothesis about response timing. State when you would switch, whether the alternative answers the same question, and whether you can complete it with the time and resources requested. If the method works and the result challenges the hypothesis, reconsider the hypothesis.

Use the applicant and environment sections to show how the work can be done. Connect your experience to the proposed tasks and identify collaborators who provide expertise you lack. Confirm access to essential facilities and distinguish agreed support from requests still awaiting an answer. These commitments should agree with the staffing and costs elsewhere in the application [3]; [2].

A focused proposal excerpt

In this fictional basic-research proposal, a researcher plans to characterize seed accession S under a defined pulsed-light protocol. The researcher has seed-handling and image-scoring experience and confirmed access to programmable light chambers, but no preliminary biological result. Red/far-red order effects are already known; the proposed question concerns this accession and protocol. A real application would need literature showing why this particular characterization is still needed, calibrated treatment parameters and the sections required by the scheme.

Objective and significance. We will characterize accession S's germination response to a defined red/far-red light protocol. We will determine whether exposure order changes germination timing when the dose of each exposure is held constant, and how the interval between exposures affects that comparison. We hypothesize that reversing the order produces different germination trajectories within this protocol. If changing the order or interval changes germination, that detail will need to be specified when the protocol is used for this accession. The project will not identify a receptor mechanism, establish a treatment for agricultural use or determine how other accessions respond.

Aim 1: Determine whether exposure order changes germination. Seeds will receive one calibrated one-hour red exposure (R) and one calibrated one-hour far-red exposure (F), in sequences R⟶F or F⟶R with a fixed dark interval. Each sequence will deliver the same dose of R and the same dose of F; temperature, hydration and seed age will be matched. All final pulses will end at the same seed age, followed by the same observation period. A dark-only group will establish germination without the pulses. We will record time to visible radicle emergence and retain the fraction that has not germinated by the observation limit. This comparison tests order under the specified doses, rather than attributing differences to total illumination.

Aim 2: Determine how the interval between exposures affects the order comparison. For both sequences, we will compare predefined short and long dark intervals. Compensating darkness before the first pulse will keep final-pulse timing and the subsequent observation period aligned. Doses will remain unchanged. We will estimate the order difference at each interval and compare those differences directly. A smaller difference after a longer interval would be consistent with a response that depends on the temporal separation of the pulses; it would not, by itself, identify molecular memory. This aim uses the predefined sequences and can proceed without selecting a favorable result from Aim 1. An imprecise order estimate would still limit what either aim can conclude.

Design and feasibility. Each chamber run will receive one complete light schedule. Schedule allocation will be randomized and balanced across chambers and run blocks; containers will be randomized across positions. Several independently sourced lots of accession S will be represented across schedules. The analysis will retain chamber and run blocks and the nesting of seeds within containers and chamber runs. Containers sharing a chamber run will not be counted as independent treatment replicates, nor will repeated runs from the same lot count

> as new independent lots. Before biological testing, we will verify delivered spectral doses and temperature stability and compare image scoring with manual scoring. Calibration variation and the precision needed for an informative order contrast will determine the replication plan. Our existing seed-handling and scoring experience supports this stage; reliability of this particular pulsed assay remains to be established.
>
> **Risks and interpretation.** Poor seed viability could make an apparent lack of response uninterpretable, so each lot will have a separate viability check. If image scoring is unreliable, we will use manual scoring at planned observation times only if it can retain the timing resolution required by the aims; otherwise we will narrow the timing question or revise the design before scaling up. If dose or temperature control fails, we will repair and requalify the treatment system before testing the hypothesis. If the checks pass and the estimates exclude an order difference large enough to matter for the stated question, we will report that constraint at the tested doses and intervals. We will not call an uncertain estimate evidence that order never matters.

Treatment calibration, a justified precision target and the resulting replication plan remain unfinished. Those details would need completion before this became a submission-ready proposal.

Structure and readability

Use the required headings. Open each section with its main point, then develop the explanation. A reviewer skimming the objective and aim statements should understand the question and how you plan to answer it. Explain terms that a scientifically trained reader from another field may not know. Give the reason for an experimental comparison where you describe it, even if the background section has introduced the problem [5]; [4].

A figure can help explain a complicated design. For a schedule with several interventions, show when each treatment occurs and when measurements will be taken. If one aim depends on another's result, a diagram can show the dependency and the alternative you would pursue. Explain the comparison in the caption so readers can interpret the figure without searching the main text. Follow the scheme's format and accessibility requirements [5]; [4].

Ask a specialist to challenge the scientific logic and a reader outside the immediate field to explain the proposal back to you. Can they say what the aims will establish and why the work is feasible? What would you learn if the expected result did not appear? Where they struggle, check whether an explanation is missing, a comparison is ambiguous or a claim lacks support. After revising, read the whole application. A change to an aim may also require changes to the figures, staffing or budget [5]; [6].

References

1. Burchardi, K. [@kbburchardi]. (2022). **Last year I had the chance to serve on the @European Research Council (ERC) Starting Grant evaluation panel.** *Twitter* thread, 7 March.
2. Burroughs Wellcome Fund and Howard Hughes Medical Institute. (2006). **Making the Right Moves: A Practical Guide to Scientific Management for Postdocs and New Faculty.** 2nd ed. Chapter 9.
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Funding interviews and preparing for an award

Prepare to discuss the proposal's most important questions and its weakest points at interview. If the application succeeds, check the award's terms with your institution before arranging the start of the work.

Prepare for the panel's questions

Read the invitation alongside the submitted proposal and assessment criteria. Confirm the presentation time, question period, permitted materials, submission arrangements and interview format. Use those instructions to choose what fits. A previous applicant's slide deck can show how they explained their project; it cannot establish this year's rules.

Prepare a concise opening that explains the question and the work proposed to answer it. Why does the question matter? Which comparison is decisive, and why can this team carry it out? Leave time for the panel to examine those choices. In his account of an ERC panel, Burchardi recommends short, focused answers about contribution and feasibility, with backup material for likely questions [1].

Reread the application for assumptions the panel may question. If a facility has not confirmed access, explain how you would obtain it and what happens if it remains unavailable. If an aim depends on an earlier result, prepare to discuss other outcomes. Check the evidence for an untested assay and the basis of a tight cost estimate. State what you have already checked and what still needs to be arranged.

Choose backup slides by the question they answer: a control, a sampling plan, a dependency or the basis of a cost. Label them so you can reach the relevant evidence without searching through the whole deck. Detailed slide construction belongs in designing a presentation.

Rehearsing the interview

Rehearse aloud with someone who can challenge the science and someone less familiar with the project. Ask them to interrupt, request a simpler explanation and question a central assumption. Notice where you answer a different question, rely on an unexplained term or take too long to reach the point. Revise those passages, then rehearse again [3].

Listen to the whole question. Clarify an ambiguity if it changes the answer, then answer directly and explain your reasoning. If the answer is unknown, say so. Explain how the proposed work would investigate it, or why it lies outside the project's scope. When discussing an alternative experiment, explain whether it answers the same question or changes what the project can establish.

Example: answer the limitation directly

This fictional proposal compares two preservation methods using paired aliquots from an existing collection of freshwater samples. The outcome is recovery of viable microbial isolates under specified culture conditions. Sample access is confirmed, but the comparison has not been performed. The applicant has experience with the culture workflow. No preliminary result establishes which preservation method works better.

Panel member: Your culture assay misses organisms that do not grow in those conditions. How will you know whether you have preserved the community?

Applicant: We will not be able to claim that we preserved the whole community. Our question is narrower: which method gives better recovery of isolates with this culture workflow? We will split each sample between the two methods and an untreated recovery

control, then keep the culture conditions the same. That makes the comparison useful for choosing how to preserve samples for this collection. It leaves uncultured organisms and other culture conditions outside the conclusion.

Panel member: What if both methods give very low recovery?

Applicant: We would first look at the untreated control and the detection limits. If the control also gives poor recovery, the experiment may not tell us much about preservation. If the control works, low recovery would show a problem under the tested conditions, but it would not establish that the methods are equivalent. We would use the paired estimates and their uncertainty to decide whether the difference is resolved. If it is not, we would report that limit rather than select a winner from noise.

Test the presentation setup, including opening the files and sharing slides. For a remote interview, check that you can hear questions and know how to reconnect or contact the organizer if the connection fails. Arrange accessibility or technical support with the organizer and keep a permitted backup copy. For pacing, voice and general question handling, see delivering a talk and answering questions [3].

Read the decision before making commitments

If the application is unsuccessful, separate the funding decision from the lessons in the review. Identify whether the main concern was fit, scientific reasoning, feasibility or explanation. Decide what evidence or redesign would address it before polishing the same text. Check the scheme's current rules before assuming that resubmission is available. Ask the funder or research office if those rules are unclear. Feedback can help improve the project even when another opportunity is the better next step [2].

If the proposal is selected, read what the notification actually confirms. It may begin a period of award preparation rather than settle every term. Bring the notification and proposed work to the institutional team responsible for research awards. Establish the agreed start date, funded activities, available resources, outstanding conditions and who has authority to make commitments. Check any proposed early spending through the applicable institutional and funder process; do not infer permission from an encouraging message.

Pina's guide to EU research grants treats preparation of the grant agreement as substantive work for the researcher and institutional support staff. Its broader lesson is to understand the conditions attached to the award, including its financial form, rather than reading only the scientific description. The specific obligations come from the applicable agreement, not from a general guide [4].

Preparing to start

Reconcile the approved work with what is available now. A facility may have changed its schedule, a collaborator's commitment may need renewal or the awarded budget may differ from the request. Identify which changes affect the scientific comparison and which affect only its timing. Bring a concrete proposal to the appropriate funder and institutional contacts when an adjustment needs agreement. Do not promise the original scope with resources that no longer support it [2].

Before the first substantive work, agree who owns the initial tasks, what records will be kept and when progress or problems will be reviewed. Include administrative colleagues where their work controls purchasing, appointments or reporting. For a collaboration, settle the partner contributions and access arrangements needed at the start; collaboration, authorship and credit covers those agreements in more detail.

Example: selection with an unresolved start condition

Later in the same fictional project, the proposal is selected for funding and award preparation begins. The core facility now reports an eight-week delay in access needed for the first experimental batch. The final start arrangements remain unconfirmed.

Decision. Hold the dependent sample-processing and purchasing commitments while the applicant and institutional grants officer seek agreement on the revised schedule. Do not substitute a different recovery assay merely because it is available sooner; that could change the comparison on which the award was assessed.

Action and owner. The applicant will obtain a revised access date from the core and show which milestones depend on it. The grants officer will establish what the award process permits and whether the proposed sequence or start date needs formal agreement. Together they will submit a specific adjustment with its scientific and budget consequences.

Work that can proceed. The team can prepare protocols and analysis documentation using resources already authorized for that preparation. This does not authorize new grant charges, appointments or processing of the unique samples.

Release condition. Begin the dependent work after the necessary award arrangements and facility access are confirmed. If access remains unavailable, assess an adequate alternative or seek an agreed change; financial authorization alone does not make the assay feasible.

Keep a short record of the agreed start arrangements. Name who will do the first tasks and which commitments are confirmed. Include reporting dates and where to raise a proposed change. The amount of coordination depends on the award: an individual fellowship usually needs less than a multi-institutional programme. Plan required outputs early and retain the records needed for reporting [4].

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Chapter 8

Publishing

Choosing a journal and submitting a paper

While planning the paper, identify the readers who would use its findings and journals that reach them. Return to that shortlist once the results and argument are settled, and compare each journal's expectations with the evidence you have.

The examples continue the fictional Brake study, inspired by Tang and colleagues (2024) [10].

Choose a journal for the contribution

Match the paper to its readers

Summarize the contribution and identify who would use it. Match both the subject and kind of advance: a journal that publishes work on your organism may have little interest in the particular question or method. [1].

Read the aims, scope and article types, then several recent papers close to yours in purpose. Compare their questions, evidence and intended readers with your own. One similar article can suggest a candidate journal; a pattern across several papers gives a firmer basis for judging fit. [1]; [3].

Use the journal's editorial criteria to judge what kind of advance it expects. A broad journal may expect relevance beyond the immediate system; a methods journal will usually look for a useful capability and evidence of its performance. [3].

Ask a colleague familiar with the journals to read the abstract and main figures. Which journal would they recommend, and what additional evidence might its editors expect? [1].

Make a realistic shortlist

Agree a first choice and a few alternatives with coauthors, weighing scientific fit, practical constraints and likely delay. An ambitious first submission can be reasonable when the case for that readership is clear and the authors accept the time cost of a possible rejection. [2]; [1].

First rule out journals whose charges you cannot cover, whose license conflicts with your requirements or whose format would force you to omit essential explanation. Then compare the remaining choices. When weighing delay, check whether advertised decision times cover only the initial editorial assessment or also external review. They are estimates, not promises. Recheck the chosen journal's current instructions before submission.

Journal comparison

Record brief reasons for keeping or rejecting each journal, including any question that must be resolved before choosing it.

Question	What to examine	Decision to record
Who would use this paper?	Scope and recent papers addressing comparable questions	The readers and the contribution that interests them
Does the evidence meet the journal's expectations?	Article type, editorial criteria and comparable studies	A good match, or a specific unresolved concern
Is this the right first choice?	Scientific fit, practical constraints, likely delay and coauthor judgment	First choice, alternative, or remove from the shortlist

Example: matching a biological finding to a journal

Brake reveals a previously unknown pathway in which an RNA template supplies a protein-coding defence gene.

Journal profile	Relevant readership	Assessment
General science, such as Science	Readers interested in gene organization, expression and cellular defence, including researchers outside microbiology.	The unexpected origin of a functional gene provides the case for broad interest. The sequence and genetic evidence must establish the proposed pathway.
Molecular microbiology	Researchers investigating bacterial defence and phage–host interactions.	A close fit for the pathway and the experiments establishing its function; also a natural audience for a narrower follow-up.
Microbial biotechnology	Researchers seeking useful traits in production strains or other applications.	The current study lacks application tests; an appealing use alone would not make this the strongest fit.

The team chooses Science as its ambitious first submission because the origin of the coding sequence could interest researchers beyond bacterial defence. A molecular-microbiology journal remains an alternative for readers focused on the pathway and its function.

Presubmission inquiries

Send a presubmission inquiry when an answer about scope, article type or expected validation would change where or how you submit. Check whether the journal accepts inquiries and use its requested route. [4].

Briefly explain the scientific question, finding and approach, adding the abstract if requested. Ask the question that affects your decision. For example, if your paper tests a method in one system, ask whether the journal would consider that evidence or expect validation in additional systems.

An encouraging reply does not guarantee external review. Keep the correspondence and refer to it accurately when submitting. You can contact an editor directly without a prior meeting or senior introduction. [4]; [1].

Check the terms and requirements

Investigate unfamiliar journals

For an unfamiliar journal, verify its identity, publisher contacts, editorial board and stated review process. Check indexing and preservation claims with the named services, and verify directory listings or professional memberships at the organization's own site. Confusingly similar journal names, unclear charges or promises of guaranteed acceptance warrant further investigation. [7].

Establish the total charges, when payment is due, available waivers, the publication license and the terms for sharing the work in a repository. Confirm that the proposed route meets your funder's and institution's requirements. A librarian or research-support office can help resolve unclear terms before you commit. Record the policy page and date consulted. [7].

Distinguish initial submission from later production

Read the instructions for the exact article type and submission stage. Some journals allow flexible initial formatting; others require specific forms or structures. Distinguish what is needed for initial assessment from what will be required later, while keeping editable text, figures and data ready for production. [5].

Check limits on text, figures and supporting material. If they prevent a clear account, reconsider the article type or journal before removing essential explanation. The reporting chapter helps decide what the scientific account needs.

Check policies on preprints, related manuscripts and previous dissemination. Disclose overlapping work where requested. Pursue alternative journals sequentially: the usual declaration requires that the manuscript is not under consideration elsewhere. [2]; [5].

Prepare the submission package

Assemble the agreed manuscript and supporting files

Assign a submission coordinator and agree on corresponding-author responsibilities. Circulate the submission version, resolve coauthor comments and confirm names, order, affiliations and contribution statements. Resolve authorship disputes before submission. [2].

Assemble the agreed manuscript, figures, legends, tables, supporting files and forms in one submission folder. Label versions clearly and assign the remaining checks to named coauthors.

Start repository preparation early because metadata and access queries can delay release. Before submission, test accession numbers and reviewer links, check access permissions, and identify the code version accompanying the paper. [8].

Check which declarations apply to the study and collect them from the responsible authors, including funding, conflicts of interest, ethics, consent, contributions and data availability as required. Include requested reporting checklists and follow anonymization instructions for blinded review.

Writing the cover letter

Use the cover letter to explain what was unresolved, what the paper establishes and why the finding matters to the journal's readers. Draw on the abstract, keeping the findings that establish the advance and omitting secondary results or unnecessary technical detail. A concise letter, often about a page, is usually sufficient unless the journal specifies otherwise. [5]; [6].

Open with the title and article type, then state the scientific advance early. For a broad readership, explain how the finding changes understanding of a wider question. For a specialist readership, identify the particular uncertainty it resolves. Mention earlier work in the journal when the connection helps explain the contribution. [5]; [1].

Include requested declarations and relevant editorial information, such as an earlier inquiry, related manuscript or potential reviewer conflict. Use designated portal fields where provided. If you claim to be first, check the relevant prior work and state exactly what is new. [5].

Example: a discovery paper and a narrower follow-up

The letters use the same genetic findings in two different publication contexts. Both assume coauthor approval and exclusive submission.

	Discovery paper	Follow-up paper
Starting knowledge	The route from an RNA template to a functional defence gene is unknown.	The pathway is known; the effector's function in Brake remains unresolved.
Main contribution	Establishes an unexpected way of producing a protein-coding gene.	Distinguishes the protein's role from a possible direct effect of the DNA product.
Reason to read	Changes how researchers can look for functional genes and defence effectors.	Resolves a mechanistic question for researchers studying bacterial defence.

Discovery paper — letter to Science

Dear Editor,

Please consider our manuscript, "An RNA-templated gene arrests bacterial growth and restricts phage propagation," as a research article in Science.

We identify a bacterial defence pathway that assembles its protein-coding gene from an RNA template. Brake contains a reverse transcriptase and an RNA template, but no recognizable effector gene in its genomic sequence. We show that repeated RNA-derived DNA creates a continuous coding sequence whose protein product arrests bacterial growth and restricts phage propagation.

Genetic tests connect gene assembly to function. Blocking translation preserves DNA production but abolishes growth arrest and phage restriction; expressing the protein separately restores both. These experiments identify the protein as the defence effector and distinguish its activity from a direct effect of the DNA product.

The discovery connects bacterial immunity to an unexpected route for producing a functional gene. It suggests that genome annotation can miss an effector whose coding sequence emerges only after DNA assembly, a possibility relevant to studies of gene organization and expression beyond bacterial defence.

All authors have approved the manuscript, and it is not under consideration elsewhere.

Thank you for considering our manuscript.

Sincerely,
[Corresponding author], on behalf of the authors

Follow-up paper — letter to a molecular-microbiology journal

Dear Editor,

Please consider our manuscript, “An RNA-derived protein mediates phage restriction in Brake,” as a research article in [Journal].

RNA-templated antiviral gene synthesis provides a route to bacterial defence, but the product responsible for defence in Brake remains unresolved. We distinguish a protein-mediated effect from a possible direct effect of its RNA-derived DNA product.

Preventing translation abolishes phage restriction while preserving DNA production. Expressing the protein separately restores restriction and growth arrest. Together, these experiments identify the Brake protein as the functional effector.

By linking the DNA product to an active protein, the work explains how a reverse-transcriptase-associated defence region restricts phage propagation. It should interest researchers investigating the function of these regions and their encoded effectors. The protein’s molecular target remains an open question.

All authors have approved the manuscript, and it is not under consideration elsewhere.

Thank you for considering our manuscript.

Sincerely,
[Corresponding author], on behalf of the authors

The same experiments make different contributions depending on what was already known. In the discovery letter, the genetic tests help establish the new pathway; in the follow-up, they resolve the identity of its effector.

Suggest reviewers for complementary expertise

Suggest reviewers with complementary expertise in the paper’s main methods and scientific questions. Give a short reason for each nomination and follow the journal’s requested number and information fields. [6].

Check potential conflicts involving institutional colleagues, recent collaborators, mentors or mentees. Follow the journal’s definitions, disclose uncertain relationships and verify contact details independently. [9].

For permitted exclusions, give a concise reason relevant to fair review. Scientific disagreement alone does not establish bias. Keep exclusions proportionate; the editor makes the appointments. [5]; [9]; [3].

Submit and follow the paper

Check the actual submission

Enter metadata from the agreed manuscript, checking author order, affiliations, funding and special characters after pasting. Assign files to the correct categories and remove superseded uploads before generating the review PDF.

Inspect the assembled review PDF for text and figure order, legibility, equations and symbols. Open supporting files and check any figures converted or compressed by the system. [5].

Once the assembled files are correct, complete the final approval step and retain the receipt, manuscript identifier and submitted files. Share them with coauthors and agree who monitors correspondence. If no receipt arrives, check the portal or contact the office before submitting again.

Following the submission

Journal staff may return a submission for missing files, declarations or unreadable material. Supply the correction with the manuscript identifier. Direct portal problems to staff and scientific questions to the handling editor. [5].

The editor assesses fit, contribution and whether the evidence warrants review. Read the reason for an early rejection before deciding what to change. After external review, the reports inform the editor's decision. [3].

Follow the journal's timing guidance. If an expected interval has passed, or a deadline makes an update necessary, send a brief inquiry with the manuscript identifier, submission date and question. Avoid parallel inquiries to several editors. [4].

If you discover a material error after submission, tell the editor promptly, identify the affected files or conclusions and ask how to supply the correction. Keep the coauthors informed. When a decision arrives, circulate the complete message and reports. For a requested revision, continue with Responding to peer review, which covers revision planning and the response letter.

For a rejection or transfer offer, see Rejection, transfer and published reviews.

Before you submit

- The journal and article type suit the contribution and intended readers; coauthors agree on the choice.
- Publication costs, rights and applicable institutional or funder requirements are resolved.
- The submitted version, author information and declarations are agreed and complete.
- The cover letter explains the contribution and journal fit; reviewer suggestions provide relevant, independent expertise.
- Figures, supporting files and data or code access work in the form supplied for review.
- The assembled submission has been checked, and the team knows who will confirm receipt and monitor correspondence.

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Reviewing a manuscript

Explain what the paper establishes and identify any problems that affect its conclusions. For each concern, tell the authors what needs to be resolved and explain to the editor how it affects your recommendation. [1].

Decide whether to accept

Read the invitation, abstract and reviewer instructions before accepting. Consider whether you can assess the central methods and conclusions within the requested time. You need not cover every technique in a multidisciplinary paper: tell the editor which parts you can evaluate and where another reviewer is needed. If you cannot take the assignment, decline promptly and, when possible, suggest suitable alternatives. [2]; [10].

Disclose relationships or interests that could affect—or reasonably appear to affect—your judgment. These can include recent collaboration, supervision, financial interests or close competition. Follow the journal's definitions and ask the editor about uncertain cases before reading further. If a conflict becomes apparent only after you receive the manuscript, pause the review and contact the editor. [3].

Check what the journal asks reviewers to judge. Some decisions depend mainly on technical soundness; others also require a particular advance or relevance to a defined readership. Distinguish disagreement with a conclusion from lack of interest in the question. A result that challenges your preferred explanation deserves the same examination of methods and evidence as one that supports it. [2]; [3].

Confidentiality and co-review

Establish whether your report will be published, whether your identity will be disclosed, and how communication with authors is handled. Published reports can be anonymous; signing a review and publishing it are separate choices. Follow the agreed process, including any later request to share a report. [3].

Keep the submission and review correspondence confidential. Do not circulate them to colleagues, use unpublished information in your own work or contact the authors directly without the journal's permission. Before involving a trainee or specialist, obtain the editor's agreement and arrange how their contribution will be acknowledged. The invited reviewer remains responsible for the submitted report. [3].

Apply the same confidentiality check to external tools. Before entering manuscript material or an identifiable draft report into an AI service, check the journal's current policy, the service's handling of the content and any required disclosure. Ask the editor if permission is unclear. Permission to use a tool does not transfer responsibility for the assessment or its accuracy.

If you can no longer meet the agreed deadline, tell the editor early and propose a date you can meet. [2].

Assess the manuscript

Read the complete submission, including relevant supporting material. Summarize the question, approach and main finding in a few sentences, then assess which weaknesses affect that finding. Check apparent omissions elsewhere in the submission before raising them. [4]; [9].

Return to the results on which the main conclusions depend. Read each figure with its legend and the relevant Methods. Note where a description is missing and where you have found evidence of a problem. "The legend does not identify the experimental unit" calls for clarification; "The analysis treats technical measurements as independent experiments" requires evidence that this is how the analysis was performed. [4]; [6].

Examine the evidence needed for the claim

Ask whether the design distinguishes the proposed explanation from plausible alternatives. Identify the role of the controls and whether the intervention changes other features that could explain the result. When important information is missing, identify the comparison you cannot assess and request the relevant detail. The whole-paper assessment chapter develops this reasoning in more detail. [6].

For quantitative conclusions, first identify the comparison and its independent units, including any pairing or repeated measurements. Check the uncertainty around the estimated effect. Examine how observations were selected and whether the analysis accounts for testing many outcomes. If an analysis was chosen after seeing the results, consider how that choice affects the conclusion. A significant result in one group and a non-significant result in another do not establish a difference between groups; non-significance alone does not establish that an effect is absent. If you cannot assess a statistical issue that could change a main conclusion, ask the editor for specialist review. [6]; [3].

Inspect the supporting data and code to the extent needed for your assessment and expected by the journal. Check that the files are accessible, identifiable and connected to the reported results. If access is restricted, assess the stated reason and available review arrangements rather than assuming unrestricted release is possible. Tell the editor when inaccessible material prevents an important check. [4].

When novelty or interpretation depends on an earlier publication, check the relevant passage in that paper. If the manuscript overlooks it, explain the consequence: perhaps a claimed discovery was already reported, or an apparent disagreement arises from different experimental conditions. Give the reference and the specific connection. A list of loosely related papers, especially your own, does not establish a substantive omission. [1]; [3].

Distinguish essential revisions from extensions

Sort your concerns by their effect on the paper's conclusions and your ability to assess them. A **major concern** challenges an important conclusion or the claimed contribution, or prevents you from judging whether they are supported. A **minor concern** is a local clarification or presentation problem that leaves the main conclusions intact. The amount of work is a separate question: an unsupported claim may be fixed by deleting a sentence, while an optional experiment could take months. [1]; [5].

State what remains unresolved before proposing a remedy. If the Methods omit information, ask for it. If the analysis does not answer the stated comparison, explain the mismatch. If the evidence supports a narrower conclusion, identify that option. Make clear when a concern prevents you from assessing the paper, rather than presenting your suspicion as an established flaw. [1]; [7].

Request experiments for a stated reason

An additional experiment is justified when its answer is needed for a claim the paper retains. Explain which alternative it would distinguish or which essential observation is missing. You can suggest a design, but allow another valid way to resolve the concern. Sometimes an existing analysis, a clearer explanation or a narrower claim is sufficient. [3]; [7].

Consider whether the requested work is needed for the paper's conclusion or asks a new question. Identifying a protein's molecular target may be valuable without being necessary to establish its role in a phenotype; label such work as an optional future direction. For an essential concern, consider whether a different analysis or narrower claim could resolve it without costly new experiments. If neither would leave the paper's conclusions supported, explain to the editor which evidence is still needed. [8]; [9].

Write the report

Use the journal's form. Where the format is open, begin with a short summary of the question, main result and overall assessment, then present numbered major concerns followed by minor points. Identify strengths precisely—for example, a comparison that separates two competing explanations—rather than adding generic praise. You do not need to manufacture a list of weaknesses in a sound paper. [1]; [5].

Writing specific comments

For each substantive concern, identify the passage or figure, explain the problem and state what would resolve it. Quote a short phrase when its wording is the issue. If a comment contains several independently answerable questions, split them. Use continuous numbering or a clear section scheme so the response can refer to each point. [1]; [5].

Write directly about the manuscript. "Figure 2 does not distinguish these explanations because…" is more useful than a judgment about the authors' competence. Be equally clear when the problem is serious: courteous wording should still let the reader recognize a conclusion you consider unsupported. When you may have misunderstood, say what you could not determine and ask the authors to clarify it. [3]; [7].

Concentrate language comments on scientific meaning and readability. Show where an ambiguous sentence changes the interpretation or a missing transition obscures the argument. Correcting every stylistic preference can bury more important concerns; widespread language problems can be described with a few representative examples. [3].

Example: a report on the Brake manuscript

Brake is the fictional bacterial defence study introduced in Planning the paper, inspired by Tang and colleagues (2024) [11]. Here a submitted draft describes an RNA-derived DNA intermediate that encodes a growth-arrest protein. The experiments connect protein production to phage restriction, but recovery of infected cells was not measured. The draft nevertheless claims survival, gives an ambiguous replication description and introduces the rationale for a key control only in the Discussion.

Comments to the authors

The manuscript describes Brake, a bacterial defence system in which reverse transcription of an RNA template produces a protein-coding DNA intermediate. Disrupting translation preserves DNA production but removes growth arrest and phage restriction; separate expression of the protein restores both. This comparison provides strong support for a protein-mediated defence mechanism. The main concerns are the inference about infected-cell survival and the description of replication in the phage-yield analysis.

Major concerns

1. The abstract states that Brake enables infected cells to survive. Figure 4 shows growth arrest during observation, but I could not find a measurement of recovery after infection. These observations do not establish survival. Please remove that conclusion and describe the observed growth arrest, or provide evidence of recovery if such measurements are available. The RNA-to-DNA-to-protein pathway remains an important finding without the survival claim.
2. Figure 1B reports a reduction in infectious phage yield, but the legend's description of replicates does not let me assess the uncertainty around that estimate. Please state how many independently started cultures were tested for each genotype, whether

genotypes were assayed together on the same days, and how technical plaque measurements were combined. The Methods should explain which values entered the comparison and how any matching by day was handled. If the current analysis already accounts for this design, a clear description would resolve the concern; otherwise, please revise the analysis and the resulting estimate and interval.

Minor point

3. The reason for preserving DNA production in the translation-disrupted construct is introduced in the Discussion. Please explain it when introducing Figure 3A. This would make it easier to see how the loss of restriction in Figure 3B, followed by complementation in Figure 3C, separates the protein's contribution from loss of the DNA intermediate.

Optional future direction

Identifying the protein's molecular target would help explain how growth arrest occurs. I do not consider this necessary for the present paper, provided the mechanistic conclusion remains at the level established by the translation and complementation experiments.

The replication concern cannot be settled until the reviewer knows how the observations were analyzed. The survival concern is already clear: it requires a change to the conclusion or additional evidence, even though removing the claim would take little work.

Confidential comments to the editor

I recommend revision. The evidence connecting an RNA-derived DNA intermediate to a defence protein is compelling, particularly the separation of DNA production from protein function in Figure 3. The survival claim should be removed unless supported by additional evidence, and I need the replication and analysis details described in point 2 before reaching a final judgment on the phage-yield comparison. I would be willing to assess a revised manuscript addressing these concerns. Identification of the protein's molecular target is not a condition of my recommendation.

Give the editor a consistent recommendation

In the editor's recommendation field, explain whether the problem can be corrected, requires a different study design or concerns the journal's scope and expected advance. State any clarification you need before deciding. Do not request a substantial revision if you would still recommend rejection after the authors resolved your concerns. [1]; [10]; [9].

The editor makes the publication decision, but may ask for your assessment of both scientific merit and journal fit. Explain which criterion underlies a negative recommendation. A sound paper may fall outside the journal's scope or offer less of an advance than it requires. A result that would interest its readers may still lack the evidence needed for publication. [2]; [9].

Keep scientific criticisms visible to the authors. A favorable author-facing report paired with a hidden negative assessment leaves them unable to respond and makes the editor's decision harder. Use confidential comments for matters requiring editorial handling, such as a competing interest, a limit to your expertise or a suspected ethical irregularity. [3]; [1].

For a suspected irregularity, describe the observation and give precise locations or a relevant reference. Do not turn a suspicion into an accusation or conduct your own investigation with the authors. Ask the editor how to proceed. A substantive problem may need correction regardless of its cause; establishing misconduct is a separate matter. [3].

Review the revision

Read the decision letter, your earlier report and the response alongside the revised manuscript. For each substantive point, assess whether the underlying concern has been resolved. The authors may have used a different analysis, supplied an explanation you overlooked or narrowed the conclusion. Judge that answer on its merits rather than requiring your preferred solution. [3]; [9].

Check the revised evidence and wording, not just the response letter. A clear reply may describe a change that has not reached the manuscript, or a new analysis may alter the interpretation elsewhere. If an experiment remains inconclusive, reassess what the retained claim requires. [9].

Avoid adding a new list of desirable experiments after the authors have addressed the original review. A revision can introduce a problem, and you may discover a serious issue missed earlier; raise it with a specific explanation of why it now affects your assessment. If your original criticism was mistaken, acknowledge that and withdraw it. [3]; [9].

If another journal asks you to review a manuscript you have seen before, tell the editor about your earlier involvement. Assess the current version afresh against the new journal's criteria. Obtain the original journal's permission before sharing your earlier report. [3].

Tell the editor which important concerns are resolved, what remains and how that affects your recommendation. When no substantive issue remains, a short report can be sufficient. [10]; [9].

Before submitting the report

- The summary identifies the paper's contribution and the evidence that matters to the assessment.
- Each major concern has a location, a reason and a clear criterion for resolution.
- Essential revisions are distinguished from optional extensions and stylistic preferences.
- Scientific criticisms in the recommendation are also visible to the authors; confidential matters are directed to the editor.
- The report states any limits to the review and meets the journal's instructions on conflicts, collaboration, confidentiality and disclosure.

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11. Tang, S., and colleagues (2024). **De novo gene synthesis by an antiviral reverse transcriptase.** *Science*, 386(6717), eadq0876.

Responding to peer review

Reviewer comments often mix questions about evidence with requests for clearer writing or additional work. Identify the concern behind each request before deciding what to change. The revised manuscript must address those concerns, and the response should explain what changed, why, and where the editor and reviewers can find it.

Start the response letter while planning the revision. A reply you cannot yet write may reveal an analysis to check or a question to discuss with coauthors. Mark proposed work clearly, then replace those notes with what was actually done as you revise the manuscript. [2].

The examples continue the fictional Brake study.

From reviewer reports to resubmission

[2]; [1]; [5].

1. **Read the decision and all the reviews.** Record the revision conditions and deadline.
2. **Map the concerns.** Separate each comment or subpoint and identify the conclusion or explanation at issue.
3. **Plan with coauthors.** Assign the work, dependencies and schedule in the response document.
4. **Clarify unresolved requirements.** Ask the editor before work that depends on an unclear condition or changes the paper's scope.
5. **Complete and assess the work.** Revise the manuscript and conclusions when the results change what can be claimed.
6. **Write the replies.** State what was done, why and where, with each reviewer's section complete on its own.
7. **Reconcile and submit.** Check the final files, locations, data and coauthor approval together.

Understanding the decision

Start with the editor's decision, which sets the conditions and deadline for revision. Read those conditions alongside the reports, and ask the editor which requests are essential if this is unclear. [4]; [5].

Read all reviews before answering individual points. Identify recurring concerns and challenges to the central result; these may require more work than a long list of minor corrections. Read the relevant manuscript passages beside the comments. A question whose answer seems obvious may reveal an unstated assumption, while several questions may point to one poorly explained idea. [1]; [2].

Work through possible responses privately with coauthors, then draft the reply around the scientific concern.

Planning the revision

Comments and tasks

Copy the full comments into a working document and number the points. Separate the issues within a compound comment so that each receives an answer. Beneath each point, sketch a response or specify the work needed. [2]; [1].

Name what to inspect and what to change: "Define the experimental unit in the legend and check that the analysis uses independent cultures." An unclear error bar might need only a better legend, or its calculation might be wrong. Inspect the analysis before choosing the remedy. Likewise, check whether an overstated claim results from an analysis error or an interpretation that goes beyond the findings.

Additional experiments

Identify the conclusion the reviewer is questioning and ask what the requested experiment would establish. If you intend to retain that conclusion, you need evidence that resolves the concern—for example, a test of a necessary assumption or a comparison that distinguishes plausible explanations. The requested experiment may provide that evidence; propose a different test if it answers the same question more directly. [6].

Other checks may strengthen a conclusion that is already supported. Weigh the concern they resolve against the time and resources needed. If you do the work, describe its result in the response and include evidence necessary for the published conclusion with the paper. [1]; [6].

A request may also open a new project. Establishing a cellular effect does not necessarily require explaining its mechanism or testing a treatment application. Whether that further work is needed depends on what the paper claims. Discuss the intended scope with the editor when a reviewer expects work beyond that question. [3]; [1].

Alternatives and narrower claims

A narrower claim may resolve the concern when the available evidence supports it. Explain what the revised paper would establish and what would remain unanswered, then change the claim wherever it appears, including the title and abstract when affected. If you propose an alternative experiment instead, explain how it answers the reviewer's question. Distinguish a scientifically unsuitable test from one blocked by access, equipment or time. [6]; [1].

Consult the editor when an alternative changes a revision condition or the paper's scope. If essential evidence is unavailable, assess whether a narrower contribution remains sound and suitable for the journal. [5]; [4].

Example: narrowing a claim after review

An early Brake draft claimed that the system helps infected cells survive. A reviewer asks for recovery measurements. The authors have measured growth arrest and phage output, but not recovery. They propose removing the survival claim and ask the editor whether the paper can be assessed on that basis. In this scenario, the editor agrees.

- *The manuscript states that Brake promotes survival after infection, but growth arrest does not establish survival. Please track whether infected cells recover and resume division.*

  We agree that our measurements do not establish recovery. We have not performed the requested recovery experiment. Following discussion with the editor, we have removed the survival claim and focused the conclusion on growth arrest and phage restriction.

  The abstract now states: "The Brake protein arrests bacterial growth and restricts phage propagation." In the Discussion, "Fate of infected cells," we now write: "Reduced infectious output establishes phage restriction without showing that individual cells survive the infection. Recovery measurements would distinguish reversible arrest from a lasting loss of growth capacity."

Recovery evidence is necessary for the original survival claim. The narrower paper can proceed only if the remaining contribution is supported and the editor agrees to assess it.

Responsibilities and timing

Agree on substantial changes early and assign the work. Account for dependencies when setting the schedule: a reanalysis must be settled before the affected figure, Results and abstract can be finalized.

Plan with the people doing the work, allowing time for equipment access, sample preparation, analysis and coauthor review alongside their other commitments. When one trainee carries most of the revision, check whether the proposed workload is feasible. Request an extension if the resulting schedule exceeds the deadline. [3].

Include those doing the work in scientific discussions and relevant exchanges with the editor. Record decisions and open questions in the revision plan. [5].

Revision worksheet

Use one entry per comment or subpoint to track the work and record the final answer. A shared document is often enough; a spreadsheet can help with a large revision. [2]; [1].

Field	What to record
Comment	Reviewer and point number, with the full comment
Issue to resolve	The question, missing explanation or disputed conclusion
Planned work	The edit, analysis, experiment or reason for retaining the current approach
Responsibility and timing	Who will do the work, what it depends on and when it is needed
Progress	Work still to do, questions awaiting a decision and completed changes
Final response	The answer supported by the work actually completed
Manuscript location	Where the reader can find the revised text, figure or analysis

Questions for the editor

Contact the editor before substantial work if the revision conditions are unclear. Describe the concern, propose the work that would address it and ask whether that plan is suitable. If the work will exceed the deadline, explain why and propose a realistic date. [5].

Carrying out the revision

After each experiment or analysis, check whether the result answers the reviewer's question. Revise the plan if the question remains unanswered or the result reveals a new problem.

New experiments and analyses

For new experiments, report the additional data and findings. For reanalyses, identify the existing data and explain what changed in the analysis and why. If you uncover an error, describe its consequences and check all affected figures and passages before stating that the conclusions are unchanged. [1].

Record which analyses were specified before examining the new results and which arose during exploration. Explain changes to the analysis plan and report relevant outcomes, including those that challenge the expected interpretation. [7].

Unexpected or inconclusive findings

When a result is inconclusive, check why. If the intervention or measurement failed, explain what failed and whether the requested test can still be completed. If the experiment worked but the estimate is imprecise, report the estimate and uncertainty; a non-significant result does not establish

absence of an effect. To claim that an effect is too small to matter, define the magnitude that would matter for the question and use an analysis that can assess that claim, with assumptions you can justify. See estimates and intervals. [7].

If new evidence weakens the interpretation, revise it throughout the paper and explain what changed your judgment. Discuss loss of support for a central conclusion with coauthors and the editor before continuing the old plan. [4]; [5].

Supporting material

Include the evidence needed to assess the published conclusions in the paper and its supporting material, including findings that qualify those conclusions. A figure or table may remain only in the response when it answers a reviewer's specific question and the published account is complete without it. Label response-only material distinctly and state whether any part was added to the paper. [1].

Writing the response

Letter structure

Thank the editor and reviewers, then summarize the major changes and the concerns they address.

Follow the overview with the reviews in order, placing your answer directly beneath each comment. If a long comment raises several issues, split it at natural boundaries and answer each part. Retain the full wording, order and meaning, including any shared context, and note that the comment has been divided for clarity. Keep the original point number when it helps the reviewer recognize the comment. [1].

Make each reviewer's section readable on its own. When concerns overlap, repeat the explanation, relevant quoted changes and manuscript locations in each section. A reviewer should be able to assess your reply without reading your answers to someone else.

Point-by-point replies

Lead with the correction, experimental result or reason for retaining the approach. Explain how it addresses the concern and identify the corresponding manuscript passage or figure. [1].

Distinguish old and new material explicitly. "This is described in the Methods" leaves the reviewer unsure whether the description was already there or has just been added. "We have added a description of the exclusion criteria to the Methods" makes the change clear.

Quote short changes directly: 'We have clarified the Figure 1B legend: "Each point represents one independently started culture."' Identify the revised figure, table or section for every change. Add page or line numbers when the passage is harder to find or the journal requests them, and state that they refer to the revised manuscript. For a long addition, summarize its purpose and give the section title. [1].

Use consistent typography, spacing and indentation to distinguish comments, replies and manuscript quotations, including in print. [1]; [2].

Example: clarifying the experimental unit

A reviewer asks about replication in the Brake phage-yield assay. The authors confirm that the analysis already uses one titre per culture and matches conditions by experimental day. They split the two-sentence comment below.

Dear Editor,

Thank you for the comments. We have clarified the sampling and analysis behind Figure 1B. We have divided the reviewer's comment into its two questions; locations and quotations refer to the revised manuscript.

- *Please define what each point represents.*

  Each point represents one independently started culture. The Figure 1B legend now states: "Each point represents one independently started culture; lines connect the three cultures assayed on the same day. Six days were measured, with technical duplicate plaque counts combined for each culture."

- *Explain how the technical duplicates and measurements on different days were handled.*

  We combined the duplicate plaque measurements into one titre estimate per culture, then compared intact Brake with empty vector within each experimental day. The six day-matched contrasts were the observations used to estimate the effect and its uncertainty. The analysis and results are unchanged.

  Methods, "Infectious-titre analysis," now states: "We divided the intact-Brake titre by the empty-vector titre within each day and took the log10 of that ratio. We estimated the mean log10 ratio and its two-sided 95% confidence interval using a Student t distribution with five degrees of freedom." The preceding paragraph explains that technical duplicates were combined into one culture-level estimate.

The analysis already used one titre per culture and matched the cultures by day, so the reply describes a reporting correction. If technical plates had entered the analysis as independent cultures, the authors would also need to correct the analysis and report its consequences.

Disagreements and unchanged text

When you disagree with a request, explain how the existing evidence answers the concern or how an alternative would resolve it. If the request concerns a different question, name that question and explain why answering it is unnecessary for the claim you retain. The phrase "beyond the scope" leaves that reasoning unstated.

Reread the relevant passage from the reviewer's perspective. If it omits an explanation, leaves an assumption unstated or makes information difficult to find, improve the manuscript as well as the reply. The analysis may be sound even when its description needs revision. [1].

If the passage already answers the concern or there is a sound reason to retain the approach, explain that reason, give the evidence and manuscript location, and explicitly say what is unchanged. Quote the passage when useful. Before finalizing the reply, check whether it contains an explanation the manuscript itself needs. [1]; [6].

Discuss unresolved disagreements with the editor when they affect a revision condition or the central claim.

Read suggested references and add those that provide relevant context or correct an omission. Briefly explain why an unrelated citation was not added. [6].

Concerns about a conflict of interest or inappropriate conduct belong in a separate communication to the editor. Describe the relevant facts and the action you are asking the editor to consider. Keep the point-by-point scientific reply focused on the manuscript. [1].

Conflicting requests

First ask whether both requests can be met. More explanation and shorter Results can often be reconciled by moving analytical detail to Methods. If requests require incompatible analyses or contradict the editor's conditions, explain which approach you propose and why, then ask the editor to clarify the requirement. [1]; [5].

Example: more explanation in a shorter Results section

Reviewer 1 wants the day-matched comparison explained. Reviewer 2 wants a shorter Results section, which currently gives the calculation over several paragraphs. The authors keep the comparison and finding in Results and move the calculation to Methods.

Reviewer 1

- *Please explain why cultures were compared within each day and how the effect and confidence interval were calculated.*

  Each experimental day included all three genotypes. Matching intact Brake to its same-day empty-vector control accounts for shared differences between experimental days. We have retained that comparison in Results, "Brake restricts phage propagation," and moved the calculation to Methods.

  The revised Results read: "We compared intact Brake with empty vector within each experimental day. Across six days, the geometric mean infectious titre was 79-fold lower with Brake (95% confidence interval, 55–116-fold; Figure 1B)."

  Methods, "Infectious-titre analysis," now states: "We divided the intact-Brake titre by the empty-vector titre within each day and took the log10 of that ratio. We estimated the mean log10 ratio and its two-sided 95% confidence interval using a Student t distribution with five degrees of freedom." It then explains how the estimate and interval were converted to a fold reduction. The calculation is unchanged.

Reviewer 2

- *The Results contain too much detail about the titre calculation. Please shorten this passage and keep the focus on the finding.*

  We have shortened the Results by replacing the calculation paragraphs with two sentences under "Brake restricts phage propagation": "We compared intact Brake with empty vector within each experimental day. Across six days, the geometric mean infectious titre was 79-fold lower with Brake (95% confidence interval, 55–116-fold; Figure 1B)."

  The calculation now appears in Methods, "Infectious-titre analysis": "We divided the intact-Brake titre by the empty-vector titre within each day and took the log10 of that ratio. We estimated the mean log10 ratio and its two-sided 95% confidence interval using a Student t distribution with five degrees of freedom." The following sentences describe conversion to a fold reduction. This retains the day matching needed to understand the result while moving the analytical detail out of Results. The analysis is unchanged.

Checking and submitting

Read the paper and response side by side to check that they describe the same revision. Verify every cited manuscript location and follow changes to results or conclusions through to the abstract and Discussion.

Ask each responsible researcher to check the relevant result and reply, then have a coauthor read the complete package for consistency. Resolve disagreements about the interpretation or response before submitting.

Update data, code and materials information affected by the revision. Test repository access and verify versions against the revised analyses. [4].

Prepare the required clean and marked manuscripts from the same final text. Check response locations against that version and remove working notes.

Before resubmitting, check the following. [1]; [2].

- Every editor and reviewer comment, including each subpoint, has an answer.
- Each reviewer's section includes the explanations, quotations and manuscript locations needed to assess the reply on its own.
- The letter reports the work completed and its actual results.
- The manuscript contains the explanations and evidence readers will need.
- Revised methods, sample counts, figure legends and supplementary files match the work reported.
- New analyses and revised conclusions agree across the text, figures and supplements.
- Inconclusive findings are described accurately, and any retained text or approach is justified in the reply.
- Data, code, materials information and access links reflect the revised work.
- All manuscript locations refer to the files being submitted.
- Coauthors have reviewed the revision, and the required clean, marked and supporting files are ready.

References

1. Noble, W. S. (2017). **Ten simple rules for writing a response to reviewers.** *PLOS Computational Biology*, 13(10), e1005730.
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5. **Let's chat.** (2021). *Nature Methods*, 18, 221.
6. COPE Council (2017). **COPE Ethical guidelines for peer reviewers.** Version 2, September 2017.
7. Makin, T. R., and Orban de Xivry, J.-J. (2019). **Science Forum: Ten common statistical mistakes to watch out for when writing or reviewing a manuscript.** *eLife*, 8, e48175.

Rejection, transfer and published reviews

Start with the editor's reasons for rejection and discuss them with coauthors. If the paper does not suit the journal's audience, another destination may be appropriate. If the main conclusion lacks support, address that problem before submitting elsewhere.

Understanding the rejection

Read the decision letter alongside the reports. Identify the concerns the editor considers decisive: the journal's audience, the contribution, a weakness in the evidence, or some combination. Reviewers can disagree, and the editor may give one concern more weight than several favorable assessments. [1].

Discuss the next step with your coauthors before writing to the journal. Separate work that would improve the paper wherever it is submitted from work intended to meet this journal's expectations. Consider the time required, available resources and whether the revised contribution would justify another attempt there. Agree on an alternative destination as well. [2].

Choosing the next route

What explains the decision?	What to do next
The contribution does not suit the journal's audience or selection criteria.	Reassess the destination. Appeal only if you can identify a consequential misunderstanding of the contribution.
A central claim lacks evidence.	Decide whether to obtain that evidence, narrow the claim or substantially rework the paper before submitting again.
A factual error appears to determine the rejection.	Check the appeal policy and show how correcting the error changes the assessment.
The editor explicitly invites a substantially revised new submission.	Clarify the expected changes, assess whether they are feasible and follow the specified resubmission route.

A rejection with an invitation to submit new work differs from a revision request: the journal may want to reconsider the project only after substantial changes. If the wording leaves this unclear, ask whether the proposed work would make a new submission appropriate and which concern it must resolve. Briefly describe the work rather than asking whether there is any chance of acceptance. [3]; [4].

Decide whether to appeal

An appeal is most useful when you can show why the reasoning behind the rejection should change. Correcting an incidental error is unlikely to help if another unresolved concern still justifies rejection. A successful appeal may lead to further revision or review. [4]; [1].

Check the journal's permitted grounds, deadline, recipient and required documents. In the PLOS One process consulted for this chapter, an appeal must be written and sent through the stated route with the appeal form. A significant factual error or documented competing interest is considered only when correcting it would change the decision. Informal appeals are not accepted; while an appeal is active, the manuscript remains under consideration and must not be submitted elsewhere. Other journals may use different procedures. Recheck the applicable instructions before acting. [5].

Explain the decisive issue

Open with the manuscript title or identifier and the request for reconsideration. State the editor's decisive concern, then explain the error or new information with precise manuscript locations. Where

several concerns determined the decision, address each; do not select only the easiest criticism. Supply a point-by-point response if the journal requires one. [4].

New results may change an editor's assessment if they resolve the reason for rejection. State what the completed experiments show and identify any work still proposed. If the concern is relevance to the journal, explain what its readers would learn that the editor's assessment overlooked. [1].

Example: appealing a factual misunderstanding

In the fictional Brake study, inspired by Tang and colleagues (2024) [6], disrupting translation leaves the RNA-derived DNA product intact but removes growth arrest and phage restriction. Separate expression of the Brake protein restores both effects. Suppose the editor rejects the paper on the mistaken premise that the translation-disrupting change also prevented DNA production, so the protein's contribution could not be distinguished from the DNA's. The appeal can point to the control already present in Figure 3A.

Dear Editor,

We would like to appeal the decision on our manuscript, *An RNA-templated gene arrests bacterial growth and restricts phage propagation.*

The decision rests on the concern that our translation-disrupting change also abolishes DNA production. Figure 3A shows that the DNA product is retained in this construct. Growth arrest and phage restriction are lost despite this retained DNA production (Figure 3B), and both are restored by expressing the Brake protein separately (Figure 3C).

These results distinguish the protein's contribution from DNA production and address the reason given for rejection. Would you reconsider the manuscript in light of this control? We can also revise the Results passage and Figure 3 legend to make the comparison more prominent.

Thank you for considering our appeal.

If the editor had also rejected the paper for insufficient novelty, the appeal would need to address that judgment too.

Resubmitting to the same journal

If the reports raise a possible conflict of interest or inappropriate conduct, describe the specific evidence through the journal's complaints or appeal procedure. Explain separately whether it affects the scientific decision. PLOS One, for example, asks authors to raise concerns about correspondence with journal staff and sets explicit grounds for an appeal. [5].

A substantially revised manuscript may be eligible for a fresh submission to the same journal, but check its rules or obtain the editor's agreement first. Disclose the previous submission and explain the changes. Wilke advocates this route as an alternative to appeal; its suitability depends on the journal and the extent of revision. [4].

If you abandon a pending appeal, notify the journal and resolve its status before submitting elsewhere. After a final rejection, return to the scientific concerns and the next journal's criteria with your coauthors. Keep an appeal and a transfer or new submission as separate routes; do not pursue them at the same time. [5]; [7]; [2].

Transfer or submit elsewhere

A transfer offer is worth considering alongside other suitable journals. Check the destination's audience, publication criteria, article type, charges and licensing arrangements using the approach in Choosing a journal and submitting a paper. Agree the choice with your coauthors. [2].

Find out whether the offer is an automated suggestion, a recommendation from the original editor, or follows consultation with the receiving editor. Nature Portfolio describes all three; the receiving journal makes its own editorial decision, and authors choose whether to transfer. [7].

Check which files and correspondence will transfer, including earlier versions, and remove confidential material you do not want to send. The Nature Portfolio guide consulted for this chapter permits file changes during transfer. It also prohibits an appeal while the paper is being transferred for consideration elsewhere. Check the receiving journal's applicable instructions before proceeding. [7].

Review transfer follows the journals' agreements about confidentiality and identity. Do not forward an editor's confidential material or reveal a reviewer on the assumption that permission to use the report also permits disclosure. COPE advises journals and reviewers to establish what can be shared in a transfer. [8].

Prepare the next submission

Correct known errors and unsupported claims in the next version. For a new submission, address valid criticisms and decide whether requests for a broader study are necessary for the new destination. If a claim depends on missing evidence, obtain it or revise the claim throughout the paper. [2].

Some transfer routes allow a plan for outstanding work. Nature Portfolio advises distinguishing completed changes from proposed work and explaining the effort required. Establish the receiving journal's interest before undertaking substantial destination-specific work. [7]. Detailed response planning is covered in Responding to peer review.

Update the cover letter and replace obsolete files. With an ordinary submission, previous reports may not reach the receiving editor automatically; follow its instructions on submission history and earlier correspondence. For transfers, check that the intended response and revised files accompany the manuscript. [7]; [2].

Read and publish review histories

Published reviews can show how a paper's evidence and interpretation were challenged and revised. Begin with the final article, then follow an important concern through the report, the authors' response and the corresponding change. Note the dates, manuscript versions and journals involved. A report written for an earlier version may criticize a claim that no longer appears in the paper; a request from one journal may reflect its particular expectations. [9].

Follow the concern into the paper

For each concern that matters to your reading, identify what the reviewer questioned and how the authors answered. Did they add evidence, correct an analysis, explain an existing control, narrow a claim or leave the point unresolved? Check the final paper rather than relying on the response's description of the change. If the corresponding manuscript version is unavailable, you may be able to assess the final claim without reconstructing exactly what the reviewer saw. [9].

Example: following a corrected claim

In this Brake example, the recorded experiment measured growth arrest; recovery was not tested.

- *The growth trajectories in Figure 4B do not establish survival. Did the arrested cells resume growth after infection?*

  We did not measure recovery after infection. We have removed the survival claim and now describe the observed growth arrest. The Results passage accompanying Figure 4B now states: "Infected cells stopped growing during observation; whether they subsequently recovered was not tested."

Final Results passage

> Infected cells stopped growing during observation; whether they subsequently recovered was not tested.

The claim is corrected; recovery remains untested.

When using published correspondence to learn about reviewing, examine how the reviewer explains a problem and whether the response resolves it in the manuscript. The number of review rounds and the severity of the language do not tell you whether a criticism was justified or a repair adequate. [10]; [9].

Check what becomes public

"Open review" can describe different arrangements. Reports may be published without reviewer names; a signed report may be visible to authors without being publicly available. Review transfer and post-publication commenting are further, separate practices. Check the journal's model rather than inferring these choices from the label. [10].

Before agreeing to publish a review history, inspect what it includes and who can consent to publication. In the PLOS One process consulted for this chapter, authors can opt into publication of decision letters, reports, responses and attachments; names appear when reviewers have signed their reports, and ethical considerations can limit publication. Recheck the journal's model and permissions for the history you intend to publish. [5].

If earlier reports came from another journal, ask how their publication is authorized and how the original context will be identified. Keep confidential correspondence private until the applicable permissions are clear. For teaching or quoting from an already published report, check its stated license and cite the particular report and version. [8]; [9].

Before taking the next step

- Coauthors agree on the route and the work required before proceeding.
- An appeal addresses a decision-relevant issue and meets the journal's instructions.
- The paper is no longer under consideration elsewhere when the next submission begins.
- The next package addresses valid criticisms and contains the intended files and correspondence.
- Any shared or published reviews follow the relevant permissions and identify their manuscript context.

References

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Chapter 9

Career development

Exploring career options

Scientific training can lead to research, teaching, publishing, technology development, policy or managing shared facilities. The titles alone tell you little about the working day. An editor evaluates manuscripts; an applications scientist helps researchers use a technology; a facility manager organizes equipment, expertise and services for many groups. Compare what you would do repeatedly in each role and who would use the work [1].

Compare the work

Think about the work you would like to do regularly. You might enjoy troubleshooting instruments but prefer less time writing papers, or want to teach while staying involved in research. Being good at an activity does not necessarily mean you want it to fill your week. Also note conditions that could rule out a role, such as frequent travel or a location you cannot move to.

Compare actual posts within those categories. Research roles differ in freedom to choose projects, expectations for publication and the importance of a commercial application. Ask who sets priorities, how success is assessed and which decisions you would make yourself [2].

Topic	Questions
Contribution	What problem does the role solve, and for whom?
Week of work	How much time goes to experiments, analysis, writing, meetings, documentation, training or coordination?
Responsibilities	Who sets priorities, and what limits the work?
Qualifications	What experience, knowledge or training does the role require?
A useful trial	What conversation, shadowing opportunity, short project or course would show whether the work fits?

Scientific training develops skills in framing questions, evaluating evidence, learning unfamiliar subjects and explaining results. Identify which of these the role uses, then assess what else you need to learn. A supervised project or placement may provide relevant experience; a course may be sufficient for a specific knowledge gap.

Example: comparing two roles

Lena enjoys troubleshooting assays and explaining results to collaborators. She wants to remain in one region and prefers working with others to spending long periods writing alone.

Applications scientist: Lena spoke with two scientists who support genomics users and tried teaching a workflow to a new colleague. She enjoyed diagnosing the problem and watching the user become independent. Both conversations, however, described recurring travel outside her region. She will look for local or remote support posts and check the travel requirements of each one.

Research project manager: Lena observed a planning meeting and coordinated one defined part of a collaboration. She liked clarifying decisions and making sure one group's delay did not block another. She missed doing the technical work herself, so she narrows the search to roles close enough to the science for her assay knowledge to matter.

Lena now gives locally based scientific project roles priority. She will accept less bench time for collaborative problem-solving, but not routine travel or work detached from the technical decisions.

Learn from people doing the work

Speak with people who hold the role in more than one organization and at different career stages. Ask them to describe a recent week and a difficult decision. Which responsibilities surprised them when they started? Find out how they learned the role and what experience they would look for in a new colleague. Ask how their work is assessed.

Ask similar questions across conversations. If people in the same role describe different weeks or constraints, ask what comes from the role and what comes from their organization or manager.

A conversation can suggest a way to try the work. Shadowing someone or taking on a short project may reveal more than another general discussion. For a postdoctoral position, ask what you expect to gain from the appointment and whether the group can provide it. Discuss how you would respond if the promised training or opportunities did not materialize. Choosing an advisor and research environment explains how to assess that support.

Experience and next steps

Keep a brief record of relevant work you have done. Describe what you did, who used it and what happened as a result. Include evidence you could show or someone who could confirm your contribution. Work beyond your research project may also be relevant: teaching a workshop can demonstrate training experience, for example. Compare these experiences with the responsibilities of the roles you are considering.

"Kept a shared project schedule up to date" shows coordination; it does not show that you led the project. "Completed a training module" may show interest in quality practice; it is not the same as having worked in that system. Check where you still need practical experience before applying.

Reconsider the options after the conversations or trials. Which work did you want more of after trying it? Did a working condition rule out an otherwise attractive role? Use what you learned to narrow the search or choose the experience you need next.

References

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Choosing an advisor and research environment

Two laboratories can work on equally interesting questions yet offer very different training. Read their work, talk with each prospective advisor, and ask current and former members how projects and supervision develop in practice. Compare the research, the support you need and the working environment before choosing. [1]; [2].

Decide what you need from the position

Start with the outcome you need from the training period. A PhD and a postdoc both involve research, but they make different demands of supervision.

Stage	Central outcome	Questions that change the decision
PhD	Learn to frame questions, conduct and interpret research, communicate results, and leave with several viable directions.	How is a new student taught? How often is feedback available? Who can advise if the first project or supervision arrangement does not work?
Postdoc	Extend or redirect expertise while developing work, visibility and support for the intended next role.	What new capability, research direction or independent contribution can this position build? What work can the postdoc lead or carry forward? How does the supervisor support the next search?

Write down your priorities and any conditions you cannot accept. A student learning to design experiments may need frequent early feedback; a postdoc preparing for an independent position may need opportunities to develop and carry forward a distinct research direction. Ask about support for the career you intend to pursue, including work outside academia. [2].

For a PhD, identify who provides advice beyond the primary supervisor, such as a committee member or programme director. Ask how that person becomes involved if supervision or progress is difficult.

Learning how the laboratory works

Keep notes for each serious option, recording who described a practice and what you observed yourself. Compare these accounts: do the advisor and trainees describe similar supervision, or do their experiences differ?

Topic	What to find out
Supervision	Meeting patterns, feedback on drafts and data, who provides day-to-day teaching. “Can you describe how a new trainee’s project is discussed in its first few months?”
Research and learning	Questions the group works on, project choice, access to methods and room to develop ideas. “How have recent trainees moved from an initial task to a question they led?”

Topic	What to find out
Credit and collaboration	How ideas, data, authorship and shared projects are discussed. “Could you describe a recent collaboration and how contributions were decided?”
Resources and feasibility	Current support, essential equipment or collaborators, and realistic project scope. “What resources are already committed to this work, and what would require a new decision?”
Trainee trajectories	Completion, publications or other outputs, and the range of next roles people have entered. “May I speak with a recent alumnus about how the group supported their next step?”
Environment	How colleagues help one another, what happens when work stalls, and whether people can ask difficult questions. “What did the group do when a recent project ran into a serious obstacle?”

Read recent papers before these conversations. Ask how a possible project would build on that work, which methods you would need to learn and who would teach them. Confirm the funding and facilities available for your work, including what depends on a pending application. [1].

Conversations and visits

Talk with people at different points in the laboratory: a newer member, someone approaching a transition, and, where possible, a former member. A visit, rotation or attendance at a lab meeting can reveal how questions are handled and who participates. Conversations away from the immediate selection setting may make it easier for people to explain ordinary practice. Ask the same core questions of several people, but leave room for what they think matters. [2]; [3].

Ask for concrete examples: “Could you describe the feedback on the last paper or project proposal?” Follow up on how the advisor helped when a trainee was stuck or a project needed to change direction. Ask how decisions about credit were made. Compare the advisor’s account with those of current and former members.

Give people room to decline a question, and keep private accounts confidential. If several people describe the same problem with supervision, treatment or credit, take it seriously. Seek independent advice before committing, especially when a prospective advisor’s explanation conflicts with the accounts of current or former members. [2].

Example: compare a laboratory through evidence

In this fictional comparison, Mina is choosing between two PhD laboratories in the same field. She needs structured feedback while learning to frame a project, but she also wants room to develop an independent question.

Laboratory A: the advisor and two trainees describe weekly individual meetings. An alumnus recalls detailed feedback on an early proposal. A student explains how a lab discussion helped narrow a stalled project to a related, feasible question. The first-year project would sit within a funded collaboration, so its initial scope is relatively narrow and day-to-day teaching would often come from a senior postdoc.

Laboratory B: the research question matches Mina's interests more closely, and two trainees value the freedom to shape their projects. The advisor schedules individual meetings every two weeks, and a staff scientist offers regular methods guidance. An alumnus describes developing a question in the group and carrying that line of work into the next position. When one project stalled, the trainee moved to a new question and another researcher took over the original work. The trainee says the advisor supported the change, although the first months required more independent planning than Mina expects to manage comfortably.

Mina chooses Laboratory A. She accepts a narrower starting project and more day-to-day dependence on a senior postdoc because weekly feedback meets her immediate training need. Laboratory B offers the closer topic, earlier ownership and a credible route to independence, but expects more planning between meetings. A student who already works confidently with that method might reasonably choose it.

Compare the opportunities

Compare the opportunities against your priorities. You might accept less freedom in exchange for closer initial supervision, but first establish what that supervision would involve. For a new group with no alumni, ask who will provide day-to-day training while it is being established. Observe scientific discussions where possible and find out what support the department offers. [1].

For a PhD, check whether the programme provides alternatives if the initial project or advisor arrangement does not work. For a postdoc, discuss how you could develop a distinct contribution and what support you would receive for the next position. Agree which projects you could take with you and which collaborations would continue. [2]; [3].

Resolve the questions that could change your choice before accepting. Record the agreed project, supervision and funding arrangements so you and the advisor can revisit them when work begins.

Before choosing an advisor or environment

- I know what training and opportunities I need from this PhD or postdoc.
- I have assessed research fit, supervision, environment, resources and trainee trajectories separately.
- I have asked several people for concrete examples and compared their accounts with direct observation where possible.
- I know which unanswered question would change my decision.
- I have not treated prestige, a familiar topic or one positive interaction as sufficient evidence.

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CVs, resumes, letters and references

Read the advertisement to establish what the employer needs and which documents, format and deadline it requires. Choose examples that show how your experience meets the selection criteria. Use the requested letter or statement to explain connections that a reader may not see from a list of appointments and achievements.

Choose the document

An academic CV presents education, research, publications, teaching, awards and service. A resume selects the experience and capabilities most relevant to a particular role. Document names and conventions vary between countries: use the employer's instructions to decide how much detail to include. [4]; [1].

Keep one private master record of roles, dates, contributions, outputs and collaborators. When preparing an application, select evidence from that record against the advertised requirements. A short working table can help you decide what to include:

What the reader needs to decide	Evidence I can show	Where it belongs
Can this person do the role's central work?	A project, decision or deliverable with an identifiable individual contribution	Resume experience entry; letter example
Can this person work with the relevant people?	A collaboration, mentoring activity or communication task with a concrete outcome	Resume experience entry; letter example
What is this person's scholarly record?	Education, appointments, publications with accurate status, talks and service	CV

Keep factual labels precise: submitted, under review, preprint and published do not mean the same thing. A stable PDF is sensible when the call permits it and layout matters; check the submitted file after export. [4].

Organize the CV or resume

Decide the section order before polishing individual entries. In an academic CV, research appointments and publications may lead an application for a postdoctoral position; teaching experience deserves more space in an application for a teaching position. Use clear section names and include grants, mentoring and service where relevant. Within dated sections, list recent work first and use consistent dates, role titles and publication details. [3]; [1].

In a faculty application, dates, project names and descriptions of your contribution must agree across the CV, letter and statements. The documents develop different aspects of the application: the CV records the work, the letter explains how your experience fits the position, and the research statement develops the proposed program. [7].

The examples follow Mira Okafor, a fictional postdoc applying for a research program-manager role involving collaborations between genomic researchers and clinical teams.

Compare the outline she might use for a research post with the one for this program-manager role:

Document and reader	Early sections	Later sections
Academic CV for a research post	Education; research appointments and contributions; publications and preprints with exact status	Teaching and mentoring; awards, service and invited presentations
Resume for the program-manager role	Selected project coordination; collaboration and review experience	Relevant analysis methods; education and concise research background

These are working outlines, not universal templates. If the advertisement emphasizes teaching, regulation or a particular method, the order and space should change accordingly.

For a resume, choose the few experiences that answer the role's questions. Explain the research in terms the intended reader can understand while retaining the technical detail needed to judge your work. Name the problem or responsibility, the action you personally took, and the output a new reader can assess. Use the vocabulary of the call when it accurately describes your work. [6]; [1].

Use short entries, visible dates and enough white space to separate sections. Quantify the scale or outcome when it helps—for example, the number of people trained or projects coordinated. Keep method names that establish a technical qualification; explain their purpose when the first reader may be a recruiter outside your field. [6].

Example: one factual contribution, two document entries

CV — Research experience

Postdoctoral researcher, Systems Genomics Group, 2024–present
Cross-laboratory perturbation screen

- Developed and documented the reproducible analysis workflow.
- Coordinated a results review with clinical and computational collaborators.
- Trained two junior researchers to run and check the workflow.

Resume — Project coordination

Postdoctoral researcher, Systems Genomics Group, 2024–present

- Coordinated a perturbation-screen project across two laboratories, using one documented analysis workflow for the joint review.
- Presented the findings and unresolved questions to clinical and computational collaborators, then recorded the checks assigned to each laboratory.
- Trained two junior researchers to run and check the workflow.

The resume entry gives more space to coordination and shared analysis because these are central to the target role. For a computational-research position, the analysis itself would need more technical detail.

Writing a cover letter

When a letter is requested, name the role, explain your interest in the organization and develop a few examples of relevant experience. Use each paragraph to address a requirement, with enough context to explain your contribution. Leave the full chronology to the CV or resume. [2].

For this example, the advertised role includes coordinating joint projects and organizing research reviews.

Example: a short letter that connects fit to evidence

Dear Hiring Committee,

I am applying for the Research Program Manager position because your joint-project program brings genomic researchers and clinical teams together around shared studies. Coordinating that kind of work has become the part of my postdoctoral research that I most want to develop.

During my postdoc, I coordinated a perturbation screen across two laboratories. I developed the shared analysis workflow and trained two junior researchers to run and check it. This gave both teams a common set of results to discuss while retaining access to the underlying analysis.

I also organized the project review with clinical and computational colleagues. I presented the findings and unresolved questions, then documented the checks each laboratory would undertake. This combination of technical understanding, a common review process and clear follow-up is what I would bring to the institute's joint projects.

Thank you for considering my application. I would be pleased to discuss the position with you.

Sincerely,

Mira Okafor

The letter connects Mira's interest in the institute's joint projects to her experience coordinating analysis and agreeing follow-up work with collaborators. The resume lists these contributions; the letter explains why she wants to make this work central to her next role.

Asking for references

Choose referees who know your work well enough to describe your contribution. Ask whether they can provide a strong reference, leaving time to approach someone else if necessary. Discuss why you are seeking this role and which parts of your work together are relevant to it. Send the position description, current application materials, those selected examples, and the submission instructions and deadline. For several applications, keep the deadlines and links in one table. [5].

If a referee asks for a draft or notes, provide factual material: the context, your role, the result and relevant examples from your work together. The referee must decide what to endorse and write the final assessment. Check submission confirmations and send a polite reminder when a deadline approaches.

Before submitting, compare the documents with the call. Check dates, role titles and publication status across the packet; ask a reader to identify your strongest qualifications from the first page; and inspect the exported files for layout problems.

References

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Finding and applying for positions

Begin with the work you want to do and the experience you can offer. Search advertisements and organizations working in that area, and ask colleagues about opportunities. Keep track of promising positions and their deadlines.

Choose the appropriate application route

The application route depends on whether you are seeking research training, a postdoctoral project or a role with defined responsibilities.

Search	What to establish	Application route
PhD	Relevant projects or supervisors, training arrangements and funding	Check whether you apply to a programme or a specific project, and whether prior supervisor contact is expected. [2]
Academic postdoc	The research direction, training and resources the group can offer	Apply to advertised positions or inquire directly with a relevant group, following any contact instructions. [1]
Non-academic scientific role	The work, required capabilities and team	Use the advertised process; conversations with people in the role can help you understand the requirements.

For each serious option, note why it interests you, which requirements you meet and what you still need to find out. Keep the requested materials and contact details with the deadline.

The selection conversation also differs. A PhD interview usually tests preparation for training and gives you evidence about supervision and the programme. A postdoc discussion asks what scientific direction you could develop together, what you already contribute and what the group can support. A non-academic employer is recruiting for defined work: an early screen may test the basic match, while later technical, behavioral or panel interviews may ask how you handled decisions, collaboration and setbacks. Ask for the format rather than assuming that one academic research talk prepares you for every route.

Finding opportunities

Combine programme and employer sites with job boards, professional networks and conversations. Ask colleagues which organizations work on the questions or problems that interest you. Follow useful leads beyond familiar institutions.

For a postdoc, list groups connected to the questions, methods or research direction you want to develop. Read recent work, then narrow the list through conversations with people in the field, current members and alumni. Include relevant groups without advertised openings: some recruit through direct inquiries. [1].

For a PhD, check how a supervisor is assigned. A programme may offer rotations or several potential supervisors; a project-based position may already have a defined question and supervisor. Confirm availability, entry requirements and the funding arrangement before preparing an application. Ask what support covers and for how long. See Choosing an advisor and research environment for assessing supervision. [2].

For a non-academic role, identify what the person hired will do, which capabilities are essential and who they will work with. Match each requirement to evidence from your experience. Separate requirements you do not yet meet from experience you have but need to explain in the employer's terms. If the advert leaves a requirement unclear, ask someone familiar with the work before deciding whether you meet it. Use CVs, resumes, letters and references to turn these matches into application material.

Write a specific inquiry

Before contacting someone, read their group's work and any application instructions. Explain the connection to your experience and ask a question the recipient can answer, such as whether the group anticipates recruiting in your intended period.

For an academic-postdoc inquiry, name the research connection, give one or two examples of relevant work and say when you hope to start. Ask whether the group expects to recruit or would be willing to discuss a possible project. Include or link materials only when requested or when they help the recipient assess the inquiry. [1].

If a direct inquiry receives no reply, send a short follow-up after allowing time for a response, then continue with other opportunities. A conversation does not replace a required formal application: keep track of that deadline separately. [1].

Example: a short, specific postdoc inquiry

Nadia is finishing a cell-biology PhD in September 2027. She has built a live-imaging workflow that segments and tracks epithelial repair fronts. Dr Imani's fictional group studies immune–epithelial interactions during tissue repair. Nadia has read two recent papers from the group and wants to develop a project that combines imaging with quantitative analysis.

Subject: Postdoctoral inquiry: live imaging of epithelial repair

Dear Dr Imani,

I expect to complete my cell-biology PhD in September 2027. I developed a live-imaging workflow to segment and track epithelial repair fronts, and would now like to study how immune cells influence that repair. Your group's work on immune–epithelial interactions is closely connected to this question.

Do you expect to recruit a postdoc in autumn 2027? I would welcome a conversation about whether my imaging and analysis experience could contribute to your research. I have attached my CV and a brief research summary.

Best wishes,
Nadia Patel

Questions about supervision and possible projects can follow if Dr Imani is interested in discussing recruitment.

Prepare for the interviews

An interview can include an initial screening call, a research talk, meetings with possible colleagues, an assessment task or several conversations over a visit. Ask for the agenda, who you will meet and any presentation or task instructions. Use this information to prepare: a recruiter may need a concise explanation of how your experience meets the role's requirements, while a scientific colleague may ask how you chose a method or interpreted a result. Prepare questions about the work and support you would need before accepting.

For a postdoc, prepare to explain your PhD question, your contribution and what the results establish. Read the prospective group's recent papers and bring ideas to discuss. If a question falls outside your knowledge, explain what you understand and ask for clarification where needed. Ask about supervision, resources, project ownership and support for your next career step. [1].

For a non-academic role, prepare brief examples that connect your experience to the problem described in the advert: a technical decision, a collaboration, a setback or a result that changed a next step. Be ready to distinguish what you did from what the team did. Ask how the team defines a successful first period in the role, what it needs from the person hired and how the work is evaluated. For a PhD, explain the experience that prepares you to learn and investigate the proposed area, then ask about the programme route, available supervisors and training environment.

> **Example: discussing a possible postdoc project**
>
> Asked how she would begin in Dr Imani's group, Nadia says: "I would first learn how the group currently measures repair and which results are reliable enough to build on. Then I would try my tracking workflow on a small experiment and compare what it shows with the existing analysis. If it adds useful information, we could plan a larger imaging project. I can bring the segmentation and reproducible analysis; I would need help learning the immune-cell assays."

After each conversation, write down what you learned, what remains uncertain and the promised next action. Send a brief follow-up when it is appropriate to thank the person, correct a material omission or confirm an agreed step. This record will also help you compare an offer with the criteria you set at the start.

Review the search and compare offers

Keep several possibilities active while awaiting decisions. Review where applications stop. If applications to apparently suitable roles rarely reach screening, ask a knowledgeable reader to assess whether your experience meets the requirements and whether the documents make that clear. If interviews repeatedly expose the same gap, distinguish difficulty explaining your experience from experience you still need to gain. The first calls for a clearer example; the second may require training or a different choice of roles. Record promised follow-ups and deadlines [3].

When an offer arrives, compare the actual work and conditions with your priorities. Resolve unanswered questions about the project, supervision, funding or employment period, location and start date. If other decisions are pending, ask when a response is needed and whether more time is possible. Use the research-environment comparison for a PhD or postdoc, and return to your career priorities for other roles. [1].

Checks as the search progresses

- I understand the position, application route and requirements.
- I can explain why the work interests me and which experience prepares me for it.
- I have followed the position's stated application or contact instructions.
- I have prepared research or work examples and questions for the interview.
- I have recorded what I learned and kept the rest of the search moving.

References

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Finding a group leader position

Leading a group changes the work as well as the job title. You would set research priorities, support people and obtain resources alongside doing science. Decide whether you want that combination, then look for positions whose research, teaching and mentoring responsibilities fit the group you intend to build. [3].

Define an independent program

Build the program around questions you intend to lead. Explain how your previous work prepares you to address them, which projects you would begin, and how their results would guide later work. Independence can come from a distinct question or approach within the same field. Discuss with your current supervisor which questions each group will pursue, what materials or data can be transferred, and where collaboration will continue. Agree who will lead the overlapping work; continued collaboration need not prevent you from developing a distinct program. [2].

Prepare a short program summary before searching widely. Include the people, facilities and funding needed to start; these needs will help you assess calls and later offers. If an essential assay can only be performed in your current lab, agree how that work could continue or develop an alternative before making it central to the proposal.

Finding suitable positions

Compare each call with your program summary. A research institute, a university department and a teaching-focused position may offer very different time, facilities and student populations. Check whether your projects can work in that setting, who you could collaborate with and how the group would be staffed. Use recent faculty appointments and conversations with colleagues to understand the local expectations. [2].

For each role, keep the advertised criteria and requested materials alongside the parts of your program that fit and the questions you still need to ask. Confirm facilities, appointment terms and local expectations with the host. A department's scientific interests may match yours even when its available space or staffing cannot support your first projects. [2].

Prepare the faculty application

Adapt the materials to the call, using CVs, resumes, letters and references. The research statement should connect your important contributions to the program you intend to build. Write for scientists outside your specialty: explain why the question matters, how your prior work prepares you to address it, and what you will lead. Give enough experimental detail for them to assess whether the first projects are feasible. The exact balance between past and future work depends on the call. [2]; [4].

Use the letter to explain your interest in the position and the contribution you would make. In a teaching or mentoring statement, connect experience to specific plans for this setting: what you have taught, how you helped someone learn, and how you would develop that work with the students or researchers there. [2].

Test readiness with the research statement

Drafting the statement can help you judge readiness. You need both an account of completed work and credible future projects. Organize the draft around four parts:

Part	What a reader should be able to find
Research trajectory	How your prior contribution prepares you to address the new question, and which work belongs to the former and proposed groups.
Program question	A question broad enough to matter to adjacent scientists, stated without assuming that the proposed answer is already known.
First projects	What each project compares, which possible explanations it can distinguish, how the findings would guide the next experiment, and whether it can run alongside another project.
Feasibility and growth	The people, facilities and access required to begin; how the first projects could be funded and how their answers create the next tractable question.

The following planning sheet uses the hypothetical epithelial-injury program introduced in Chalk talks.

Example: turn a program into a research-statement plan

Opening question and significance
How does resolved inflammation change the next epithelial injury response? Understanding whether cells retain an injury history could explain why the same tissue responds differently to a later insult.

Preparation and independent boundary
Prior work on acute inflammation supplies the biological starting point and culture experience. The former lab continues acute inflammation-signalling studies. The new group leads the epithelial re-injury comparisons and functional assays; any shared protocol or collaboration will be agreed explicitly.

First project and decision — Year 1
After removing the inflammatory cue, compare exposed and control airway organoids. Measure cell composition, expression and accessibility to assess whether a retained difference occurs within the same cell types or reflects a changed mixture of cells. These comparisons guide which explanation to investigate further; the re-injury project tests whether exposure history changes function.

Parallel project and dependency — Months 9–24
Re-injure matched organoids and measure barrier repair and inflammatory output. This can begin alongside the first project. A mechanistic project starts only after a reproducible functional difference is established.

Conditions for starting
Confirm access to primary airway organoid cultures, a cue-clearance assay, sufficient single-cell facility capacity and a first researcher able to run the functional assay. State an alternative analysis or facility route if the preferred access is unavailable.

Give the draft to readers with different roles: one who knows the field, one who can assess a related scientific question, and one who understands the prospective setting. Ask them to mark the first point where they lose the question, cannot distinguish the projects, or cannot see how the work starts. Revise the plan behind the prose when their answer exposes a missing comparison, dependency or boundary.

Share the program summary and application materials with your referees, including the agreed boundary with your current lab. See reference requests for coordinating their submissions.

Prepare for the interview

Before a visit or remote interview, ask for the itinerary, audiences and presentation formats. Read recent work by the people you will meet. Prepare concise explanations of your previous contributions, initial projects, independence, possible funding routes and teaching experience. Discuss genuine scientific connections with prospective colleagues, and use meetings with recent hires to ask how the department supports a new group. [2]; [4].

Match questions to the person who can answer them. A facility director can explain capacity, fees and sample preparation; a department chair can clarify teaching, review and space commitments; a recent hire can describe what support was available in practice; trainees can describe supervision and local research life. Record the answer, the person who gave it and what still needs written confirmation. For a resource your first project depends on, ask when you could use it and what would have to be arranged first.

For the job talk, use Explaining research to an audience, Designing a presentation and Delivering a talk and answering questions to develop and rehearse a scientific explanation. For a chalk talk, use the board-map method: show the program overview, a comparison that distinguishes your proposed explanations, and how different findings would change the next experiment. Rehearse interruptions and returning to the map.

Comparing and negotiating offers

Once an offer exists, cost the first projects and compare their needs with the proposed support. Include staffing and running costs as well as equipment and space. Check when the resources become available and what restrictions apply: do salary amounts include benefits, for example, and must startup funds be spent by a particular date? Also discuss the teaching and service expected while you establish the group. Use these details to explain each request. [1]; [4].

For the proposed epithelial program, access to an established single-cell facility could reduce the need to purchase equipment. Check its capacity, charges and sample-preparation support before removing those items from the budget. Airway-organoid culture, cue-clearance measurements, staffing and facility access must also be available when each project is due to begin.

Use the same categories to compare offers, including the start date, teaching load, appointment or review timetable and personal constraints. Prioritize the requests that determine whether you can establish the program. Discuss alternatives when an item is unavailable—for example, funded facility access instead of equipment—and ask for the agreed resources and conditions in writing. Seek advice from recent hires and an experienced colleague who understands the local appointment system. [1].

For the hypothetical epithelial program, a request sheet could look like this:

Program need	What to establish from the offer	Acceptable alternative or follow-up
Airway-organoid culture	Suitable space, access date and responsibility for any preparation	Temporary shared space with a written date for the permanent arrangement
Single-cell analysis	Facility capacity, access charges, sample-preparation support and booking lead time	Another facility only if its cost and turnaround fit the first-project schedule
First researcher	Amount, duration, eligible role and whether salary costs include benefits	A different staffing mix that still supplies the required assay expertise
Time to establish the group	First-year teaching and service assignments	A specified lighter schedule or delayed course responsibility, where the setting permits
Startup funds	Included cost categories, purchasing process and any spending deadline	Written extension or staged funds when procurement or hiring will outlast the initial period

The written follow-up should lead with appreciation, then separate essential clarifications from ranked requests. Each request should name the work it enables. Avoid copying another applicant's list: equipment access, teaching arrangements, appointment terms and negotiable items vary by institution and jurisdiction.

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3. Tregoning, J. S., & McDermott, J. E. (2020). **Ten simple rules to becoming a principal investigator.** *PLOS Computational Biology*, 16(2), e1007448.
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Career transitions

Before making a career move, work out what you want to change. A new position may leave the same workload or funding uncertainty in place. Consider whether a change of project, supervisor or role would address the problem.

Decide what needs to change

Think about what you would want to keep as well as what you want to leave. You may still enjoy research but need more predictable hours or a longer contract. If so, investigate whether another research setting offers those conditions before deciding whether to change the kind of work you do.

Compare the available options against the conditions you want to change and those you want to preserve. Include changing project or supervisor as well as leaving a degree or institution. Note what you still need to find out before deciding [4].

Discuss the options with people who know the work and can help assess its prospects. Working conditions, mentorship and alternatives to an academic career were recurring concerns in a survey of postdoctoral researchers [1]. When a supervisor is part of the problem, seek advice elsewhere before deciding how to raise it.

Example: deciding what to change

Amir is a postdoc who still enjoys microscopy and troubleshooting. He no longer wants his next position to depend mainly on building an independent publication programme, and he needs to remain in the same region.

Staying for another year could complete an active paper, but it would not change the type of work or the uncertain funding horizon. A second postdoc would offer more bench research, yet preserve the same features he wants to leave. After visiting a microscopy core and speaking with two staff members, Amir learns that the role would replace much of his own-project time with user training, method development and responsibility for reliable shared workflows.

Amir accepts the core-facility role. The decision is a trade-off: he gives up some control over research questions in return for locally based, collaborative technical work. He agrees to a six-week start date only after identifying what the active paper needs and what can be handed over.

Plan the move

Consider applying directly when you have investigated the new role and could start when needed. An intermediate step may help if something specific prevents the move or you still need to try the work before committing. A short placement could help you judge whether the work suits you. A course could teach a required skill; a short contract could provide income before the next position starts. Relocation or caring responsibilities may also affect the route. Decide what the intermediate step needs to accomplish and when you would be ready to move on. Compare it with applying directly before committing to another course or position.

Work backwards from the intended start date. List unfinished work and any materials or access that need to be transferred, then agree who will handle each item and when. Discuss any contribution you expect to continue after departure.

Allow time for delays and agree which unfinished tasks could be handed over if they cannot be completed before departure. Consider the workload during the transition alongside family and caring responsibilities. Agree the limits of any temporary overlap between roles [5].

Handing over projects

For each active project, explain the question, what the work has established and what remains to be decided. Record where a colleague can find the data, notebooks, code and samples, with the relevant versions and permissions. Include instructions for regenerating the current result. Label provisional analyses, identify assumptions still to be tested and describe approaches that failed. Flag supplies that may run out or access that will expire, and identify who can decide how to continue.

Ask a colleague to retrieve the material and reproduce a result from these instructions before you leave. Arrange continued institutional storage and access with the group; follow its rules for notebooks, data and samples [3].

Example: a project handover

Amir will start in a microscopy core facility in six weeks. His handover record assigns the remaining work and identifies where to find the material.

Manuscript: the main analysis is complete. The collaborating researcher will check a control by 14 May and discuss its implications with Amir on 16 May. Repository release `v1.3` and the figure-to-data table are in the lab project space.

Pilot project: no feasible next experiment remains with the available materials and time. The notebook, raw data and a note explaining the stopping decision are indexed in the project space.

Access check: a colleague will regenerate Figure 2 using the README before Amir leaves. Amir will correct any missing instructions. The lab administrator will confirm which group account retains access.

Continuing responsibilities

Tell the people who need to plan around the change once the timing is settled. Explain what you can finish before leaving and agree any continuing responsibilities, such as contributing to manuscript revisions. If you are taking a project to a new group, discuss which questions, materials and collaborations will move with it [2].

Discuss authorship on unfinished papers with the contributors and record the agreement. Ask prospective referees whether they can support your application. Arrange any continuing access to data or systems with the institution.

If a relationship is difficult, reduce unnecessary dependence: keep clear records, obtain advice from more than one mentor and use the appropriate local route for a matter that cannot be resolved directly.

Begin the next role

Before starting, discuss your initial responsibilities and who can give feedback. Ask colleagues how decisions and routine communication work, and agree when to review your progress [2].

Agree what to work on before the first review, allowing time to learn unfamiliar procedures. A research scientist might learn a quality-system workflow and complete a defined analysis; an editor might assess a supervised submission while learning the journal's decision process and audience. Find out whose work depends on yours and what they need from you. Adjust the plan with your supervisor or manager as you learn more about the role.

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Chapter 10

Work Culture & Management

Mentoring

Discuss what the researcher wants to learn and what support the mentor can provide. A short mentoring agreement can record those expectations and give both people something to revisit as the work and the researcher's needs change [6].

Agreeing expectations

Prepare separately before the first substantial conversation or when revisiting the arrangement. The researcher identifies current work, decisions they want help with, capabilities they want to develop and circumstances that affect what they can undertake. The mentor identifies what support they can reliably provide, where another person is better placed to help and which constraints are fixed. Begin with the researcher's account, then compare it with the mentor's expectations. Preparing independently gives both people a chance to identify priorities before either adopts the other's view [7].

Record the decisions in language that both people can use. A concise agreement might cover:

- **Purpose:** Which research decisions and development goals should this relationship support now? What lies outside its scope?
- **Contact:** Which meetings are protected? What belongs in a meeting, a short message or a written draft? What counts as urgent?
- **Preparation:** What should the researcher send or decide before a meeting? What will the mentor read, check or arrange?
- **Decisions:** Which choices can the researcher make independently? Which require discussion because of cost, safety, ethics, shared resources or project commitments?
- **Feedback:** How will each person ask for feedback on work, reasoning and the relationship? How quickly can each realistically respond?
- **Boundaries:** What working-time, privacy and leave expectations apply? Which matters should go to a programme, manager, committee or specialist service?
- **Development:** Which capability or opportunity will be developed next, and how will practice and feedback occur?
- **Review:** When will the agreement be revisited, and what event should trigger an earlier conversation?

Choose a meeting interval that fits the current work and test whether it produces useful decisions. A weekly meeting that repeatedly ends without a choice, action or feedback is not automatically better than a less frequent meeting with good preparation. Conversely, a new method, a difficult transition or a stalled project may justify temporarily closer contact. Early-career researchers surveyed in a 2026 preprint placed high value on useful regular meetings, open communication, boundaries and participation in decisions; the study does not imply one schedule for every relationship [5].

Do not require disclosure of private circumstances. A researcher can state the work consequence—"I cannot attend before 10:00 for the next six weeks" or "I need to change the travel plan"—without explaining a diagnosis, family matter or other personal detail. The mentor should explain which constraints can be changed, which require another office and how to reach that office.

Keep individual supervision and development in this agreement. Put practices shared across the group in the lab manual, and keep safety, employment, data and programme requirements in their authoritative records.

Use meetings to develop judgment

Bring the material needed for one or two consequential questions. A useful agenda says what has changed, what decision is needed and what the researcher currently recommends. Attach a figure, short rationale or draft when the evidence itself matters; routine status can be brief so the meeting has time for the decision.

Ask the researcher to propose the next experiment and explain what its possible results would tell them. Discuss the reasoning before supplying your own answer. If they cannot choose a next step, find out why. They may need help interpreting a control, training in the method or a decision about access that only you can make. Asking them to be more independent will not resolve a decision they lack the authority to make.

Give a novice a task on which they can practise without putting irreplaceable material or a major project commitment at risk. Demonstrate the technique, explain common failure modes and begin with a task whose expected behavior is already known. As performance becomes reliable, shorten the instructions and move responsibility for planning, execution and interpretation to the researcher [4]. When choosing a research problem together, consider whether it interests the researcher, whether it is feasible with the available expertise and what they would learn by pursuing it [2].

Example: mentor a decision without taking it over

In this fictional example, Aruna is a second-year doctoral researcher studying a signaling reporter in cultured cells. She can run the established assay and controls reproducibly and can plan and document a complete run without supervision. Two later pilot runs after a perturbation show different reporter timing. The second also has lower viability and uses a new reagent lot. There is time for one more run this week: either a limited diagnostic comparison or the first large downstream assay.

Aruna's mentor begins: "Which run would you do this week?"

"Delay the downstream assay and compare the old and new lots in the same culture batch," Aruna says. "I'll use the established controls and record viability."

"What would that let you conclude?"

"If the timing follows the lot and the controls and viability are acceptable, I'll replace that lot and start the downstream assay next week. If it does not, or viability still differs, the comparison has not resolved the earlier inconsistency and I'll bring back the options before the next run."

The mentor asks what delaying the downstream assay costs. They agree to use this week's run to check the reagent-lot explanation before committing to the larger assay. The mentor confirms access to the old lot and control material and offers to review Aruna's one-paragraph rationale before she commits the material. Aruna owns the design, run and first interpretation. Their follow-up will examine the controls and viability before deciding whether to proceed.

If Aruna had never run the assay, the same handover would be premature. The mentor or an experienced colleague should first demonstrate the workflow, observe a practice run and check that Aruna can recognize failed controls. If the choice could irreversibly consume rare material, the mentor should join the decision even after Aruna has mastered the technique. Match support to demonstrated capability for this task and the cost of error.

Peer and specialist mentors can provide day-to-day methods knowledge that the primary mentor lacks. Agree which training they will provide and recognize that contribution. A junior researcher should not become solely responsible for another person's training. The primary mentor must check that the teacher has time and credit, and that the learner knows who will review their progress and help when the teacher cannot.

Feedback

Be clear about what needs to change. "The conclusion exceeds these controls" concerns scientific reasoning. "I need the figure two days before the meeting if you want line-by-line comments" concerns workflow. "I spoke for most of our last three meetings; I want you to present your

recommendation first next time" concerns the mentoring relationship. Mixing these can turn a correctable issue into a judgment about the person.

Point to the part of the work or the interaction that needs to change, explain the problem and agree what will happen differently next time. The researcher should also be able to say which support was useful, where advice conflicted or arrived too late and what they need next. Reciprocal feedback does not make the power relationship equal; it gives the senior person information they are responsible for acting on.

When the same problem recurs, compare it with the agreement. Ask whether the expectation was unclear, the promised support was unavailable, the capability was overestimated or circumstances changed. Revise the arrangement and set a review point. Repeated credit-taking, retaliation, blocked access to other advice, coercive disclosure or unsafe practice requires help beyond an ordinary feedback conversation. Use a committee, programme lead, manager, ombud or the institution's current formal route. Do not require the more vulnerable person to repair the relationship alone.

Plan development beyond the immediate project

Hold a broader planning conversation periodically and at transitions. Both people prepare independently; the researcher speaks first. Review recent work and discuss what they want to learn or explore next, including career questions that their current project will not answer. Identify opportunities to do that work, the support needed and any limits on time or access. Agree a small number of actions and dates so that development receives time alongside project tasks.

Look for opportunities to practise the skill in real work. A researcher learning to write could draft part of a paper and discuss the revisions with a coauthor. Someone learning how publication decisions are made could join an editorial conversation or work through a review with an experienced colleague [1]; [8]. Obtain permission before involving a trainee in confidential communication, and agree their role.

A useful plan names one near-term opportunity and what evidence will show learning. "Improve presentations" is difficult to act on. "Give the next 10-minute project update; rehearse once with the postdoc; ask two colleagues whether the decision and uncertainty were clear; review the recording with the mentor on 12 November" creates practice and feedback.

Career development may reveal options that the primary mentor does not know well. Use exploring career options to compare paths, then identify someone who can supply current experience. The mentor's responsibility is to support an informed choice, including when it differs from their own path.

Build a mentoring network

One person rarely provides deep technical guidance, project authority, career knowledge, sponsorship and confidential perspective. List the roles the researcher needs now and where each can come from: primary supervisor, methods expert, collaborator, committee member, peer, career contact or institutional adviser.

The primary mentor should welcome complementary mentors, make introductions where useful and state that seeking other advice is expected. Multiple relationships can reduce dependence and make conflicting advice visible [3]. When advice conflicts, return to the decision: which adviser has relevant knowledge or authority, what assumptions differ and who ultimately owns the choice?

Review a mentoring relationship

- Both people can state the current purpose, boundaries and next review date.
- Meetings end with a decision, action or clearly named unresolved question.
- The researcher explains reasoning before the mentor supplies an answer.
- Support matches capability for the particular task and the cost of error.
- Feedback can travel in both directions without requiring private disclosure.

- The development plan includes real practice, feedback and a date, not only aspirations.
- Other mentors cover needs that the primary relationship cannot meet.
- Repeated power, safety, credit or conduct problems have a route beyond the direct supervisor.

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Lab manuals and shared expectations

A lab manual helps members find the group's shared practices without having to learn every convention by watching others. Explain who is responsible for routine work and whom to ask when an arrangement does not fit.

The manual need not contain every protocol, personal agreement or university rule. Operational instructions and checklists can sit in the same document when a small group can keep it navigable; a larger group may link separate protocol, safety and data records. Choose the arrangement that lets a member find the current instruction and its owner [5].

Decide what needs documenting

List the recurring questions first. Ask current members what they had to learn by watching others, which questions receive different answers and which unstated expectations have delayed work or made them reluctant to ask for help. Review onboarding questions, repeated booking or handover problems and decisions that depend on seniority. Include technicians, managers, students, postdocs and research staff so that the manual covers the responsibilities and support each needs, including the work of maintaining shared resources.

Write down a practice when it recurs across members and knowing it in advance changes what someone does. Common candidates include communication and response times, meetings, shared equipment, routine responsibilities, time away, data and code, mistakes, feedback, joining and leaving. Choose among these topics according to the group's work: field seasons, clinical data and shared instruments raise different questions from those of a small theoretical group. Published manuals illustrate those choices; they do not establish a universal policy.

Keep shared practices in the **lab manual**: routine workflow, reciprocal expectations, responsible roles and routes for questions or concerns. Keep an individual's goals, supervision, feedback and development in a **personal mentoring agreement**; see mentoring. Keep employment conditions, safety rules, ethics approvals, data-management plans, access controls and collaboration-specific decisions in the responsible **institutional or project record**. The lab manual can link to those records and explain what members need to do.

Keeping these records separate prevents a local preference from appearing to be an institutional requirement and keeps private matters out of a group document. Mentor–mentee expectations should be tailored and revisited as the relationship changes [4].

Decide which material can be public, which belongs on an internal group site and which should remain in an access-controlled project or institutional system. Prospective members can benefit from seeing genuine expectations. Keep sensitive contact details, restricted data procedures and individual accommodations in the appropriate protected record.

Writing the policies

A value such as "respect people's time" can introduce a section, but it does not settle a booking clash. State when the policy applies, what people should normally do and who is responsible. Then explain what changes in a foreseeable exception and where someone can take a question or concern. Name who keeps the policy current. These points can fit into a short paragraph; they do not each need a separate heading.

Use reciprocal language where responsibility is reciprocal. If members should flag a delayed analysis, state when and to whom; if the project lead promises a response or decision, state that too. Include a route that does not depend solely on the person whose conduct might be questioned. A manual can name an institutional ombud, graduate programme, staff representative or reporting office using the organization's current terminology. It should not promise confidentiality, remedies or authority that the group cannot provide.

The SAFE Labs authors similarly turn broad aspirations into actions that can be documented while leaving groups to choose policies suited to their setting [3].

Example: a shared-equipment policy

Northbank Imaging Group manual — shared microscope bookings, version 1.1

Finish at the booked end time whenever possible. If an acquisition is likely to run over, the current user updates the calendar and contacts the next user as soon as they know. The next user decides whether moving their start is workable; silence does not mean agreement. If the users cannot agree, each tells the lab manager how time-critical the session is, what preparation is already committed and whether the work can be repeated. The manager decides, records the reason and arranges a replacement slot when one is needed. If the manager is involved in the clash, the facility director makes the booking decision. Concerns about unfair treatment can also go to the institution's staff-advice office.

Live-cell acquisitions may need a longer window because the useful endpoint depends on cell behaviour. Label this when booking and include a realistic buffer. This exception does not change institutional access or lone-working rules.

The current user remains responsible for the acquisition log, shutdown or handover and cleaning. The lab manager owns the booking policy. Repeated overruns or an exception that the policy does not handle trigger an earlier review; otherwise the next review is 1 March 2027.

This is one fictional group's arrangement. Another facility could reserve fixed changeover periods, assign priority by study stage or prohibit overruns.

Draft the manual with the people affected

The person accountable for the group should first state the constraints and promises they can actually uphold. Then collect cases from the people affected. A meeting can expose differing interpretations; a short form or private route lets members raise concerns they may not voice in front of a supervisor. Asking members to contribute does not transfer the leader's responsibility for employment, safety or fair treatment.

Scale the drafting process to the group. A small lab may revise a short draft together. A larger group can assign topic teams, exchange drafts and use several meetings to resolve inconsistent language. The WIN Physics group used volunteer teams, cross-review and advice from departmental and human-resources specialists for its multi-PI group of about 30 people [6].

For each section, compare an ordinary case with an awkward one. Check whether the responsible person has the authority and capacity to act and whether the exception is clear. Ask the appropriate institutional office to review policy-sensitive sections. The responsible leader then confirms what the group can adopt. Record unresolved concerns and keep a route outside the direct supervisor for a member who cannot safely resolve a matter within the group.

Keep the first version short enough to finish. Mehr selected topics around his group's asynchronous collaborations, left specialist platform detail elsewhere and observed which sections members ignored or could not understand [5].

Introduce, use and revise it

Publish a version, effective date and named owner. Walk through the document with current members, then discuss the relevant sections with a prospective or new member rather than sending a link without context. An acknowledgement can record that the person received the current version

and had a chance to ask questions. It should not be presented as blanket consent, a waiver of rights or proof that every expectation is fair.

Refer to the manual when the described situation occurs. If practice differs from the text, either correct the practice or revise the document. Mariam Aly describes using her manual during onboarding, holding herself to its promises and sharing maintenance of a separate practical wiki [1].

Review the manual on a set date, and revise it sooner after a new instrument, changed institutional rule, repeated exception or unclear responsibility. Communicate material changes directly and keep a short change log so that members can tell which expectation applied when. A review of how the group works can reveal when the practice itself needs to change before its description is updated.

Example: adopting and revising the booking policy

The Northbank lab manager begins with two recurring cases: unannounced overruns and unrealistically short bookings made to secure scarce time. Current users discuss both at a lab meeting and can add examples through an anonymous form. The manager drafts two options, checks the facility's access rules and asks the affected users to test each option against a normal fixed-sample session and a variable live-cell acquisition.

The group leader confirms which arrangement the lab can support. The manager publishes version 1.0, explains it at a meeting and includes the booking section in onboarding. Members are asked to report unclear cases to the manager; a person concerned about unfair treatment can also use the institution's staff or student route.

During the six-week trial, members interpret "a short shift" differently. Version 1.1 removes that phrase: affected users must agree directly, and the lab manager decides when they cannot. The change log records the reason. The next review is scheduled for six months later, with an earlier review if the same exception recurs.

Data and AI practices

The data section should help a member find an approved place to store and work with their data and the person who can arrange access. Explain which kinds of data each environment can hold and how to record versions or changes during active work. Point to the project record used when someone joins or leaves. Keep dataset-specific retention, consent, transfer and access conditions in the data-management or ethics record that governs the project. Link to that record and explain what members need to do, rather than creating a second copy of rules that may change.

For generative AI, check the current policies that apply to the institution, funder, project partners, data and intended publication. Record the version consulted, approved tools or environments, excluded inputs, uses that require a project record or disclosure, and who verifies the output. A trainee may be able to check generated syntax while still needing to develop the judgment required to choose an analysis or interpret a result. Discuss concrete tasks before writing the rule, and connect permitted use to what the member can explain and verify [2]. Detailed questions about contribution and authorship belong in collaboration, authorship and credit.

Check whether the manual works

Give one section to a member who did not draft it. Ask them what they would do in the ordinary case, what would change in the stated exception and whom they would contact if the responsible person were involved in the concern. If those answers require unwritten knowledge, revise the section.

Check that links open, named roles still exist, public and internal material are separated, and each promised response has an owner with capacity to provide it. Compare the written rule with recent

practice. Where they differ, establish whether the practice needs correcting or the manual needs updating before revising the text.

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Leading and reviewing a research group

Before taking on new work, consider what the group would need to do it. A promising project may require a method nobody knows yet or more supervision than you can provide. Plan those commitments alongside existing projects and the support already promised to group members.

Priorities and planning

Translate the group's longer-term questions into near-term priorities. To understand recovery from stress, for example, you may first need to establish an assay. Decide whether that takes priority this month over completing another analysis or recruiting a specialist, and explain the choice to the group [2].

For each priority, agree who will do the work and what support they need. Check it against existing obligations, including thesis completion and the maintenance of shared material. Decide when to review progress before adding further work.

Keep near-term goals concrete while leaving the longer direction open to discovery. Three to six months for immediate priorities and one to three years for scientific growth can be useful starting horizons; shorten them when access or funding is uncertain. Discuss the plan with the group so that people can question the assumptions and explain what their part will require [4].

Scientific goals should specify what can be learned, not demand a preferred result. A milestone can be a working assay with interpretable controls, a complete dataset or a decision about feasibility. If evidence undermines the original idea, completing that decision can be good work even when the project changes direction.

Match resources to the work

Review what is available, what is committed and what is only hoped for. A submitted grant, anticipated fellowship or informal offer of instrument access should remain a separate scenario until it is confirmed. Ask the local finance or research-support team to help translate the plan into costs over time. Include the full appointment cost of people, required student costs where applicable, consumables, facility charges and continuing maintenance. Spending restrictions and end dates can make a nominal balance unavailable for the work you intend [4].

Check what would happen if a grant were unsuccessful, an instrument arrived late or a key person left. Which promised work could no longer be completed? Decide what to reduce or postpone before the group runs out of the time or resources needed for it. Buying facility access may let a project continue while an instrument is unavailable; another project may need to wait.

Equipment decisions also need an operating plan. Compare ownership with facility access using anticipated use, booking constraints, maintenance, technical support and delivery time. A purchase that depends on an unavailable room, missing component or untrained user cannot yet support an incoming researcher. Ask who can help with procurement, installation and access before promising a start date [3].

Time is another limited resource. Estimate the training, supervision and coordination that a new commitment requires, including the demands it places on senior researchers. Count the leader's teaching, writing, administration and existing supervision. A salary can fit the budget while the person's training cannot fit the week. When the problem concerns how workload or access is distributed, use sustainable and inclusive research workplaces to redesign the arrangement with the people affected.

Hiring

Identify the work that needs someone to do it. A group may need reliable daily culture maintenance, help with a specialist analysis or a researcher to develop a new question. Consider whether a facility service or collaboration could meet the need, or whether the group should reduce its planned work. If hiring is the appropriate choice, describe the responsibilities and required skills. State what the person can learn in the role and who will train and supervise them.

There is no universal first hire. An experienced technician may establish continuity and support training but cost more. A novice may be an excellent choice when the leader has time to teach and the work can wait for that learning. A postdoc needs scope for scientific judgment and development as well as funded time. Compare the role's contribution with the support it needs, rather than treating one career stage as a cheaper substitute for another [1].

Check the appointment route with the relevant programme, department or human-resources team before advertising. Student placements, fellowships and staff appointments can have different procedures and obligations. Follow the current local process for classification, pay, recruitment and offers.

Evaluate evidence for the role

Choose the criteria before reviewing applicants. Separate what someone must already be able to do from what you can teach. Use the same core questions and a comparable opportunity to demonstrate the required work. A short, relevant scenario might ask how a candidate would investigate an inconsistent control or document a handover. Explain what it assesses and avoid asking applicants to perform an unpaid piece of the group's research.

Ask about the candidate's actual contribution, how they handled uncertainty and how they worked with others. For someone without research experience, examples from coursework or other work can demonstrate careful records, learning and coordination. Evaluate those examples against the role rather than requiring an experience the position is meant to provide [5].

Record evidence against the criteria and ask other interviewers to do the same. "The candidate described checking and correcting a sample-label error" is useful evidence; "they seemed like one of us" is difficult to inspect. Collaboration matters, but similarity, confidence, eye contact or enthusiasm are poor substitutes for what the work requires. Give candidates time to think and opportunities to ask questions. Follow the agreed reference-check process, including permission before contacting a current supervisor [2].

The candidate is also choosing an environment. Explain the available funding, current limitations, supervision, responsibilities and scope for development. Arrange a conversation with current members in which the leader is absent. Answer what you can honestly provide; separate an intention from an authorized commitment.

Example: choose a hire and revise the operating plan

Nadia leads a fictional cell-biology group with two doctoral researchers. One is completing a thesis; the other recently joined. The technician who maintains cultures and reagent records will leave in eight weeks. Nadia wants to replace that role and begin a second assay, but she can currently provide six hours a week of direct training. The new doctoral researcher needs four of those hours.

Nadia and the department's finance adviser cost two appointments over their full proposed duration, including the associated running costs. Both are affordable from committed funds. A novice appointment would leave enough money for the second assay; an experienced appointment would require postponing it. They do not include a pending grant in either budget.

Work required: Maintain cultures and stocks, identify routine quality problems, keep complete records and provide a usable handover. Developing the second assay is a separate aim.

Support needed: Nadia estimates that a novice would initially need four hours of direct training each week, in addition to the doctoral researcher's four. An experienced hire would still need local training, estimated at two hours. These are assumptions for this group, to be checked after appointment. The finishing researcher has no unallocated training capacity.

Decision: Recruit for the experienced role and postpone the second assay. Six hours of planned training fits the six available; eight does not. The group retains an important existing activity and protects the student's training. The trade-off is a slower expansion of the programme. Core-facility imaging already meets the present study's needs, so Nadia also defers a proposed instrument purchase.

Continuity: Nadia checks the hiring timetable with the department. If the appointment starts after the technician leaves, she will pause new experimental runs and maintain only essential cultures herself. The expansion will remain deferred during that gap.

Selection: Candidates receive the same culture-record scenario and explain what they would check before declaring a batch usable. Interviewers assess the required technical judgment, documentation and communication. The selected candidate supplies concrete examples of these capabilities and asks for training on the group's local system. Nadia and the candidate agree that this training will precede independent responsibility.

Start and review: The outgoing technician prepares the inventory and handover. Nadia arranges access and required training, protects the two weekly training hours and reviews the first complete work records with the new member. Independence requires reliable execution and recognition of failed controls, rather than a desired experimental result. If local training takes longer than estimated, Nadia revises the work allocation before adding tasks.

Reconsideration: At the next resource review, Nadia will revisit the second assay if funding and supervision capacity are both available. A grant award alone will not establish that the group can support another trainee.

The experienced appointment is justified by this group's bottleneck and capacity. With sufficient teaching time and a less urgent continuity need, the novice route could be preferable. Experience also does not remove the need to check readiness for the particular task.

Onboarding

Arrange access, workspace and training before the new member arrives. Name who can answer everyday questions and who is responsible for supervision. Discuss the relevant shared expectations, then make a separate mentoring agreement for the person's support and development.

Choose an early task that teaches the local workflow and has an interpretable outcome. Supply the protocol or starting material, expected controls, record location and a person to consult when something differs. Observe enough of the work to check that the new member recognizes errors and knows when to pause. Include technical training and the practical work of finding records, sharing progress and making a handover [5].

Set a review point around that first task. Ask what remained unclear, which promised support was available and what the person can now do independently. Adjust the next assignment accordingly. If a senior researcher provides training, agree the time and its effect on their own project; do not quietly add responsibility because they happen to be approachable.

Delegating responsibility

When delegating work, agree what the person is expected to accomplish and which decisions they can make. Specify the available resources and the changes they must discuss with you first. Check that they understand and can accept the responsibility, and agree when to review progress. Tell others whose cooperation is needed, and support the person's decisions within the agreed scope [2].

Match independence to capability for the task and the consequence of error. Someone can independently maintain an established workflow while needing close discussion for a new method. Review at meaningful points without reclaiming every choice because you prefer a different style. Retain decisions involving commitments or authority that cannot be delegated by the group itself.

Example: a bounded handover of booking decisions

For an experienced researcher coordinating a fictional imaging pilot, a written handover could say:

> You will coordinate imaging for the current pilot within the approved facility allocation. You can arrange slots with the users, change the order of runs within the agreed design and ask the facility to resolve an access problem. Keep the schedule and reasons for changes in the project record. Bring any change to sampling, extra expenditure or a clash with another project's agreed time to me before committing it. We will review the schedule after the first acquisition cycle. I will tell the team that these booking decisions are yours and remain responsible for resolving a resource conflict you cannot settle.

If the coordinator has never used the facility's booking system, first provide training and supervised practice.

Detailed agreements between collaborating groups belong in collaboration, authorship and credit.

Meetings and review

Use different conversations for different decisions. An individual meeting can examine a researcher's reasoning and support needs. A project meeting can resolve a shared dependency. A group meeting can compare interpretations and discuss scientific direction. Give each a purpose, useful preparation and a record of decisions, owners and next checks; choose frequency around the work rather than filling the calendar [2].

When work stalls, first establish what happened. A valid negative result, a failed procedure, a blocked resource and a missed agreed action need different responses. Compare the expectation with training, access, workload and supervision before deciding that the person lacks motivation. Discuss the specific observation privately, hear their account and agree the support, action and review needed. The leader may need to change a promise or priority as well as asking the member to change their work [5].

Keep enough contact with methods, controls and original records to understand the science being produced. Invite concerns before results are polished, give people room to disagree and recognize work that enables others: maintenance, curation, training and coordination. A recurring problem should produce a decision about support or responsibility, not an indefinitely repeated expression of disappointment. Detailed feedback and difficult conversations are treated in feedback and conflict. A safety or integrity concern needs the appropriate institutional route promptly; it should not wait for the next routine project review.

Give your own leadership work time and support. Protect thinking, writing and supervision in the calendar; tell the group how to reach someone when you are unavailable. Seek advice from experienced leaders, peers and administrative specialists. To improve a particular practice, examine

where it causes difficulty. For example, record which decisions wait for your approval for a fortnight, then delegate a suitable recurring class of decision and check whether the team can act within its scope [2].

Preparing the review

Choose a period that allows you to compare the last agreed priorities with what actually happened. An annual discussion can provide a broad view, with shorter reviews after a substantial change in staffing, funding or workload. Do not wait for that meeting to address an urgent concern. [7]; [8].

Prepare a brief account of what the group has learned and can now do. Include progress that has not yet led to a paper, such as establishing a method or completing an analysis. Recognize training and maintenance that allow others to work. Then identify commitments that have changed and obstacles that recur. If a paper is delayed, explain what is holding it up: an unresolved scientific question may require a different response from work that repeatedly loses its allotted time. [8].

Bring enough information to discuss resources honestly. Which work is already promised? Which skills, facilities and supervision does it require? Where will a departure, a funding end date or a new responsibility change what is feasible? Distinguish confirmed support from an application or an informal offer. Share the constraints needed for the decision without circulating individual salary, health or personnel information. Beltrao (2022) [6] offers one example of considering people, research and funding together; its particular costs and group size are not benchmarks for other laboratories.

Invite members to prepare their own observations before seeing the leader's conclusions. Ask which arrangements help them do good work, where work repeatedly gets stuck and what change would make a meaningful difference. A specific example gives the discussion something to examine. People should also have a way to raise concerns outside the group meeting, especially when they involve a supervisor or colleague on whom they depend. [8]; [7].

Agree who will read written responses, what will be retained and how concerns requiring another route will be handled. In a small group, a supposedly anonymous comment can identify its author through its details. Do not promise anonymity that the collection method cannot provide. The feedback and conflict chapter explains how to distinguish a discussion of shared practice from a concern that needs individual or institutional attention.

Discussing recurring problems

Begin with what is working well enough to preserve. A new procedure can unintentionally remove something valuable, such as informal technical help or time to explore a surprising observation. Then examine the concerns, allowing members to describe what they see before the leader offers an explanation. Differences between accounts are useful information: someone who receives quick help may not see how much work it creates for the person providing it. [7].

Follow a concern far enough to distinguish plausible explanations. If analyses are late, ask whether the data are ready, the method is established, the necessary expertise is available and the assigned researcher has the time the plan assumed. A problem shared by several projects may require changing how the group allocates work. A difficulty specific to one person's training may be better addressed in a mentoring conversation. Neither conclusion follows simply from the missed date. [8].

Include the effects of routine work on access and development. Who repeatedly trains newcomers, maintains equipment or resolves urgent requests? Who has opportunities to learn, present and take responsibility? Use the workplace review to examine these patterns in more detail. The purpose is to understand how the arrangements operate, not to ask individuals to justify private circumstances in front of colleagues.

Keep the discussion specific when views differ. Record the observation, the competing explanations and what would help distinguish them. A vote cannot determine why an assay is failing, and a majority's comfortable experience does not settle another member's report of poor treatment.

Address concerns about conduct through the appropriate route; do not use a group review as an improvised investigation.

Choosing changes

Compare a proposed change with the work it would displace. Training a backup, improving documentation or changing a meeting format takes effort now, even when it will save effort later. State where that time will come from. Asking everyone to fit it around existing commitments leaves the capacity problem unresolved. [8].

Members can help compare alternatives without being made responsible for securing a budget they do not control. Explain which changes the group can agree itself and which require a decision elsewhere. Do not promise additional funding or support until the person with that authority has agreed.

Choose a small enough change that the group can support it and tell whether it helped. The following example postpones an attractive collaboration because the existing work already depends on support that is spread too thinly.

Worked example: protect an existing project before adding another

This fictional microscopy group has a PI, two postdocs, two doctoral researchers and a technician. Two projects use the same imaging assay and analysis workflow. A collaborator has proposed a third pilot. One postdoc will leave in four months, and the technician handles most troubleshooting.

The PI initially thinks more frequent progress meetings would help a delayed analysis. Preparing for the group review changes the picture: sample processing is on schedule, but the researcher doing the analysis is repeatedly interrupted to help resolve assay problems. The technician also lacks a trained backup. Another meeting would reveal the delay without supplying the missing time.

Decision record — 2 October

Priority. Bring the existing analysis to its agreed checkpoint and establish a second person who can troubleshoot the assay before the postdoc leaves.

Change. Defer the collaborator's pilot. Protect two analysis sessions each week by routing routine assay questions through the technician during agreed support periods. Urgent problems that could lose samples still receive immediate attention; repeated urgent requests trigger a review of the arrangement.

Training and resources. One doctoral researcher will complete a supervised troubleshooting run with the technician. The PI will remove a planned exploratory experiment from that researcher's schedule and postpone a nonessential imaging batch assigned to the technician. Those changes provide time for both people. The PI will also ask the core facility whether it can provide continuing support.

Responsibility. The PI will explain the delay to the collaborator this week. The technician and doctoral researcher will agree the training run. The analyst will bring the completed analysis, or a specific remaining obstacle, to the next review.

Review — six weeks later. Check whether the analysis reached its checkpoint, the backup can complete the troubleshooting run and interruptions have become manageable. Reconsider the pilot only with an explicit estimate of its workload and available support.

At the six-week review, the analysis has reached its checkpoint and the trainee has completed the supervised run. Independent troubleshooting still needs to be demonstrated. The core

> facility cannot offer continuing support. The group retains the protected analysis time and keeps the pilot deferred while the PI and collaborator reduce its scope. Progress on the existing work has improved, but the extra capacity needed for a third project has not yet been established.

The collaborator could reasonably be disappointed: postponement may cost a scientific opportunity. Accepting the pilot would also have a cost, borne partly by the people whose current work would slow down. The decision therefore depends on the value and timing of both projects, not on a general rule against new work. Here, the unresolved support problem favors postponement. Check that the new arrangement has not quietly transferred every interruption to the technician.

Following up

Distribute a concise account of the agreed decisions and invite corrections. Keep personal disclosures and sensitive case details out of the shared record. If a change affects a standing practice, update the relevant lab-manual section. If it changes a collaborator's responsibilities or delivery date, revise the collaboration agreement.

At the next review, distinguish an action completed from a problem resolved. A training session may have happened without making the trainee ready to work independently. A revised schedule may meet a deadline while shifting the same workload into evenings. Ask the people affected whether the arrangement works, using an appropriate private conversation when a public answer would be difficult. Compare that account with the agreed evidence of improvement. [7]; [8].

When a change has not helped, revisit the explanation and the resources before repeating the instruction more firmly. Keep the reason for changing or retaining the arrangement with the decision, so a later review can build on what the group learned.

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Collaboration, authorship and credit

Discuss the shared question and what each collaborator hopes to gain from the work. Agree the contributions and resources needed to begin, then revisit responsibilities and credit as the project develops.

Start from the shared question

Describe the question and how combining your expertise or resources will help answer it. Ask what each participant contributes and hopes to learn. Decide whether the work calls for a consultation, a standard service or continuing scientific collaboration. A specialist invited only after data collection may be unable to repair the design. When their judgment can change the question, measurements or interpretation, involve them during design [6].

Check capacity as well as interest. Estimate each contribution and identify the time, staff, computing, samples or access that support it. Routine analysis, data curation and software maintenance still require skilled labor. If one group must absorb unfunded work, say so and decide whether the project remains mutually workable [6].

The starting conversation should leave room for different aims. One collaborator may want a biological finding; another may need a method that advances their own research. Agree whether developing that method is part of the project and whether the available time and resources support it. Otherwise, one person may expect a finished analysis while the other expects time to develop and test a new approach.

The collaboration agreement

Record the agreed scope, responsibilities and resources while everyone can check that they understood the conversation in the same way. The agreement need not anticipate every event. Team-science guides recommend clarifying goals, roles, resources, communication, data practices and credit early, then revisiting them as the project evolves [2]; [3].

Include:

- the shared question, current scope and evidence that would answer it;
- each person's work, dependencies, protected time and resources;
- decisions that one person can make and decisions that require the group;
- handoffs, quality checks and the records needed to inspect the result;
- meeting or update rhythm, including how a concern can be raised;
- data, code, sample and document locations, access and stewardship;
- likely outputs and a provisional credit process, including when it will be reviewed;
- conditions that trigger a change in scope, roles, timing or budget; and
- what happens to unfinished outputs, records and consultation if someone leaves or the collaboration ends.

Roles say who does work; decision rights say who can change it. A quantitative lead might choose the model implementation, while the group decides whether the analysis supports the biological claim. A facility scientist might set acquisition-quality requirements, while a project lead coordinates the schedule. Name both. Otherwise, a person can carry responsibility for a result without having authority over the choices that determine it.

The following fictional team revisits its plan as the work develops; its eventual credit decision appears later in the chapter.

Example: a collaboration plan

Question and scope: Does transient hypoxia change the recovery of an endothelial barrier over 24 hours? This project covers the exposure experiment, time-course imaging and a prespecified recovery analysis. A mechanism study would require a new decision.

Scientific value: Aisha will test a biological hypothesis and develop the exposure system. Leo will develop a time-course model that is useful beyond this dataset. Mei will establish an imaging protocol whose limits are reported with the result. Rao will help connect the design and interpretation.

Work and resources: Aisha leads cell experiments, biological quality control and the integrated record. Leo joins before data collection to choose sampling times and replicate structure, then leads analysis, code and quantitative figures. Mei leads acquisition settings, segmentation checks and image interpretation. Rao has budget authority and contributes to design and interpretation. Leo has two protected days per week and named computing support.

Decisions and handoffs: Aisha can reject a culture batch against the agreed quality rules. Leo can reject an analysis run that fails the diagnostic checks. Mei can require reacquisition when image quality falls outside the protocol. Changing the exposure, sampling schedule or main claim requires Aisha, Leo and Mei to review the evidence together; Rao resolves a budget constraint after hearing its scientific consequences.

Records and communication: Data, code and decisions are stored in the shared project space. The team meets every two weeks while data are being collected and records actions and owners. A failed handoff is raised at the next meeting or sooner if it blocks another person's work.

Outputs and credit review: Aisha is the likely first author and Leo the likely second author at project start. Mei's position remains open because the imaging-method and manuscript contributions are not yet known. These are expectations, not promises. The group checks the anticipated target journal's policy now, updates contributions after the pilot and complete dataset, and verifies the current authorship and equal-contribution policies before manuscript drafting.

Change and exit: A new mechanistic aim, more than one additional experiment cycle or loss of Leo's funded time triggers a scope review. If a collaborator leaves, the team records which outputs remain planned, who stewards the shared materials and how the departing contributor will review work arising from their contribution, subject to institutional rules.

Coordinate across expertise

Ask collaborators which choices make their part of the work trustworthy and which limitations affect the shared claim. Before handing data over for analysis, discuss how measurements were acquired, which observations were excluded or missing, and what transformations and model assumptions the analysis requires. The analyst needs to understand how the measurements arose, and the experimentalist needs to understand what the analysis can distinguish.

Before a meeting, circulate the protocol change, diagnostic plot, discrepant result or draft claim that needs discussion. Record what the group decides, why, who will act and what they will check next. Invite disagreement while the group can still change its interpretation or planned work. If a collaborator cannot raise concerns about the design or interpretation, their expertise has little opportunity to shape the result [2].

Review the plan when the evidence changes the work. A new analysis may become a central intellectual contribution; a planned method may become routine; a contributor may leave or join. Revise responsibilities, resources and credit expectations together.

Contributions, authorship and order

First describe each person's contribution. Then apply the relevant authorship criteria, agree author order and decide how to acknowledge other work. A contribution record informs these decisions but does not automatically determine them.

Describe contributions. Record what each person actually did. CRediT offers 14 roles, including conceptualization, investigation, methodology, data curation, software, formal analysis, visualization, supervision and two writing roles. Use the vocabulary when it fits the work; do not turn the roles into points or assume that every role has equal weight [5].

Decide authorship eligibility and responsibility. Check the target journal and relevant disciplinary or institutional guidance. CRediT is descriptive and is not designed to determine authorship. For journals that follow ICMJE, authors must meet four requirements. In summary, each author must:

1. Have contributed substantially to developing the study's ideas, planning it, or obtaining, analyzing or interpreting its data.
2. Have helped write the manuscript or critically assessed its intellectual content.
3. Have approved the version being published.
4. Accept responsibility for helping resolve concerns about the work, including parts carried out by others. [4].

If someone questions the work's reliability, authors must help ensure that the concern is investigated and any problem addressed. They should know who carried out the other parts of the study and have confidence in those contributions. This does not mean personally repeating every specialist check. Consult the full ICMJE criteria when deciding eligibility, and confirm that every author has reviewed and approved the version being submitted. [4].

Corresponding authorship involves managing communication with the journal and ensuring that submission requirements are met; it does not by itself establish a larger scientific contribution. [4].

Decide order collectively. Name the convention being used and the rationale. Order conventions differ across fields and journals. Shared-first or shared-senior designations also depend on journal policy. A contribution statement can clarify the work, but displayed order may still affect evaluation [6].

Recognize other work. Name non-author contributions as the journal permits, specify the contribution and obtain any required consent. Recognition can also include named presentations, software or data citations and a contribution record for evaluation when those outputs reflect the work.

For an ICMJE-following journal, do not exclude a person who made the substantial contribution in the first criterion by withholding the chance to review or draft the manuscript and approve it. Offer a real opportunity with access and time to participate. Authorship follows only if the person then meets all four criteria [4].

Example: a completed contribution and credit decision

At the final dataset review, the hypoxia project no longer matches its opening expectations. Leo's diagnostics reveal a batch effect; he redesigns the model and works with Aisha on an additional balanced collection. The change alters the claim. Leo also drafts the analysis and findings and creates the main quantitative figures. Aisha continues to lead the biological work

and writes the first complete manuscript. Their central intellectual and practical leadership is now comparable in scope and duration, although the work differs.

Mei's acquisition design and interpretation are also substantial. Because the selected journal follows ICMJE recommendations, the team gives her the opportunity to complete its manuscript responsibilities. She revises the imaging methods and interpretation, critically reviews the complete paper, approves the submitted version and accepts the journal's accountability requirement. Rao's documented conception, interpretation and substantive manuscript review qualify him under the selected policy; funding and supervision alone would not.

Target rule: The journal permits a shared-first statement and follows ICMJE authorship recommendations. All four authors have critically reviewed or drafted the paper, approved this version and accepted the stated accountability obligation.

Contribution record: Aisha—conceptualization, investigation, validation and writing the original draft. Leo—methodology, software, formal analysis, visualization and writing the analysis and findings. Mei—methodology, investigation, validation and critical review and editing. Rao—conceptualization, supervision, interpretation and critical review and editing.

Byline decision: Aisha and Leo are shared first authors. Aisha is displayed first because the group agrees that her integrated study leadership and complete first draft justify that sequence. Leo's comparable quantitative leadership is stated in the equal-contribution note and contribution statement. Mei follows them. Rao is last under the group's field convention and his documented contribution.

Trade-off: The shared-first note represents the team's judgment, but displayed order may still influence evaluation.

Other credit: The technician performed routine stock maintenance and trained Aisha in an established assay, without contributing to the study's conception, design, data acquisition, analysis or interpretation. The team records the actual work and acknowledges it by name with permission. Aisha and Leo are assigned the biological and quantitative portions of two planned presentations, and each contributor receives this record for evaluation.

If the journal did not allow a shared-first designation, the contribution and eligibility decisions would remain the same. The group would need to choose one displayed order, document its rationale and record the unchosen alternative and trade-off. CRediT roles would still describe the work; they would not calculate the answer.

Address a credit disagreement with evidence

Start with the disputed decision. "My analysis is missing from the contribution statement," "I meet the journal's authorship criteria," and "I should be listed first" are different claims. Reconstruct the agreed plan and actual work, including changes in scope, drafts, decisions and approvals. Then apply the current target policy and ask what proposed outcome follows from that evidence.

Discuss the record with everyone whose credit would change and write down the revised decision. Early discussion and written updates reduce surprises, but the first agreement is not conclusive when contributions change [1]. Avoid bargaining with unrelated resources or future favors. When a power difference makes direct discussion difficult, use the neutral route named in the agreement or seek confidential procedural advice from the institution.

Journal editors usually lack the evidence and authority to adjudicate who qualifies. Under current ICMJE recommendations, unresolved authorship-eligibility questions should be investigated by the institution or institutions where the work was performed, while author order is decided collectively by the author group [4]. Other journals and disciplines may use different procedures. Broader

relationship conflict, retaliation concerns or formal allegations need their appropriate local route and more support than an authorship worksheet can provide; see feedback and conflict.

End or transform the collaboration

A project can end while a collaboration continues, and a person can leave while papers, software or datasets remain active. Name which event is happening. Review the outputs still expected, stewardship of shared records and materials, permitted access, remaining consultation, credit review and the date when obligations end. Follow the applicable institutional agreements for data, confidentiality, intellectual property and employment; a collaboration plan does not decide those rights.

State what was learned, what remains unresolved and who can answer questions about each part. If the team begins a new question, write a new scope and contribution record [2].

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Sustainable and inclusive research workplaces

A research group can offer flexible hours and still make them difficult to use. Someone who leaves at the agreed time may miss decisions made over evening drinks; someone who works remotely may spend their day waiting for access to a shared computer. A colleague who reliably trains newcomers may have less time for the research on which their progress is judged.

Start with a recurring difficulty and discuss what would help with the people affected. If someone has too much assigned work, the person assigning it needs to help decide what can wait. Advice about personal planning will have little effect while the same commitments remain [3].

Workload and availability

Before adding a deadline or responsibility, look at the work already committed. Include preparation, documentation, analysis, meetings, teaching, training and shared duties as well as experiments. Ask which tasks have a fixed window and which can move. An afternoon sampling endpoint may be biologically constrained; a meeting scheduled after it may simply be convenient for one person.

Compare the plan with the time, equipment and qualified help actually available. When it does not fit, choose what changes: reduce the scope, move a deadline, stop a lower-priority task or arrange suitable cover. Members can identify which commitments compete for their time, but the person who assigns the work must help decide what gives way. Review planned and actual work after a representative period, including unexpected duties, rather than treating a missed target as evidence of poor motivation [3].

Separate tasks that require presence from those that allow choice of time or location. An operator must attend an acquisition they have agreed to run; an analyst may be able to work elsewhere. Agree when people need to overlap, how handovers happen and when a response can be expected. Flexibility still needs supervision, access to information and a way to resolve a blocked task [5].

Make communication match that agreement. If a leader sends messages at unusual hours, state when a reply is expected and use scheduled delivery where useful. Establish a separate route for genuinely time-sensitive operational problems, with a named person responsible during the relevant window. Routine questions should not turn every group member into permanent backup. Someone's preference for late work does not set everyone else's availability.

Occasional field campaigns, sampling windows or submission deadlines may need a temporary arrangement. Before committing, agree what it must accomplish, when it ends, who can provide qualified cover and what other work will be reduced or deferred. Check applicable working-time and safety arrangements with the responsible people. Plan the return to ordinary work as part of the decision. If the same demand keeps recurring, change the project, staffing or deadline to account for it in the regular workload [6]; [5].

Access and participation

Look at what someone must be able to do to contribute. Can they obtain the necessary equipment without paying personally? Reach the meeting space, use the online platform and read the material? Attend when decisions are made? Understand enough of the discussion to ask a question? Ask about practical needs privately and offer a route for raising them with the appropriate institutional service. Do not expect one member to represent an entire identity group or teach colleagues about every barrier [4].

Give people more than one way to contribute to important decisions. Circulate the question and essential material in advance, allow written input, and record the decision where everyone involved can find it. During discussion, make room for questions and contributions from people who have not

spoken. A quiet person may have useful evidence; fluent or rapid speech is not the only sign of understanding. Low-stakes opportunities to ask about a method or work through a coding problem can help people learn without turning every uncertainty into a performance assessment [4].

Keep optional social gatherings optional in practice. Offer different ways to meet colleagues, including some during ordinary working hours. If a conversation over dinner produces a research decision, bring that decision back to the relevant work discussion before relying on it. People who cannot attend should retain access to opportunities and information. Ask whether the arrangements work for the people affected; attendance alone cannot tell you whether they feel included.

Make the work of maintaining the group visible. Training, organizing visitors, taking notes and arranging celebrations all take time. Check who repeatedly does them, how much capacity they consume and whether that person is also getting development opportunities. Distribute duties with attention to role, competence and workload; identical rotations may be unsuitable for specialist tasks. Acknowledge contributions and include them when discussing workload and progress. Raffington describes tracking and discussing this often overlooked work so that it can be shared more fairly [6].

Document the shared arrangement in the lab manual. Keep personal circumstances and individual support agreements in the appropriate private record. A written expectation is useful only when the group follows it and the responsible person can act on it.

Assessing contributions

Explain what good work looks like for the person's role and stage. A developing researcher may need to show sound experimental decisions and explain uncertainty. A technician may contribute reliable operation, troubleshooting and training. An analyst may deliver a checked, documented analysis that others can use. Discuss the work itself, the evidence that will demonstrate progress and the support needed to produce it [6]; [5].

Use concrete evidence rather than impressions of commitment. Long presence in the lab, quick replies and confident language do not establish that a result is trustworthy. A journal's average citation count also cannot establish the quality of a particular paper or its author's contribution; the underlying papers vary substantially [1]. Read the work and identify what the person actually did.

Distinguish what the researcher can influence from what remains uncertain. They can choose appropriate controls, keep records, check an analysis and respond to feedback; they cannot promise a desired biological result or journal acceptance. A result that challenges the hypothesis differs from a failed procedure or a measurement that cannot answer the question. Assess what the work permits the researcher to conclude and how they followed it up. When progress differs from the plan, also examine the support: was required training available, did a shared duty expand, or was an instrument unavailable? Agree the next action and record why the target or support changed. Recognizing careful work does not require continuing a project when no feasible next experiment can answer its question.

Use the mentoring agreement for individual development and the collaboration plan for project contributions and credit. Shared support work deserves recognition, while authorship still requires an appropriate contribution to the particular publication.

Practical support

If someone says an arrangement is becoming difficult, offer a private conversation about what would help. They might need a changed task, a clearer priority, time away, equipment access or advice from another person. Listen before choosing the remedy. An ordinary conversation about workload need not begin with a diagnosis or an account of family circumstances. Formal support or accommodation processes may have their own information requirements; help the person find the responsible service rather than collecting that information in a group meeting [2].

A supervisor can change work within their authority and help someone reach suitable support. That role does not make them a therapist or medical assessor. Keep current staff and student support routes easy to find, including an option outside the direct supervisor. Check who a service covers, how to contact it and what its privacy terms actually say. Do not promise confidentiality or an institutional outcome on behalf of another office.

Peer groups, personal interests and optional wellbeing activities may be useful resources. They cannot settle an impossible deadline, inaccessible workplace or unresolved mistreatment. Ayres's discussion of university programmes pairs support for individuals with changes to the conditions creating difficulty [2]. If a problem lies beyond the group leader's authority, identify the institutional decision needed and who will pursue it. Avoid leaving the person who raised the problem responsible for obtaining every remedy. For concerns about treatment or retaliation, use the support and power considerations before choosing a conversation or disclosure route.

Planning a change

Choose a change the group can carry out with the people and resources available. State what will continue, what will change and what work will be displaced. Give each action an owner who has the authority and time to carry it out. Agree how you will check whether it addressed the problem, when to review it and what would require an earlier decision. Discuss individual needs privately; record the availability and work arrangements needed for coordination in the shared plan.

Example: a four-week sampling and analysis plan

This fictional group has six afternoon sampling endpoints over four weeks. Two people are trained and authorized to operate the assay. One cannot stay into the evening; the other already maintains stocks and trains a new member. The analyst lacks a suitable workstation. Some research decisions have been made at optional evening gatherings.

The group considers training a third operator immediately, which would delay this assay, or keeping the two qualified operators while postponing a new pilot and reducing their other duties. The established assay can finish within the operators' agreed hours if preparation begins earlier. The group chooses the second option for this run and writes the following plan.

Purpose and period. Complete the six established sampling sessions and a documented initial analysis over the next four weeks. This plan applies to those sessions; it does not create an expectation of evening availability.

Sampling and other duties. Before the first session, the two operators and group leader confirm the six dates, preparation times and handovers. The leader postpones the new pilot until after this run. A third colleague, already trained in stock care, covers that duty for the four weeks; the leader reduces that colleague's other assigned work accordingly. New-member assay training resumes after the run. During the pause, the leader agrees a suitable starter task and a named supervisor with available time; the new member is not assigned unsupervised assay work. The sampling operators retain responsibility for the assay records.

Access and decisions. The analyst and computing support agree a suitable shared-workstation booking before the first dataset is ready. If access cannot be provided, the leader revises the analysis date rather than expecting the analyst to buy equipment. Project decisions take place at the scheduled daytime meeting, with material circulated beforehand and a written-input route for anyone who cannot attend. The meeting chair records decisions. Evening social conversations may suggest ideas, but do not commit absent colleagues to work.

Progress and support. The leader and each member agree their relevant contribution: complete assay records, checked analysis with unresolved issues identified, stock-care

coverage or defined training progress after the run. Each assay run is reviewed against the protocol, controls and observations. A result that does not support the hypothesis is not itself an assay failure. Individual support arrangements are discussed separately; the shared plan records only the availability needed for coordination.

Review and exceptions. At the end of week two, the leader checks with the members whether the workload and access arrangements are working. An absent qualified operator, repeated overrun or missing workstation access triggers an earlier decision. The leader reschedules a session or changes scope if safe, authorized cover is unavailable. Nobody is treated as automatic evening backup. At the end of week four, the group confirms which duties return and whether the pilot can start; it does not assume that deferred work now fits.

Check whether cover has merely moved an overload to another colleague. Another group may need to delay the experiment, arrange different resources or abandon a lower-priority aim. When similar difficulties recur across projects, include them in a review of how the group works.

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Feedback and conflict

A disagreement about a figure needs a different response from a dispute about working conditions. Before arranging a conversation, identify what needs to change and who can help. If speaking directly could expose someone to retaliation, seek independent advice first.

Identify the issue and the right route

Separate what happened from your interpretation of why it happened. “The sample sheet arrived on Wednesday after the Tuesday deadline” identifies an event. “They do not care about the project” supplies a motive you have not established. Ask what expectation existed, who knew it, what resources were available and who had authority to change the plan. Conflict can come from overlapping responsibilities, incompatible demands or unclear decisions as well as interactions between people [2].

The following questions help choose a starting point. More than one may apply.

What needs to be resolved?	Useful starting point
A claim, inference or proposed research approach	Identify the evidence and assumptions in dispute; agree how to assess them.
Work that differs from a communicated expectation	Compare the agreed task with the observed work, hear the explanation and decide what support or change is needed.
Competing demands, responsibilities or access to resources	Identify each person’s needs, the constraints and who can authorize a workable arrangement.
Threats, coercion, humiliation, discrimination, harassment or retaliation	Seek appropriate support and assess risk; direct confrontation is not a prerequisite.

An unsuccessful experiment is not by itself evidence of poor performance. Discuss the quality of the reasoning, conduct, records and follow-up rather than demanding a preferred result. Conversely, a productive researcher still needs to meet standards of conduct. A promising dataset does not resolve a concern about how a colleague is treated.

For ordinary misunderstandings, allow for incomplete information without deciding that either person’s account is wrong. An explanation of intent may change your understanding, but it does not remove the impact or the need for a different arrangement. Repeated harmful conduct and immediate safety or integrity concerns require a route beyond this initial conversation [7].

Giving feedback

Prepare a small number of relevant examples. Name the work or behavior you observed, its consequence and the expectation or requested change. Include specific positive feedback when it helps someone recognize what to continue. “Your summary separated the observed result from the proposed explanation, so we could decide which control was missing” is more useful than “Good job.” Avoid making a personality judgment stand in for an explanation [2].

Discuss an individual’s performance privately, at a time when both people can give the issue attention. Explain the concern directly, then invite their account. Check whether they had the training, time, access and guidance the task required. If you are the supervisor, examine your own commitments as part of that comparison. Missing a deadline does not establish lack of effort when the person was assigned incompatible priorities or could not obtain necessary access [3]; [7].

Agree a feasible next attempt, the support each person will provide and when you will review it. Check that both people understand the same task; invite corrections to the written summary. For recurring problems, review whether the agreed support was delivered and whether the expectation remains realistic. Formal performance management, disciplinary action or a change of role needs the institution's current process for that person's status. An informal agreement does not replace it [7].

An original example: a missed submission deadline

In this fictional laboratory, a research assistant prepares sample sheets for weekly sequencing submissions. Two sheets due Tuesday at noon arrived Wednesday morning; one lacked the required batch field. The analyst could not begin the agreed check on Tuesday. The supervisor schedules a private discussion with the assistant.

Supervisor: The last two sheets arrived Wednesday rather than Tuesday noon, and one was missing the batch field. That delayed the analyst's check. I want to understand what happened and agree how we will handle the next submission.

Assistant: I thought the new template was optional because the old one still opens from the shared folder. I have not been shown how the batch field should be filled. I was also preparing for the imaging work you assigned for Wednesday through Friday, and I did not know which task came first.

Supervisor: I assigned both without clarifying the priority, and I did not arrange training on the new template. I need to fix that. For next week, the sample sheet is the Tuesday priority; the imaging preparation can follow. What would you need to complete it by noon?

Assistant: A short walkthrough with the analyst and a completed sheet to check against. If I cannot resolve a missing field, I can flag it before the deadline rather than hold the whole sheet back.

Supervisor: I will arrange the walkthrough for Monday and confirm that the current template is the one linked from the submission instructions. Please prepare the next sheet by Tuesday noon and tell the analyst and me if a field or another assignment blocks you. Let us review the next two submissions together.

Assistant: That works. I will send the sheet Tuesday noon, flag anything unresolved and check the fields with the analyst. Could we put the priority change in the summary so the imaging team knows?

Supervisor: Yes. I will confirm the change with them today.

Written summary agreed after the meeting

For the next two submissions, the assistant will prepare the sample sheet by Tuesday noon using the current template and flag unresolved fields or competing assignments before the deadline. The supervisor will arrange Monday's walkthrough, correct the template link and confirm the revised priority with the imaging team. The analyst will check the required fields. The assistant and supervisor will review both submissions in two weeks.

At the review

Both sheets arrived by the agreed time with the required fields completed. An equipment delay affecting one sample was flagged in advance, allowing the analyst and supervisor to decide how to handle it. They retain the template and priority arrangement, with a further discussion if the workload changes.

If the sheets remain late, check whether the training occurred and whether competing assignments still prevent timely preparation. Review the agreed work and communication; the outcome of the experiment is a separate question.

Scientific disagreement

When receiving criticism, ask which result, method or inference the person is questioning and what evidence supports their concern. Take time to examine it before answering if needed. Feeling upset does not establish that the criticism is wrong, and a criticism's scientific value does not excuse personal humiliation. Lakens's account of reconsidering his own published work illustrates that distinction; it is a personal reflection, not a prescription to make criticism hurt [6].

State the competing interpretations and the assumptions behind them. Ask what observation would distinguish them. Sometimes the missing step is an analysis or a control; sometimes the available evidence cannot decide. Agree what can be concluded now and whether further work is justified by the importance of the question and the resources it would require. Establish who can authorize that work. Senior expertise deserves consideration, but seniority cannot settle an empirical claim [1].

> **An original example: a challenge during lab meeting**
>
> **Researcher:** The figure shows a lower signal after treatment, so I think the treatment reduces the process we are studying.
>
> **Colleague:** Could lower cell numbers explain that signal? I cannot see a cell-number control in this figure.
>
> **Researcher:** This figure alone does not distinguish those explanations. Let us check whether we have an appropriate control before interpreting the decrease. If we do not, we need to decide whether to obtain one or limit the claim.

Group members should be able to question an interpretation during a scientific discussion without first obtaining the project leader's private approval. The person chairing a meeting can keep criticism focused on the claim, let the presenter finish an answer and acknowledge a useful challenge, including one that changes their own view.

As a senior researcher receiving feedback about your decisions or behavior, listen before rebutting. Ask for clarification, reflect on the examples and return with the action you will take or the reason a proposed change is not feasible. An invitation to disagree carries little weight if disagreement is followed by ridicule, exclusion or loss of opportunity [2].

Resolve competing needs and unclear responsibilities

For an ordinary work conflict, ask each person what they need and what is preventing it. Two colleagues disputing a microscope booking may both need access, but one needs a particular time because a live experiment has a fixed window while the other's samples can wait. Learning that difference creates options that a repeated demand for the same slot cannot reveal. Check the booking rules, actual capacity and alternatives before deciding.

Explore feasible options and judge them against agreed criteria: experimental timing, safety, available staffing, costs and existing commitments. Establish who has authority to resolve the remaining trade-off. A temporary compromise may be adequate for a low-cost scheduling choice; a recurring resource shortage may need a different allocation or additional capacity. Joint problem-solving is useful when people can participate safely and the decision is negotiable. It is not a reason to compromise a safety or integrity requirement [2].

End with a record of the arrangement, owners, unresolved points and review date. If the same problem returns, inspect the system as well as the last conversation. Did two supervisors promise the same person's time? Was project authority ambiguous? Did the arrangement depend on a resource that was never supplied? Use the group's shared expectations to clarify recurring practices and the collaboration agreement for project responsibilities and credit.

Seeking independent support

Direct discussion is not always a safe or effective first step. Threats about graduation, employment, references, funding or residence-related arrangements can make an apparently simple conversation consequential. Repeated humiliation, coercion, harassment, discrimination or retaliation warrants support beyond an ordinary feedback exchange. You do not need to establish a legal label or confront the person before asking for advice [1].

Identify an adviser whose role fits the concern and whose position is sufficiently independent of the person involved. Possible routes include a programme or committee contact, another manager, an ombud service, an employee or student representative, an equality or harassment adviser, a research-integrity office or a safety contact. Availability and authority vary. A trusted mentor can help you think through a decision, but may not be able to initiate protective or formal action [4].

Before sharing identifying details, ask what the adviser can do, what they must report, who may receive information and what confidentiality means in that service. An informal conversation is not automatically off the record. An anonymous channel may also reveal a person's identity through the details of a small group. Clear routes help, but a lab policy or feedback channel cannot itself remove the barriers to challenging a powerful person [1]; [5].

> **An original example: first contact for advice**
>
> I would like advice about a supervision concern. Before I share names or details, could you explain your role, what you would have to report and whether information could reach my supervisor? I would also like to understand what support or formal routes are available, because my completion timetable and reference depend on this relationship.

Keep a factual chronology where it is appropriate and safe: dates, observed words or actions, decisions, relevant messages, witnesses, effects on work and previous requests for help. Distinguish your account from facts you learned from others. Preserve records through permitted channels; ask how to retain relevant evidence without taking restricted research data or exposing someone else's personal information. A written summary sent to the other person may clarify an ordinary work agreement, as in the example above. If contacting them may trigger retaliation, seek advice about the record and communication first [1].

If a colleague approaches you for help, listen without demanding that they confront the person or immediately join a meeting with them. Explain the limits of your role and any information-sharing obligations; do not promise confidentiality you cannot provide. Ask what support they are seeking and help them identify a suitable route. A supervisor implicated in the concern cannot serve as its independent facilitator. If there is an immediate danger, prioritize safety and the relevant emergency or institutional response rather than completing a conversation plan.

Discuss with the adviser how each option could affect your work and position. Changing supervisors or leaving the group may affect funding, degree progress or access to records. Papers and references can leave you dependent on the former supervisor after a move. Where relevant, ask about consequences for affiliation or residence arrangements. Consider supported discussion, independent facilitation and formal reporting alongside changes of supervision, assignment or group. Seek current advice before relying on an assumed outcome.

Check the next step

- The concern names an observed event, claim or pattern, with uncertainty kept explicit.
- The proposed route fits the scientific, performance, resource or treatment issue.
- A work agreement includes support, owners, a feasible task and a review point.
- A person with less power can seek advice without first confronting the other party.
- Information-sharing limits and the adviser's authority are understood before detailed disclosure.
- Follow-up checks what actually changed, including whether promised support was delivered.

References

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4. Chopra, V., Edelson, D. P., & Saint, S. (2016). **Mentorship malpractice.** *JAMA*, 315(14), 1453–1454.
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